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- (54) **CELL DEATH-INDUCING DFFA-LIKE EFFECTOR B (CIDEB) IRNA COMPOSITIONS AND METHODS OF USE THEREOF**
- (71) Applicant: **ALNYLAM PHARMACEUTICALS, INC.**, Cambridge, MA (US)
- (72) Inventors: **Jeffrey Zuber**, Somerville, MA (US); **James D. McNinch**, Burlington, MA (US); **Mark K. Schlegel**, Lexington, MA (US); **Adam Castoreno**, Framingham, MA (US); **Lucas Bondurant**, Brookline, MA (US); **Dhaval Oza**, Boston, MA (US)

- (73) Assignee: **Alnylam Pharmaceuticals, Inc.**, Cambridge, MA (US)

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- (58) **Field of Classification Search**
None
See application file for complete search history.

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Primary Examiner — Nancy J Leith

Assistant Examiner — Ruth Sophia Arieti

(74) *Attorney, Agent, or Firm* — McCarter & English, LLP; Maria Laccotripe Zacharakis; Deborah L. Nagle

(57) **ABSTRACT**

The invention relates to double-stranded ribonucleic acid (dsRNA) compositions targeting the CIDEB gene, as well as methods of inhibiting expression of CIDEB, and methods of treating subjects that would benefit from reduction in expression of CIDEB, such as subjects having a CIDEB-associated disease, disorder, or condition, using such dsRNA compositions.

23 Claims, 2 Drawing Sheets

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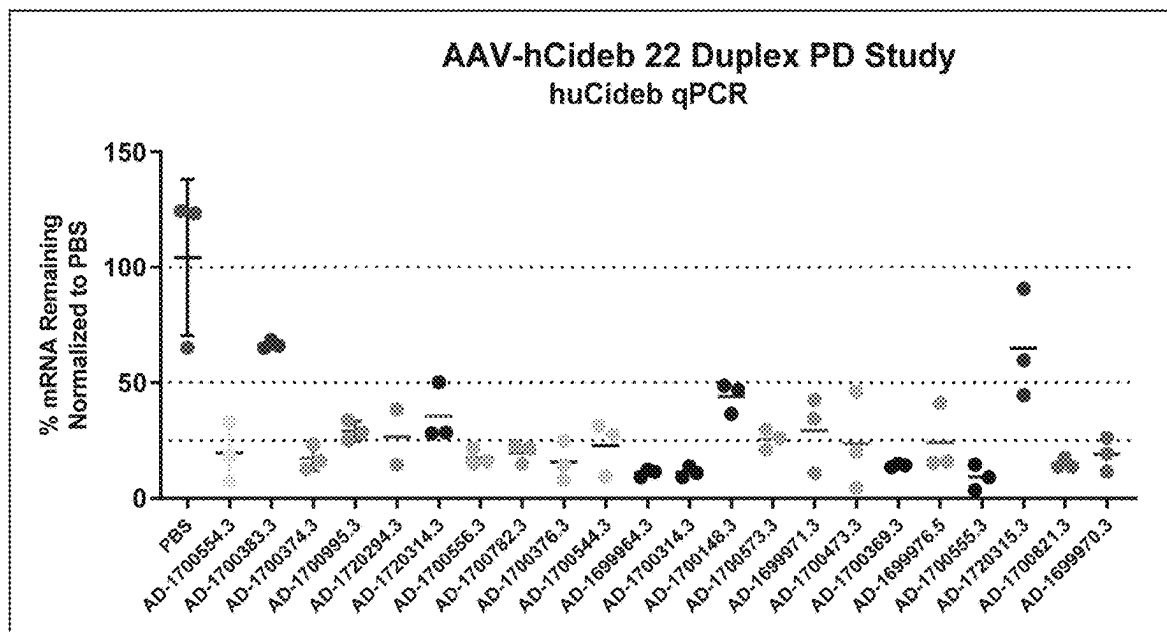


FIG. 1

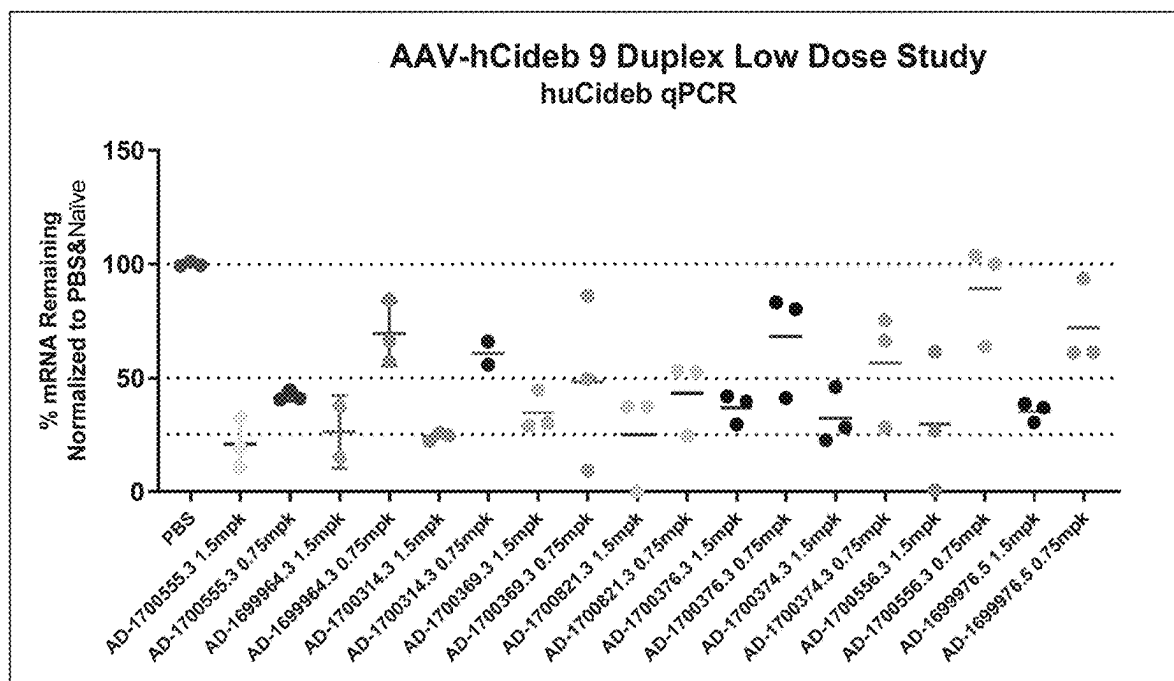


FIG. 2

**CELL DEATH-INDUCING DFFA-LIKE
EFFECTOR B (CIDEB) iRNA
COMPOSITIONS AND METHODS OF USE
THEREOF**

CROSS-REFERENCE TO RELATED
APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/687,251 filed Feb. 27, 2024, which is a national phase application based on PCT/US2022/075715 filed Aug. 21, 2022, which claims the benefit of priority to U.S. Provisional Application No. 63/239,271, filed on Aug. 31, 2021, and claims the benefit of priority to U.S. Provisional Application No. 63/341,848, filed on May 13, 2022. The entire contents of the foregoing applications are hereby incorporated herein by reference.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. The XML copy, created on Jun. 14, 2023, is named A108868_1300WO_SL.xml and is 18,162,483 bytes in size.

BACKGROUND OF THE INVENTION

Cell death-inducing DFFA-like effector B (CIDEB), a member of the CIDE protein family, is primarily expressed in liver and small intestine. CIDEB is an endoplasmic reticulum (ER)- and lipid droplet (LD)-associated protein. Overexpression of CIDEB protein induces cell death, but the physiological function of CIDEB is more closely related to various lipid metabolic pathways, particularly the VLDL pathway.

Non-alcoholic fatty liver disease (NAFLD) is the commonest form of liver disease in all regions of the world with modern industrialized economies, including Korea and many other Asian countries. Patients usually present without symptoms or clinical features are non-specific. Instead, liver abnormalities are found incidentally by hepatic imaging, particularly ultrasonography, and/or there are raised liver enzymes (alanine aminotransferase [ALT] and gamma-glutamyltranspeptidase). The diagnosis of NAFLD requires exclusion of other disorders, particularly viral hepatitis, significant alcohol intake, and exposure to potentially hepatotoxic medications. By agreements such as the Asia-Pacific Guidelines on NAFLD, the term NAFLD is now retained for cases of fatty liver associated with metabolic complications of over-nutrition, usually with central obesity and overweight.

Non-alcoholic steatohepatitis (NASH) is considered the progressive form of nonalcoholic fatty liver disease (NAFLD) and is characterized by liver steatosis, inflammation, hepatocellular injury and different degrees of fibrosis. Adipose tissue dysfunction and the hepatic inflammatory response have a fundamental role during NASH development. Cellular and molecular response mechanisms also promote liver inflammation in the absence of a fatty liver by inducing a chronic inflammatory response that results in hepatocyte damage.

Accordingly, there is a need for improved methods of treating chronic inflammatory diseases of the liver, such as NASH, including agents that can selectively and efficiently inhibit the CIDEB gene.

BRIEF SUMMARY OF THE INVENTION

There is a need for improved methods of treating chronic inflammatory diseases of the liver, such as NASH, including agents that can selectively and efficiently inhibit the CIDEB gene. Current standards of care for subjects with chronic inflammatory diseases include lifestyle modifications (diet and exercise, cessation of smoking, drinking, etc.), steroidal and/or nonsteroidal anti-inflammatory medications, and management of associated comorbidities, e.g., hypertension, hyperlipidemia, diabetes, etc. Once established, chronic inflammatory conditions can maintain a self-perpetuating cycle of inflammation, tissue damage, release of proinflammatory damage-associated molecular patterns (DAMPs) from injured cells, and cytokine release leading to further inflammation. Abrogation of liver inflammation could be achieved by exploiting active, physiological pro-resolving mechanisms instead of the classical passive blockade of pro-inflammatory mediators. (Schuster et al., *Nature Reviews Gastroenterology & Hepatology*, volume 15, pages 349-364, 2018).

The present invention provides iRNA compositions which effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of a cell death-inducing DFFA-like effector b (CIDEB) gene. The CIDEB gene may be within a cell, e.g., a cell within a subject, such as a human. The present invention also provides methods of using the iRNA compositions of the invention for inhibiting the expression of a CIDEB gene and/or for treating a subject who would benefit from inhibiting or reducing the expression of a CIDEB gene, e.g., a subject suffering or prone to suffering from a CIDEB-associated disease, for example, a chronic inflammatory disease.

Accordingly, in one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEB) in a cell. The dsRNA agent includes a sense strand and an antisense strand, wherein the sense strand comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO: 1 and the antisense strand comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO: 2. In some embodiments, the dsRNA agent includes a sense strand and an antisense strand, wherein the sense strand comprises at least 15 contiguous nucleotides from the nucleotide sequence of SEQ ID NO: 1 and the antisense strand comprises at least 15 contiguous nucleotides from the nucleotide sequence of SEQ ID NO: 2.

In another aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEB) in a cell. The dsRNA agent includes a sense strand and an antisense strand forming a double stranded region, wherein said antisense strand comprises a region of complementarity to an mRNA encoding CIDEB which comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from any one of the antisense sequences listed in Tables 3-6. In some embodiments, the dsRNA agent includes a sense strand and an antisense strand forming a double stranded region, wherein said antisense strand comprises a region of complementarity to an mRNA encoding CIDEB which comprises at least 15 contiguous nucleotides from any one of the antisense sequences listed in Tables 3-6.

In one embodiment, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b

(CIDEB), wherein the dsRNA agent comprises (a) an anti-sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the antisense strand nucleotide sequence of duplex AD-1700555 and a sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the sense strand nucleotide sequence of duplex AD-1700555; (b) an anti-sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the antisense strand nucleotide sequence of duplex AD-1700821 and a sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the sense strand nucleotide sequence of duplex AD-1700821; (c) an antisense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the antisense strand nucleotide sequence of duplex AD-1700369 and a sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the sense strand nucleotide sequence of duplex AD-1700369; (d) an anti-sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the antisense strand nucleotide sequence of duplex AD-1699976 and a sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the sense strand nucleotide sequence of duplex AD-1699976; (e) an antisense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the antisense strand nucleotide sequence of duplex AD-1700374 and a sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the sense strand nucleotide sequence of duplex AD-1700374; (f) an antisense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the antisense strand nucleotide sequence of duplex AD-1700314 and a sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the sense strand nucleotide sequence of duplex AD-1700314; (g) an anti-sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the antisense strand nucleotide sequence of duplex AD-1700376 and a sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the sense strand nucleotide sequence of duplex AD-1700376; (h) an anti-sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the antisense strand nucleotide sequence of duplex AD-1699964 and a sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the sense strand nucleotide sequence of duplex AD-1699964; or (i) an anti-sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the antisense strand nucleotide sequence of duplex AD-1700556 and a sense strand comprising a nucleotide sequence comprising at least 15 contiguous nucleotides, with 0, 1, 2, or 3 mismatches, from the sense strand nucleotide sequence of duplex AD-1700556.

In another embodiment, the region of complementarity comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from any one of nucleotides

29-51, 67-89, 154-176, 163-185, 173-195, 184-206, 196-218, 206-228, 257-279, 270-292, 446-468, 459-481, 468-490, 518-540, 530-552, 641-663, 687-709, 702-724, 711-733, 727-749, 758-780, 769-791, 781-803, 790-812, 807-829, 839-861, 850-872, 874-896, 907-929, 917-939, 958-980, 974-996, 983-1005, 999-1021, 1009-1031, 1018-1040, 1027-1049, 1036-1058, 1045-1067, 1054-1076, 1080-1102, 1089-1111, 1098-1120, 1108-1130, 1140-1162, 1156-1178, 1174-1196, 1183-1205, 1192-1214, 1205-1227, 1214-1236, 1225-1247, 1254-1276, 1263-1285, 1273-1295, 1282-1304, 1292-1314, 1316-1338, 1337-1359, 1350-1372, 1364-1386, 1375-1397, 1408-1430, 1417-1439, 1429-1451, 1454-147, 1478-1500, 1487-1509, 1496-1518, 1507-1529, 1519-1541, 1542-1564, 1552-1574, 1562-1584, 1573-1595, 1585-1607, 1597-1619, 1607-1629, 1623-1645, 1633-1655, 1642-1664, 1651-1673, 1668-1690, 1677-1699, 1691-1713, 1700-1722, 1712-1734, 1749-1771, 1764-1786, 1773-1795, 1784-1806, 1796-1818, 1807-1829, 1824-1846, 1833-1855, 1847-1869, 1856-1878, 1865-1887, 1881-1903, 1896-1918, 1905-1927, 1921-1943, 1938-1960, 1948-1970, 1961-1983, 1970-1992, 1994-2016, 2008-2030, 2017-2039, 2056-2078, 2066-2088, 2075-2097, 2087-2109, 2096-2118, 2106-2128, 2116-2138, 2129-2151, 2176-2198, 2185-2207, 2196-2218, 2207-2229, 2220-2242, 2236-2258, 2247-2269, 2256-2278, 2265-2287, 2274-2296, 2299-2321, 2309-2331, 2318-2340, 2349-2371, 2371-2393, 2382-2404, 2391-2413, 2401-2423, 1267-1289, 1270-1292, 1271-1293, 1272-1294, 1273-1295, 1274-1296, 1275-1297, 1276-1298, 1278-1300, 1285-1307, 1294-1316, 1295-1317, 1327-1349, 1330-1352, 1371-1393, 1372-1394, 1374-1396, 1407-1429, 1410-1432, 1413-1435, 1414-1436, 1415-1437, 1416-1438, 1419-1441, 1420-1442, 1421-1443, 1422-1444, 1425-1447, 1426-1448, 1427-1449, 1428-1450, 1429-1451, 1430-1452, 1431-1453, 1432-1454, 1433-1455, 1478-1500, 1498-1520, 1500-1522, 1501-1523, 1502-1524, 1503-1525, 1504-1526, 1545-1567, 1548-1570, 1549-1571, 1550-1572, 1551-1573, 1559-1581, 1560-1582, 1562-1584, 1565-1587, 1567-1589, 1568-1590, 1569-1591, 1572-1594, 1577-1599, 1580-1602, 1581-1603, 1582-1604, 1583-1605, 1584-1606, 1589-1611, 1590-1612, 1593-1615, 1616-1638, 1617-1639, 1624-1646, 1626-1648, 1627-1649, 1628-1650, 1634-1656, 1635-1657, 1648-1670, 1655-1677, 1656-1678, 1657-1679, 1658-1680, 1659-1681, 1661-1683, 1681-1703, 1710-1732, 1711-1733, 1712-1734, 1713-1735, 1716-1738, 1717-1739, 1718-1740, 1720-1742, 1744-1766, 1751-1773, 1752-1774, 1775-1797, 1781-1803, 1784-1806, 1786-1808, 1787-1809, 1788-1810, 1789-1811, 1790-1812, 1795-1817, 1796-1818, 1797-1819, 1799-1821, 1800-1822, 1801-1823, 1808-1830, 1811-1833, 1816-1838, 1822-1844, 1824-1846, 1825-1847, 1826-1848, 1827-1849, 1828-1850, 1829-1851, 1830-1852, 1831-1853, 1837-1859, 1838-1860, 1840-1862, 1841-1863, 1842-1864, 1843-1865, 1844-1866, 1846-1868, 1847-1869, 1848-1870, 1850-1872, 1855-1877, 1856-1878, 1857-1879, 1858-1880, 1859-1881, 1860-1882, 1880-1902, 1882-1904, 1883-1905, 1885-1907, 1886-1908, 1894-1916, 1895-1917, 1896-1918, 1897-1919, 1898-1920, 1899-1921, 1900-1922, 1911-1933, 1933-1955, 1934-1956, 1936-1958, 1937-1959, 1940-1962, 1945-1967, 1946-1968, 1948-1970, 1949-1971, 1951-1973, 1954-1976, 1957-1979, 1958-1980, 1959-1981, 1960-1982, 1961-1983, 1962-1984, 2011-2033, 2013-2035, 2014-2036, 2016-2038, 2074-2096, 2076-2098, 2082-2104, 2085-2107, 2086-2108, 2087-2109, 2088-2110, 2089-2111, 2090-2112, 2092-2114, 2095-2117, 2098-2120, 2105-2127, 2107-2129, 2108-2130, 2110-2132, 2112-2134, 2114-2136, 2192-2214, 2239-2261, 2240-2262, 2249-2271, 2250-2272, 2253-2275, 2300-2322, 2346-2368, 2347-2369, 2348-2370, 2432-2454, 2433-2455, 2434-2456, 1267-1289, 1276-1298, 1277-1299, 1279-1301, 1283-1305, 1284-1306,

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1285-1307, 1286-1308, 1292-1314, 1295-1317, 1319-1341, 1328-1350, 1329-1351, 1330-1352, 1331-1353, 1332-1354, 1340-1362, 1341-1363, 1342-1364, 1343-1365, 1344-1366, 1345-1367, 1346-1368, 1368-1390, 1371-1393, 1373-1395, 1375-1397, 1408-1430, 1417-1439, 1418-1440, 1419-1441, 1423-1445, 1424-1446, 1430-1452, 1431-1453, 1437-1459, 1443-1465, 1478-1500, 1503-1525, 1512-1534, 1544-1566, 1545-1567, 1546-1568, 1547-1569, 1552-1574, 1553-1575, 1560-1582, 1561-1583, 1563-1585, 1566-1588, 1567-1589, 1570-1592, 1571-1593, 1572-1594, 1573-1595, 1574-1596, 1578-1600, 1579-1601, 1580-1602, 1584-1606, 1585-1607, 1586-1608, 1593-1615, 1595-1617, 1599-1621, 1600-1622, 1603-1625, 1609-1631, 1611-1633, 1612-1634, 1613-1635, 1614-1636, 1616-1638, 1618-1640, 1619-1641, 1620-1642, 1621-1643, 1622-1644, 1623-1645, 1625-1647, 1629-1651, 1632-1654, 1633-1655, 1635-1657, 1640-1662, 1645-1667, 1647-1669, 1651-1673, 1656-1678, 1657-1679, 1660-1682, 1680-1702, 1704-1726, 1705-1727, 1707-1729, 1709-1731, 1713-1735, 1714-1736, 1715-1737, 1716-1738, 1719-1741, 1720-1742, 1749-1771, 1773-1795, 1774-1796, 1775-1797, 1776-1798, 1778-1800, 1782-1804, 1783-1805, 1784-1806, 1785-1807, 1791-1813, 1792-1814, 1793-1815, 1805-1827, 1809-1831, 1810-1832, 1812-1834, 1813-1835, 1815-1837, 1817-1839, 1818-1840, 1819-1841, 1826-1848, 1833-1855, 1834-1856, 1836-1858, 1838-1860, 1839-1861, 1841-1863, 1844-1866, 1846-1868, 1847-1869, 1851-1873, 1852-1874, 1853-1875, 1854-1876, 1856-1878, 1860-1882, 1861-1883, 1880-1902, 1881-1903, 1882-1904, 1884-1906, 1887-1909, 1888-1910, 1889-1911, 1893-1915, 1900-1922, 1902-1924, 1910-1932, 1912-1934, 1917-1939, 1936-1958, 1941-1963, 1942-1964, 1944-1966, 1945-1967, 1949-1971, 1955-1977, 1962-1984, 2011-2033, 2015-2037, 2016-2038, 2074-2096, 2075-2097, 2076-2098, 2083-2105, 2091-2113, 2093-2115, 2094-2116, 2095-2117, 2096-2118, 2102-2124, 2103-2125, 2106-2128, 2109-2131, 2111-2133, 2112-2134, 2113-2135, 2115-2137, 2117-2139, 2241-2263, 2250-2272, 2253-2275, 2300-2322, 2301-2323, 2343-2365, 2347-2369, 2349-2371, 1796-1818, 1270-1292, 1624-1646, 1795-1817, 1822-1844, 1847-1869, 1582-1604, 1582-1604, 1582-1604, 1570-1592, 2110-2132, 1954-1976, 2013-2035, 1840-1862, 2114-2136, 1565-1587, 1808-1830, 2249-2271, 1614-1636, 1655-1677, 1581-1603, 1717-1739, 1717-1739, 1717-1739, 1883-1905, 1894-1916, 1957-1979, 1957-1979, 1957-1979, 1786-1808, 1681-1703, 1589-1611, 2098-2120, 1327-1349, 2432-2454, 1374-1396, 1911-1933, 1372-1394, 1816-1838, 2240-2262, 1577-1599, 2105-2127, 2092-2114, 1951-1973, 2346-2368, 1781-1803, 2014-2036, 1410-1432, 1940-1962, 1718-1740, 1859-1881, 1744-1766, 1787-1809, 1787-1809, 1787-1809, 1855-1877, or 2082-2104 of SEQ ID NO: 1. In some embodiments, the region of complementarity comprises at least 15 contiguous nucleotides from any one of nucleotides 29-51, 67-89, 154-176, 163-185, 173-195, 184-206, 196-218, 206-228, 257-279, 270-292, 446-468, 459-481, 468-490, 518-540, 530-552, 641-663, 687-709, 702-724, 711-733, 727-749, 758-780, 769-791, 781-803, 790-812, 807-829, 839-861, 850-872, 874-896, 907-929, 917-939, 958-980, 974-996, 983-1005, 999-1021, 1009-1031, 1018-1040, 1027-1049, 1036-1058, 1045-1067, 1054-1076, 1080-1102, 1089-1111, 1098-1120, 1108-1130, 1140-1162, 1156-1178, 1174-1196, 1183-1205, 1192-1214, 1205-1227, 1214-1236, 1225-1247, 1254-1276, 1263-1285, 1273-1295, 1282-1304,

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1292-1314, 1316-1338, 1337-1359, 1350-1372, 1364-1386, 1375-1397, 1408-1430, 1417-1439, 1429-1451, 1454-147, 1478-1500, 1487-1509, 1496-1518, 1507-1529, 1519-1541, 1542-1564, 1552-1574, 1562-1584, 1573-1595, 1585-1607, 1597-1619, 1607-1629, 1623-1645, 1633-1655, 1642-1664, 1651-1673, 1668-1690, 1677-1699, 1691-1713, 1700-1722, 1712-1734, 1749-1771, 1764-1786, 1773-1795, 1784-1806, 1796-1818, 1807-1829, 1824-1846, 1833-1855, 1847-1869, 1856-1878, 1865-1887, 1881-1903, 1896-1918, 1905-1927, 1921-1943, 1938-1960, 1948-1970, 1961-1983, 1970-1992, 1994-2016, 2008-2030, 2017-2039, 2056-2078, 2066-2088, 2075-2097, 2087-2109, 2096-2118, 2106-2128, 2116-2138, 2129-2151, 2176-2198, 2185-2207, 2196-2218, 2207-2229, 2220-2242, 2236-2258, 2247-2269, 2256-2278, 2265-2287, 2274-2296, 2299-2321, 2309-2331, 2318-2340, 2349-2371, 2371-2393, 2382-2404, 2391-2413, 2401-2423, 1267-1289, 1270-1292, 1271-1293, 1272-1294, 1273-1295, 1274-1296, 1275-1297, 1276-1298, 1278-1300, 1285-1307, 1294-1316, 1295-1317, 1327-1349, 1330-1352, 1371-1393, 1372-1394, 1374-1396, 1407-1429, 1410-1432, 1413-1435, 1414-1436, 1415-1437, 1416-1438, 1419-1441, 1420-1442, 1421-1443, 1422-1444, 1425-1447, 1426-1448, 1427-1449, 1428-1450, 1429-1451, 1430-1452, 1431-1453, 1432-1454, 1433-1455, 1478-1500, 1498-1520, 1500-1522, 1501-1523, 1502-1524, 1503-1525, 1504-1526, 1545-1567, 1548-1570, 1549-1571, 1550-1572, 1551-1573, 1559-1581, 1560-1582, 1562-1584, 1565-1587, 1567-1589, 1568-1590, 1569-1591, 1572-1594, 1577-1599, 1580-1602, 1581-1603, 1582-1604, 1583-1605, 1584-1606, 1589-1611, 1590-1612, 1593-1615, 1616-1638, 1617-1639, 1624-1646, 1626-1648, 1627-1649, 1628-1650, 1634-1656, 1635-1657, 1648-1670, 1655-1677, 1656-1678, 1657-1679, 1658-1680, 1659-1681, 1661-1683, 1681-1703, 1710-1732, 1711-1733, 1712-1734, 1713-1735, 1716-1738, 1717-1739, 1718-1740, 1720-1742, 1744-1766, 1751-1773, 1752-1774, 1775-1797, 1781-1803, 1784-1806, 1786-1808, 1787-1809, 1788-1810, 1789-1811, 1790-1812, 1795-1817, 1796-1818, 1797-1819, 1799-1821, 1800-1822, 1801-1823, 1808-1830, 1811-1833, 1816-1838, 1822-1844, 1824-1846, 1825-1847, 1826-1848, 1827-1849, 1828-1850, 1829-1851, 1830-1852, 1831-1853, 1837-1859, 1838-1860, 1840-1862, 1841-1863, 1842-1864, 1843-1865, 1844-1866, 1846-1868, 1847-1869, 1848-1870, 1850-1872, 1855-1877, 1856-1878, 1857-1879, 1858-1880, 1859-1881, 1860-1882, 1880-1902, 1882-1904, 1883-1905, 1885-1907, 1886-1908, 1894-1916, 1895-1917, 1896-1918, 1897-1919, 1898-1920, 1899-1921, 1900-1922, 1911-1933, 1933-1955, 1934-1956, 1936-1958, 1937-1959, 1940-1962, 1945-1967, 1946-1968, 1948-1970, 1949-1971, 1951-1973, 1954-1976, 1957-1979, 1958-1980, 1959-1981, 1960-1982, 1961-1983, 1962-1984, 2011-2033, 2013-2035, 2014-2036, 2016-2038, 2074-2096, 2076-2098, 2082-2104, 2085-2107, 2086-2108, 2087-2109, 2088-2110, 2089-2111, 2090-2112, 2092-2114, 2095-2117, 2098-2120, 2105-2127, 2107-2129, 2108-2130, 2110-2132, 2112-2134, 2114-2136, 2192-2214, 2239-2261, 2240-2262, 2249-2271, 2250-2272, 2253-2275, 2300-2322, 2346-2368, 2347-2369, 2348-2370, 2432-2454, 2433-2455, 2434-2456, 1267-1289, 1276-1298, 1277-1299, 1279-1301, 1283-1305, 1284-1306, 1285-1307, 1286-1308, 1292-1314, 1295-1317, 1319-1341, 1328-1350, 1329-1351, 1330-1352, 1331-1353, 1332-1354, 1340-1362, 1341-1363, 1342-1364, 1343-1365, 1344-1366, 1345-1367, 1346-1368, 1368-1390, 1371-1393, 1373-1395,

1375-1397, 1408-1430, 1417-1439, 1418-1440, 1419-1441, 1423-1445, 1424-1446, 1430-1452, 1431-1453, 1437-1459, 1443-1465, 1478-1500, 1503-1525, 1512-1534, 1544-1566, 1545-1567, 1546-1568, 1547-1569, 1552-1574, 1553-1575, 1560-1582, 1561-1583, 1563-1585, 1566-1588, 1567-1589, 1570-1592, 1571-1593, 1572-1594, 1573-1595, 1574-1596, 1578-1600, 1579-1601, 1580-1602, 1584-1606, 1585-1607, 1586-1608, 1593-1615, 1595-1617, 1599-1621, 1600-1622, 1603-1625, 1609-1631, 1611-1633, 1612-1634, 1613-1635, 1614-1636, 1616-1638, 1618-1640, 1619-1641, 1620-1642, 1621-1643, 1622-1644, 1623-1645, 1625-1647, 1629-1651, 1632-1654, 1633-1655, 1635-1657, 1640-1662, 1645-1667, 1647-1669, 1651-1673, 1656-1678, 1657-1679, 1660-1682, 1680-1702, 1704-1726, 1705-1727, 1707-1729, 1709-1731, 1713-1735, 1714-1736, 1715-1737, 1716-1738, 1719-1741, 1720-1742, 1749-1771, 1773-1795, 1774-1796, 1775-1797, 1776-1798, 1778-1800, 1782-1804, 1783-1805, 1784-1806, 1785-1807, 1791-1813, 1792-1814, 1793-1815, 1805-1827, 1809-1831, 1810-1832, 1812-1834, 1813-1835, 1815-1837, 1817-1839, 1818-1840, 1819-1841, 1826-1848, 1833-1855, 1834-1856, 1836-1858, 1838-1860, 1839-1861, 1841-1863, 1844-1866, 1846-1868, 1847-1869, 1851-1873, 1852-1874, 1853-1875, 1854-1876, 1856-1878, 1860-1882, 1861-1883, 1880-1902, 1881-1903, 1882-1904, 1884-1906, 1887-1909, 1888-1910, 1889-1911, 1893-1915, 1900-1922, 1902-1924, 1910-1932, 1912-1934, 1917-1939, 1936-1958, 1941-1963, 1942-1964, 1944-1966, 1945-1967, 1949-1971, 1955-1977, 1962-1984, 2011-2033, 2015-2037, 2016-2038, 2074-2096, 2075-2097, 2076-2098, 2083-2105, 2091-2113, 2093-2115, 2094-2116, 2095-2117, 2096-2118, 2102-2124, 2103-2125, 2106-2128, 2109-2131, 2111-2133, 2112-2134, 2113-2135, 2115-2137, 2117-2139, 2241-2263, 2250-2272, 2253-2275, 2300-2322, 2301-2323, 2343-2365, 2347-2369, 2349-2371, 1796-1818, 1270-1292, 1624-1646, 1795-1817, 1822-1844, 1847-1869, 1582-1604, 1582-1604, 1582-1604, 1570-1592, 2110-2132, 1954-1976, 2013-2035, 1840-1862, 2114-2136, 1565-1587, 1808-1830, 2249-2271, 1614-1636, 1655-1677, 1581-1603, 1717-1739, 1717-1739, 1717-1739, 1883-1905, 1894-1916, 1957-1979, 1957-1979, 1957-1979, 1786-1808, 1681-1703, 1589-1611, 2098-2120, 1327-1349, 2432-2454, 1374-1396, 1911-1933, 1372-1394, 1816-1838, 2240-2262, 1577-1599, 2105-2127, 2092-2114, 1951-1973, 2346-2368, 1781-1803, 2014-2036, 1410-1432, 1940-1962, 1718-1740, 1859-1881, 1744-1766, 1787-1809, 1787-1809, 1787-1809, 1855-1877, or 2082-2104 of SEQ ID NO: 1.

In one embodiment, the dsRNA agent comprises at least one modified nucleotide.

In one embodiment, substantially all of the nucleotides of the sense strand comprise a modification. In another embodiment, substantially all of the nucleotides of the antisense strand comprise a modification. In yet another embodiment, substantially all of the nucleotides of the sense strand and substantially all of the nucleotides of the antisense strand comprise a modification.

In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell. The dsRNA agent includes a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO:1 and the

antisense strand comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO:2, wherein substantially all of the nucleotides of the sense strand and substantially all of the nucleotides of the antisense strand are modified nucleotides, and wherein the sense strand is conjugated to a ligand attached at the 3'-terminus. In some embodiments, the dsRNA agent includes a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15 contiguous nucleotides from the nucleotide sequence of SEQ ID NO:1 and the antisense strand comprises at least 15 contiguous nucleotides from the nucleotide sequence of SEQ ID NO:2, wherein substantially all of the nucleotides of the sense strand and substantially all of the nucleotides of the antisense strand are modified nucleotides, and wherein the sense strand is conjugated to a ligand attached at the 3'-terminus.

In one embodiment, all of the nucleotides of the sense strand comprise a modification. In another embodiment, all of the nucleotides of the antisense strand comprise a modification. In yet another embodiment, all of the nucleotides of the sense strand and all of the nucleotides of the antisense strand comprise a modification.

In one embodiment, at least one of said modified nucleotides is selected from the group consisting of a deoxynucleotide, a 3'-terminal deoxy-thymine (dT) nucleotide, a 2'-O-methyl modified nucleotide, a 2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, a locked nucleotide, an unlocked nucleotide, a conformationally restricted nucleotide, a constrained ethyl nucleotide, an abasic nucleotide, a 2'-amino-modified nucleotide, a 2'-O-allyl-modified nucleotide, 2'-C-alkyl-modified nucleotide, 2'-hydroxyl-modified nucleotide, a 2'-methoxyethyl modified nucleotide, a 2'-O-alkyl-modified nucleotide, a morpholino nucleotide, a phosphoramidate, a non-natural base comprising nucleotide, a tetrahydropyran modified nucleotide, a 1,5-anhydrohexitol modified nucleotide, a cyclohexenyl modified nucleotide, a nucleotide comprising a phosphorothioate group, a nucleotide comprising a methylphosphonate group, a nucleotide comprising a 5'-phosphate, a nucleotide comprising a 5'-phosphate mimic, a glycol modified nucleotide, and a 2-O-(N-methylacetamide) modified nucleotide, and combinations thereof.

In one embodiment, the nucleotide modifications are 2'-O-methyl and/or 2'-fluoro modifications.

The region of complementarity may be at least 17 nucleotides in length; 19 to 30 nucleotides in length; 19-25 nucleotides in length; or 21 to 23 nucleotides in length.

Each strand may be no more than 30 nucleotides in length, e.g., each strand is independently 19-30 nucleotides in length; each strand is independently 19-25 nucleotides in length; each strand is independently 21-23 nucleotides in length.

The dsRNA may include at least one strand that comprises a 3' overhang of at least 1 nucleotide; or at least one strand that comprises a 3' overhang of at least 2 nucleotides.

In some embodiment, the dsRNA agent further comprises a ligand.

In one embodiment, the ligand is conjugated to the 3' end of the sense strand of the dsRNA agent.

In one embodiment, the ligand is an N-acetylgalactosamine (GalNAc) derivative.

either modified or unmodified or combinations thereof, each sequence comprising at least two differently modified nucleotides;

each N_b and N_b' independently represents an oligonucleotide sequence comprising 0-10 nucleotides which are either modified or unmodified or combinations thereof;

each n_p , n_p' , n_q , and n_q' , each of which may or may not be present, independently represents an overhang nucleotide;

XXX, YYY, ZZZ, X'X'X', Y'Y'Y', and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides;

modifications on N_b differ from the modification on Y and modifications on N_b' differ from the modification on Y'; and

wherein the sense strand is conjugated to at least one ligand.

In one embodiment, i is 0; j is 0; i is 1; j is 1; both i and j are 0; or both i and j are 1. In another embodiment, k is 0; l is 0; k is 1; l is 1; both k and l are 0; or both k and l are 1.

In one embodiment, XXX is complementary to X'X'X', YYY is complementary to Y'Y'Y', and ZZZ is complementary to Z'Z'Z'.

In one embodiment, the YYY motif occurs at or near the cleavage site of the sense strand, e.g., the Y'Y'Y' motif occurs at the 11, 12 and 13 positions of the antisense strand from the 5'-end.

In one embodiment, formula (Ij) is represented by formula (Ik):

(Ik)
sense:
5' n_p - N_a -Y Y Y - N_a - n_q 3'

antisense:
3' n_p' - N_a' -Y'Y'Y' - N_a' - n_q' 5'.

In another embodiment, formula (Ij) is represented by formula (Il):

(Il)
sense:
5' n_p - N_a -Y Y Y - N_b -Z Z Z - N_a - n_q 3'

antisense:
3' n_p' - N_a' -Y'Y'Y' - N_b' -Z'Z'Z' - N_a' - n_q' 5'

wherein each N_b and N_b' independently represents an oligonucleotide sequence comprising 1-5 modified nucleotides.

In yet another embodiment, formula (Ij) is represented by formula (Im):

(Im)
sense:
5' n_p - N_a -X X X - N_b -Y Y Y - N_a - n_q 3'

antisense:
3' n_p' - N_a' -X'X'X' - N_b' -Y'Y'Y' - N_a' - n_q' 5'

wherein each N_b and N_b' independently represents an oligonucleotide sequence comprising 1-5 modified nucleotides.

In another embodiment, formula (Ij) is represented by formula (In):

(In)
sense:
5' n_p - N_a -X X X - N_b -Y Y Y - N_b -Z Z Z - N_a - n_q 3'

antisense:
3' n_p' - N_a' -X'X'X' - N_b' -Y'Y'Y' - N_b' -Z'Z'Z' - N_a' - n_q' 5'

wherein each N_b and N_b' independently represents an oligonucleotide sequence comprising 1-5 modified nucleotides and each N_a and N_a' independently represents an oligonucleotide sequence comprising 2-10 modified nucleotides.

The region of complementarity may be at least 17 nucleotides in length; 19 to 30 nucleotides in length; 19-25 nucleotides in length; or 21 to 23 nucleotides in length.

Each strand may be no more than 30 nucleotides in length, e.g., each strand is independently 19-30 nucleotides in length.

In one embodiment, the modifications on the nucleotides are selected from the group consisting of LNA, HNA, CeNA, 2'-methoxyethyl, 2'-O-alkyl, 2'-O-allyl, 2'-C-allyl, 2'-fluoro, 2'-O-methyl, 2'-deoxy, 2'-hydroxyl, and combinations thereof.

In one embodiment, the modifications on the nucleotides are 2'-O-methyl or 2'-fluoro modifications.

In one embodiment, the Y' is a 2'-O-methyl or 2'-fluoro modified nucleotide.

In one embodiment, at least one strand of the dsRNA agent may comprise a 3' overhang of at least 1 nucleotide; or a 3' overhang of at least 2 nucleotides.

In one embodiment, the dsRNA agent further comprises at least one phosphorothioate or methylphosphonate internucleotide linkage.

In one embodiment, the phosphorothioate or methylphosphonate internucleotide linkage is at the 3'-terminus of one strand. In one embodiment, the strand is the antisense strand.

In another embodiment, the strand is the sense strand.

In one embodiment, the phosphorothioate or methylphosphonate internucleotide linkage is at the 5'-terminus of one strand. In one embodiment, the strand is the antisense strand. In another embodiment, the strand is the sense strand.

In one embodiment, the strand is the antisense strand. In another embodiment, the strand is the sense strand.

In one embodiment, the phosphorothioate or methylphosphonate internucleotide linkage is at both the 5'- and 3'-terminus of one strand.

In one embodiment, the base pair at the 1 position of the 5'-end of the antisense strand of the duplex is an AU base pair.

In one embodiment, $p' > 0$. In another embodiment, $p' = 2$.

In one embodiment, $q' = 0$, $p = 0$, $q = 0$, and p' overhang nucleotides are complementary to the target mRNA. In another embodiment, $q' = 0$, $p = 0$, $q = 0$, and p' overhang nucleotides are non-complementary to the target mRNA.

In one embodiment, the sense strand has a total of 21 nucleotides and the antisense strand has a total of 23 nucleotides.

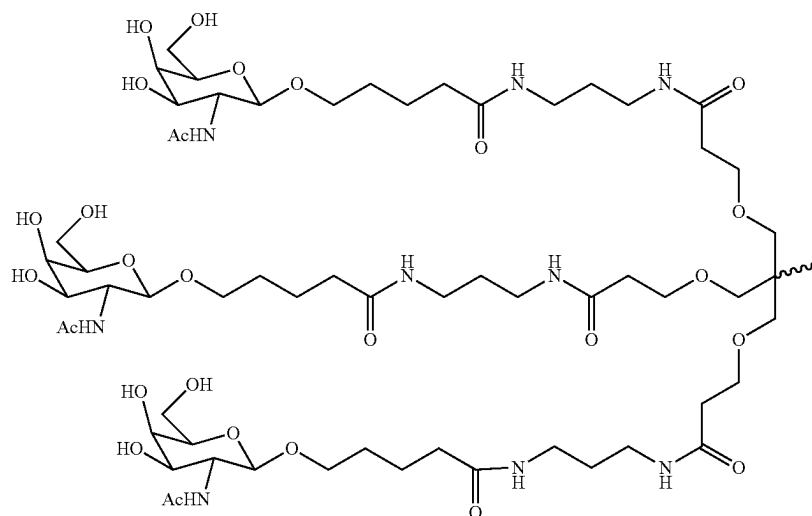
In one embodiment, at least one n_p is linked to a neighboring nucleotide via a phosphorothioate linkage. In another embodiment, wherein all n_p are linked to neighboring nucleotides via phosphorothioate linkages.

In one embodiment, all of the nucleotides of the sense strand and all of the nucleotides of the antisense strand comprise a modification.

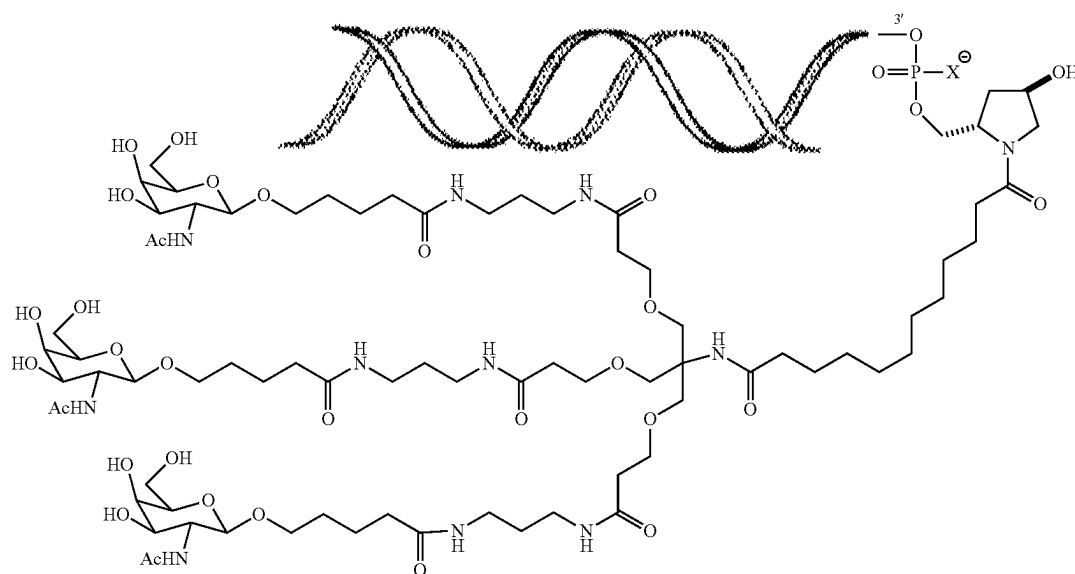
In one embodiment, the ligand is conjugated to the 3' end of the sense strand of the dsRNA agent.

In one embodiment, the ligand is one or more N-acetyl-galactosamine (GalNAc) derivatives attached through a monovalent, bivalent, or trivalent branched linker.

In one embodiment, the ligand is



In one embodiment, the dsRNA agent is conjugated to the ligand as shown in the following schematic



and, wherein X is O or S.

In one embodiment, the X is O.

In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting the expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell. The dsRNA agent includes a sense strand complementary to an antisense strand, wherein the antisense strand comprises a region complementary to part of an mRNA encoding CIDEb, wherein each strand is about 14 to about 30 nucleotides in length, wherein the dsRNA agent is represented by formula (Ij):

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(Ij)

sense:

$$5' \quad n_p \quad -N_a \quad - (X \ X \ X)_{i-N_b} \quad -Y \ Y \ Y \quad -N_b \quad - (Z \ Z \ Z)_{i-}$$
$$N_a - n_a \quad 3'$$

antisense:

$$3' \text{ } n_p' \text{ } -N_a' \text{ } - (X'X'X')_k \text{ } -N_b' \text{ } -Y'Y'Y' \text{ } -N_b' \text{ } - (Z'Z'Z')_l \text{ }$$
$$N_a' - n_a' \quad 5'$$

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wherein:

i, j, k, and l are each independently 0 or 1;

p, p', q, and q' are each independently 0-6;

each N_a and N_a' independently represents an oligonucleotide sequence comprising 0-25 nucleotides which are either modified or unmodified or combinations thereof, each sequence comprising at least two differently modified nucleotides;

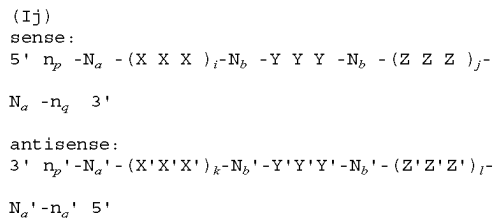
each N_b and N_b' independently represents an oligonucleotide sequence comprising 0-10 nucleotides which are either modified or unmodified or combinations thereof; each n_p , n_p' , n_q , and n_q' , each of which may or may not be present independently represents an overhang nucleotide;

XXX, YYY, ZZZ, X'X'X', Y'Y'Y', and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides, and wherein the modifications are 2'-O-methyl or 2'-fluoro modifications;

modifications on N_b differ from the modification on Y and modifications on N_b' differ from the modification on Y'; and

wherein the sense strand is conjugated to at least one ligand.

In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting the expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell. The dsRNA agent includes a sense strand complementary to an antisense strand, wherein the antisense strand comprises a region complementary to part of an mRNA encoding CIDEb, wherein each strand is about 14 to about 30 nucleotides in length, wherein the dsRNA agent is represented by formula (Ij):



wherein:

i, j, k, and l are each independently 0 or 1;

each n_p , n_q , and n_q' , each of which may or may not be present, independently represents an overhang nucleotide;

p, q, and q' are each independently 0-6;

$n_p > 0$ and at least one n_p' is linked to a neighboring nucleotide via a phosphorothioate linkage;

each N_a and N_a' independently represents an oligonucleotide sequence comprising 0-25 nucleotides which are either modified or unmodified or combinations thereof, each sequence comprising at least two differently modified nucleotides;

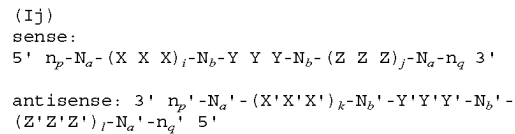
each N_b and N_b' independently represents an oligonucleotide sequence comprising 0-10 nucleotides which are either modified or unmodified or combinations thereof; XXX, YYY, ZZZ, X'X'X', Y'Y'Y', and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides, and wherein the modifications are 2'-O-methyl or 2'-fluoro modifications;

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modifications on N_b differ from the modification on Y and modifications on N_b' differ from the modification on Y'; and

wherein the sense strand is conjugated to at least one ligand.

In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting the expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell. The dsRNA agent includes a sense strand complementary to an antisense strand, wherein the antisense strand comprises a region complementary to part of an mRNA encoding CIDEb, wherein each strand is about 14 to about 30 nucleotides in length, wherein the dsRNA agent is represented by formula (Ij):



wherein:

i, j, k, and l are each independently 0 or 1;

each n_p , n_q , and n_q' , each of which may or may not be present, independently represents an overhang nucleotide;

p, q, and q' are each independently 0-6;

$n_p > 0$ and at least one n_p' is linked to a neighboring nucleotide via a phosphorothioate linkage;

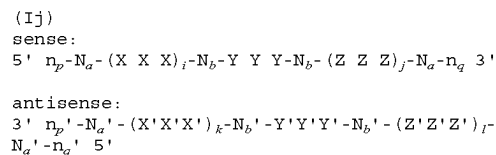
each N_a and N_a' independently represents an oligonucleotide sequence comprising 0-25 nucleotides which are either modified or unmodified or combinations thereof, each sequence comprising at least two differently modified nucleotides;

each N_b and N_b' independently represents an oligonucleotide sequence comprising 0-10 nucleotides which are either modified or unmodified or combinations thereof; XXX, YYY, ZZZ, X'X'X', Y'Y'Y', and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides, and wherein the modifications are 2'-O-methyl or 2'-fluoro modifications;

modifications on N_b differ from the modification on Y and modifications on N_b' differ from the modification on Y'; and

wherein the sense strand is conjugated to at least one ligand, wherein the ligand is one or more GalNAc derivatives attached through a monovalent, bivalent, or trivalent branched linker.

In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting the expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell. The dsRNA agent includes a sense strand complementary to an antisense strand, wherein the antisense strand comprises a region complementary to part of an mRNA encoding CIDEb, wherein each strand is about 14 to about 30 nucleotides in length, wherein the dsRNA agent is represented by formula (Ij):



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wherein:

i, j, k, and l are each independently 0 or 1;

each n_p , n_q , and n_q' , each of which may or may not be present, independently represents an overhang nucleotide;

p, q, and q' are each independently 0-6;

$n_p' > 0$ and at least one n_p' is linked to a neighboring nucleotide via a phosphorothioate linkage;

each N_a and N_a' independently represents an oligonucleotide sequence comprising 0-25 nucleotides which are either modified or unmodified or combinations thereof, each sequence comprising at least two differently modified nucleotides;

each N_b and N_b' independently represents an oligonucleotide sequence comprising 0-10 nucleotides which are either modified or unmodified or combinations thereof; XXX, YYY, ZZZ, X'X'X', Y'Y'Y', and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides, and wherein the modifications are 2'-O-methyl or 2'-fluoro modifications;

modifications on N_b differ from the modification on Y and modifications on N_b' differ from the modification on Y';

wherein the sense strand comprises at least one phosphorothioate linkage; and

wherein the sense strand is conjugated to at least one ligand, wherein the ligand is one or more GalNAc derivatives attached through a monovalent, bivalent, or trivalent branched linker.

In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting the expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell. The dsRNA agent includes a sense strand complementary to an antisense strand, wherein the antisense strand comprises a region complementary to part of an mRNA encoding CIDEb, wherein each strand is about 14 to about 30 nucleotides in length, wherein the dsRNA agent is represented by formula (Ij):

(Ik)
sense:
5' n_p - N_a -Y Y Y- N_a - n_q 3'

antisense:
3' n_p' - N_a' -Y'Y'Y'- N_a' - n_q' 5'

wherein:

each n_p , n_q , and n_q' , each of which may or may not be present, independently represents an overhang nucleotide;

p, q, and q' are each independently 0-6; $n_p' > 0$ and at least one n_p' is linked to a neighboring nucleotide via a phosphorothioate linkage;

each N_a and N_a' independently represents an oligonucleotide sequence comprising 0-25 nucleotides which are either modified or unmodified or combinations thereof, each sequence comprising at least two differently modified nucleotides;

YYY and Y'Y'Y' each independently represent one motif of three identical modifications on three consecutive nucleotides, and wherein the modifications are 2'-O-methyl and/or 2'-fluoro modifications;

wherein the sense strand comprises at least one phosphorothioate linkage; and

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wherein the sense strand is conjugated to at least one ligand, wherein the ligand is one or more GalNAc derivatives attached through a monovalent, bivalent, or trivalent branched linker.

In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting the expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell. The dsRNA agent includes a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO:1 and the antisense strand comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO:2, wherein substantially all of the nucleotides of the sense strand comprise a modification selected from the group consisting of a 2'-O-methyl modification and a 2'-fluoro modification, wherein the sense strand comprises two phosphorothioate internucleotide linkages at the 5'-terminus, wherein substantially all of the nucleotides of the antisense strand comprise a modification selected from the group consisting of a 2'-O-methyl modification and a 2'-fluoro modification, wherein the antisense strand comprises two phosphorothioate internucleotide linkages at the 5'-terminus and two phosphorothioate internucleotide linkages at the 3'-terminus, and wherein the sense strand is conjugated to one or more GalNAc derivatives attached through a monovalent, bivalent or trivalent branched linker at the 3'-terminus. In some embodiments, the dsRNA agent includes a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15 contiguous nucleotides from the nucleotide sequence of SEQ ID NO:1 and the antisense strand comprises at least 15 contiguous nucleotides from the nucleotide sequence of SEQ ID NO:2, wherein substantially all of the nucleotides of the sense strand comprise a modification selected from the group consisting of a 2'-O-methyl modification and a 2'-fluoro modification, wherein the sense strand comprises two phosphorothioate internucleotide linkages at the 5'-terminus, wherein substantially all of the nucleotides of the antisense strand comprise a modification selected from the group consisting of a 2'-O-methyl modification and a 2'-fluoro modification, wherein the antisense strand comprises two phosphorothioate internucleotide linkages at the 5'-terminus and two phosphorothioate internucleotide linkages at the 3'-terminus, and wherein the sense strand is conjugated to one or more GalNAc derivatives attached through a monovalent, bivalent or trivalent branched linker at the 3'-terminus.

In one embodiment, all of the nucleotides of the sense strand and all of the nucleotides of the antisense strand are modified nucleotides.

In one embodiment, the region of complementarity comprises any one of the antisense sequences listed in Tables 3-6.

In one embodiment, the sense strand and the antisense strand comprise nucleotide sequences selected from the group consisting of the nucleotide sequences of any one of the agents listed in Tables 3-6.

In various embodiments of the aforementioned dsRNA agents, the dsRNA agent targets a hotspot region of an mRNA encoding CIDEb.

In another aspect, the present invention provides a dsRNA agent that targets a hotspot region of a cell death-inducing DFFA-like effector B (CIDEb) mRNA.

The present invention also provides cells, vectors, and pharmaceutical compositions which include any of the

dsRNA agents of the invention. The dsRNA agents may be formulated in an unbuffered solution, e.g., saline or water, or in a buffered solution, e.g., a solution comprising acetate, citrate, prolamine, carbonate, or phosphate or any combination thereof. In one embodiment, the buffered solution is phosphate buffered saline (PBS).

In one aspect, the present invention provides a method of inhibiting cell death-inducing DFFA-like effector b (CIDEB) expression in a cell. The method includes contacting the cell with a dsRNA agent or a pharmaceutical composition of the invention, thereby inhibiting expression of CIDEB in the cell.

The cell may be within a subject, such as a human subject.

In one embodiment, the CIDEB expression is inhibited by at least 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, or to below the level of detection of CIDEB expression.

In one embodiment, the human subject suffers from a CIDEB-associated disease, disorder, or condition. In one embodiment, the CIDEB-associated disease, disorder, or condition is a chronic inflammatory disease, such as a chronic inflammatory disease of the liver and other tissues. In one embodiment, the chronic inflammatory disease is chronic inflammatory liver disease. In one embodiment, the chronic inflammatory liver disease is selected from the group consisting of accumulation of fat in the liver, inflammation of the liver, liver fibrosis, fatty liver disease (steatosis), nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD) and cirrhosis of the liver.

In one aspect, the present invention provides a method of inhibiting the expression of CIDEB in a subject. The methods include administering to the subject a therapeutically effective amount of a dsRNA agent or a pharmaceutical composition of the invention, thereby inhibiting the expression of CIDEB in the subject.

In another aspect, the present invention provides a method of treating a subject suffering from a CIDEB-associated disease, disorder, or condition. The method includes administering to the subject a therapeutically effective amount of a dsRNA agent or a pharmaceutical composition of the invention, thereby treating the subject suffering from a CIDEB-associated disease, disorder, or condition.

In another aspect, the present invention provides a method of preventing at least one symptom in a subject having a disease, disorder or condition that would benefit from reduction in expression of a CIDEB gene. The method includes administering to the subject a prophylactically effective amount of the agent of a dsRNA agent or a pharmaceutical composition of the invention, thereby preventing at least one symptom in a subject having a disease, disorder or condition that would benefit from reduction in expression of a CIDEB gene.

In another aspect, the present invention provides a method of reducing the risk of developing chronic liver disease in a subject having steatosis. The method includes administering to the subject a therapeutically effective amount of a dsRNA agent or a pharmaceutical composition of the invention, thereby reducing the risk of developing chronic liver disease in the subject having steatosis.

In one aspect, the present invention provides a method of inhibiting the accumulation of lipid droplets in the liver of a subject suffering from a CIDEB-associated disease, disorder, or condition. The method includes administering to the subject a therapeutically effective amount of a dsRNA agent or a pharmaceutical composition of the invention, and a dsRNA agent targeting a CIDEB gene or a pharmaceutical composition comprising a dsRNA agent targeting a CIDEB

gene, thereby inhibiting the accumulation of fat in the liver of the subject suffering from a CIDEB-associated disease, disorder, or condition.

In another aspect, the present invention provides a method of treating a subject suffering from a CIDEB-associated disease, disorder, or condition. The method includes administering to the subject a therapeutically effective amount of a dsRNA agent or a pharmaceutical composition of the invention, and a dsRNA agent targeting a CIDEB gene or a pharmaceutical composition comprising a dsRNA agent targeting a CIDEB gene, thereby treating the subject suffering from a CIDEB-associated disease, disorder, or condition.

In another aspect, the present invention provides a method of preventing at least one symptom in a subject having a disease, disorder or condition that would benefit from reduction in expression of a CIDEB gene. The method includes administering to the subject a therapeutically effective amount of a dsRNA agent or a pharmaceutical composition of the invention, and a dsRNA agent targeting a CIDEB gene or a pharmaceutical composition comprising a dsRNA agent targeting a CIDEB gene, thereby preventing at least one symptom in a subject having a disease, disorder or condition that would benefit from reduction in expression of a CIDEB gene.

In another aspect, the present invention provides a method of reducing the risk of developing chronic liver disease in a subject having steatosis. The method includes administering to the subject a therapeutically effective amount of a dsRNA agent or a pharmaceutical composition of the invention, and a dsRNA agent targeting a CIDEB gene or a pharmaceutical composition comprising a dsRNA agent targeting a CIDEB gene, thereby reducing the risk of developing chronic liver disease in the subject having steatosis.

In another aspect, the present invention provides a method of inhibiting the progression of steatosis to steatohepatitis in a subject suffering from steatosis. The method includes administering to the subject a therapeutically effective amount of a dsRNA agent or a pharmaceutical composition of the invention, and a dsRNA agent targeting a CIDEB gene or a pharmaceutical composition comprising a dsRNA agent targeting a CIDEB gene, thereby inhibiting the progression of steatosis to steatohepatitis in the subject.

In one embodiment, the administration of the dsRNA agent or the pharmaceutical composition to the subject causes a decrease in CIDEB protein activity, e.g., a decrease in the interaction of CIDEB with ApoB and/or a decrease in lipid maturation in liver; a decrease in CIDEB protein accumulation, a decrease in CIDEB enzymatic activity, a decrease in CIDEB protein accumulation, and/or a decrease in accumulation of fat and/or expansion of lipid droplets in the liver of a subject.

In one embodiment, the CIDEB-associated disease, disorder, or condition is a chronic inflammatory disease.

In one embodiment, the chronic inflammatory disease is chronic inflammatory liver disease.

In one embodiment, the chronic inflammatory liver disease is selected from the group consisting of accumulation of fat in the liver, inflammation of the liver, liver fibrosis, fatty liver disease (steatosis), nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD) and cirrhosis of the liver.

In one embodiment, the chronic inflammatory liver disease is nonalcoholic steatohepatitis (NASH).

In one embodiment, the subject is obese.

In one embodiment, the methods and uses of the invention further include administering an additional therapeutic to the subject.

In one embodiment, the dsRNA agent is administered to the subject at a dose of about 0.01 mg/kg to about 10 mg/kg or about 0.5 mg/kg to about 50 mg/kg.

The agent may be administered to the subject intravenously, intramuscularly, or subcutaneously. In one embodiment, the agent is administered to the subject subcutaneously.

In one embodiment, the methods and uses of the invention further include determining, the level of CIDEB in the subject.

In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEB) in a cell, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises a nucleotide sequence of any one of the agents in Tables 3-6 and the antisense strand comprises a nucleotide sequence of any one of the agents in Tables 3-6, wherein substantially all of the nucleotide of the sense strand and substantially all of the nucleotides of the antisense strand are modified nucleotides, and wherein the dsRNA agent is conjugated to a ligand.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 depicts the qPCR results for CIDEB mRNA in a mouse in vivo single dose (3 mg/kg) study using exemplary human CIDEB dsRNA duplexes. The results are presented as the percent mRNA remaining normalized to PBS.

FIG. 2 depicts the qPCR results for CIDEB mRNA in a mouse in vivo multi-dose (1.5 mg/kg and 0.75 mg/kg) study using exemplary human CIDEB dsRNA duplexes. The results are presented as the percent mRNA remaining normalized to PBS.

DETAILED DESCRIPTION OF THE INVENTION

The present invention provides iRNA compositions, which effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of a CIDEB gene. The CIDEB gene may be within a cell, e.g., a cell within a subject, such as a human. The present invention also provides methods of using the iRNA compositions of the invention for inhibiting the expression of a CIDEB gene, and for treating a subject who would benefit from inhibiting or reducing the expression of a CIDEB gene, e.g., a subject that would benefit from a reduction in inflammation, e.g., a subject suffering or prone to suffering from a CIDEB-associated disease disorder, or condition, such as a subject suffering or prone to suffering from chronic inflammatory diseases of the liver and other tissues, e.g., a subject suffering from chronic inflammatory liver disease, such as liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), cirrhosis of the liver, HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, insulin insensitivity and diabetes.

The iRNAs of the invention targeting CIDEB may include an RNA strand (the antisense strand) having a region which is about 30 nucleotides or less in length, e.g., 15-30, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27,

21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length, which region is substantially complementary to at least part of an mRNA transcript of a CIDEB gene.

In some embodiments, one or both of the strands of the double stranded RNAi agents of the invention is up to 66 nucleotides in length, e.g., 36-66, 26-36, 25-36, 31-60, 22-43, 27-53 nucleotides in length, with a region of at least 19 contiguous nucleotides that is substantially complementary to at least a part of an mRNA transcript of a CIDEB gene. In some embodiments, such iRNA agents having longer length antisense strands may include a second RNA strand (the sense strand) of 20-60 nucleotides in length wherein the sense and antisense strands form a duplex of 18-30 contiguous nucleotides.

The use of the iRNA agents described herein enables the targeted degradation of mRNAs of a CIDEB gene in mammals.

Very low dosages of the iRNAs, in particular, can specifically and efficiently mediate RNA interference (RNAi), resulting in significant inhibition of expression of a CIDEB gene. Thus, methods and compositions including these iRNAs are useful for treating a subject who would benefit from inhibiting or reducing the expression of a CIDEB gene, e.g., a subject that would benefit from a reduction of inflammation, e.g., a subject suffering or prone to suffering from a CIDEB-associated disease disorder, or condition, such as a subject suffering or prone to suffering from chronic inflammatory diseases of the liver and other tissues, e.g., a subject suffering from chronic inflammatory liver disease, such as liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), cirrhosis of the liver, HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, insulin insensitivity and diabetes.

The following detailed description discloses how to make and use compositions containing iRNAs to inhibit the expression of a CIDEB gene, as well as compositions and methods for treating subjects having diseases and disorders that would benefit from inhibition and/or reduction of the expression of this gene.

I. Definitions

In order that the present invention may be more readily understood, certain terms are first defined. In addition, it should be noted that whenever a value or range of values of a parameter are recited, it is intended that values and ranges intermediate to the recited values are also intended to be part of this invention.

The articles "a" and "an" are used herein to refer to one or to more than one (i.e., to at least one) of the grammatical object of the article. By way of example, "an element" means one element or more than one element, e.g., a plurality of elements.

The term "including" is used herein to mean, and is used interchangeably with, the phrase "including but not limited to".

The term "or" is used herein to mean, and is used interchangeably with, the term "and/or," unless context clearly indicates otherwise.

The term "about" is used herein to mean within the typical ranges of tolerances in the art. For example, "about" can be understood as about 2 standard deviations from the mean. In certain embodiments, about means $\pm 10\%$. In certain embodiments, about means $\pm 5\%$. When about is present

before a series of numbers or a range, it is understood that “about” can modify each of the numbers in the series or range.

The term “CIDEB,” also known as “cell death-inducing DFFA-like effector b”, “Cell Death Activator CIDE-B” or “cell death-inducing DFF45-like effector B”, refers to the well-known gene encoding a CIDEB protein from any vertebrate or mammalian source, including, but not limited to, human, bovine, chicken, rodent, mouse, rat, porcine, ovine, primate, monkey, and guinea pig, unless specified otherwise.

The term also refers to fragments and variants of native CIDEB that maintain at least one in vivo or in vitro activity of a native CIDEB.

CIDEB, a member of the CIDE protein family, is primarily expressed in liver tissues as well as in the small intestine, for example, the jejunum and ileum sections of the small intestine (Zhang et al., *Lipid Res.*; 55(7):1279-87; 2014). CIDEB is an endoplasmic reticulum (ER)- and lipid droplet (LD)-associated protein. As a CIDE family member, over-expression of CIDEB protein induces cell death, but the physiological function of CIDEB is more closely related to various lipid metabolic pathways, particularly the VLDL pathway. For instance, CIDEB mediates VLDL lipidation and maturation by interacting with ApoB; CIDEB is also required for the biogenesis of VLDL transport vesicles and for chylomicron lipidation in the small intestine. CIDEB mutant mice displayed significantly increased insulin sensitivity and enhanced rate of whole-body metabolism and hepatic fatty acid oxidation (Li et al., *Diabetes*. 56(10): 2523-32. 2007). Therefore, CIDEB may represent a new therapeutic target for the treatment of obesity, diabetes, and liver steatosis (Li et al., *Diabetes*. 56(10):2523-32. 2007). CIDEB is transcriptionally regulated by hepatocyte nuclear factor 4 α (HNF4 α), the most abundant transcription factor in the liver; HNF4 α is crucial for VLDL-mediated lipid transport and participates in HCV assembly/release. As a key transcriptional coactivator of HNF4 α , peroxisome proliferator-activated receptor- γ coactivator-1 α (PGC-1 α) also regulates HCV production and PGC-1 α stimulates VLDL assembly in a CIDEB-dependent manner (Cai et al., *Scientific Reports*, volume 6, Article number: 27778, 2016).

CIDEB is required for HCV entry into hepatocytes and HCV assembly (Xu et al., *J Virol.*, 88, 8433-8444, 2014). CIDEB interacts with the HCV NS5A protein and regulates the association of HCV particles with ApoE. CIDEB also regulates the post-entry stages of the dengue virus (DENV) lifecycle (Cai et al., *Scientific Reports*, volume 6, Article number: 27778, 2016).

Exemplary nucleotide and amino acid sequences of CIDEB can be found, for example, at GenBank Accession No. NM_001393338.1 (SEQ ID NO: 1; reverse complement SEQ ID NO: 2) for *Homo sapiens*.

Additional examples of CIDEB mRNA sequences are readily available using publicly available databases, e.g., GenBank, UniProt, and OMIM.

Further information on CIDEB is provided, for example in the NCBI Gene database at ncbi.nlm.nih.gov/gene/27141.

In some embodiments, the iRNAs that are substantially complementary to a region of a mouse or rat CIDEB mRNA cross-react with human CIDEB mRNA and represent potential candidates for human targeting.

The term “CIDEB” as used herein also refers to a particular polypeptide expressed in a cell by naturally occurring DNA sequence variations of the CIDEB gene, such as a single nucleotide polymorphism in the CIDEB gene. Numerous Single Nucleotide Polymorphisms (SNPs) within

the CIDEB gene have been identified and may be found at, for example, NCBI dbSNP (see, e.g., ncbi.nlm.nih.gov/snp).

As used herein, “target sequence” refers to a contiguous portion of the nucleotide sequence of an mRNA molecule formed during the transcription of a CIDEB gene, including mRNA that is a product of RNA processing of a primary transcription product. In one embodiment, the target portion of the sequence will be at least long enough to serve as a substrate for iRNA-directed cleavage at or near that portion of the nucleotide sequence of an mRNA molecule formed during the transcription of a CIDEB gene.

The target sequence of a CIDEB gene may be from about 9-36 nucleotides in length, e.g., about 15-30 nucleotides in length. For example, the target sequence can be from about 15-30 nucleotides, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the invention.

As used herein, the term “strand comprising a sequence” refers to an oligonucleotide comprising a chain of nucleotides that is described by the sequence referred to using the standard nucleotide nomenclature.

“G,” “C,” “A,” “T” and “U” each generally stand for a nucleotide that contains guanine, cytosine, adenine, thymine and uracil as a base, respectively. However, it will be understood that the term “ribonucleotide” or “nucleotide” can also refer to a modified nucleotide, as further detailed below, or a surrogate replacement moiety (see, e.g., Table 2). The skilled person is well aware that guanine, cytosine, adenine, and uracil can be replaced by other moieties without substantially altering the base pairing properties of an oligonucleotide comprising a nucleotide bearing such replacement moiety. For example, without limitation, a nucleotide comprising inosine as its base can base pair with nucleotides containing adenine, cytosine, or uracil. Hence, nucleotides containing uracil, guanine, or adenine can be replaced in the nucleotide sequences of dsRNA featured in the invention by a nucleotide containing, for example, inosine. In another example, adenine and cytosine anywhere in the oligonucleotide can be replaced with guanine and uracil, respectively to form G-U Wobble base pairing with the target mRNA. Sequences containing such replacement moieties are suitable for the compositions and methods featured in the invention.

The terms “iRNA”, “RNAi agent”, “iRNA agent”, “RNA interference agent” as used interchangeably herein, refer to an agent that contains RNA as that term is defined herein, and which mediates the targeted cleavage of an RNA transcript via an RNA-induced silencing complex (RISC) pathway. iRNA directs the sequence-specific degradation of mRNA through a process known as RNA interference (RNAi). The iRNA modulates, e.g., inhibits, the expression of CIDEB gene in a cell, e.g., a cell within a subject, such as a mammalian subject.

In one embodiment, an RNAi agent of the invention includes a single stranded RNA that interacts with a target RNA sequence, e.g., a CIDEB target mRNA sequence, to direct the cleavage of the target RNA. Without wishing to be bound by theory it is believed that long double stranded RNA introduced into cells is broken down into siRNA by a Type III endonuclease known as Dicer (Sharp et al. (2001)

Genes Dev. 15:485). Dicer, a ribonuclease-III-like enzyme, processes the dsRNA into 19-23 base pair short interfering RNAs with characteristic two base 3' overhangs (Bernstein, et al., (2001) *Nature* 409:363). The siRNAs are then incorporated into an RNA-induced silencing complex (RISC) where one or more helicases unwind the siRNA duplex, enabling the complementary antisense strand to guide target recognition (Nykanen, et al., (2001) *Cell* 107:309). Upon binding to the appropriate target mRNA, one or more endonucleases within the RISC cleave the target to induce silencing (Elbashir, et al., (2001) *Genes Dev.* 15:188). Thus, in one aspect the invention relates to a single stranded RNA (ssRNA) generated within a cell and which promotes the formation of a RISC complex to effect silencing of the target gene, i.e., a CIDEB gene. Accordingly, the term "siRNA" is also used herein to refer to an RNAi as described above.

In another embodiment, the RNAi agent may be a single-stranded RNAi agent that is introduced into a cell or organism to inhibit a target mRNA. Single-stranded RNAi agents (ssRNAi) bind to the RISC endonuclease, Argonaute 2, which then cleaves the target mRNA. The single-stranded siRNAs are generally 15-30 nucleotides and are chemically modified. The design and testing of single-stranded RNAi agents are described in U.S. Pat. No. 8,101,348 and in Lima et al., (2012) *Cell* 150: 883-894, the entire contents of each of which are hereby incorporated herein by reference. Any of the antisense nucleotide sequences described herein may be used as a single-stranded siRNA as described herein or as chemically modified by the methods described in Lima et al., (2012) *Cell* 150: 883-894.

In another embodiment, an "iRNA" for use in the compositions and methods of the invention is a double-stranded RNA and is referred to herein as a "double stranded RNAi agent," "double-stranded RNA (dsRNA) molecule," "dsRNA agent," or "dsRNA". The term "dsRNA", refers to a complex of ribonucleic acid molecules, having a duplex structure comprising two anti-parallel and substantially complementary nucleic acid strands, referred to as having "sense" and "antisense" orientations with respect to a target RNA, i.e., a CIDEB gene. In some embodiments of the invention, a double-stranded RNA (dsRNA) triggers the degradation of a target RNA, e.g., an mRNA, through a post-transcriptional gene-silencing mechanism referred to herein as RNA interference or RNAi.

In general, the majority of nucleotides of each strand of a dsRNA molecule are ribonucleotides, but as described in detail herein, each or both strands can also include one or more non-ribonucleotides, e.g., a deoxyribonucleotide and/or a modified nucleotide. In addition, as used in this specification, an "RNAi agent" may include ribonucleotides with chemical modifications; an RNAi agent may include substantial modifications at multiple nucleotides. As used herein, the term "modified nucleotide" refers to a nucleotide having, independently, a modified sugar moiety, a modified internucleotide linkage, and/or a modified nucleobase. Thus, the term modified nucleotide encompasses substitutions, additions or removal of, e.g., a functional group or atom, to internucleotide linkages, sugar moieties, or nucleobases. The modifications suitable for use in the agents of the invention include all types of modifications disclosed herein or known in the art. Any such modifications, as used in a siRNA type molecule, are encompassed by "RNAi agent" for the purposes of this specification and claims.

The duplex region may be of any length that permits specific degradation of a desired target RNA through a RISC pathway, and may range from about 9 to 36 base pairs in length, e.g., about 15-30 base pairs in length, for example,

about 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, or 36 base pairs in length, such as about 15-30, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the invention.

The two strands forming the duplex structure may be different portions of one larger RNA molecule, or they may be separate RNA molecules. Where the two strands are part of one larger molecule, and therefore are connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming the duplex structure, the connecting RNA chain is referred to as a "hairpin loop." A hairpin loop can comprise at least one unpaired nucleotide. In some embodiments, the hairpin loop can comprise at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 20, at least 23 or more unpaired nucleotides.

Where the two substantially complementary strands of a dsRNA are comprised by separate RNA molecules, those molecules need not, but can be covalently connected. Where the two strands are connected covalently by means other than an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming the duplex structure, the connecting structure is referred to as a "linker". The RNA strands may have the same or a different number of nucleotides. The maximum number of base pairs is the number of nucleotides in the shortest strand of the dsRNA minus any overhangs that are present in the duplex. In addition to the duplex structure, an RNAi may comprise one or more nucleotide overhangs.

In one embodiment, an RNAi agent of the invention is a dsRNA, each strand of which comprises less than 30 nucleotides, e.g., 17-27, 19-27, 17-25, 19-25, or 19-23, that interacts with a target RNA sequence, e.g., a CIDEB target mRNA sequence, to direct the cleavage of the target RNA. In another embodiment, an RNAi agent of the invention is a dsRNA, each strand of which comprises 19-23 nucleotides, that interacts with a target RNA sequence, e.g., a CIDEB target mRNA sequence, to direct the cleavage of the target RNA. In one embodiment, the sense strand is 21 nucleotides in length. In another embodiment, the antisense strand is 23 nucleotides in length.

As used herein, the term "nucleotide overhang" refers to at least one unpaired nucleotide that protrudes from the duplex structure of an iRNA, e.g., a dsRNA. For example, when a 3-end of one strand of a dsRNA extends beyond the 5-end of the other strand, or vice versa, there is a nucleotide overhang. A dsRNA can comprise an overhang of at least one nucleotide; alternatively, the overhang can comprise at least two nucleotides, at least three nucleotides, at least four nucleotides, at least five nucleotides or more. A nucleotide overhang can comprise or consist of a nucleotide/nucleoside analog, including a deoxynucleotide/nucleoside. The overhang(s) can be on the sense strand, the antisense strand or any combination thereof. Furthermore, the nucleotide(s) of an overhang can be present on the 5-end, 3-end or both ends of either an antisense or sense strand of a dsRNA.

In one embodiment, the antisense strand of a dsRNA has a 1-10 nucleotide, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end and/or the 5'-end. In one

embodiment, the sense strand of a dsRNA has a 1-10 nucleotide, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end and/or the 5'-end. In another embodiment, one or more of the nucleotides in the overhang is replaced with a nucleoside thiophosphate.

In certain embodiments, the overhang on the sense strand or the antisense strand, or both, can include extended lengths longer than 10 nucleotides, e.g., 10-30 nucleotides, 10-25 nucleotides, 10-20 nucleotides or 10-15 nucleotides in length. In certain embodiments, an extended overhang is on the sense strand of the duplex. In certain embodiments, an extended overhang is present on the 3' end of the sense strand of the duplex. In certain embodiments, an extended overhang is present on the 5' end of the sense strand of the duplex. In certain embodiments, an extended overhang is on the antisense strand of the duplex. In certain embodiments, an extended overhang is present on the 3' end of the antisense strand of the duplex. In certain embodiments, an extended overhang is present on the 5' end of the antisense strand of the duplex. In certain embodiments, one or more of the nucleotides in the extended overhang is replaced with a nucleoside thiophosphate.

The terms "blunt" or "blunt ended" as used herein in reference to a dsRNA mean that there are no unpaired nucleotides or nucleotide analogs at a given terminal end of a dsRNA, i.e., no nucleotide overhang. One or both ends of a dsRNA can be blunt. Where both ends of a dsRNA are blunt, the dsRNA is said to be blunt ended. To be clear, a "blunt ended" dsRNA is a dsRNA that is blunt at both ends, i.e., no nucleotide overhang at either end of the molecule. Most often such a molecule will be double-stranded over its entire length.

The term "antisense strand" or "guide strand" refers to the strand of an iRNA, e.g., a dsRNA, which includes a region that is substantially complementary to a target sequence, e.g., a CIDEB mRNA.

As used herein, the term "region of complementarity" refers to the region on the antisense strand that is substantially complementary to a sequence, for example a target sequence, e.g., a CIDEB nucleotide sequence, as defined herein. Where the region of complementarity is not fully complementary to the target sequence, the mismatches can be in the internal or terminal regions of the molecule. Generally, the most tolerated mismatches are in the terminal regions, e.g., within 5, 4, 3, or 2 nucleotides of the 5'- and/or 3'-terminus of the iRNA.

The term "sense strand" or "passenger strand" as used herein, refers to the strand of an iRNA that includes a region that is substantially complementary to a region of the antisense strand as that term is defined herein.

As used herein, the term "cleavage region" refers to a region that is located immediately adjacent to the cleavage site. The cleavage site is the site on the target at which cleavage occurs. In some embodiments, the cleavage region comprises three bases on either end of, and immediately adjacent to, the cleavage site. In some embodiments, the cleavage region comprises two bases on either end of, and immediately adjacent to, the cleavage site. In some embodiments, the cleavage site specifically occurs at the site bound by nucleotides 10 and 11 of the antisense strand, and the cleavage region comprises nucleotides 11, 12 and 13.

As used herein, and unless otherwise indicated, the term "complementary," when used to describe a first nucleotide sequence in relation to a second nucleotide sequence, refers to the ability of an oligonucleotide or polynucleotide comprising the first nucleotide sequence to hybridize and form a duplex structure under certain conditions with an oligo-

nucleotide or polynucleotide comprising the second nucleotide sequence, as will be understood by the skilled person. Such conditions can, for example, be stringent conditions, where stringent conditions can include: 400 mM NaCl, 40 mM PIPES pH 6.4, 1 mM EDTA, 50° C. or 70° C. for 12-16 hours followed by washing (see, e.g., "Molecular Cloning: A Laboratory Manual, Sambrook, et al. (1989) Cold Spring Harbor Laboratory Press). Other conditions, such as physiologically relevant conditions as can be encountered inside an organism, can apply. The skilled person will be able to determine the set of conditions most appropriate for a test of complementarity of two sequences in accordance with the ultimate application of the hybridized nucleotides.

Complementary sequences within an iRNA, e.g., within a dsRNA as described herein, include base-pairing of the oligonucleotide or polynucleotide comprising a first nucleotide sequence to an oligonucleotide or polynucleotide comprising a second nucleotide sequence over the entire length of one or both nucleotide sequences. Such sequences can be referred to as "fully complementary" with respect to each other herein. However, where a first sequence is referred to as "substantially complementary" with respect to a second sequence herein, the two sequences can be fully complementary, or they can form one or more, but generally not more than 5, 4, 3 or 2 mismatched base pairs upon hybridization for a duplex up to 30 base pairs, while retaining the ability to hybridize under the conditions most relevant to their ultimate application, e.g., inhibition of gene expression via a RISC pathway. However, where two oligonucleotides are designed to form, upon hybridization, one or more single stranded overhangs, such overhangs shall not be regarded as mismatches with regard to the determination of complementarity. For example, a dsRNA comprising one oligonucleotide 21 nucleotides in length and another oligonucleotide 23 nucleotides in length, wherein the longer oligonucleotide comprises a sequence of 21 nucleotides that is fully complementary to the shorter oligonucleotide, can yet be referred to as "fully complementary" for the purposes described herein.

"Complementary" sequences, as used herein, can also include, or be formed entirely from, non-Watson-Crick base pairs and/or base pairs formed from non-natural and modified nucleotides, in so far as the above requirements with respect to their ability to hybridize are fulfilled. Such non-Watson-Crick base pairs include, but are not limited to, G:U Wobble or Hoogsteen base pairing.

The terms "complementary," "fully complementary" and "substantially complementary" herein can be used with respect to the base matching between the sense strand and the antisense strand of a dsRNA, or between the antisense strand of an iRNA agent and a target sequence, as will be understood from the context of their use.

As used herein, a polynucleotide that is "substantially complementary to at least part of" a messenger RNA (mRNA) refers to a polynucleotide that is substantially complementary to a contiguous portion of the mRNA of interest (e.g., an mRNA encoding CIDEB). For example, a polynucleotide is complementary to at least a part of a CIDEB mRNA if the sequence is substantially complementary to a non-interrupted portion of an mRNA encoding CIDEB.

Accordingly, in some embodiments, the antisense strand polynucleotides disclosed herein are fully complementary to the target CIDEB sequence. In other embodiments, the antisense strand polynucleotides disclosed herein are substantially complementary to the target CIDEB sequence and comprise a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to the

equivalent region of the nucleotide sequence of SEQ ID NO:1, or a fragment of SEQ ID NO:1, such as about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

In one embodiment, an RNAi agent of the invention includes a sense strand that is substantially complementary to an antisense polynucleotide which, in turn, is complementary to a target CIDEB sequence, and wherein the sense strand polynucleotide comprises a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to the equivalent region of the nucleotide sequence of SEQ ID NOs: 2, or a fragment of any one of SEQ ID NOs: 2, such as about 85%, about 86%, about 87%, about 88%, about 89%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

In some embodiments, an iRNA of the invention includes an antisense strand that is substantially complementary to the target CIDEB sequence and comprises a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to the equivalent region of the nucleotide sequence of any one of the sense strands in Tables 3-6, or a fragment of any one of the sense strands in Tables 3-6, such as about 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% complementary, or 100% complementary.

The term "inhibiting," as used herein, is used interchangeably with "reducing," "silencing," "downregulating," "suppressing" and other similar terms, and includes any level of inhibition.

The phrase "inhibiting expression of a CIDEB gene," as used herein, includes inhibition of expression of any CIDEB gene (such as, e.g., a mouse CIDEB gene, a rat CIDEB gene, a monkey CIDEB gene, or a human CIDEB gene) as well as variants or mutants of a CIDEB gene that encode a CIDEB protein.

"Inhibiting expression of a CIDEB gene" includes any level of inhibition of a CIDEB gene, e.g., at least partial suppression of the expression of a CIDEB gene, such as an inhibition by at least about 20%. In certain embodiments, inhibition is by at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99%.

The expression of a CIDEB gene may be assessed based on the level of any variable associated with CIDEB gene expression, e.g., CIDEB mRNA level or CIDEB protein level. The expression of a CIDEB gene may also be assessed indirectly based on, for example, a decrease in CIDEB protein activity, e.g., a decrease in the interaction of CIDEB with ApoB and/or a decrease in lipid maturation in a tissue sample, such as a liver sample. Inhibition may be assessed by a decrease in an absolute or relative level of one or more of these variables compared with a control level. The control level may be any type of control level that is utilized in the art, e.g., a pre-dose baseline level, or a level determined from a similar subject, cell, or sample that is untreated or treated with a control (such as, e.g., buffer only control or inactive agent control).

In one embodiment, at least partial suppression of the expression of a CIDEB gene, is assessed by a reduction of

the amount of CIDEB mRNA which can be isolated from, or detected, in a first cell or group of cells in which a CIDEB gene is transcribed and which has or have been treated such that the expression of a CIDEB gene is inhibited, as compared to a second cell or group of cells substantially identical to the first cell or group of cells but which has or have not been so treated (control cells).

The degree of inhibition may be expressed in terms of:

$$\frac{(mRNA \text{ in control cells}) - (mRNA \text{ in treated cells})}{(mRNA \text{ in control cells})} \cdot 100\%$$

The phrase "contacting a cell with an RNAi agent," such as a dsRNA, as used herein, includes contacting a cell by any possible means. Contacting a cell with an RNAi agent includes contacting a cell in vitro with the iRNA or contacting a cell in vivo with the iRNA. The contacting may be done directly or indirectly. Thus, for example, the RNAi agent may be put into physical contact with the cell by the individual performing the method, or alternatively, the RNAi agent may be put into a situation that will permit or cause it to subsequently come into contact with the cell.

Contacting a cell in vitro may be done, for example, by incubating the cell with the RNAi agent. Contacting a cell in vivo may be done, for example, by injecting the RNAi agent into or near the tissue where the cell is located, or by injecting the RNAi agent into another area, e.g., the bloodstream or the subcutaneous space, such that the agent will subsequently reach the tissue where the cell to be contacted is located. For example, the RNAi agent may contain and/or be coupled to a ligand, e.g., GalNAc3, that directs the RNAi agent to a site of interest, e.g., the liver. Combinations of in vitro and in vivo methods of contacting are also possible. For example, a cell may also be contacted in vitro with an RNAi agent and subsequently transplanted into a subject.

In one embodiment, contacting a cell with an iRNA includes "introducing" or "delivering the iRNA into the cell" by facilitating or effecting uptake or absorption into the cell. Absorption or uptake of an iRNA can occur through unaided diffusive or active cellular processes, or by auxiliary agents or devices. Introducing an iRNA into a cell may be in vitro and/or in vivo. For example, for in vivo introduction, iRNA can be injected into a tissue site or administered systemically. In vivo delivery can also be done by a beta-glucan delivery system, such as those described in U.S. Pat. Nos. 5,032,401 and 5,607,677, and U.S. Publication No. 2005/0281781, the entire contents of which are hereby incorporated herein by reference. In vitro introduction into a cell includes methods known in the art such as electroporation and lipofection. Further approaches are described herein below and/or are known in the art.

The term "lipid nanoparticle" or "LNP" is a vesicle comprising a lipid layer encapsulating a pharmaceutically active molecule, such as a nucleic acid molecule, e.g., an iRNA or a plasmid from which an iRNA is transcribed. LNPs are described in, for example, U.S. Pat. Nos. 6,858,225, 6,815,432, 8,158,601, and 8,058,069, the entire contents of which are hereby incorporated herein by reference.

As used herein, a "subject" is an animal, such as a mammal, including a primate (such as a human, a non-human primate, e.g., a monkey, and a chimpanzee), a non-primate (such as a cow, a pig, a camel, a llama, a horse, a goat, a rabbit, a sheep, a hamster, a guinea pig, a cat, a dog, a rat, a mouse, a horse, and a whale), or a bird (e.g., a duck or a goose).

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In an embodiment, the subject is a human, such as a human being treated or assessed for a disease, disorder or condition that would benefit from reduction in CIDEB expression; a human at risk for a disease, disorder or condition that would benefit from reduction in CIDEB expression; a human having a disease, disorder or condition that would benefit from reduction in CIDEB expression; and/or human being treated for a disease, disorder or condition that would benefit from reduction in CIDEB expression as described herein.

As used herein, the terms “treating” or “treatment” refer to a beneficial or desired result including, but not limited to, alleviation or amelioration of one or more symptoms associated with CIDEB gene expression and/or CIDEB protein production, e.g., a CIDEB-associated disease, such as a chronic inflammatory disease of the liver and other tissues. In one embodiment, the chronic inflammatory disease is chronic inflammatory liver disease, e.g., inflammation of the liver, liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, hepatocellular carcinoma, insulin insensitivity and/or diabetes. “Treatment” can also mean prolonging survival as compared to expected survival in the absence of treatment.

The term “lower” in the context of a CIDEB-associated disease refers to a statistically significant decrease in such level. The decrease can be, for example, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or more. In certain embodiments, a decrease is at least 20%. “Lower” in the context of the level of CIDEB in a subject is preferably down to a level accepted as within the range of normal for an individual without such disorder.

As used herein, “prevention” or “preventing,” when used in reference to a disease, disorder or condition thereof, that would benefit from a reduction in expression of a CIDEB gene, refers to a reduction in the likelihood that a subject will develop a symptom associated with such disease, disorder, or condition, e.g., a symptom of CIDEB gene expression, such as inflammation of the liver, liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, hepatocellular carcinoma, insulin insensitivity and/or diabetes. The failure to develop a disease, disorder or condition, or the reduction in the development of a symptom associated with such a disease, disorder or condition (e.g., by at least about 10% on a clinically accepted scale for that disease or disorder), or the exhibition of delayed symptoms (e.g., reduction in inflammation, or reduction in lipid accumulation in the liver and/or lipid droplet expansion in the liver) delayed (e.g., by days, weeks, months or years) is considered effective prevention.

As used herein, the term “CIDEB-associated disease,” is a disease or disorder that is caused by, or associated with, CIDEB gene expression or CIDEB protein production. The term “CIDEB-associated disease” includes a disease, disorder or condition that would benefit from a decrease in CIDEB gene expression or protein activity.

In one embodiment, an “CIDEB-associated disease” is a chronic inflammatory disease. A “chronic inflammatory disease” is any disease, disorder, or condition associated with

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chronic inflammation. Non-limiting examples of a chronic inflammatory disease include, for example, inflammation of the liver and other tissues. Non-limiting examples of chronic inflammatory liver disease include, for example, fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, hepatocellular carcinoma, insulin insensitivity and/or diabetes.

“Therapeutically effective amount,” as used herein, is intended to include the amount of an RNAi agent that, when administered to a subject having a CIDEB-associated disease, disorder, or condition, is sufficient to effective treatment of the disease (e.g., by diminishing, ameliorating or maintaining the existing disease or one or more symptoms of disease). The “therapeutically effective amount” may vary depending on the RNAi agent, how the agent is administered, the disease and its severity and the history, age, weight, family history, genetic makeup, the types of preceding or concomitant treatments, if any, and other individual characteristics of the subject to be treated.

“Prophylactically effective amount,” as used herein, is intended to include the amount of an iRNA that, when administered to a subject having a CIDEB-associated disease, disorder, or condition, is sufficient to prevent or ameliorate the disease or one or more symptoms of the disease. Ameliorating the disease includes slowing the course of the disease or reducing the severity of later-developing disease. The “prophylactically effective amount” may vary depending on the iRNA, how the agent is administered, the degree of risk of disease, and the history, age, weight, family history, genetic makeup, the types of preceding or concomitant treatments, if any, and other individual characteristics of the patient to be treated.

A “therapeutically-effective amount” or “prophylactically effective amount” also includes an amount of an RNAi agent that produces some desired local or systemic effect at a reasonable benefit/risk ratio applicable to any treatment. iRNA employed in the methods of the present invention may be administered in a sufficient amount to produce a reasonable benefit/risk ratio applicable to such treatment.

The phrase “pharmaceutically acceptable” is employed herein to refer to those compounds, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human subjects and animal subjects without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

The phrase “pharmaceutically-acceptable carrier” as used herein means a pharmaceutically-acceptable material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, manufacturing aid (e.g., lubricant, talc magnesium, calcium or zinc stearate, or steric acid), or solvent encapsulating material, involved in carrying or transporting the subject compound from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be “acceptable” in the sense of being compatible with the other ingredients of the formulation and not injurious to the subject being treated. Some examples of materials which can serve as pharmaceutically-acceptable carriers include: (1) sugars, such as lactose, glucose and sucrose; (2) starches, such as corn starch and potato starch; (3) cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; (4) powdered tragacanth; (5) malt; (6) gelatin; (7) lubricating agents, such as magnesium

stearate, sodium lauryl sulfate and tale; (8) excipients, such as cocoa butter and suppository waxes; (9) oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; (10) glycols, such as propylene glycol; (11) polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; (12) esters, such as ethyl oleate and ethyl laurate; (13) agar; (14) buffering agents, such as magnesium hydroxide and aluminum hydroxide; (15) alginate; (16) pyrogen-free water; (17) isotonic saline; (18) Ringer's solution; (19) ethyl alcohol; (20) pH buffered solutions; (21) polyesters, polycarbonates and/or polyanhydrides; (22) bulking agents, such as polypeptides and amino acids (23) serum component, such as serum albumin, HDL and LDL; and (22) other non-toxic compatible substances employed in pharmaceutical formulations.

The term "sample," as used herein, includes a collection of similar fluids, cells, or tissues isolated from a subject, as well as fluids, cells, or tissues present within a subject. Examples of biological fluids include blood, serum and serosal fluids, plasma, cerebrospinal fluid, ocular fluids, lymph, urine, saliva, and the like. Tissue samples may include samples from tissues, organs or localized regions. For example, samples may be derived from particular organs, parts of organs, or fluids or cells within those organs. In certain embodiments, samples may be derived from the liver (e.g., whole liver or certain segments of liver or certain types of cells in the liver, such as, e.g., hepatocytes). In some embodiments, a "sample derived from a subject" refers to blood or plasma drawn from the subject.

II. iRNAs of the Invention

Described herein are iRNAs which inhibit the expression of a target gene. In one embodiment, the iRNAs inhibit the expression of a CIDEA gene. In one embodiment, the iRNA agent includes double stranded ribonucleic acid (dsRNA) molecules for inhibiting the expression of a CIDEA gene in a cell, such as a liver cell, such as a liver cell within a subject, e.g., a mammal, such as a human having a chronic inflammatory disease, disorder, or condition.

The dsRNA includes an antisense strand having a region of complementarity which is complementary to at least a part of an mRNA formed in the expression of a CIDEA gene. The region of complementarity is about 30 nucleotides or less in length (e.g., about 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, or 18 nucleotides or less in length). Upon contact with a cell expressing the target gene, the iRNA inhibits the expression of the target gene (e.g., a human, a primate, a non-primate, or a rodent target gene) by at least about 10% as assayed by, for example, a PCR or branched DNA (bDNA)-based method, or by a protein-based method, such as by immunofluorescence analysis, using, for example, Western Blotting or flowcytometric techniques.

A dsRNA includes two RNA strands that are complementary and hybridize to form a duplex structure under conditions in which the dsRNA will be used. One strand of a dsRNA (the antisense strand) includes a region of complementarity that is substantially complementary, and generally fully complementary, to a target sequence. The target sequence can be derived from the sequence of an mRNA formed during the expression of a CIDEA gene. The other strand (the sense strand) includes a region that is complementary to the antisense strand, such that the two strands hybridize and form a duplex structure when combined under suitable conditions. As described elsewhere herein and as known in the art, the complementary sequences of a dsRNA

can also be contained as self-complementary regions of a single nucleic acid molecule, as opposed to being on separate oligonucleotides.

Generally, the duplex structure is between 15 and 30 base pairs in length, e.g., between, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the invention.

Similarly, the region of complementarity to the target sequence is between 15 and 30 nucleotides in length, e.g., between 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the invention.

In some embodiments, the sense and antisense strands of the dsRNA are each independently about 15 to about 30 nucleotides in length, or about 25 to about 30 nucleotides in length, e.g., each strand is independently between 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length. In some embodiments, the dsRNA is between about 15 and about 23 nucleotides in length, or between about 25 and about 30 nucleotides in length. In general, the dsRNA is long enough to serve as a substrate for the Dicer enzyme. For example, it is well known in the art that dsRNAs longer than about 21-23 nucleotides can serve as substrates for Dicer. As the ordinarily skilled person will also recognize, the region of an RNA targeted for cleavage will most often be part of a larger RNA molecule, often an mRNA molecule. Where relevant, a "part" of an mRNA target is a contiguous sequence of an mRNA target of sufficient length to allow it to be a substrate for RNAi-directed cleavage (i.e., cleavage through a RISC pathway).

One of skill in the art will also recognize that the duplex region is a primary functional portion of a dsRNA, e.g., a duplex region of about 9 to 36 base pairs, e.g., about 10-36, 11-36, 12-36, 13-36, 14-36, 15-36, 9-35, 10-35, 11-35, 12-35, 13-35, 14-35, 15-35, 9-34, 10-34, 11-34, 12-34, 13-34, 14-34, 15-34, 9-33, 10-33, 11-33, 12-33, 13-33, 14-33, 15-33, 9-32, 10-32, 11-32, 12-32, 13-32, 14-32, 15-32, 9-31, 10-31, 11-31, 12-31, 13-31, 14-31, 15-31, 15-30, 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs. Thus, in one embodiment, to the extent that it becomes processed to a functional duplex, of e.g., 15-30 base pairs,

that targets a desired RNA for cleavage, an RNA molecule or complex of RNA molecules having a duplex region greater than 30 base pairs is a dsRNA. Thus, an ordinarily skilled artisan will recognize that in one embodiment, a MicroRNA (miRNA) is a dsRNA. In another embodiment, a dsRNA is not a naturally occurring miRNA. In another embodiment, an iRNA agent useful to target CIDEB expression is not generated in the target cell by cleavage of a larger dsRNA.

A dsRNA as described herein can further include one or more single-stranded nucleotide overhangs e.g., 1, 2, 3, or 4 nucleotides. dsRNAs having at least one nucleotide overhang can have unexpectedly superior inhibitory properties relative to their blunt-ended counterparts. A nucleotide overhang can comprise or consist of a nucleotide/nucleoside analog, including a deoxynucleotide/nucleoside. The overhang(s) can be on the sense strand, the antisense strand or any combination thereof. Furthermore, the nucleotide(s) of an overhang can be present on the 5'-end, 3'-end or both ends of either an antisense or sense strand of a dsRNA.

A dsRNA can be synthesized by standard methods known in the art as further discussed below, e.g., by use of an automated DNA synthesizer, such as are commercially available from, for example, Biosearch, Applied Biosystems, Inc.

iRNA compounds of the invention may be prepared using a two-step procedure. First, the individual strands of the double-stranded RNA molecule are prepared separately. Then, the component strands are annealed. The individual strands of the siRNA compound can be prepared using solution-phase or solid-phase organic synthesis or both. Organic synthesis offers the advantage that the oligonucleotide strands comprising unnatural or modified nucleotides can be easily prepared. Single-stranded oligonucleotides of the invention can be prepared using solution-phase or solid-phase organic synthesis or both.

In one aspect, a dsRNA of the invention includes at least two nucleotide sequences, a sense sequence and an antisense sequence. The sense strand sequence is selected from the group of sequences provided in Tables 3-6, and the corresponding nucleotide sequence of the antisense strand of the sense strand is selected from the group of sequences of Tables 3-6. In this aspect, one of the two sequences is complementary to the other of the two sequences, with one of the sequences being substantially complementary to a sequence of an mRNA generated in the expression of a CIDEB gene. As such, in this aspect, a dsRNA will include two oligonucleotides, where one oligonucleotide is described as the sense strand (passenger strand) in Tables 3-6, and the second oligonucleotide is described as the corresponding antisense strand (guide strand) of the sense strand in Tables 3-6. In one embodiment, the substantially complementary sequences of the dsRNA are contained on separate oligonucleotides. In another embodiment, the substantially complementary sequences of the dsRNA are contained on a single oligonucleotide.

It will be understood that, although the sequences in Tables 3-6 are described as modified, unmodified, unconjugated, and/or conjugated sequences, the RNA of the iRNA of the invention e.g., a dsRNA of the invention, may comprise any one of the sequences set forth Tables 3-6 that is unmodified, unconjugated, and/or modified and/or conjugated differently than described therein.

The skilled person is well aware that dsRNAs having a duplex structure of between about 20 and 23 base pairs, e.g., 21, base pairs have been hailed as particularly effective in inducing RNA interference (Elbashir et al., (2001) *EMBO J.*,

20:6877-6888). However, others have found that shorter or longer RNA duplex structures can also be effective (Chu and Rana (2007) *RNA* 14:1714-1719; Kim et al. (2005) *Nat Biotech* 23:222-226). In the embodiments described above, by virtue of the nature of the oligonucleotide sequences provided herein, dsRNAs described herein can include at least one strand of a length of minimally 21 nucleotides. It can be reasonably expected that shorter duplexes minus only a few nucleotides on one or both ends can be similarly effective as compared to the dsRNAs described above. Hence, dsRNAs having a sequence of at least 15, 16, 17, 18, 19, 20, or more contiguous nucleotides derived from one of the sequences provided herein, and differing in their ability to inhibit the expression of a CIDEB gene by not more than about 5, 10, 15, 20, 25, or 30% inhibition from a dsRNA comprising the full sequence, are contemplated to be within the scope of the present invention.

In addition, the RNAs described in Tables 3-6 identify a site(s) in a CIDEB transcript that is susceptible to RISC-mediated cleavage. As such, the present invention further features iRNAs that target within this site(s). As used herein, an iRNA is said to target within a particular site of an RNA transcript if the iRNA promotes cleavage of the transcript anywhere within that particular site. Such an iRNA will generally include at least about 15 contiguous nucleotides from one of the sequences provided herein coupled to additional nucleotide sequences taken from the region contiguous to the selected sequence in the gene.

While a target sequence is generally about 15-30 nucleotides in length, there is wide variation in the suitability of particular sequences in this range for directing cleavage of any given target RNA. Various software packages and the guidelines set out herein provide guidance for the identification of optimal target sequences for any given gene target, but an empirical approach can also be taken in which a "window" or "mask" of a given size (as a non-limiting example, 21 nucleotides) is literally or figuratively (including, e.g., in silico) placed on the target RNA sequence to identify sequences in the size range that can serve as target sequences. By moving the sequence "window" progressively one nucleotide upstream or downstream of an initial target sequence location, the next potential target sequence can be identified, until the complete set of possible sequences is identified for any given target size selected. This process, coupled with systematic synthesis and testing of the identified sequences (using assays as described herein or as known in the art) to identify those sequences that perform optimally can identify those RNA sequences that, when targeted with an iRNA agent, mediate the best inhibition of target gene expression. Thus, while the sequences identified herein represent effective target sequences, it is contemplated that further optimization of inhibition efficiency can be achieved by progressively "walking the window" one nucleotide upstream or downstream of the given sequences to identify sequences with equal or better inhibition characteristics.

Further, it is contemplated that for any sequence identified herein, further optimization could be achieved by systematically either adding or removing nucleotides to generate longer or shorter sequences and testing those sequences generated by walking a window of the longer or shorter size up or down the target RNA from that point. Again, coupling this approach to generating new candidate targets with testing for effectiveness of iRNAs based on those target sequences in an inhibition assay as known in the art and/or as described herein can lead to further improvements in the efficiency of inhibition. Further still, such optimized

sequences can be adjusted by, e.g., the introduction of modified nucleotides as described herein or as known in the art, addition or changes in overhang, or other modifications as known in the art and/or discussed herein to further optimize the molecule (e.g., increasing serum stability or circulating half-life, increasing thermal stability, enhancing transmembrane delivery, targeting to a particular location or cell type, increasing interaction with silencing pathway enzymes, increasing release from endosomes) as an expression inhibitor.

An iRNA agent as described herein can contain one or more mismatches to the target sequence. In one embodiment, an iRNA as described herein contains no more than 3 mismatches. If the antisense strand of the iRNA contains mismatches to a target sequence, it is preferable that the area of mismatch is not located in the center of the region of complementarity. If the antisense strand of the iRNA contains mismatches to the target sequence, it is preferable that the mismatch be restricted to be within the last 5 nucleotides from either the 5'- or 3'-end of the region of complementarity. For example, for a 23 nucleotide iRNA agent the strand which is complementary to a region of a CIDEB gene, generally does not contain any mismatch within the central 13 nucleotides. The methods described herein or methods known in the art can be used to determine whether an iRNA containing a mismatch to a target sequence is effective in inhibiting the expression of a CIDEB gene. Consideration of the efficacy of iRNAs with mismatches in inhibiting expression of a CIDEB gene is important, especially if the particular region of complementarity in a CIDEB gene is known to have polymorphic sequence variation within the population.

An RNA target may have regions, or spans of the target RNA's nucleotide sequence, which are relatively more susceptible or amenable than other regions of the RNA target to mediating cleavage of the RNA target via RNA interference induced by the binding of an RNAi agent to that region. The increased susceptibility to RNA interference within such "hotspot regions" (or simply "hotspots") means that iRNA agents targeting the region will likely have higher efficacy in inducing iRNA interference than iRNA agents which target other regions of the target RNA. For example, without being bound by theory, the accessibility of a target region of a target RNA may influence the efficacy of iRNA agents which target that region, with some hotspot regions having increased accessibility. Secondary structures, for instance, that form in the RNA target (e.g., within or proximate to hotspot regions) may affect the ability of the iRNA agent to bind the target region and induce RNA interference.

According to certain aspects of the invention, an iRNA agent may be designed to target a hotspot region of any of the target RNAs described herein, including any identified portions of a target RNA (e.g., a particular exon). As used herein, a hotspot region may refer to an approximately 19-200, 19-150, 19-100, 19-75, 19-50, 21-200, 21-150, 21-100, 21-75, 21-50, 50-200, 50-150, 50-100, 50-75, 75-200, 75-150, 75-100, 100-200, or 100-150 nucleotide region of a target RNA sequence for which targeting using RNAi agents provides an observably higher probability of efficacious silencing relative to targeting other regions of the same target RNA. According to certain aspects of the invention, a hotspot region may comprise a limited region of the target RNA, and in some cases, a substantially limited region of the target, including for example, less than half of the length of the target RNA, such as about 5%, 10%, 15%, 20%, 25%, or 30% of the length of the target RNA. Conversely, the other regions against which a hotspot is

compared may cumulatively comprise at least a majority of the length of the target RNA. For example, the other regions may cumulatively comprise at least about 60%, or at least about 70%, or at least about 80%, or at least about 90%, or at least about 95% of the length of the target RNA.

Compared regions of the target RNA may be empirically evaluated for identification of hotspots using efficacy data obtained from in vitro or in vivo screening assays. For example, RNAi agents targeting various regions that span a target RNA may be compared for frequency of efficacious iRNA agents (e.g., the amount by which target gene expression is inhibited, such as measured by mRNA expression or protein expression) that bind each region. In general, a hotspot can be recognized by observing clustering of multiple efficacious RNAi agents that bind to a limited region of the RNA target. A hotspot may be sufficiently characterized as such by observing efficacy of iRNA agents which cumulatively span at least about 60% of the target region identified as a hotspot, such as about 70%, about 80%, about 90%, or about 95% or more of the length of the region, including both ends of the region (i.e. at least about 60%, 70%, 80%, 90%, or 95% or more of the nucleotides within the region, including the nucleotides at each end of the region, were targeted by an iRNA agent). According to some aspects of the invention, an iRNA agent which demonstrates at least about 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% inhibition over the region (e.g., no more than about 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, or 5% mRNA remaining) may be identified as efficacious.

Amenability to targeting of RNA regions may also be assessed using quantitative comparison of inhibition measurements across different regions of a defined size (e.g., 25, 30, 40, 50, 60, 70, 80, 90, or 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, or 200 nts). For example, an average level of inhibition may be determined for each region and the averages of each region may be compared. The average level of inhibition within a hotspot region may be substantially higher than the average of averages for all evaluated regions. According to some aspects, the average level of inhibition in a hotspot region may be at least about 10%, 20%, 30%, 40%, or 50% higher than the average of averages. According to some aspects, the average level of inhibition in a hotspot region may be at least about 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, or 2.0 standard deviations above the average of averages. The average level of inhibition may be higher by a statistically significant (e.g., $p < 0.05$) amount. According to some aspects, each inhibition measurement within a hotspot region may be above a threshold amount (e.g., at or below a threshold amount of mRNA remaining). According to some aspects, each inhibition measurement within the region may be substantially higher than an average of all inhibition measurements across all the measured regions. For example, each inhibition measurement in a hotspot region may be at least about 10%, 20%, 30%, 40%, or 50% higher than the average of all inhibition measurements. According to some aspects, each inhibition measurement may be at least about 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, or 2.0 standard deviations above the average of all inhibition measurements. Each inhibition measurement may be higher by a statistically significant (e.g., $p < 0.05$) amount than the average of all inhibition measurements. A standard for evaluating a hotspot may comprise various combinations of the above standards where compatible (e.g., an average level of inhibition of at least about a first amount and having no inhibition measurements below a threshold level of a second amount, lesser than the first amount).

It is therefore expressly contemplated that any iRNA agent, including the specific exemplary iRNA agents described herein, which targets a hotspot region of a target RNA, may be preferably selected for inducing RNA interference of the target mRNA as targeting such a hotspot region is likely to exhibit a robust inhibitory response relative to targeting a region which is not a hotspot region. RNAi agents targeting target sequences that substantially overlap (e.g., by at least about 70%, 75%, 80%, 85%, 90%, 95% of the target sequence length) or, preferably, that reside fully within the hotspot region may be considered to target the hotspot region. Hotspot regions of the RNA target(s) of the instant invention may include any region for which the data disclosed herein demonstrates higher frequency of targeting by efficacious RNAi agents, including by any of the standards described elsewhere herein, whether or not the range(s) of such hotspot region(s) are explicitly specified.

In various embodiments, a dsRNA agent of the present invention targets a hotspot region of an mRNA encoding CIDEB.

III. Modified iRNAs of the Invention

In one embodiment, the RNA of the iRNA of the invention e.g., a dsRNA, is un-modified, and does not comprise, e.g., chemical modifications and/or conjugations known in the art and described herein. In another embodiment, the RNA of an iRNA of the invention, e.g., a dsRNA, is chemically modified to enhance stability or other beneficial characteristics. In certain embodiments of the invention, substantially all of the nucleotides of an iRNA of the invention are modified. In other embodiments of the invention, all of the nucleotides of an iRNA of the invention are modified. iRNAs of the invention in which "substantially all of the nucleotides are modified" are largely but not wholly modified and can include not more than 5, 4, 3, 2, or 1 unmodified nucleotides.

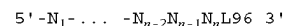
In some aspects of the invention, substantially all of the nucleotides of an iRNA of the invention are modified and the iRNA agents comprise no more than 10 nucleotides comprising 2'-fluoro modifications (e.g., no more than 9 2'-fluoro modifications, no more than 8 2'-fluoro modifications, no more than 7 2'-fluoro modifications, no more than 6 2'-fluoro modifications, no more than 5 2'-fluoro modifications, no more than 4 2'-fluoro modifications, no more than 3 2'-fluoro modifications, or no more than 2 2'-fluoro modifications). For example, in some embodiments, the sense strand comprises no more than 4 nucleotides comprising 2'-fluoro modifications (e.g., no more than 3 2'-fluoro modifications, or no more than 2 2'-fluoro modifications). In other embodiments, the antisense strand comprises no more than 6 nucleotides comprising 2'-fluoro modifications (e.g., no more than 5 2'-fluoro modifications, no more than 4 2'-fluoro modifications, no more than 3 2'-fluoro modifications, or no more than 2 2'-fluoro modifications).

In other aspects of the invention, all of the nucleotides of an iRNA of the invention are modified and the iRNA agents comprise no more than 10 nucleotides comprising 2'-fluoro modifications (e.g., no more than 9 2'-fluoro modifications, no more than 8 2'-fluoro modifications, no more than 7 2'-fluoro modifications, no more than 6 2'-fluoro modifications, no more than 5 2'-fluoro modifications, no more than 4 2'-fluoro modifications, no more than 3 2'-fluoro modifications, or no more than 2 2'-fluoro modifications).

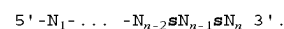
cations, no more than 4 2'-fluoro modifications, no more than 3 2'-fluoro modifications, or no more than 2 2'-fluoro modifications).

In one embodiment, the double stranded RNAi agent of the invention further comprises a 5'-phosphate or a 5'-phosphate mimic at the 5' nucleotide of the antisense strand. In another embodiment, the double stranded RNAi agent further comprises a 5'-phosphate mimic at the 5' nucleotide of the antisense strand. In a specific embodiment, the 5'-phosphate mimic is a 5'-vinyl phosphonate (5'-VP). In one embodiment, the phosphate mimic is a 5'-cyclopropyl phosphonate. In some embodiments, the 5'-end of the antisense strand of the double-stranded iRNA agent does not contain a 5'-vinyl phosphonate (VP).

In one embodiment, at least one of the modified nucleotides is selected from the group consisting of a deoxy-nucleotide, a 2'-O-methyl modified nucleotide, a 2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, a glycol modified nucleotide (GNA), e.g., Ggn, Cgn, Tgn, or Agn, a nucleotide with a 2' phosphate, e.g., G2p, C2p, A2p or U2p, and, a vinyl-phosphonate nucleotide; and combinations thereof. In other embodiments, each of the duplexes of Tables 4 and 6 may be particularly modified to provide another double-stranded iRNA agent of the present disclosure. In one example, the 3'-terminus of each sense duplex may be modified by removing the 3'-terminal L96 ligand and exchanging the two phosphodiester internucleotide linkages between the three 3'-terminal nucleotides with phosphorothioate internucleotide linkages. That is, the three 3'-terminal nucleotides (N) of a sense sequence of the formula:



may be replaced with



That is, for example, AD-1685156, the sense sequence:

(SEQ ID NO: 277)
csusgcagAfaGfGfUfugacugc**guuL96**

may be replaced with

(SEQ ID NO: 2922)
csusgcagAfaGfGfUfugacugc**gsusu**

while the antisense sequence remains unchanged to provide another double-stranded iRNA agent of the present disclosure.

The nucleic acids featured in the invention can be synthesized and/or modified by methods well established in the art, such as those described in "Current protocols in nucleic acid chemistry," Beaucage, S. L. et al. (Edrs.), John Wiley & Sons, Inc., New York, NY, USA, which is hereby incorporated herein by reference. Modifications include, for example, end modifications, e.g., 5'-end modifications (phosphorylation, conjugation, inverted linkages) or 3'-end modifications (conjugation, DNA nucleotides, inverted linkages, etc.); base modifications, e.g., replacement with stabilizing bases, destabilizing bases, or bases that base pair with an expanded repertoire of partners, removal of bases (abasic nucleotides), or conjugated bases; sugar modifications (e.g., at the 2'-position or 4'-position) or replacement of the sugar; and/or backbone modifications, including modi-

fication or replacement of the phosphodiester linkages. Specific examples of iRNA compounds useful in the embodiments described herein include, but are not limited to RNAs containing modified backbones or no natural internucleoside linkages. RNAs having modified backbones include, among others, those that do not have a phosphorus atom in the backbone. For the purposes of this specification, and as sometimes referenced in the art, modified RNAs that do not have a phosphorus atom in their internucleoside backbone can also be considered to be oligonucleosides. In some embodiments, a modified iRNA will have a phosphorus atom in its internucleoside backbone.

Modified RNA backbones include, for example, phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates including 3'-alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates including 3'-amino phosphoramidate and aminoalkylphosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, and boranophosphates having normal 3'-5' linkages, 2'-5'-linked analogs of these, and those having inverted polarity wherein the adjacent pairs of nucleoside units are linked 3'-5' to 5'-3' or 2'-5' to 5'-2'. Various salts, mixed salts and free acid forms are also included. In some embodiments of the invention, the dsRNA agents of the invention are in a free acid form. In other embodiments of the invention, the dsRNA agents of the invention are in a salt form. In one embodiment, the dsRNA agents of the invention are in a sodium salt form. In certain embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for substantially all of the phosphodiester and/or phosphorothioate groups present in the agent. Agents in which substantially all of the phosphodiester and/or phosphorothioate linkages have a sodium counterion include not more than 5, 4, 3, 2, or 1 phosphodiester and/or phosphorothioate linkages without a sodium counterion. In some embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for all of the phosphodiester and/or phosphorothioate groups present in the agent.

Representative U.S. patents that teach the preparation of the above phosphorus-containing linkages include, but are not limited to, U.S. Pat. Nos. 3,687,808; 4,469,863; 4,476,301; 5,023,243; 5,177,195; 5,188,897; 5,264,423; 5,276,019; 5,278,302; 5,286,717; 5,321,131; 5,399,676; 5,405,939; 5,453,496; 5,455,233; 5,466,677; 5,476,925; 5,519,126; 5,536,821; 5,541,316; 5,550,111; 5,563,253; 5,571,799; 5,587,361; 5,625,050; 6,028,188; 6,124,445; 6,160,109; 6,169,170; 6,172,209; 6,239,265; 6,277,603; 6,326,199; 6,346,614; 6,444,423; 6,531,590; 6,534,639; 6,608,035; 6,683,167; 6,858,715; 6,867,294; 6,878,805; 7,015,315; 7,041,816; 7,273,933; 7,321,029; and U.S. Pat. RE39464, the entire contents of each of which are hereby incorporated herein by reference.

Modified RNA backbones that do not include a phosphorus atom therein have backbones that are formed by short chain alkyl or cycloalkyl internucleoside linkages, mixed heteroatoms and alkyl or cycloalkyl internucleoside linkages, or one or more short chain heteroatomic or heterocyclic internucleoside linkages. These include those having morpholino linkages (formed in part from the sugar portion of a nucleoside); siloxane backbones; sulfide, sulfoxide and sulfone backbones; formacetyl and thioformacetyl backbones; methylene formacetyl and thioformacetyl backbones; alkene containing backbones; sulfamate backbones; methyleneimino and methylenehydrazino backbones; sulfonate

and sulfonamide backbones; amide backbones; and others having mixed N, O, S and CH₂ component parts.

Representative U.S. patents that teach the preparation of the above oligonucleosides include, but are not limited to, U.S. Pat. Nos. 5,034,506; 5,166,315; 5,185,444; 5,214,134; 5,216,141; 5,235,033; 5,64,562; 5,264,564; 5,405,938; 5,434,257; 5,466,677; 5,470,967; 5,489,677; 5,541,307; 5,561,225; 5,596,086; 5,602,240; 5,608,046; 5,610,289; 5,618,704; 5,623,070; 5,663,312; 5,633,360; 5,677,437; and, 5,677,439, the entire contents of each of which are hereby incorporated herein by reference.

In other embodiments, suitable RNA mimetics are contemplated for use in iRNAs, in which both the sugar and the internucleoside linkage, i.e., the backbone, of the nucleotide units are replaced with novel groups. The base units are maintained for hybridization with an appropriate nucleic acid target compound. One such oligomeric compound, an RNA mimetic that has been shown to have excellent hybridization properties, is referred to as a peptide nucleic acid (PNA). In PNA compounds, the sugar backbone of an RNA is replaced with an amide containing backbone, in particular an aminoethylglycine backbone. The nucleobases are retained and are bound directly or indirectly to aza nitrogen atoms of the amide portion of the backbone. Representative U.S. patents that teach the preparation of PNA compounds include, but are not limited to, U.S. Pat. Nos. 5,539,082; 5,714,331; and 5,719,262, the entire contents of each of which are hereby incorporated herein by reference. Additional PNA compounds suitable for use in the iRNAs of the invention are described in, for example, in Nielsen et al., *Science*, 1991, 254, 1497-1500.

Some embodiments featured in the invention include RNAs with phosphorothioate backbones and oligonucleosides with heteroatom backbones, and in particular —CH₂—NH—CH₂—, —CH₂—N(CH₃)—O—CH₂—[known as a methylene (methylimino) or MMI backbone], —CH₂—O—N(CH₃)—CH₂—, —CH₂—N(CH₃)—N(CH₃)—CH₂— and —N(CH₃)—CH₂—CH₂—[wherein the native phosphodiester backbone is represented as —O—P—O—CH₂—] of the above-referenced U.S. Pat. No. 5,489,677, and the amide backbones of the above-referenced U.S. Pat. No. 5,602,240. In some embodiments, the RNAs featured herein have morpholino backbone structures of the above-referenced U.S. Pat. No. 5,034,506.

Modified RNAs can also contain one or more substituted sugar moieties. The iRNAs, e.g., dsRNAs, featured herein can include one of the following at the 2'-position: OH; F; O—, S—, or N-alkyl; O—, S—, or N-alkenyl; O—, S- or N-alkynyl; or O-alkyl-O-alkyl, wherein the alkyl, alkenyl and alkynyl can be substituted or unsubstituted C₁ to C₁₀ alkyl or C₂ to C₁₀ alkenyl and alkynyl. Exemplary suitable modifications include O[(CH₂)_nO]_mCH₃, O(CH₂)_nOCH₃, O(CH₂)_nNH₂, O(CH₂)_nCH₃, O(CH₂)_nONH₂, and O(CH₂)_nON[(CH₂)_nCH₃]₂, where n and m are from 1 to about 10. In other embodiments, dsRNAs include one of the following at the 2' position: C₁ to C₁₀ lower alkyl, substituted lower alkyl, alkaryl, aralkyl, O-alkaryl or O-aralkyl, SH, SCH₃, OCN, Cl, Br, CN, CF₃, OCF₃, SOCH₃, SO₂CH₃, ONO₂, NO₂, N₃, NH₂, heterocycloalkyl, heterocycloalkaryl, aminoalkylamino, polyalkylamino, substituted silyl, an RNA cleaving group, a reporter group, an intercalator, a group for improving the pharmacokinetic properties of an iRNA, or a group for improving the pharmacodynamic properties of an iRNA, and other substituents having similar properties. In some embodiments, the modification includes a 2'-methoxyethoxy (2'-O—CH₂CH₂OCH₃, also known as 2'-O-(2-methoxyethyl) or 2'-MOE) (Martin et al., *Helv.*

Chim. Acta, 1995, 78:486-504) i.e., an alkoxy-alkoxy group. Another exemplary modification is 2'-dimethylaminoethoxyethoxy, i.e., a $O(CH_2)_2ON(CH_3)_2$ group, also known as 2'-DMAOE, as described in examples herein below, and 2'-dimethylaminoethoxyethoxy (also known in the art as 2'-O-dimethylaminoethoxyethyl or 2'-DMAEOE), i.e., $2'-O-CH_2-O-CH_2-N(CH_3)_2$. Further exemplary modifications include: 5'-Me-2'-F nucleotides, 5'-Me-2'-OME nucleotides, 5'-Me-2'-deoxynucleotides, (both R and S isomers in these three families); 2'-alkoxyalkyl; and 2'-NMA (N-methylacetamide).

Other modifications include 2'-methoxy (2'-OCH₃), 2'-aminopropoxy (2'-OCH₂CH₂CH₂NH₂) and 2'-fluoro (2'-F). Similar modifications can also be made at other positions on the RNA of an iRNA, particularly the 3' position of the sugar on the 3' terminal nucleotide or in 2'-5' linked dsRNAs and the 5' position of 5' terminal nucleotide. iRNAs can also have sugar mimetics such as cyclobutyl moieties in place of the pentofuranosyl sugar. Representative U.S. patents that teach the preparation of such modified sugar structures include, but are not limited to, U.S. Pat. Nos. 4,981,957; 5,118,800; 5,319,080; 5,359,044; 5,393,878; 5,446,137; 5,466,786; 5,514,785; 5,519,134; 5,567,811; 5,576,427; 5,591,722; 5,597,909; 5,610,300; 5,627,053; 5,639,873; 5,646,265; 5,658,873; 5,670,633; and 5,700,920, certain of which are commonly owned with the instant application. The entire contents of each of the foregoing are hereby incorporated herein by reference.

An iRNA of the invention can also include nucleobase (often referred to in the art simply as "base") modifications or substitutions. As used herein, "unmodified" or "natural" nucleobases include the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C) and uracil (U). Modified nucleobases include other synthetic and natural nucleobases such as 5-methylcytosine (5-me-C), 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-halouracil and cytosine, 5-propynyl uracil and cytosine, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl and other 8-substituted adenines and guanines, 5-halo, particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine and 7-daazaadenine and 3-deazaguanine and 3-deazaadenine. Further nucleobases include those disclosed in U.S. Pat. No. 3,687,808, those disclosed in Modified Nucleosides in Biochemistry, Biotechnology and Medicine, Herdewijn, P. ed. Wiley-VCH, 2008; those disclosed in The Concise Encyclopedia Of Polymer Science And Engineering, pages 858-859, Kroschwitz, J. L., ed. John Wiley & Sons, 1990, those disclosed by Englisch et al., (1991) *Angewandte Chemie, International Edition*, 30:613, and those disclosed by Sanghvi, Y. S., Chapter 15, dsRNA Research and Applications, pages 289-302, Crooke, S. T. and Lebleu, B., Ed., CRC Press, 1993. Certain of these nucleobases are particularly useful for increasing the binding affinity of the oligomeric compounds featured in the invention. These include 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and O-6 substituted purines, including 2-aminopropyladenine, 5-propynyluracil and 5-propynylcytosine. 5-methylcytosine substitutions have been shown to increase nucleic acid duplex stability by 0.6-1.2° C. (Sanghvi, Y. S., Crooke, S. T. and Lebleu, B., Eds., dsRNA Research and Applications, CRC Press, Boca

Raton, 1993, pp. 276-278) and are exemplary base substitutions, even more particularly when combined with 2'-O-methoxyethyl sugar modifications.

Representative U.S. patents that teach the preparation of certain of the above noted modified nucleobases as well as other modified nucleobases include, but are not limited to, the above noted U.S. Pat. Nos. 3,687,808, 4,845,205; 5,130,30; 5,134,066; 5,175,273; 5,367,066; 5,432,272; 5,457,187; 5,459,255; 5,484,908; 5,502,177; 5,525,711; 5,552,540; 5,587,469; 5,594,121, 5,596,091; 5,614,617; 5,681,941; 5,750,692; 6,015,886; 6,147,200; 6,166,197; 6,222,025; 6,235,887; 6,380,368; 6,528,640; 6,639,062; 6,617,438; 7,045,610; 7,427,672; and 7,495,088, the entire contents of each of which are hereby incorporated herein by reference.

An iRNA of the invention can also be modified to include one or more locked nucleic acids (LNA). A locked nucleic acid is a nucleotide having a modified ribose moiety in which the ribose moiety comprises an extra bridge connecting the 2' and 4' carbons. This structure effectively "locks" the ribose in the 3'-endo structural conformation. The addition of locked nucleic acids to siRNAs has been shown to increase siRNA stability in serum, and to reduce off-target effects (Elmen, J. et al., (2005) *Nucleic Acids Research* 33(1):439-447; Mook, OR. et al., (2007) *Mol Canc Ther* 6(3):833-843; Grunweller, A. et al., (2003) *Nucleic Acids Research* 31(12):3185-3193).

An iRNA of the invention can also be modified to include one or more bicyclic sugar moieties. A "bicyclic sugar" is a furanosyl ring modified by the bridging of two atoms. A "bicyclic nucleoside" ("BNA") is a nucleoside having a sugar moiety comprising a bridge connecting two carbon atoms of the sugar ring, thereby forming a bicyclic ring system. In certain embodiments, the bridge connects the 4'-carbon and the 2'-carbon of the sugar ring. Thus, in some embodiments an agent of the invention may include one or more locked nucleic acids (LNA). A locked nucleic acid is a nucleotide having a modified ribose moiety in which the ribose moiety comprises an extra bridge connecting the 2' and 4' carbons. In other words, an LNA is a nucleotide comprising a bicyclic sugar moiety comprising a 4'-CH₂-O-2' bridge. This structure effectively "locks" the ribose in the 3'-endo structural conformation. The addition of locked nucleic acids to siRNAs has been shown to increase siRNA stability in serum, and to reduce off-target effects (Elmen, J. et al., (2005) *Nucleic Acids Research* 33(1):439-447; Mook, OR. et al., (2007) *Mol Canc Ther* 6(3):833-843; Grunweller, A. et al., (2003) *Nucleic Acids Research* 31(12):3185-3193). Examples of bicyclic nucleosides for use in the polynucleotides of the invention include without limitation nucleosides comprising a bridge between the 4' and the 2' ribosyl ring atoms. In certain embodiments, the antisense polynucleotide agents of the invention include one or more bicyclic nucleosides comprising a 4' to 2' bridge. Examples of such 4' to 2' bridged bicyclic nucleosides, include but are not limited to 4'-(CH₂)-O-2' (LNA); 4'-(CH₂)-S-2'; 4'-(CH₂)-O-2' (ENA); 4'-CH(CH₃)-O-2' (also referred to as "constrained ethyl" or "cEt") and 4'-CH(CH₂OCH₃)-O-2' (and analogs thereof; see, e.g., U.S. Pat. No. 7,399,845); 4'-C(CH₃)(CH₃)-O-2' (and analogs thereof; see e.g., U.S. Pat. No. 8,278,283); 4'-CH₂-N(OCH₃)-2' (and analogs thereof; see e.g., U.S. Pat. No. 8,278,425); 4'-CH₂-O—N(CH₃)-2' (see, e.g., U.S. Patent Publication No. 2004/0171570); 4'-CH₂-N(R)-O-2', wherein R is H, C1-C12 alkyl, or a protecting group (see, e.g., U.S. Pat. No. 7,427,672); 4'-CH₂-C(H)(CH₃)-2' (see, e.g., Chattopadhyaya et al., *J. Org. Chem.*, 2009, 74, 118-134); and 4'-CH₂-C(=CH₂)-2'

(and analogs thereof; see, e.g., U.S. Pat. No. 8,278,426). The entire contents of each of the foregoing are hereby incorporated herein by reference.

Additional representative U.S. Patents and US Patent Publications that teach the preparation of locked nucleic acid nucleotides include, but are not limited to, the following: U.S. Pat. Nos. 6,268,490; 6,525,191; 6,670,461; 6,770,748; 6,794,499; 6,998,484; 7,053,207; 7,034,133; 7,084,125; 7,399,845; 7,427,672; 7,569,686; 7,741,457; 8,022,193; 8,030,467; 8,278,425; 8,278,426; 8,278,283; US 2008/0039618; and US 2009/0012281, the entire contents of each of which are hereby incorporated herein by reference.

Any of the foregoing bicyclic nucleosides can be prepared having one or more stereochemical sugar configurations including for example α -L-ribofuranose and β -D-ribofuranose (see WO 99/14226).

An iRNA of the invention can also be modified to include one or more constrained ethyl nucleotides. As used herein, a "constrained ethyl nucleotide" or "cEt" is a locked nucleic acid comprising a bicyclic sugar moiety comprising a 4'-CH (CH₃)-O-2' bridge. In one embodiment, a constrained ethyl nucleotide is in the S conformation referred to herein as "S-cEt."

An iRNA of the invention may also include one or more "conformationally restricted nucleotides" ("CRN"). CRN are nucleotide analogs with a linker connecting the C2' and C4' carbons of ribose or the C3 and —C5' carbons of ribose. CRN lock the ribose ring into a stable conformation and increase the hybridization affinity to mRNA. The linker is of sufficient length to place the oxygen in an optimal position for stability and affinity resulting in less ribose ring puckering.

Representative publications that teach the preparation of certain of the above noted CRN include, but are not limited to, US Patent Publication No. 2013/0190383; and PCT publication WO 2013/036868, the entire contents of each of which are hereby incorporated herein by reference.

In some embodiments, an iRNA of the invention comprises one or more monomers that are UNA (unlocked nucleic acid) nucleotides. UNA is unlocked acyclic nucleic acid, wherein any of the bonds of the sugar has been removed, forming an unlocked "sugar" residue. In one example, UNA also encompasses monomer with bonds between C1'-C4' have been removed (i.e. the covalent carbon-oxygen-carbon bond between the C1' and C4' carbons). In another example, the C2'-C3' bond (i.e. the covalent carbon-carbon bond between the C2' and C3' carbons) of the sugar has been removed (see *Nuc. Acids Symp. Series*, 52, 133-134 (2008) and Fluiter et al., *Mol. Biosyst.*, 2009, 10, 1039 hereby incorporated by reference).

Representative U.S. publications that teach the preparation of UNA include, but are not limited to, U.S. Pat. No. 8,314,227; and US Patent Publication Nos. 2013/0096289; 2013/0011922; and 2011/0313020, the entire contents of each of which are hereby incorporated herein by reference.

An RNAi agent of the disclosure may also include one or more "cyclohexene nucleic acids" or ("CeNA"). CeNA are nucleotide analogs with a replacement of the furanose moiety of DNA by a cyclohexene ring. Incorporation of cyclohexenyl nucleosides in a DNA chain increases the stability of a DNA/RNA hybrid. CeNA is stable against degradation in serum and a CeNA/RNA hybrid is able to activate *E. Coli* RNase H, resulting in cleavage of the RNA strand. (see Wang et al., *Am. Chem. Soc.* 2000, 122, 36, 8595-8602, hereby incorporated by reference).

Potentially stabilizing modifications to the ends of RNA molecules can include N-(acetylaminocaproyl)-4-hydroxy-

prolinol (Hyp-C6-NHAc), N-(caproyl-4-hydroxyprolinol (Hyp-C6), N-(acetyl-4-hydroxyprolinol (Hyp-NHAc), thymidine-2'-O-deoxythymidine (ether), N-(aminocaproyl)-4-hydroxyprolinol (Hyp-C6-amino), 2-docosanoyl-uridine-3"-phosphate, inverted base dT(idT) and others. Disclosure of this modification can be found in PCT Publication No. WO 2011/005861.

Other modifications of an iRNA of the invention include a 5' phosphate or 5' phosphate mimic, e.g., a 5'-terminal phosphate or phosphate mimic on the antisense strand of an RNAi agent. Suitable phosphate mimics are disclosed in, for example US Patent Publication No. 2012/0157511, the entire contents of which are incorporated herein by reference.

In certain specific embodiments, an RNAi agent of the present invention is an agent that inhibits the expression of a CIDEB gene which is selected from the group of agents listed in Tables 3-6. Any of these agents may further comprise a ligand.

A. Modified iRNAs Comprising Motifs of the Invention

In certain aspects of the invention, the double stranded RNAi agents of the invention include agents with chemical modifications as disclosed, for example, in WO 2013/075035, filed on Nov. 16, 2012, the entire contents of which are incorporated herein by reference.

Accordingly, the invention provides double stranded RNAi agents capable of inhibiting the expression of a target gene (i.e., a CIDEB gene) in vivo. The RNAi agent comprises a sense strand and an antisense strand. Each strand of the RNAi agent may range from 12-30 nucleotides in length. For example, each strand may be between 14-30 nucleotides in length, 17-30 nucleotides in length, 25-30 nucleotides in length, 27-30 nucleotides in length, 17-23 nucleotides in length, 17-21 nucleotides in length, 17-19 nucleotides in length, 19-25 nucleotides in length, 19-23 nucleotides in length, 19-21 nucleotides in length, 21-25 nucleotides in length, or 21-23 nucleotides in length. In one embodiment, the sense strand is 21 nucleotides in length. In one embodiment, the antisense strand is 23 nucleotides in length.

The sense strand and antisense strand typically form a duplex double stranded RNA ("dsRNA"), also referred to herein as an "RNAi agent." The duplex region of an RNAi agent may be 12-30 nucleotide pairs in length. For example, the duplex region can be between 14-30 nucleotide pairs in length, 17-30 nucleotide pairs in length, 27-30 nucleotide pairs in length, 17-23 nucleotide pairs in length, 17-21 nucleotide pairs in length, 17-19 nucleotide pairs in length, 19-25 nucleotide pairs in length, 19-23 nucleotide pairs in length, 19-21 nucleotide pairs in length, 21-25 nucleotide pairs in length, or 21-23 nucleotide pairs in length. In another example, the duplex region is selected from 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, and 27 nucleotides in length.

In one embodiment, the RNAi agent may contain one or more overhang regions and/or capping groups at the 3'-end, 5'-end, or both ends of one or both strands. The overhang can be 1-6 nucleotides in length, for instance 2-6 nucleotides in length, 1-5 nucleotides in length, 2-5 nucleotides in length, 1-4 nucleotides in length, 2-4 nucleotides in length, 1-3 nucleotides in length, 2-3 nucleotides in length, or 1-2 nucleotides in length. The overhangs can be the result of one strand being longer than the other, or the result of two strands of the same length being staggered. The overhang can form a mismatch with the target mRNA or it can be complementary to the gene sequences being targeted or can

be another sequence. The first and second strands can also be joined, e.g., by additional bases to form a hairpin, or by other non-base linkers.

In one embodiment, the nucleotides in the overhang region of the RNAi agent can each independently be a modified or unmodified nucleotide including, but not limited to 2'-sugar modified, such as, 2'-F, 2'-O-methyl, thymidine (T), 2'-O-methoxyethyl-5-methyluridine (Teo), 2'-O-methoxyethyladenosine (Aeo), 2'-O-methoxyethyl-5-methylcytidine (m5Ceo), and any combinations thereof. For example, TT can be an overhang sequence for either end on either strand. The overhang can form a mismatch with the target mRNA or it can be complementary to the gene sequences being targeted or can be another sequence.

The 5'- or 3'-overhangs at the sense strand, antisense strand or both strands of the RNAi agent may be phosphorylated. In some embodiments, the overhang region(s) contains two nucleotides having a phosphorothioate between the two nucleotides, where the two nucleotides can be the same or different. In one embodiment, the overhang is present at the 3'-end of the sense strand, antisense strand, or both strands. In one embodiment, this 3'-overhang is present in the antisense strand. In one embodiment, this 3'-overhang is present in the sense strand.

The RNAi agent may contain only a single overhang, which can strengthen the interference activity of the RNAi, without affecting its overall stability. For example, the single-stranded overhang may be located at the 3'-terminal end of the sense strand or, alternatively, at the 3'-terminal end of the antisense strand. The RNAi may also have a blunt end, located at the 5'-end of the antisense strand (or the 3'-end of the sense strand) or vice versa. Generally, the antisense strand of the RNAi has a nucleotide overhang at the 3'-end, and the 5'-end is blunt. While not wishing to be bound by theory, the asymmetric blunt end at the 5'-end of the antisense strand and 3'-end overhang of the antisense strand favor the guide strand loading into RISC process.

In one embodiment, the RNAi agent is a double ended bluntmer of 19 nucleotides in length, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 7, 8, 9 from the 5' end. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, 13 from the 5' end.

In another embodiment, the RNAi agent is a double ended bluntmer of 20 nucleotides in length, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 8, 9, 10 from the 5' end. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, 13 from the 5' end.

In yet another embodiment, the RNAi agent is a double ended bluntmer of 21 nucleotides in length, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 9, 10, 11 from the 5' end. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, 13 from the 5' end.

In one embodiment, the RNAi agent comprises a 21 nucleotide sense strand and a 23 nucleotide antisense strand, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 9, 10, 11 from the 5' end; the antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, 13 from the 5' end, wherein one end of the RNAi agent is blunt, while

the other end comprises a 2 nucleotide overhang. Preferably, the 2 nucleotide overhang is at the 3'-end of the antisense strand.

When the 2 nucleotide overhang is at the 3'-end of the antisense strand, there may be two phosphorothioate internucleotide linkages between the terminal three nucleotides, wherein two of the three nucleotides are the overhang nucleotides, and the third nucleotide is a paired nucleotide next to the overhang nucleotide. In one embodiment, the RNAi agent additionally has two phosphorothioate internucleotide linkages between the terminal three nucleotides at both the 5'-end of the sense strand and at the 5'-end of the antisense strand. In one embodiment, every nucleotide in the sense strand and the antisense strand of the RNAi agent, including the nucleotides that are part of the motifs are modified nucleotides. In one embodiment each residue is independently modified with a 2'-O-methyl or 3'-fluoro, e.g., in an alternating motif. Optionally, the RNAi agent further comprises a ligand (preferably GalNAc3).

In one embodiment, the RNAi agent comprises a sense and an antisense strand, wherein the sense strand is 25-30 nucleotide residues in length, wherein starting from the 5' terminal nucleotide (position 1) positions 1 to 23 of the first strand comprise at least 8 ribonucleotides; the antisense strand is 36-66 nucleotide residues in length and, starting from the 3' terminal nucleotide, comprises at least 8 ribonucleotides in the positions paired with positions 1-23 of sense strand to form a duplex; wherein at least the 3' terminal nucleotide of antisense strand is unpaired with sense strand, and up to 6 consecutive 3' terminal nucleotides are unpaired with sense strand, thereby forming a 3' single stranded overhang of 1-6 nucleotides; wherein the 5' terminus of antisense strand comprises from 10-30 consecutive nucleotides which are unpaired with sense strand, thereby forming a 10-30 nucleotide single stranded 5' overhang; wherein at least the sense strand 5' terminal and 3' terminal nucleotides are base paired with nucleotides of antisense strand when sense and antisense strands are aligned for maximum complementarity, thereby forming a substantially duplexed region between sense and antisense strands; and antisense strand is sufficiently complementary to a target RNA along at least 19 ribonucleotides of antisense strand length to reduce target gene expression when the double stranded nucleic acid is introduced into a mammalian cell; and wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides, where at least one of the motifs occurs at or near the cleavage site. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at or near the cleavage site.

In one embodiment, the RNAi agent comprises sense and antisense strands, wherein the RNAi agent comprises a first strand having a length which is at least 25 and at most 29 nucleotides and a second strand having a length which is at most 30 nucleotides with at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at position 11, 12, 13 from the 5' end; wherein the 3' end of the first strand and the 5' end of the second strand form a blunt end and the second strand is 1-4 nucleotides longer at its 3' end than the first strand, wherein the duplex region which is at least 25 nucleotides in length, and the second strand is sufficiently complementary to a target mRNA along at least 19 nucleotide of the second strand length to reduce target gene expression when the RNAi agent is introduced into a mammalian cell, and wherein dicer cleavage of the RNAi agent preferentially results in an siRNA comprising the 3'

end of the second strand, thereby reducing expression of the target gene in the mammal. Optionally, the RNAi agent further comprises a ligand.

In one embodiment, the sense strand of the RNAi agent contains at least one motif of three identical modifications on three consecutive nucleotides, where one of the motifs occurs at the cleavage site in the sense strand.

In one embodiment, the antisense strand of the RNAi agent can also contain at least one motif of three identical modifications on three consecutive nucleotides, where one of the motifs occurs at or near the cleavage site in the antisense strand.

For an RNAi agent having a duplex region of 17-23 nucleotide in length, the cleavage site of the antisense strand is typically around the 10, 11 and 12 positions from the 5'-end. Thus the motifs of three identical modifications may occur at the 9, 10, 11 positions; 10, 11, 12 positions; 11, 12, 13 positions; 12, 13, 14 positions; or 13, 14, 15 positions of the antisense strand, the count starting from the 1st nucleotide from the 5'-end of the antisense strand, or, the count starting from the 1st paired nucleotide within the duplex region from the 5'-end of the antisense strand. The cleavage site in the antisense strand may also change according to the length of the duplex region of the RNAi from the 5'-end.

The sense strand of the RNAi agent may contain at least one motif of three identical modifications on three consecutive nucleotides at the cleavage site of the strand; and the antisense strand may have at least one motif of three identical modifications on three consecutive nucleotides at or near the cleavage site of the strand. When the sense strand and the antisense strand form a dsRNA duplex, the sense strand and the antisense strand can be so aligned that one motif of the three nucleotides on the sense strand and one motif of the three nucleotides on the antisense strand have at least one nucleotide overlap, i.e., at least one of the three nucleotides of the motif in the sense strand forms a base pair with at least one of the three nucleotides of the motif in the antisense strand. Alternatively, at least two nucleotides may overlap, or all three nucleotides may overlap.

In one embodiment, the sense strand of the RNAi agent may contain more than one motif of three identical modifications on three consecutive nucleotides. The first motif may occur at or near the cleavage site of the strand and the other motifs may be a wing modification. The term "wing modification" herein refers to a motif occurring at another portion of the strand that is separated from the motif at or near the cleavage site of the same strand. The wing modification is either adjacent to the first motif or is separated by at least one or more nucleotides. When the motifs are immediately adjacent to each other then the chemistry of the motifs are distinct from each other and when the motifs are separated by one or more nucleotide than the chemistries can be the same or different. Two or more wing modifications may be present. For instance, when two wing modifications are present, each wing modification may occur at one end relative to the first motif which is at or near cleavage site or on either side of the lead motif.

Like the sense strand, the antisense strand of the RNAi agent may contain more than one motifs of three identical modifications on three consecutive nucleotides, with at least one of the motifs occurring at or near the cleavage site of the strand. This antisense strand may also contain one or more wing modifications in an alignment similar to the wing modifications that may be present on the sense strand.

In one embodiment, the wing modification on the sense strand or antisense strand of the RNAi agent typically does

not include the first one or two terminal nucleotides at the 3'-end, 5'-end or both ends of the strand.

In another embodiment, the wing modification on the sense strand or antisense strand of the RNAi agent typically does not include the first one or two paired nucleotides within the duplex region at the 3'-end, 5'-end or both ends of the strand.

When the sense strand and the antisense strand of the RNAi agent each contain at least one wing modification, the wing modifications may fall on the same end of the duplex region, and have an overlap of one, two or three nucleotides.

When the sense strand and the antisense strand of the RNAi agent each contain at least two wing modifications, the sense strand and the antisense strand can be so aligned that two modifications each from one strand fall on one end of the duplex region, having an overlap of one, two or three nucleotides; two modifications each from one strand fall on the other end of the duplex region, having an overlap of one, two or three nucleotides; two modifications one strand fall on each side of the lead motif, having an overlap of one, two or three nucleotides in the duplex region.

In one embodiment, every nucleotide in the sense strand and antisense strand of the RNAi agent, including the nucleotides that are part of the motifs, may be modified. Each nucleotide may be modified with the same or different modification which can include one or more alteration of one or both of the non-linking phosphate oxygens and/or of one or more of the linking phosphate oxygens; alteration of a constituent of the ribose sugar, e.g., of the 2' hydroxyl on the ribose sugar; wholesale replacement of the phosphate moiety with "dephospho" linkers; modification or replacement of a naturally occurring base; and replacement or modification of the ribose-phosphate backbone.

As nucleic acids are polymers of subunits, many of the modifications occur at a position which is repeated within a nucleic acid, e.g., a modification of a base, or a phosphate moiety, or a non-linking O of a phosphate moiety. In some cases, the modification will occur at all of the subject positions in the nucleic acid but in many cases it will not. By way of example, a modification may only occur at a 3' or 5' terminal position, may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand. A modification may occur in a double strand region, a single strand region, or in both. A modification may occur only in the double strand region of a RNA or may only occur in a single strand region of a RNA. For example, a phosphorothioate modification at a non-linking O position may only occur at one or both termini, may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand, or may occur in double strand and single strand regions, particularly at termini. The 5' end or ends can be phosphorylated.

It may be possible, e.g., to enhance stability, to include particular bases in overhangs, or to include modified nucleotides or nucleotide surrogates, in single strand overhangs, e.g., in a 5' or 3' overhang, or in both. For example, it can be desirable to include purine nucleotides in overhangs. In some embodiments all or some of the bases in a 3' or 5' overhang may be modified, e.g., with a modification described herein. Modifications can include, e.g., the use of modifications at the 2' position of the ribose sugar with modifications that are known in the art, e.g., the use of deoxyribonucleotides, 2'-deoxy-2'-fluoro (2'-F) or 2'-O-methyl modified instead of the ribosugar of the nucleobase,

and modifications in the phosphate group, e.g., phosphorothioate modifications. Overhangs need not be homologous with the target sequence.

In one embodiment, each residue of the sense strand and antisense strand is independently modified with LNA, CRN, cET, UNA, HNA, CeNA, 2'-methoxyethyl, 2'-O-methyl, 2'-O-allyl, 2'-C-allyl, 2'-deoxy, 2'-hydroxyl, or 2'-fluoro. The strands can contain more than one modification. In one embodiment, each residue of the sense strand and antisense strand is independently modified with 2'-O-methyl or 2'-fluoro. The term "HNA" refers to hexitol or hexose nucleic acid.

At least two different modifications are typically present on the sense strand and antisense strand. Those two modifications may be the 2'-O-methyl or 2'-fluoro modifications, or others.

In one embodiment, the N_a and/or N_b comprise modifications of an alternating pattern. The term "alternating motif" as used herein refers to a motif having one or more modifications, each modification occurring on alternating nucleotides of one strand. The alternating nucleotide may refer to one per every other nucleotide or one per every three nucleotides, or a similar pattern. For example, if A, B and C each represent one type of modification to the nucleotide, the alternating motif can be "ABABABABABAB . . .," "AABBAABBAAB . . .," "AABAABAABAAB . . .," "AAA-BAAABAAAB . . .," "AAABBBAAABBB . . .," or "ABCABCABCABC . . .," etc.

The type of modifications contained in the alternating motif may be the same or different. For example, if A, B, C, D each represent one type of modification on the nucleotide, the alternating pattern, i.e., modifications on every other nucleotide, may be the same, but each of the sense strand or antisense strand can be selected from several possibilities of modifications within the alternating motif such as "ABABAB . . .," "ACACAC . . .," "BDBDBD . . .," or "CDCDCD . . .," etc.

In one embodiment, the RNAi agent of the invention comprises the modification pattern for the alternating motif on the sense strand relative to the modification pattern for the alternating motif on the antisense strand is shifted. The shift may be such that the modified group of nucleotides of the sense strand corresponds to a differently modified group of nucleotides of the antisense strand and vice versa. For example, the sense strand when paired with the antisense strand in the dsRNA duplex, the alternating motif in the sense strand may start with "ABABAB" from 5'-3' of the strand and the alternating motif in the antisense strand may start with "BABABA" from 5'-3' of the strand within the duplex region. As another example, the alternating motif in the sense strand may start with "AABBAABB" from 5'-3' of the strand and the alternating motif in the antisense strand may start with "BBAABBAA" from 5'-3' of the strand within the duplex region, so that there is a complete or partial shift of the modification patterns between the sense strand and the antisense strand.

In one embodiment, the RNAi agent comprises the pattern of the alternating motif of 2'-O-methyl modification and 2'-F modification on the sense strand initially has a shift relative to the pattern of the alternating motif of 2'-O-methyl modification and 2'-F modification on the antisense strand initially, i.e., the 2'-O-methyl modified nucleotide on the sense strand base pairs with a 2'-F modified nucleotide on the antisense strand and vice versa. The 1 position of the sense strand may start with the 2'-F modification, and the 1 position of the antisense strand may start with the 2'-O-methyl modification.

The introduction of one or more motifs of three identical modifications on three consecutive nucleotides to the sense strand and/or antisense strand interrupts the initial modification pattern present in the sense strand and/or antisense strand. This interruption of the modification pattern of the sense and/or antisense strand by introducing one or more motifs of three identical modifications on three consecutive nucleotides to the sense and/or antisense strand surprisingly enhances the gene silencing activity to the target gene.

In one embodiment, when the motif of three identical modifications on three consecutive nucleotides is introduced to any of the strands, the modification of the nucleotide next to the motif is a different modification than the modification of the motif. For example, the portion of the sequence containing the motif is ". . . N_a YYY N_b . . .," where "Y" represents the modification of the motif of three identical modifications on three consecutive nucleotide, and " N_a " and " N_b " represent a modification to the nucleotide next to the motif "YYY" that is different than the modification of Y, and where N_a and N_b can be the same or different modifications. Alternatively, N_a and/or N_b may be present or absent when there is a wing modification present.

The RNAi agent may further comprise at least one phosphorothioate or methylphosphonate internucleotide linkage. The phosphorothioate or methylphosphonate internucleotide linkage modification may occur on any nucleotide of the sense strand or antisense strand or both strands in any position of the strand. For instance, the internucleotide linkage modification may occur on every nucleotide on the sense strand and/or antisense strand; each internucleotide linkage modification may occur in an alternating pattern on the sense strand and/or antisense strand; or the sense strand or antisense strand may contain both internucleotide linkage modifications in an alternating pattern. The alternating pattern of the internucleotide linkage modification on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the internucleotide linkage modification on the sense strand may have a shift relative to the alternating pattern of the internucleotide linkage modification on the antisense strand. In one embodiment, a double-stranded RNAi agent comprises 6-8 phosphorothioate internucleotide linkages. In one embodiment, the antisense strand comprises two phosphorothioate internucleotide linkages at the 5'-terminus and two phosphorothioate internucleotide linkages at the 3'-terminus, and the sense strand comprises at least two phosphorothioate internucleotide linkages at either the 5'-terminus or the 3'-terminus.

In one embodiment, the RNAi comprises a phosphorothioate or methylphosphonate internucleotide linkage modification in the overhang region. For example, the overhang region may contain two nucleotides having a phosphorothioate or methylphosphonate internucleotide linkage between the two nucleotides. Internucleotide linkage modifications also may be made to link the overhang nucleotides with the terminal paired nucleotides within the duplex region. For example, at least 2, 3, 4, or all the overhang nucleotides may be linked through phosphorothioate or methylphosphonate internucleotide linkages linking the overhang nucleotide with a paired nucleotide that is next to the overhang nucleotide. For instance, there may be at least two phosphorothioate internucleotide linkages between the terminal three nucleotides, in which two of the three nucleotides are overhang nucleotides, and the third is a paired nucleotide next to the overhang nucleotide. These terminal three nucleotides may

be at the 3'-end of the antisense strand, the 3'-end of the sense strand, the 5'-end of the antisense strand, and/or the 5'-end of the antisense strand.

In one embodiment, the 2 nucleotide overhang is at the 3'-end of the antisense strand, and there are two phosphorothioate internucleotide linkages between the terminal three nucleotides, wherein two of the three nucleotides are the overhang nucleotides, and the third nucleotide is a paired nucleotide next to the overhang nucleotide. Optionally, the RNAi agent may additionally have two phosphorothioate internucleotide linkages between the terminal three nucleotides at both the 5'-end of the sense strand and at the 5'-end of the antisense strand.

In one embodiment, the RNAi agent comprises mismatch(es) with the target, within the duplex, or combinations thereof. The mismatch may occur in the overhang region or the duplex region. The base pair may be ranked on the basis of their propensity to promote dissociation or melting (e.g., on the free energy of association or dissociation of a particular pairing, the simplest approach is to examine the pairs on an individual pair basis, though next neighbor or similar analysis can also be used). In terms of promoting dissociation: A:U is preferred over G:C; G:U is preferred over G:C; and I:C is preferred over G:C (I=inosine). Mismatches, e.g., non-canonical or other than canonical pairings (as described elsewhere herein) are preferred over canonical (A:T, A:U, G:C) pairings; and pairings which include a universal base are preferred over canonical pairings.

In one embodiment, the RNAi agent comprises at least one of the first 1, 2, 3, 4, or 5 base pairs within the duplex regions from the 5'-end of the antisense strand independently selected from the group of: A:U, G:U, I:C, and mismatched pairs, e.g., non-canonical or other than canonical pairings or pairings which include a universal base, to promote the dissociation of the antisense strand at the 5'-end of the duplex.

In one embodiment, the nucleotide at the 1 position within the duplex region from the 5'-end in the antisense strand is selected from the group consisting of A, dA, dU, U, and dT. Alternatively, at least one of the first 1, 2 or 3 base pair within the duplex region from the 5'-end of the antisense strand is an AU base pair. For example, the first base pair within the duplex region from the 5'-end of the antisense strand is an AU base pair.

In another embodiment, the nucleotide at the 3'-end of the sense strand is deoxy-thymine (dT). In another embodiment, the nucleotide at the 3'-end of the antisense strand is deoxy-thymine (dT). In one embodiment, there is a short sequence of deoxy-thymine nucleotides, for example, two dT nucleotides on the 3'-end of the sense and/or antisense strand.

In one embodiment, the sense strand sequence may be represented by formula (I):



wherein:

i and j are each independently 0 or 1;

p and q are each independently 0-6;

each N_a independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides;

each N_b independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides;

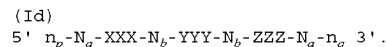
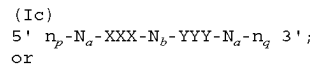
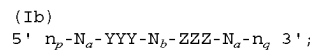
each n_p and n_q independently represent an overhang nucleotide;

wherein N_b and Y do not have the same modification; and XXX, YYY and ZZZ each independently represent one motif of three identical modifications on three consecutive nucleotides. Preferably YYY is all 2'-F modified nucleotides.

In one embodiment, the N_a and/or N_b comprise modifications of alternating pattern.

In one embodiment, the YYY motif occurs at or near the cleavage site of the sense strand. For example, when the RNAi agent has a duplex region of 17-23 nucleotides in length, the YYY motif can occur at or the vicinity of the cleavage site (e.g.: can occur at positions 6, 7, 8, 7, 8, 9, 8, 9, 10, 9, 10, 11, 10, 11, 12 or 11, 12, 13) of the sense strand, the count starting from the 1st nucleotide, from the 5'-end; or optionally, the count starting at the 1st paired nucleotide within the duplex region, from the 5'-end.

In one embodiment, i is 1 and j is 0, or i is 0 and j is 1, or both i and j are 1. The sense strand can therefore be represented by the following formulas:



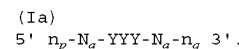
When the sense strand is represented by formula (Ib), N_b represents an oligonucleotide sequence comprising 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N_a independently can represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

When the sense strand is represented as formula (Ic), N_b represents an oligonucleotide sequence comprising 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N_a can independently represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

When the sense strand is represented as formula (Id), each N_b independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Preferably, N_b is 0, 1, 2, 3, 4, 5 or 6. Each N_a can independently represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

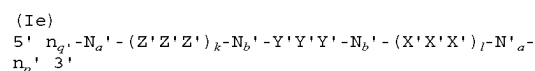
Each of X, Y and Z may be the same or different from each other.

In other embodiments, i is 0 and j is 0, and the sense strand may be represented by the formula:



When the sense strand is represented by formula (Ia), each N_a independently can represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

In one embodiment, the antisense strand sequence of the RNAi may be represented by formula (Ie):



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wherein:

k and l are each independently 0 or 1;

p' and q' are each independently 0-6;

each N_a' independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides;

each N_b' independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides;

each n_p' and n_q' independently represent an overhang nucleotide;

wherein N_b' and Y' do not have the same modification; and

X'X'X', Y'Y'Y' and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides.

In one embodiment, the N_a' and/or N_b' comprise modifications of alternating pattern.

The Y'Y'Y' motif occurs at or near the cleavage site of the antisense strand. For example, when the RNAi agent has a duplex region of 17-23 nucleotide in length, the Y'Y'Y' motif can occur at positions 9, 10, 11; 10, 11, 12; 11, 12, 13; 12, 13, 14; or 13, 14, 15 of the antisense strand, with the count starting from the 1st nucleotide, from the 5'-end; or optionally, the count starting at the 1st paired nucleotide within the duplex region, from the 5'-end. Preferably, the Y'Y'Y' motif occurs at positions 11, 12, 13.

In one embodiment, Y'Y'Y' motif is all 2'-OMe modified nucleotides.

In one embodiment, k is 1 and l is 0, or k is 0 and l is 1, or both k and l are 1.

The antisense strand can therefore be represented by the following formulas:

(Ig)
5' n_q' - N_a' -Z'Z'Z'- N_b' -Y'Y'Y'- N_a' - n_p' 3';

(Ih)
5' n_q' - N_a' -Y'Y'Y'- N_b' -X'X'X'- n_p' 3';
or

(Ii)
5' n_q' - N_a' -Z'Z'Z'- N_b' -Y'Y'Y'- N_b' -X'X'X'- N_a' - n_p' 3'.

When the antisense strand is represented by formula (Ig), N_b' represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N_a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

When the antisense strand is represented as formula (Ih), N_b' represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N_a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

When the antisense strand is represented as formula (Ii), each N_b' independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N_a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides. Preferably, N_b is 0, 1, 2, 3, 4, 5 or 6.

In other embodiments, k is 0 and l is 0 and the antisense strand may be represented by the formula:

5' n_p' - N_a' -Y'Y'Y'- N_a' - n_q' 3' (If).

When the antisense strand is represented as formula (If), each N_a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

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Each of X', Y' and Z' may be the same or different from each other.

Each nucleotide of the sense strand and antisense strand may be independently modified with LNA, CRN, UNA, cEt, HNA, CeNA, 2'-methoxyethyl, 2'-O-methyl, 2'-O-allyl, 2'-C-allyl, 2'-hydroxyl, or 2'-fluoro. For example, each nucleotide of the sense strand and antisense strand is independently modified with 2'-O-methyl or 2'-fluoro. Each X, Y, Z, X', Y' and Z', in particular, may represent a 2'-O-methyl modification or a 2'-fluoro modification.

In one embodiment, the sense strand of the RNAi agent may contain YYY motif occurring at 9, 10 and 11 positions of the strand when the duplex region is 21 nt, the count starting from the 1st nucleotide from the 5'-end, or optionally, the count starting at the 1st paired nucleotide within the duplex region, from the 5'-end; and Y represents 2'-F modification. The sense strand may additionally contain XXX motif or ZZZ motifs as wing modifications at the opposite end of the duplex region; and XXX and ZZZ each independently represents a 2'-OMe modification or 2'-F modification.

In one embodiment the antisense strand may contain Y'Y'Y' motif occurring at positions 11, 12, 13 of the strand, the count starting from the 1st nucleotide from the 5' end, or optionally, the count starting at the 1st paired nucleotide within the duplex region, from the 5'-end; and Y' represents 2'-O-methyl modification. The antisense strand may additionally contain X'X'X' motif or Z'Z'Z' motifs as wing modifications at the opposite end of the duplex region; and X'X'X' and Z'Z'Z' each independently represents a 2'-OMe modification or 2'-F modification.

The sense strand represented by any one of the above formulas (Ia), (Ib), (Ic), and (Id) forms a duplex with an antisense strand being represented by any one of formulas (If), (Ig), (Ih), and (Ii), respectively.

Accordingly, the RNAi agents for use in the methods of the invention may comprise a sense strand and an antisense strand, each strand having 14 to 30 nucleotides, the RNAi duplex represented by formula (Ij):

(Ij)
sense:
5' n_p - N_a -(X X X)i- N_b -Y Y Y- N_b -(Z Z Z)j- N_a - n_q 3'
antisense:
3' n_p' - N_a' -(X'X'X')k- N_b' -Y'Y'Y'- N_b' -(Z'Z'Z')l- N_a' - n_q' 5'

wherein:

i, j, k, and l are each independently 0 or 1;

p, p', q, and q' are each independently 0-6;

each N_a and N_a' independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides;

each N_b and N_b' independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides;

wherein each n_p , n_p' , n_q , and n_q' , each of which may or may not be present, independently represents an overhang nucleotide; and

XXX, YYY, ZZZ, X'X'X', Y'Y'Y', and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides.

In one embodiment, i is 0 and j is 0; or i is 1 and j is 0; or i is 0 and j is 1; or both i and j are 0; or both i and j are

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1. In another embodiment, k is 0 and l is 0; or k is 1 and l is 0; k is 0 and l is 1; or both k and l are 0; or both k and l are 1.

Exemplary combinations of the sense strand and antisense strand forming a RNAi duplex include the formulas below:

(Ik)

5' np-Na-Y Y Y-Na-nq 3'
3' np'-Na'-Y'Y'Y'-Na'nq' 5'

(Il)

5' np-Na-Y Y Y-Nb-Z Z Z-Na-nq 3'
3' np'-Na'-Y'Y'Y'-Nb'-Z'Z'Z'-Na'nq' 5'

(Im)

5' np-Na-X X X-Nb-Y Y Y-Na-nq 3'
3' np'-Na'-X'X'X'-Nb'-Y'Y'Y'-Na'nq' 5'

(In)

5' np-Na-X X X-Nb-Y Y Y-Nb-Z Z Z-Na-nq 3'
3' np'-Na'-X'X'X'-Nb'-Y'Y'Y'-Nb'-Z'Z'Z'-Na-nq' 5'

When the RNAi agent is represented by formula (Ik), each N_a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

When the RNAi agent is represented by formula (Il), each N_b independently represents an oligonucleotide sequence comprising 1-10, 1-7, 1-5 or 1-4 modified nucleotides. Each N_a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

When the RNAi agent is represented as formula (Im), each N_b , N_b' independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N_a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

When the RNAi agent is represented as formula (In), each N_b , N_b' independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2 or 0 modified nucleotides. Each N_a , N_a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides. Each of N_a , N_a' , N_b and N_b' independently comprises modifications of alternating pattern.

Each of X, Y and Z in formulas (Ij), (Ik), (Il), (Im), and (In) may be the same or different from each other.

When the RNAi agent is represented by formula (Ij), (Ik), (Il), (Im), and (In), at least one of the Y nucleotides may form a base pair with one of the Y' nucleotides. Alternatively, at least two of the Y nucleotides form base pairs with the corresponding Y' nucleotides; or all three of the Y nucleotides all form base pairs with the corresponding Y' nucleotides.

When the RNAi agent is represented by formula (Il) or (In), at least one of the Z nucleotides may form a base pair with one of the Z' nucleotides. Alternatively, at least two of the Z nucleotides form base pairs with the corresponding Z' nucleotides; or all three of the Z nucleotides all form base pairs with the corresponding Z' nucleotides.

When the RNAi agent is represented as formula (Im) or (In), at least one of the X nucleotides may form a base pair with one of the X' nucleotides. Alternatively, at least two of the X nucleotides form base pairs with the corresponding X' nucleotides; or all three of the X nucleotides all form base pairs with the corresponding X' nucleotides.

In one embodiment, the modification on the Y nucleotide is different than the modification on the Y' nucleotide, the modification on the Z nucleotide is different than the modi-

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fication on the Z' nucleotide, and/or the modification on the X nucleotide is different than the modification on the X' nucleotide.

In one embodiment, when the RNAi agent is represented by formula (In), the Na modifications are 2'-O-methyl or 2'-fluoro modifications. In another embodiment, when the RNAi agent is represented by formula (In), the Na modifications are 2'-O-methyl or 2'-fluoro modifications and $np' > 0$ and at least one np' is linked to a neighboring nucleotide a via phosphorothioate linkage. In yet another embodiment, when the RNAi agent is represented by formula (In), the Na modifications are 2'-O-methyl or 2'-fluoro modifications, $np' > 0$ and at least one np' is linked to a neighboring nucleotide via phosphorothioate linkage, and the sense strand is conjugated to one or more GalNAc derivatives attached through a bivalent or trivalent branched linker (described below). In another embodiment, when the RNAi agent is represented by formula (In), the Na modifications are 2'-O-methyl or 2'-fluoro modifications, $np' > 0$ and at least one np' is linked to a neighboring nucleotide via phosphorothioate linkage, the sense strand comprises at least one phosphorothioate linkage, and the sense strand is conjugated to one or more GalNAc derivatives attached through a bivalent or trivalent branched linker.

In one embodiment, when the RNAi agent is represented by formula (Ik), the Na modifications are 2'-O-methyl or 2'-fluoro modifications, $np' > 0$ and at least one np' is linked to a neighboring nucleotide via phosphorothioate linkage, the sense strand comprises at least one phosphorothioate linkage, and the sense strand is conjugated to one or more GalNAc derivatives attached through a bivalent or trivalent branched linker.

In one embodiment, the RNAi agent is a multimer containing at least two duplexes represented by formula (Ij), (Ik), (Il), (Im), and (In), wherein the duplexes are connected by a linker. The linker can be cleavable or non-cleavable. Optionally, the multimer further comprises a ligand. Each of the duplexes can target the same gene or two different genes; or each of the duplexes can target same gene at two different target sites.

In one embodiment, the RNAi agent is a multimer containing three, four, five, six or more duplexes represented by formula (Ij), (Ik), (Il), (Im), and (In), wherein the duplexes are connected by a linker. The linker can be cleavable or non-cleavable. Optionally, the multimer further comprises a ligand. Each of the duplexes can target the same gene or two different genes; or each of the duplexes can target same gene at two different target sites.

In one embodiment, two RNAi agents represented by formula (Ij), (Ik), (Il), (Im), and (In) are linked to each other at the 5' end, and one or both of the 3' ends and are optionally conjugated to a ligand. Each of the agents can target the same gene or two different genes; or each of the agents can target same gene at two different target sites.

In certain embodiments, an RNAi agent of the invention may contain a low number of nucleotides containing a 2'-fluoro modification, e.g., 10 or fewer nucleotides with 2'-fluoro modification. For example, the RNAi agent may contain 10, 9, 8, 7, 6, 5, 4, 3, 2, 1 or 0 nucleotides with a 2'-fluoro modification. In a specific embodiment, the RNAi agent of the invention contains 10 nucleotides with a 2'-fluoro modification, e.g., 4 nucleotides with a 2'-fluoro modification in the sense strand and 6 nucleotides with a 2'-fluoro modification in the antisense strand. In another specific embodiment, the RNAi agent of the invention contains 6 nucleotides with a 2'-fluoro modification, e.g., 4

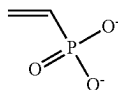
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nucleotides with a 2'-fluoro modification in the sense strand and 2 nucleotides with a 2'-fluoro modification in the anti-sense strand.

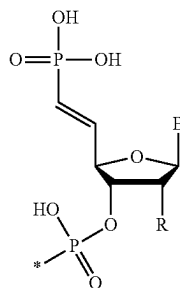
In other embodiments, an RNAi agent of the invention may contain an ultra-low number of nucleotides containing a 2'-fluoro modification, e.g., 2 or fewer nucleotides containing a 2'-fluoro modification. For example, the RNAi agent may contain 2, 1 or 0 nucleotides with a 2'-fluoro modification. In a specific embodiment, the RNAi agent may contain 2 nucleotides with a 2'-fluoro modification, e.g., 0 nucleotides with a 2'-fluoro modification in the sense strand and 2 nucleotides with a 2'-fluoro modification in the anti-sense strand.

Various publications describe multimeric RNAi agents that can be used in the methods of the invention. Such publications include WO2007/091269, U.S. Pat. No. 7,858,769, WO2010/141511, WO2007/117686, WO2009/014887 and WO2011/031520 the entire contents of each of which are hereby incorporated herein by reference.

In certain embodiments, the compositions and methods of the disclosure include a vinyl phosphonate (VP) modification of an RNAi agent as described herein. In exemplary embodiments, a vinyl phosphonate of the disclosure has the following structure:



For example, when the phosphate mimic is a 5'-vinyl phosphonate (VP), the 5'-terminal nucleotide can have the following structure,



wherein * indicates the location of the bond to 5'-position of the adjacent nucleotide;

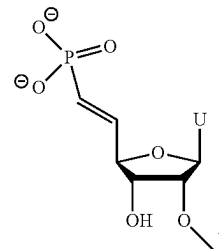
R is hydrogen, hydroxy, methoxy, fluoro (e.g., hydroxy or methoxy), or another modification described herein; and

B is a nucleobase or a modified nucleobase, optionally where B is adenine, guanine, cytosine, thymine or uracil.

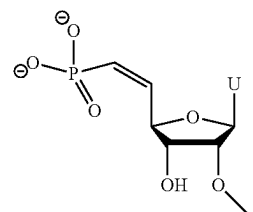
A vinyl phosphonate of the instant disclosure may be attached to either the antisense or the sense strand of a dsRNA of the disclosure. In certain embodiments, a vinyl phosphonate of the instant disclosure is attached to the antisense strand of a dsRNA, optionally at the 5' end of the antisense strand of the dsRNA. The dsRNA agent can comprise a phosphorus-containing group at the 5'-end of the sense strand or antisense strand. The 5'-end phosphorus-containing group can be 5'-end phosphate (5'-P), 5'-end phosphorothioate (5'-PS), 5'-end phosphorodithioate (5'-PS2), 5'-end vinylphosphonate (5'-VP), 5'-end methylphos-

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phonate (MePhos), or 5'-deoxy-5'-C-malonyl. When the 5'-end phosphorus-containing group is 5'-end vinylphosphonate (5'-VP), the 5'-VP can be either 5'-E-VP isomer (i.e., trans-vinylphosphonate,

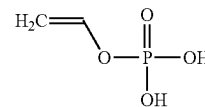


5'-Z-VP isomer (i.e., cis-vinylphosphonate,



or mixtures thereof.

Vinyl phosphate modifications are also contemplated for the compositions and methods of the instant disclosure. An exemplary vinyl phosphate structure is:



For example, when the phosphate mimic is a 5'-vinyl phosphate, the 5'-terminal nucleotide can have the immediately structure, where the phosphonate group is replaced by a phosphate.

As described in more detail below, the RNAi agent that contains conjugations of one or more carbohydrate moieties to a RNAi agent can optimize one or more properties of the RNAi agent. In many cases, the carbohydrate moiety will be attached to a modified subunit of the RNAi agent. For example, the ribose sugar of one or more ribonucleotide subunits of a dsRNA agent can be replaced with another moiety, e.g., a non-carbohydrate (preferably cyclic) carrier to which is attached a carbohydrate ligand. A ribonucleotide subunit in which the ribose sugar of the subunit has been so replaced is referred to herein as a ribose replacement modification subunit (RRMS). A cyclic carrier may be a carbocyclic ring system, i.e., all ring atoms are carbon atoms, or a heterocyclic ring system, i.e., one or more ring atoms may be a heteroatom, e.g., nitrogen, oxygen, sulfur. The cyclic carrier may be a monocyclic ring system, or may contain two or more rings, e.g. fused rings. The cyclic carrier may be a fully saturated ring system, or it may contain one or more double bonds.

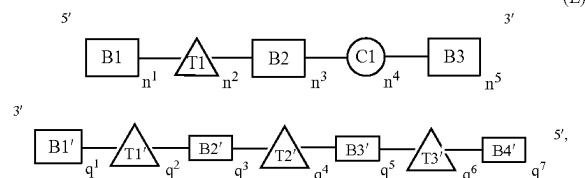
The ligand may be attached to the polynucleotide via a carrier. The carriers include (i) at least one "backbone attachment point," preferably two "backbone attachment

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points” and (ii) at least one “tethering attachment point.” A “backbone attachment point” as used herein refers to a functional group, e.g. a hydroxyl group, or generally, a bond available for, and that is suitable for incorporation of the carrier into the backbone, e.g., the phosphate, or modified phosphate, e.g., sulfur containing, backbone, of a ribonucleic acid. A “tethering attachment point” (TAP) in some embodiments refers to a constituent ring atom of the cyclic carrier, e.g., a carbon atom or a heteroatom (distinct from an atom which provides a backbone attachment point), that connects a selected moiety. The moiety can be, e.g., a carbohydrate, e.g. monosaccharide, disaccharide, trisaccharide, tetrasaccharide, oligosaccharide and polysaccharide. Optionally, the selected moiety is connected by an intervening tether to the cyclic carrier. Thus, the cyclic carrier will often include a functional group, e.g., an amino group, or generally, provide a bond, that is suitable for incorporation or tethering of another chemical entity, e.g., a ligand to the constituent ring.

The RNAi agents may be conjugated to a ligand via a carrier, wherein the carrier can be cyclic group or acyclic group; preferably, the cyclic group is selected from pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3]dioxolane, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuryl and decalin; preferably, the acyclic group is selected from serinol backbone or diethanolamine backbone.

In another embodiment of the invention, an iRNA agent comprises a sense strand and an antisense strand, each strand having 14 to 40 nucleotides. The RNAi agent may be represented by formula (L):

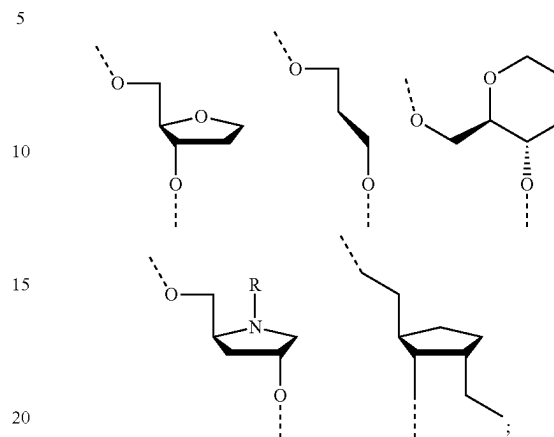


In formula (L), B1, B2, B3, B1', B2', B3', and B4' each are independently a nucleotide containing a modification selected from the group consisting of 2'-O-alkyl, 2'-substituted alkoxy, 2'-substituted alkyl, 2'-halo, ENA, and BNA/LNA. In certain embodiments, B1, B2, B3, B1', B2', B3', and B4' each contain 2'-OMe modifications. In certain embodiments, B1, B2, B3, B1', B2', B3', and B4' each contain 2'-OMe or 2'-F modifications. In certain embodiments, at least one of B1, B2, B3, B1', B2', B3', and B4' contain 2'-O—N-methylacetamido (2'-O-NMA) modification.

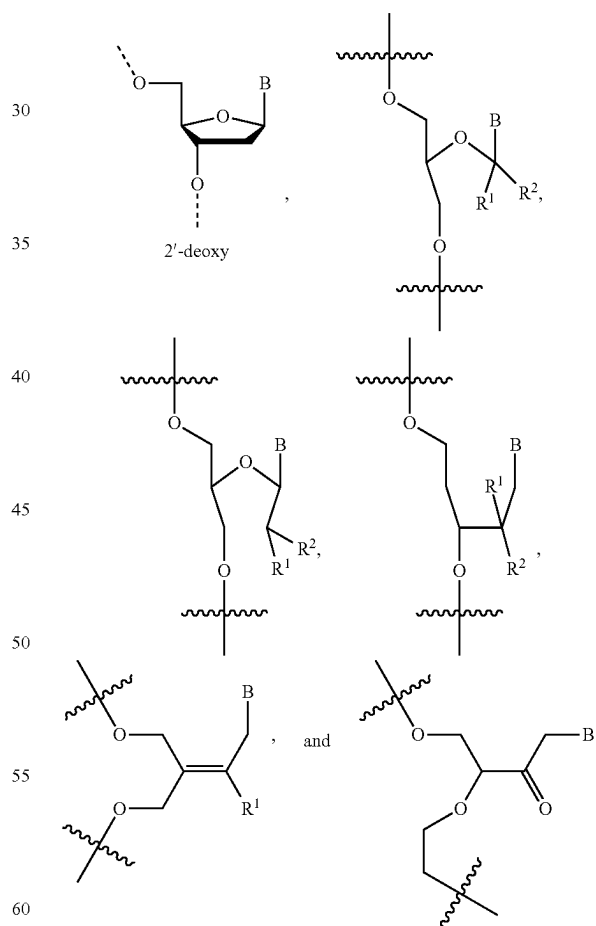
C1 is a thermally destabilizing nucleotide placed at a site opposite to the seed region of the antisense strand (i.e., at positions 2-8 of the 5'-end of the antisense strand). For example, C1 is at a position that pairs with a nucleotide at positions 2-8 of the 5'-end of the antisense strand. In one example, C1 is at position 15 from the 5'-end of the sense strand. C1 nucleotide bears the thermally destabilizing modification which can include abasic modification; mismatch with the opposing nucleotide in the duplex; and sugar modification such as 2'-deoxy modification or acyclic nucleotide e.g., unlocked nucleic acids (UNA) or glycerol nucleic acid (GNA). In certain embodiments, C1 has thermally destabilizing modification selected

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from the group consisting of: i) mismatch with the opposing nucleotide in the antisense strand; ii) abasic modification selected from the group consisting of:



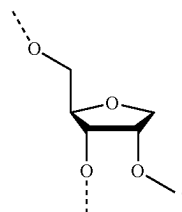
and iii) sugar modification selected from the group consisting of:



wherein B is a modified or unmodified nucleobase, R¹ and R² independently are H, halogen, OR₃, or alkyl; and R₃ is H, alkyl, cycloalkyl, aryl, aralkyl, heteroaryl or sugar. In certain embodiments, the thermally destabilizing modification in C1 is a mismatch selected from the group consisting of G:G,

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G:A, G:U, G:T, A:A, A:C, C:C, C:U, C:T, U:U, T:T, and U:T; and optionally, at least one nucleobase in the mismatch pair is a 2'-deoxy nucleobase. In one example, the thermally destabilizing modification in C1 is GNA or



T1, T1', T2', and T3' each independently represent a nucleotide comprising a modification providing the nucleotide a steric bulk that is less or equal to the steric bulk of a 2'-OMe modification. A steric bulk refers to the sum of steric effects of a modification. Methods for determining steric effects of a modification of a nucleotide are known to one skilled in the art. The modification can be at the 2' position of a ribose sugar of the nucleotide, or a modification to a non-ribose nucleotide, acyclic nucleotide, or the backbone of the nucleotide that is similar or equivalent to the 2' position of the ribose sugar, and provides the nucleotide a steric bulk that is less than or equal to the steric bulk of a 2'-OMe modification. For example, T1, T1', T2', and T3' are each independently selected from DNA, RNA, LNA, 2'-F, and 2'-F-5'-methyl. In certain embodiments, T1 is DNA. In certain embodiments, T1' is DNA, RNA or LNA. In certain embodiments, T2' is DNA or RNA. In certain embodiments, T3' is DNA or RNA.

n^1 , n^3 , and q^1 are independently 4 to 15 nucleotides in length.

n^5 , q^5 , and q^7 are independently 1-6 nucleotide(s) in length.

n^4 , q^2 , and q^6 are independently 1-3 nucleotide(s) in length; alternatively, n^4 is 0.

q^5 is independently 0-10 nucleotide(s) in length.

n^2 and q^4 are independently 0-3 nucleotide(s) in length.

Alternatively, n^4 is 0-3 nucleotide(s) in length.

In certain embodiments, n^4 can be 0. In one example, n^4 is 0, and q^2 and q^6 are 1. In another example, n^4 is 0, and q^2 and q^6 are 1, with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand).

In certain embodiments, n^4 , q^2 , and q^6 are each 1.

In certain embodiments, n^2 , n^4 , q^2 , q^4 , and q^6 are each 1.

In certain embodiments, C1 is at position 14-17 of the 5'-end of the sense strand, when the sense strand is 19-22 nucleotides in length, and n^4 is 1. In certain embodiments, C1 is at position 15 of the 5'-end of the sense strand.

In certain embodiments, T3' starts at position 2 from the 5' end of the antisense strand. In one example, T3' is at position 2 from the 5' end of the antisense strand and q^6 is equal to 1.

In certain embodiments, T1' starts at position 14 from the 5' end of the antisense strand. In one example, T1' is at position 14 from the 5' end of the antisense strand and q^2 is equal to 1.

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In an exemplary embodiment, T3' starts from position 2 from the 5' end of the antisense strand and T1' starts from position 14 from the 5' end of the antisense strand. In one example, T3' starts from position 2 from the 5' end of the antisense strand and q^6 is equal to 1 and T1' starts from position 14 from the 5' end of the antisense strand and q^2 is equal to 1.

In certain embodiments, T1' and T3' are separated by 11 nucleotides in length (i.e. not counting the T1' and T3' nucleotides).

In certain embodiments, T1' is at position 14 from the 5' end of the antisense strand. In one example, T1' is at position 14 from the 5' end of the antisense strand and q^2 is equal to 1, and the modification at the 2' position or positions in a non-ribose, acyclic or backbone that provide less steric bulk than a 2'-OMe ribose.

In certain embodiments, T3' is at position 2 from the 5' end of the antisense strand. In one example, T3' is at position 2 from the 5' end of the antisense strand and q^6 is equal to 1, and the modification at the 2' position or positions in a non-ribose, acyclic or backbone that provide less than or equal to steric bulk than a 2'-OMe ribose.

In certain embodiments, T1 is at the cleavage site of the sense strand. In one example, T1 is at position 11 from the 5' end of the sense strand, when the sense strand is 19-22 nucleotides in length, and n^2 is 1. In an exemplary embodiment, T1 is at the cleavage site of the sense strand at position 11 from the 5' end of the sense strand, when the sense strand is 19-22 nucleotides in length, and n^2 is 1.

In certain embodiments, T2' starts at position 6 from the 5' end of the antisense strand. In one example, T2' is at positions 6-10 from the 5' end of the antisense strand, and q^4 is 1.

In an exemplary embodiment, T1 is at the cleavage site of the sense strand, for instance, at position 11 from the 5' end of the sense strand, when the sense strand is 19-22 nucleotides in length, and n^2 is 1; T1' is at position 14 from the 5' end of the antisense strand, and q^2 is equal to 1, and the modification to T1' is at the 2' position of a ribose sugar or at positions in a non-ribose, acyclic or backbone that provide less steric bulk than a 2'-OMe ribose; T2' is at positions 6-10 from the 5' end of the antisense strand, and q^4 is 1; and T3' is at position 2 from the 5' end of the antisense strand, and q^6 is equal to 1, and the modification to T3' is at the 2' position or at positions in a non-ribose, acyclic or backbone that provide less than or equal to steric bulk than a 2'-OMe ribose.

In certain embodiments, T2' starts at position 8 from the 5' end of the antisense strand. In one example, T2' starts at position 8 from the 5' end of the antisense strand, and q^4 is 2.

In certain embodiments, T2' starts at position 9 from the 5' end of the antisense strand. In one example, T2' is at position 9 from the 5' end of the antisense strand, and q^4 is 1.

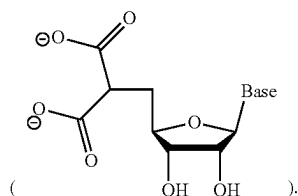
In certain embodiments, B1' is 2'-OMe or 2'-F, q^1 is 9, T1' is 2'-F, q^2 is 1, B2' is 2'-OMe or 2'-F, q^3 is 4, T2' is 2'-F, q^4 is 1, B3' is 2'-OMe or 2'-F, q^5 is 6, T3' is 2'-F, q^6 is 1, B4' is 2'-OMe, and q^7 is 1; with two phosphorothioate internucleotide linkage modifications within positions 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand).

In certain embodiments, B1 is 2'-OME or 2'-F, n¹ is 8, T1 is 2'F, n² is 3, B2 is 2'-OME, n³ is 7, n⁴ is 0, B3 is 2'-OME, n⁵ is 3, B1' is 2'-OME or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OME or 2'-F, q³ is 4, q⁴ is 0, B3' is 2'-OME or 2'-F, q⁵ is 7, T3' is 2'-F, q⁶ is 1, B4' is 2'-F, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within positions 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications

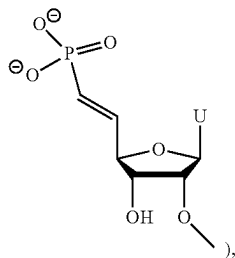
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within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand).

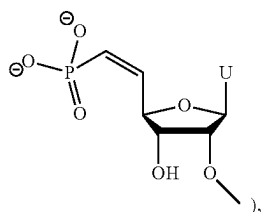
The RNAi agent can comprise a phosphorus-containing group at the 5'-end of the sense strand or antisense strand. The 5'-end phosphorus-containing group can be 5'-end phosphate (5'-P), 5'-end phosphorothioate (5'-PS), 5'-end phosphorodithioate (5'-PS₂), 5'-end vinylphosphonate (5'-VP), 5'-end methylphosphonate (MePhos), or 5'-deoxy-5'-C-malonyl



When the 5'-end phosphorus-containing group is 5'-end vinylphosphonate (5'-VP), the 5'-VP can be either 5'-E-VP isomer (i.e., trans-vinylphosphonate,



5'-Z-VP isomer (i.e., cis-vinylphosphonate,



or mixtures thereof.

In certain embodiments, the RNAi agent comprises a phosphorus-containing group at the 5'-end of the sense strand. In certain embodiments, the RNAi agent comprises a phosphorus-containing group at the 5'-end of the antisense strand.

In certain embodiments, the RNAi agent comprises a 5'-P. In certain embodiments, the RNAi agent comprises a 5'-P in the antisense strand.

In certain embodiments, the RNAi agent comprises a 5'-PS. In certain embodiments, the RNAi agent comprises a 5'-PS in the antisense strand.

In certain embodiments, the RNAi agent comprises a 5'-VP. In certain embodiments, the RNAi agent comprises a 5'-VP in the antisense strand. In certain embodiments, the RNAi agent comprises a 5'-E-VP in the antisense strand. In certain embodiments, the RNAi agent comprises a 5'-Z-VP in the antisense strand.

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In certain embodiments, the RNAi agent comprises a 5'-PS₂. In certain embodiments, the RNAi agent comprises a 5'-PS₂ in the antisense strand.

In certain embodiments, the RNAi agent comprises a 5'-PS₂. In certain embodiments, the RNAi agent comprises a 5'-deoxy-5'-C-malonyl in the antisense strand.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-OMe, and q⁷ is 1. The RNAi agent also comprises a 5'-PS.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-OMe, and q⁷ is 1. The RNAi agent also comprises a 5'-P.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-OMe, and q⁷ is 1. The RNAi agent also comprises a 5'-VP. The 5'-VP may be 5'-E-VP, 5'-Z-VP, or combination thereof.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-OMe, and q⁷ is 1. The RNAi agent also comprises a 5'-PS₂.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-OMe, and q⁷ is 1. The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-OMe, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-P.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-OMe, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe

In certain embodiments, B1 is 2'-OMe or 2'-F, n^1 is 8, T1 is 2'F, n^2 is 3, B2 is 2'-OMe, n^3 is 7, n^4 is 0, B3 is 2'OMe, n^5 is 3, B1' is 2'-OMe or 2'-F, q^1 is 9, T1' is 2'-F, q^2 is 1, B2' is 2'-OMe or 2'-F, q^3 is 4, T2' is 2'-F, q^4 is 2, B3' is 2'-OMe or 2'-F, q^5 is 5, T3' is 2'-F, q^6 is 1, B4' is 2'-F, and q^7 is 1. The RNAi agent also comprises a 5'-P.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, q⁴ is 0, B3' is 2'-OMe or 2'-F, q⁵ is 7, T3' is 2'-F, q⁶ is 1, B4' is 2'-F, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-OMe, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl and a targeting ligand. In certain embodiments, the 5'-deoxy-5'-C-ma-

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-F, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications

within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS₂ and a targeting ligand. In certain embodiments, the 5'-PS₂ is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, T2' is 2'-F, q⁴ is 2, B3' is 2'-OMe or 2'-F, q⁵ is 5, T3' is 2'-F, q⁶ is 1, B4' is 2'-F, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl and a targeting ligand. In certain embodiments, the 5'-deoxy-5'-C-malonyl is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, q⁴ is 0, B3' is 2'-OMe or 2'-F, q⁵ is 7, T3' is 2'-F, q⁶ is 1, B4' is 2'-F, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-P and a targeting ligand. In certain embodiments, the 5'-P is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, q⁴ is 0, B3' is 2'-OMe or 2'-F, q⁵ is 7, T3' is 2'-F, q⁶ is 1, B4' is 2'-F, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS and a targeting ligand. In certain embodiments, the 5'-PS is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, q⁴ is 0, B3' is 2'-OMe or 2'-F, q⁵ is 7, T3' is 2'-F, q⁶ is 1, B4' is 2'-F, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-VP (e.g., a 5'-E-VP, 5'-Z-VP, or combination thereof) and a targeting ligand. In certain embodi-

ments, the 5'-VP is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, q⁴ is 0, B3' is 2'-OMe or 2'-F, q⁵ is 7, T3' is 2'-F, q⁶ is 1, B4' is 2'-F, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS₂ and a targeting ligand. In certain embodiments, the 5'-PS₂ is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

In certain embodiments, B1 is 2'-OMe or 2'-F, n¹ is 8, T1 is 2'-F, n² is 3, B2 is 2'-OMe, n³ is 7, n⁴ is 0, B3 is 2'-OMe, n⁵ is 3, B1' is 2'-OMe or 2'-F, q¹ is 9, T1' is 2'-F, q² is 1, B2' is 2'-OMe or 2'-F, q³ is 4, q⁴ is 0, B3' is 2'-OMe or 2'-F, q⁵ is 7, T3' is 2'-F, q⁶ is 1, B4' is 2'-F, and q⁷ is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl and a targeting ligand. In certain embodiments, the 5'-deoxy-5'-C-malonyl is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

In a particular embodiment, an RNAi agent of the present invention comprises:

- (a) a sense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; and
 - (iii) 2'-F modifications at positions 1, 3, 5, 7, 9 to 11, 13, 17, 19, and 21, and 2'-OMe modifications at positions 2, 4, 6, 8, 12, 14 to 16, 18, and 20 (counting from the 5' end); and
- (b) an antisense strand having:
 - (i) a length of 23 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 3, 5, 9, 11 to 13, 15, 17, 19, 21, and 23, and 2'-F modifications at positions 2, 4, 6 to 8, 10, 14, 16, 18, 20, and 22 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end); wherein the dsRNA agents have a two-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

In another particular embodiment, an RNAi agent of the present invention comprises:

- (a) a sense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker;

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- (iii) 2'-F modifications at positions 1, 3, 5, 7, 9 to 11, 13, 15, 17, 19, and 21, and 2'-OMe modifications at positions 2, 4, 6, 8, 12, 14, 16, 18, and 20 (counting from the 5' end); and
 - (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and
 - (b) an antisense strand having:
 - (i) a length of 23 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 3, 5, 7, 9, 11 to 13, 15, 17, 19, and 21 to 23, and 2'-F modifications at positions 2, 4, 6, 8, 10, 14, 16, 18, and 20 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);
- wherein the RNAi agents have a two-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.
- In another particular embodiment, an RNAi agent of the present invention comprises:
- (a) a sense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker;
 - (iii) 2'-OMe modifications at positions 1 to 6, 8, 10, and 12 to 21, 2'-F modifications at positions 7, and 9, and a desoxy-nucleotide (e.g. dT) at position 11 (counting from the 5' end); and
 - (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and
 - (b) an antisense strand having:
 - (i) a length of 23 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 3, 7, 9, 11, 13, 15, 17, and 19 to 23, and 2'-F modifications at positions 2, 4 to 6, 8, 10, 12, 14, 16, and 18 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);
- wherein the RNAi agents have a two-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.
- In another particular embodiment, an RNAi agent of the present invention comprises:
- (a) a sense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker;
 - (iii) 2'-OMe modifications at positions 1 to 6, 8, 10, 12, 14, and 16 to 21, and 2'-F modifications at positions 7, 9, 11, 13, and 15; and
 - (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end);

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- and
 - (b) an antisense strand having:
 - (i) a length of 23 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 5, 7, 9, 11, 13, 15, 17, 19, and 21 to 23, and 2'-F modifications at positions 2 to 4, 6, 8, 10, 12, 14, 16, 18, and 20 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);
- wherein the RNAi agents have a two-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.
- In another particular embodiment, an RNAi agent of the present invention comprises:
- (a) a sense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker;
 - (iii) 2'-OMe modifications at positions 1 to 9, and 12 to 21, and 2'-F modifications at positions 10, and 11; and
 - (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end);
 - and
 - (b) an antisense strand having:
 - (i) a length of 23 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 3, 5, 7, 9, 11 to 13, 15, 17, 19, and 21 to 23, and 2'-F modifications at positions 2, 4, 6, 8, 10, 14, 16, 18, and 20 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);
- wherein the RNAi agents have a two-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.
- In another particular embodiment, a RNAi agent of the present invention comprises:
- (a) a sense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker;
 - (iii) 2'-F modifications at positions 1, 3, 5, 7, 9 to 11, and 13, and 2'-OMe modifications at positions 2, 4, 6, 8, 12, and 14 to 21; and
 - (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end);
 - and
 - (b) an antisense strand having:
 - (i) a length of 23 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 3, 5 to 7, 9, 11 to 13, 15, 17 to 19, and 21 to 23, and 2'-F modifications at positions 2, 4, 8, 10, 14, 16, and 20 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);

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wherein the RNAi agents have a two-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

In another particular embodiment, an RNAi agent of the present invention comprises:

- (a) a sense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker;
 - (iii) 2'-OMe modifications at positions 1, 2, 4, 6, 8, 12, 14, 15, 17, and 19 to 21, and 2'-F modifications at positions 3, 5, 7, 9 to 11, 13, 16, and 18; and
 - (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end);

and

- (b) an antisense strand having:
 - (i) a length of 25 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 4, 6, 7, 9, 11 to 13, 15, 17, and 19 to 23, 2'-F modifications at positions 2, 3, 5, 8, 10, 14, 16, and 18, and desoxynucleotides (e.g. dT) at positions 24 and 25 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);

wherein the RNAi agents have a four-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

In another particular embodiment, a RNAi agent of the present invention comprises:

- (a) a sense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker;
 - (iii) 2'-OMe modifications at positions 1 to 6, 8, and 12 to 21, and 2'-F modifications at positions 7, and 9 to 11; and
 - (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end);

and

- (b) an antisense strand having:
 - (i) a length of 23 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 3 to 5, 7, 8, 10 to 13, 15, and 17 to 23, and 2'-F modifications at positions 2, 6, 9, 14, and 16 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);

wherein the RNAi agents have a two-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

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In another particular embodiment, a RNAi agent of the present invention comprises:

- (a) a sense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker;
 - (iii) 2'-OMe modifications at positions 1 to 6, 8, and 12 to 21, and 2'-F modifications at positions 7, and 9 to 11; and
 - (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end);

and

- (b) an antisense strand having:
 - (i) a length of 23 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 3 to 5, 7, 10 to 13, 15, and 17 to 23, and 2'-F modifications at positions 2, 6, 8, 9, 14, and 16 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);

wherein the RNAi agents have a two-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

In another particular embodiment, a RNAi agent of the present invention comprises:

- (a) a sense strand having:
 - (i) a length of 19 nucleotides;
 - (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker;
 - (iii) 2'-OMe modifications at positions 1 to 4, 6, and 10 to 19, and 2'-F modifications at positions 5, and 7 to 9; and
 - (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end);

and

- (b) an antisense strand having:
 - (i) a length of 21 nucleotides;
 - (ii) 2'-OMe modifications at positions 1, 3 to 5, 7, 10 to 13, 15, and 17 to 21, and 2'-F modifications at positions 2, 6, 8, 9, 14, and 16 (counting from the 5' end); and
 - (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 19 and 20, and between nucleotide positions 20 and 21 (counting from the 5' end);

wherein the RNAi agents have a two-nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

In certain embodiments, the iRNA for use in the methods of the invention is an agent selected from agents listed in Tables 3-6. These agents may further comprise a ligand.

IV. iRNAs Conjugated to Ligands

Another modification of the RNA of an iRNA of the invention involves chemically linking to the RNA one or more ligands, moieties or conjugates that enhance the activity, cellular distribution or cellular uptake of the iRNA. Such moieties include but are not limited to lipid moieties such as a cholesterol moiety (Letsinger et al., (1989) *Proc. Natl. Acad. Sci. USA*, 86: 6553-6556), cholic acid (Manoharan et

al., (1994) *Biorg. Med. Chem. Let.*, 4:1053-1060), a thioether, e.g., beryl-S-tritylthiol (Manoharan et al., (1992) *Ann. N.Y. Acad. Sci.*, 660:306-309; Manoharan et al., (1993) *Biorg. Med. Chem. Let.*, 3:2765-2770), a thiocholesterol (Oberhauser et al., (1992) *Nucl. Acids Res.*, 20:533-538), an aliphatic chain, e.g., dodecandiol or undecyl residues (Saison-Behmoaras et al., (1991) *EMBO J.*, 10:1111-1118; Kabanov et al., (1990) *FEBS Lett.*, 259:327-330; Svinarchuk et al., (1993) *Biochimie*, 75:49-54), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethyl-ammonium 1,2-di-O-hexadecyl-rac-glycero-3-phosphonate (Manoharan et al., (1995) *Tetrahedron Lett.*, 36:3651-3654; Shea et al., (1990) *Nucl. Acids Res.*, 18:3777-3783), a polyamine or a polyethylene glycol chain (Manoharan et al., (1995) *Nucleosides & Nucleotides*, 14:969-973), or adamantane acetic acid (Manoharan et al., (1995) *Tetrahedron Lett.*, 36:3651-3654), a palmitoyl moiety (Mishra et al., (1995) *Biochim. Biophys. Acta*, 1264:229-237), or an octadecylamine or hexylaminocarbonyloxycholesterol moiety (Crooke et al., (1996) *J. Pharmacol. Exp. Ther.*, 277:923-937).

In one embodiment, a ligand alters the distribution, targeting or lifetime of an iRNA agent into which it is incorporated. In preferred embodiments a ligand provides an enhanced affinity for a selected target, e.g., molecule, cell or cell type, compartment, e.g., a cellular or organ compartment, tissue, organ or region of the body, as, e.g., compared to a species absent such a ligand. Preferred ligands will not take part in duplex pairing in a duplexed nucleic acid.

Ligands can include a naturally occurring substance, such as a protein (e.g., human serum albumin (HSA), low-density lipoprotein (LDL), or globulin); carbohydrate (e.g., a dextran, pullulan, chitin, chitosan, inulin, cyclodextrin, N-acetylglucosamine, N-acetylgalactosamine or hyaluronic acid); or a lipid. The ligand can also be a recombinant or synthetic molecule, such as a synthetic polymer, e.g., a synthetic polyamino acid. Examples of polyamino acids include polyamino acid is a polylysine (PLL), poly L-aspartic acid, poly L-glutamic acid, styrene-maleic acid anhydride copolymer, poly(L-lactide-co-glycolide) copolymer, divinyl ether-maleic anhydride copolymer, N-(2-hydroxypropyl)methacrylamide copolymer (HMPA), polyethylene glycol (PEG), polyvinyl alcohol (PVA), polyurethane, poly(2-ethylacrylic acid), N-isopropylacrylamide polymers, or polyphosphazine. Example of polyamines include: polyethylenimine, polylysine (PLL), spermine, spermidine, polyamine, pseudopeptide-polyamine, peptidomimetic polyamine, dendrimer polyamine, arginine, amidine, pro-tamine, cationic lipid, cationic porphyrin, quaternary salt of a polyamine, or an alpha helical peptide.

Ligands can also include targeting groups, e.g., a cell or tissue targeting agent, e.g., a lectin, glycoprotein, lipid or protein, e.g., an antibody, that binds to a specified cell type such as a kidney cell. A targeting group can be a thyrotropin, melanotropin, lectin, glycoprotein, surfactant protein A, Mucin carbohydrate, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-gulucosamine multivalent mannose, multivalent fucose, glycosylated polyaminoacids, multivalent galactose, transferrin, bisphosphonate, polyglutamate, polyaspartate, a lipid, cholesterol, a steroid, bile acid, folate, vitamin B12, vitamin A, biotin, or an RGD peptide or RGD peptide mimetic.

Other examples of ligands include dyes, intercalating agents (e.g. acridines), cross-linkers (e.g. psoralene, mitomycin C), porphyrins (TPPC4, texaphyrin, Sapphyrin), polycyclic aromatic hydrocarbons (e.g., phenazine, dihydrophenazine), artificial endonucleases (e.g. EDTA), lipophilic molecules, e.g., cholesterol, cholic acid, adamantane acetic

acid, 1-pyrene butyric acid, dihydrotestosterone, 1,3-Bis-O (hexadecyl)glycerol, geranyloxyhexyl group, hexadecylglycerol, borneol, menthol, 1,3-propanediol, heptadecyl group, palmitic acid, myristic acid, O3-(oleoyl) lithocholic acid, O3-(oleoyl)cholenic acid, dimethoxytrityl, or phenoxazine) and peptide conjugates (e.g., antennapedia peptide, Tat peptide), alkylating agents, phosphate, amino, mercapto, PEG (e.g., PEG-40K), MPEG, [MPEG]₂, polyamino, alkyl, substituted alkyl, radiolabeled markers, enzymes, haptens (e.g. biotin), transport/absorption facilitators (e.g., aspirin, vitamin E, folic acid), synthetic ribonucleases (e.g., imidazole, bisimidazole, histamine, imidazole clusters, acridine-imidazole conjugates, Eu3+ complexes of tetraazamacrocycles), dinitrophenyl, HRP, or AP.

Ligands can be proteins, e.g., glycoproteins, or peptides, e.g., molecules having a specific affinity for a co-ligand, or antibodies e.g., an antibody, that binds to a specified cell type such as a hepatic cell. Ligands can also include hormones and hormone receptors. They can also include non-peptidic species, such as lipids, lectins, carbohydrates, vitamins, cofactors, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-gulucosamine multivalent mannose, or multivalent fucose. The ligand can be, for example, a lipopolysaccharide, an activator of p38 MAP kinase, or an activator of NF- κ B.

The ligand can be a substance, e.g., a drug, which can increase the uptake of the iRNA agent into the cell, for example, by disrupting the cell's cytoskeleton, e.g., by disrupting the cell's microtubules, microfilaments, and/or intermediate filaments. The drug can be, for example, taxon, vincristine, vinblastine, cytochalasin, nocodazole, japlakinolide, latrunculin A, phalloidin, swinholide A, indanocine, or myoservin.

In some embodiments, a ligand attached to an iRNA as described herein acts as a pharmacokinetic modulator (PK modulator). PK modulators include lipophiles, bile acids, steroids, phospholipid analogues, peptides, protein binding agents, PEG, vitamins etc. Exemplary PK modulators include, but are not limited to, cholesterol, fatty acids, cholic acid, lithocholic acid, dialkylglycerides, diacylglyceride, phospholipids, sphingolipids, naproxen, ibuprofen, vitamin E, biotin etc. Oligonucleotides that comprise a number of phosphorothioate linkages are also known to bind to serum protein, thus short oligonucleotides, e.g., oligonucleotides of about 5 bases, 10 bases, 15 bases or 20 bases, comprising multiple of phosphorothioate linkages in the backbone are also amenable to the present invention as ligands (e.g. as PK modulating ligands). In addition, aptamers that bind serum components (e.g. serum proteins) are also suitable for use as PK modulating ligands in the embodiments described herein.

Ligand-conjugated oligonucleotides of the invention may be synthesized by the use of an oligonucleotide that bears a pendant reactive functionality, such as that derived from the attachment of a linking molecule onto the oligonucleotide (described below). This reactive oligonucleotide may be reacted directly with commercially-available ligands, ligands that are synthesized bearing any of a variety of protecting groups, or ligands that have a linking moiety attached thereto.

The oligonucleotides used in the conjugates of the present invention may be conveniently and routinely made through the well-known method of solid-phase synthesis. Equipment for such synthesis is sold by several vendors including, for example, Applied Biosystems (Foster City, Calif.). Any other means for such synthesis known in the art may additionally or alternatively be employed. It is also known

to use similar techniques to prepare other oligonucleotides, such as the phosphorothioates and alkylated derivatives.

In the ligand-conjugated oligonucleotides and ligand-molecule bearing sequence-specific linked nucleosides of the present invention, the oligonucleotides and oligonucleosides may be assembled on a suitable DNA synthesizer utilizing standard nucleotide or nucleoside precursors, or nucleotide or nucleoside conjugate precursors that already bear the linking moiety, ligand-nucleotide or nucleoside-conjugate precursors that already bear the ligand molecule, or non-nucleoside ligand-bearing building blocks.

When using nucleotide-conjugate precursors that already bear a linking moiety, the synthesis of the sequence-specific linked nucleosides is typically completed, and the ligand molecule is then reacted with the linking moiety to form the ligand-conjugated oligonucleotide. In some embodiments, the oligonucleotides or linked nucleosides of the present invention are synthesized by an automated synthesizer using phosphoramidites derived from ligand-nucleoside conjugates in addition to the standard phosphoramidites and non-standard phosphoramidites that are commercially available and routinely used in oligonucleotide synthesis.

A. Lipid Conjugates

In one embodiment, the ligand or conjugate is a lipid or lipid-based molecule. Such a lipid or lipid-based molecule preferably binds a serum protein, e.g., human serum albumin (HSA). An HSA binding ligand allows for distribution of the conjugate to a target tissue, e.g., a non-kidney target tissue of the body. For example, the target tissue can be the liver, including parenchymal cells of the liver. Other molecules that can bind HSA can also be used as ligands. For example, neproxin or aspirin can be used. A lipid or lipid-based ligand can (a) increase resistance to degradation of the conjugate, (b) increase targeting or transport into a target cell or cell membrane, and/or (c) can be used to adjust binding to a serum protein, e.g., HSA.

A lipid based ligand can be used to inhibit, e.g., control the binding of the conjugate to a target tissue. For example, a lipid or lipid-based ligand that binds to HSA more strongly will be less likely to be targeted to the kidney and therefore less likely to be cleared from the body. A lipid or lipid-based ligand that binds to HSA less strongly can be used to target the conjugate to the kidney.

In a preferred embodiment, the lipid based ligand binds HSA. Preferably, it binds HSA with a sufficient affinity such that the conjugate will be preferably distributed to a non-kidney tissue. However, it is preferred that the affinity not be so strong that the HSA-ligand binding cannot be reversed.

In another preferred embodiment, the lipid based ligand binds HSA weakly or not at all, such that the conjugate will be preferably distributed to the kidney. Other moieties that target to kidney cells can also be used in place of or in addition to the lipid based ligand.

In another aspect, the ligand is a moiety, e.g., a vitamin, which is taken up by a target cell, e.g., a proliferating cell. These are particularly useful for treating disorders characterized by unwanted cell proliferation, e.g., of the malignant or non-malignant type, e.g., cancer cells. Exemplary vitamins include vitamin A, E, and K. Other exemplary vitamins include are B vitamin, e.g., folic acid, B12, riboflavin, biotin, pyridoxal or other vitamins or nutrients taken up by target cells such as liver cells. Also included are HSA and low density lipoprotein (LDL).

B. Cell Permeation Agents

In another aspect, the ligand is a cell-permeation agent, preferably a helical cell-permeation agent. Preferably, the agent is amphipathic. An exemplary agent is a peptide such

as tat or antennopedia. If the agent is a peptide, it can be modified, including a peptidylmimetic, invertomers, non-peptide or pseudopeptide linkages, and use of D-amino acids. The helical agent is preferably an alpha-helical agent, which preferably has a lipophilic and a lipophobic phase.

The ligand can be a peptide or peptidomimetic. A peptidomimetic (also referred to herein as an oligopeptidomimetic) is a molecule capable of folding into a defined three-dimensional structure similar to a natural peptide. The attachment of peptide and peptidomimetics to iRNA agents can affect pharmacokinetic distribution of the iRNA, such as by enhancing cellular recognition and absorption. The peptide or peptidomimetic moiety can be about 5-50 amino acids long, e.g., about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 amino acids long.

A peptide or peptidomimetic can be, for example, a cell permeation peptide, cationic peptide, amphipathic peptide, or hydrophobic peptide (e.g., consisting primarily of Tyr, Trp or Phe). The peptide moiety can be a dendrimer peptide, constrained peptide or cross-linked peptide. In another alternative, the peptide moiety can include a hydrophobic membrane translocation sequence (MTS). An exemplary hydrophobic MTS-containing peptide is RFGF having the amino acid sequence AAVALLPAVLLALLAP (SEQ ID NO: 3). An RFGF analogue (e.g., amino acid sequence AALLPVL-LAAP (SEQ ID NO: 4) containing a hydrophobic MTS can also be a targeting moiety. The peptide moiety can be a "delivery" peptide, which can carry large polar molecules including peptides, oligonucleotides, and protein across cell membranes. For example, sequences from the HIV Tat protein (GRKKRRQRRPPQ (SEQ ID NO: 5) and the *Drosophila* Antennapedia protein (RQIKIWFQNRRMKWKK (SEQ ID NO: 6) have been found to be capable of functioning as delivery peptides. A peptide or peptidomimetic can be encoded by a random sequence of DNA, such as a peptide identified from a phage-display library, or one-bead-one-compound (OBOC) combinatorial library (Lam et al., Nature, 354:82-84, 1991). Examples of a peptide or peptidomimetic tethered to a dsRNA agent via an incorporated monomer unit for cell targeting purposes is an arginine-glycine-aspartic acid (RGD)-peptide, or RGD mimic. A peptide moiety can range in length from about 5 amino acids to about 40 amino acids. The peptide moieties can have a structural modification, such as to increase stability or direct conformational properties. Any of the structural modifications described below can be utilized.

An RGD peptide for use in the compositions and methods of the invention may be linear or cyclic, and may be modified, e.g., glycosylated or methylated, to facilitate targeting to a specific tissue(s). RGD-containing peptides and peptidomimetics may include D-amino acids, as well as synthetic RGD mimics. In addition to RGD, one can use other moieties that target the integrin ligand. Preferred conjugates of this ligand target PECAM-1 or VEGF.

A "cell permeation peptide" is capable of permeating a cell, e.g., a microbial cell, such as a bacterial or fungal cell, or a mammalian cell, such as a human cell. A microbial cell-permeating peptide can be, for example, a α -helical linear peptide (e.g., LL-37 or Ceropin P1), a disulfide bond-containing peptide (e.g., α -defensin, β -defensin or bactericidin), or a peptide containing only one or two dominating amino acids (e.g., PR-39 or indolicidin). A cell permeation peptide can also include a nuclear localization signal (NLS). For example, a cell permeation peptide can be a bipartite amphipathic peptide, such as MPG, which is

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derived from the fusion peptide domain of HIV-1 gp41 and the NLS of SV40 large T antigen (Simeoni et al., Nucl. Acids Res. 31:2717-2724, 2003).

C. Carbohydrate Conjugates

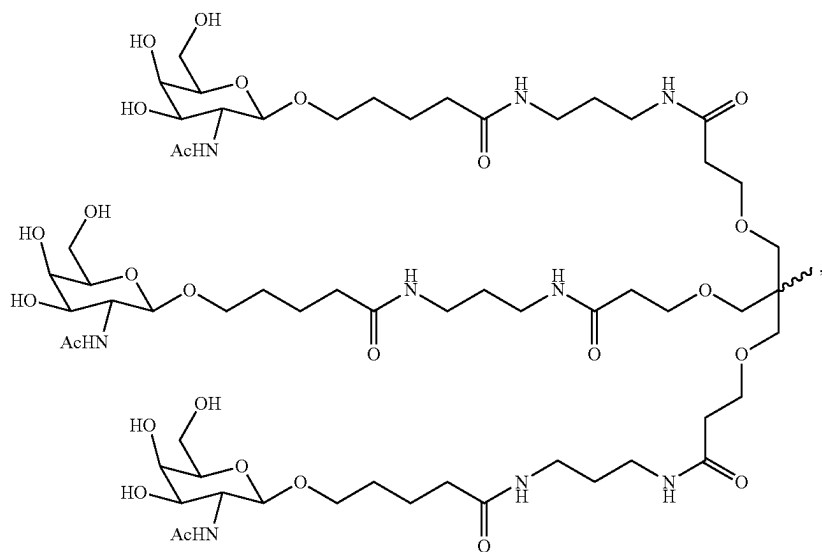
In some embodiments of the compositions and methods of the invention, an iRNA oligonucleotide further comprises a carbohydrate. The carbohydrate conjugated iRNA are advantageous for the in vivo delivery of nucleic acids, as well as compositions suitable for in vivo therapeutic use, as described herein. As used herein, "carbohydrate" refers to a compound which is either a carbohydrate per se made up of one or more monosaccharide units having at least 6 carbon atoms (which can be linear, branched or cyclic) with an oxygen, nitrogen or sulfur atom bonded to each carbon atom; or a compound having as a part thereof a carbohydrate

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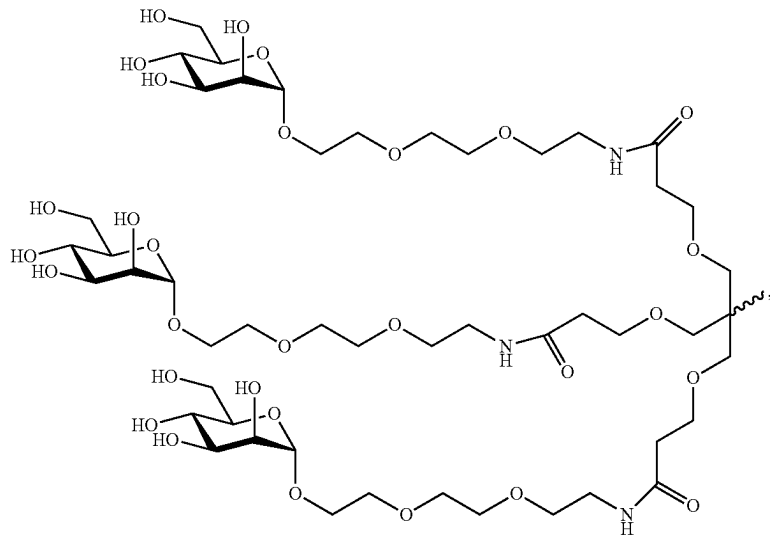
moiety made up of one or more monosaccharide units each having at least six carbon atoms (which can be linear, branched or cyclic), with an oxygen, nitrogen or sulfur atom bonded to each carbon atom. Representative carbohydrates include the sugars (mono-, di-, tri- and oligosaccharides containing from about 4, 5, 6, 7, 8, or 9 monosaccharide units), and polysaccharides such as starches, glycogen, cellulose and polysaccharide gums. Specific monosaccharides include C5 and above (e.g., C5, C6, C7, or C8) sugars; di- and trisaccharides include sugars having two or three monosaccharide units (e.g., C5, C6, C7, or C8).

In one embodiment, a carbohydrate conjugate for use in the compositions and methods of the invention is selected from the group consisting of:

Formula II



Formula III



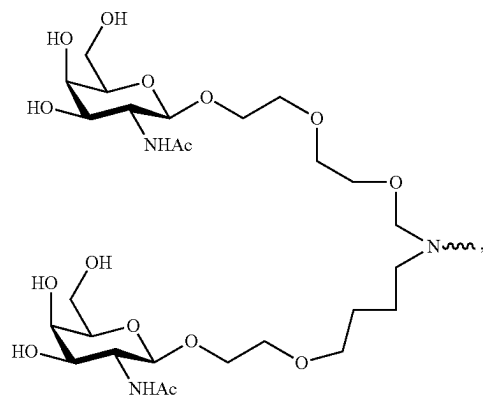
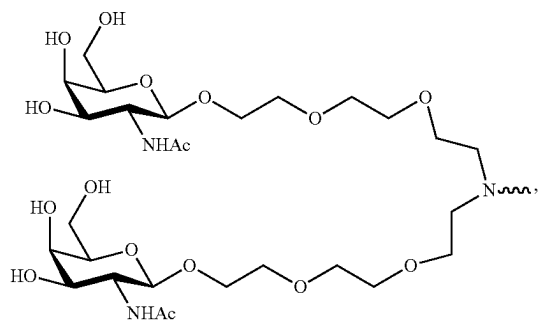
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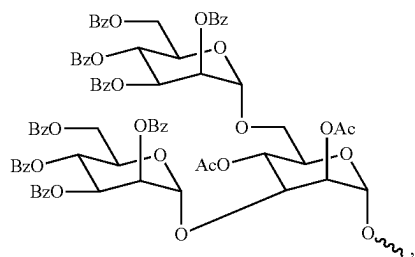
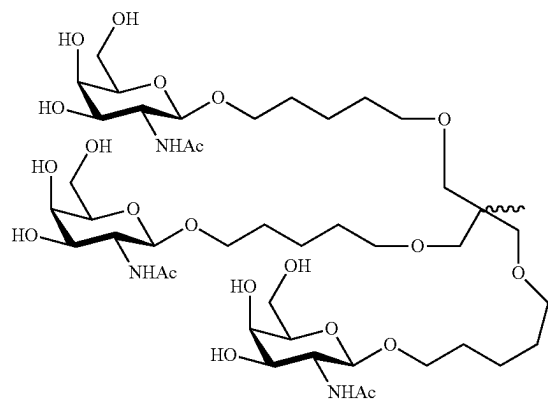
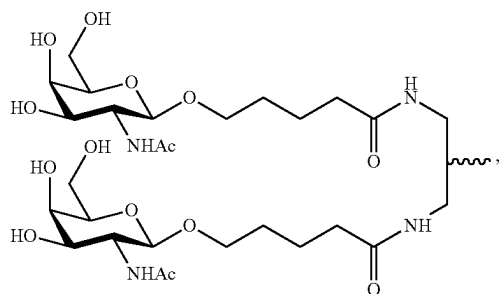
Formula IV

Formula V

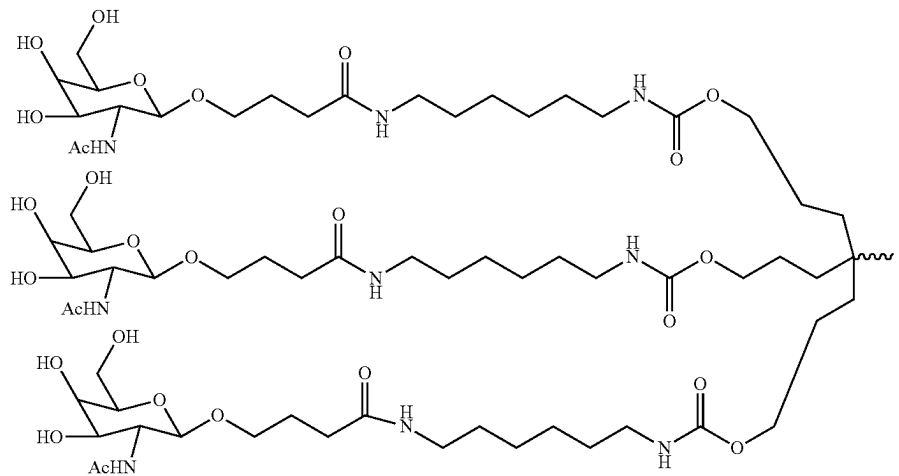


Formula VI

Formual VII



Formula VIII



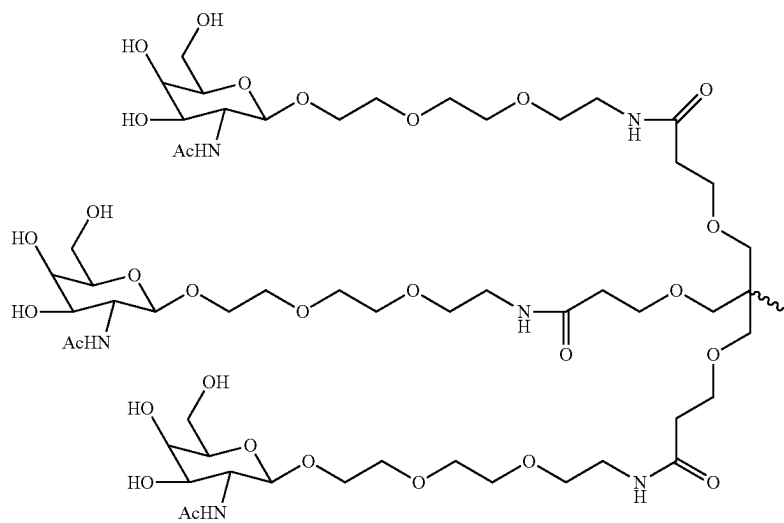
Formula IX

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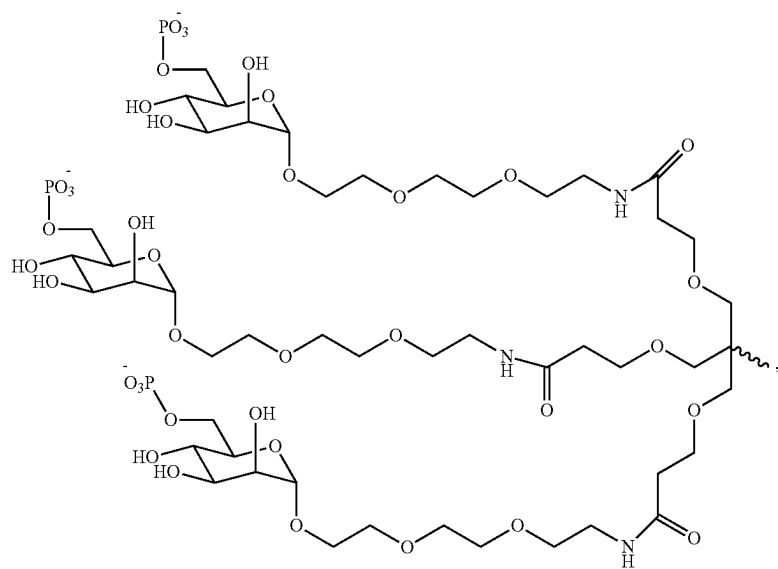
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Formula X

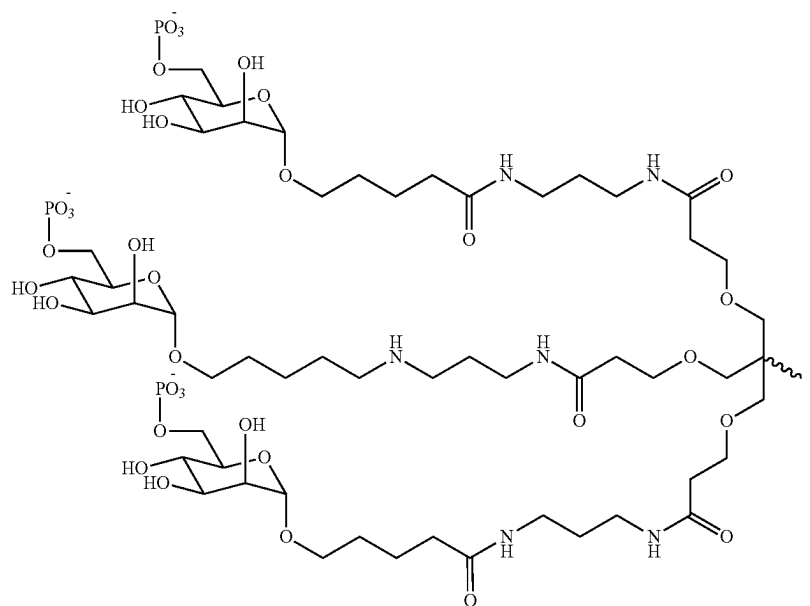


Formula XI

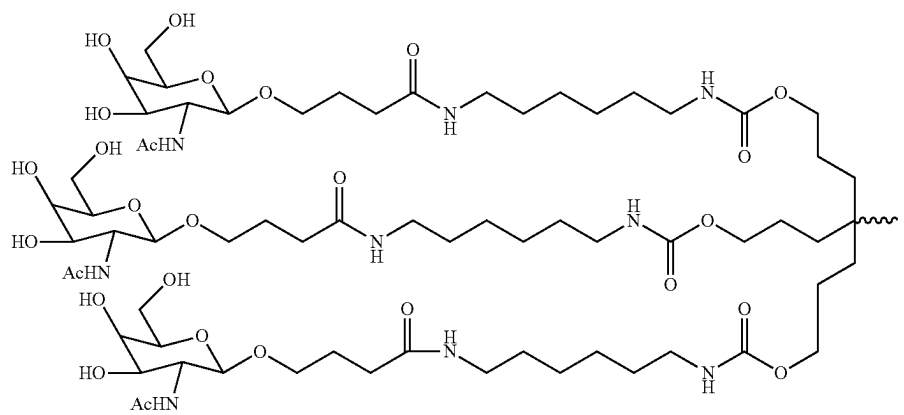


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Formula XII

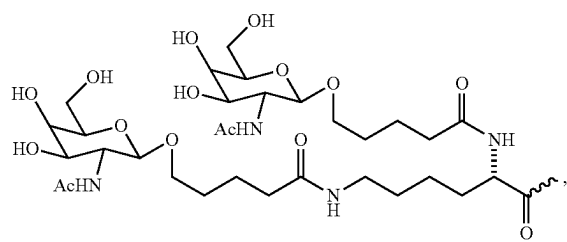


Formula XIII

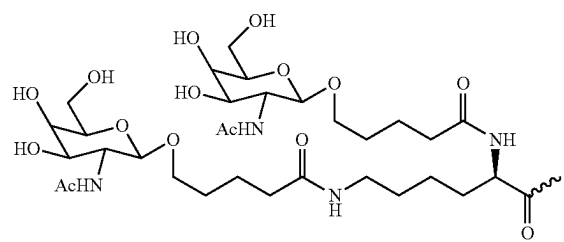


Formula XIV

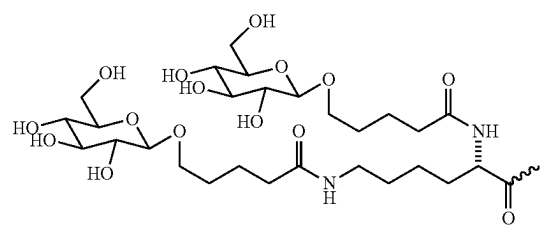
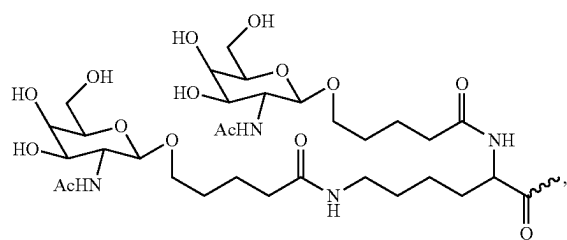
Formula XV



Formula XVI



Formula XVII



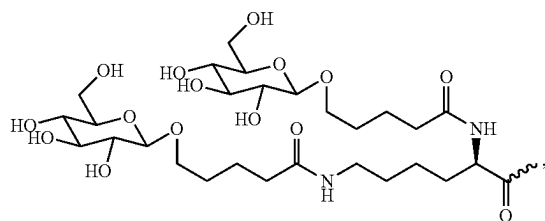
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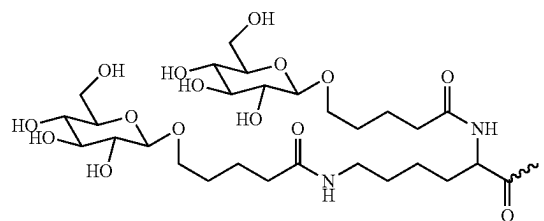
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Formula XVIII

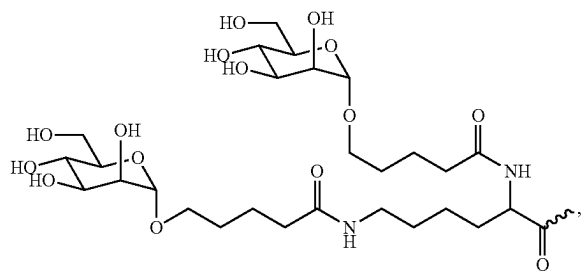
Formula XIX



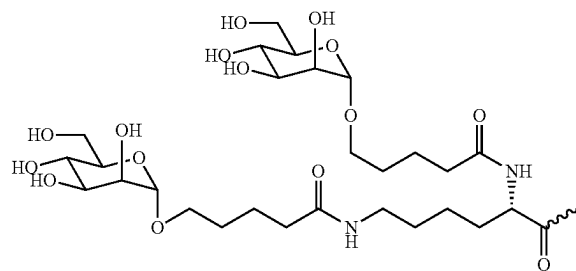
Formula XX



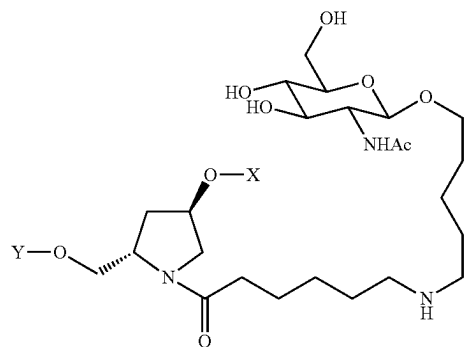
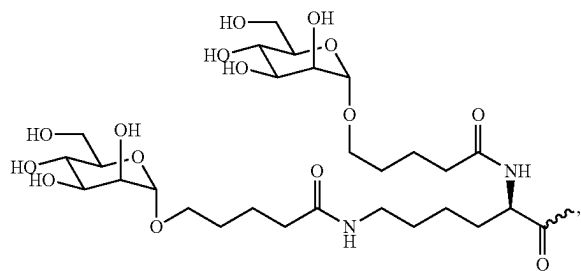
Formula XXI



Formula XXII

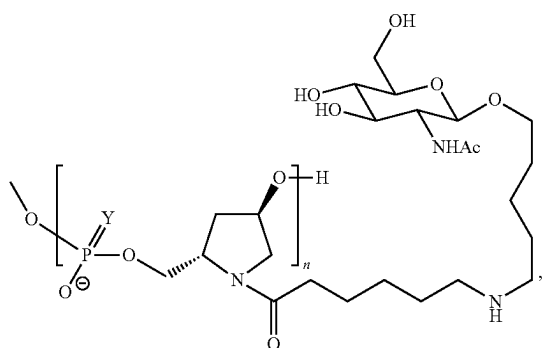


Formula XXIII

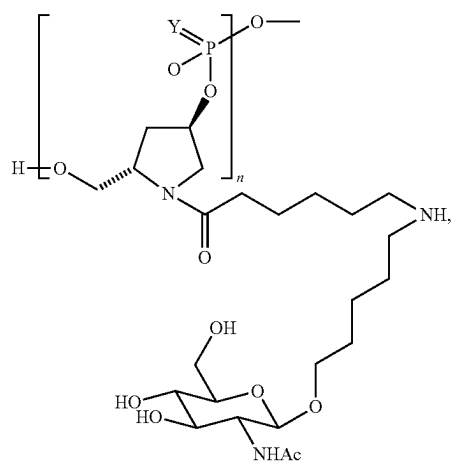


(Formula XXIV)

(Formula XXV)



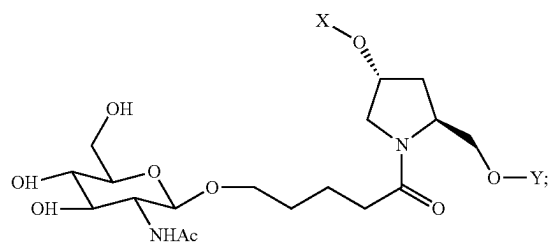
wherein Y is O or S and n is 3-6;



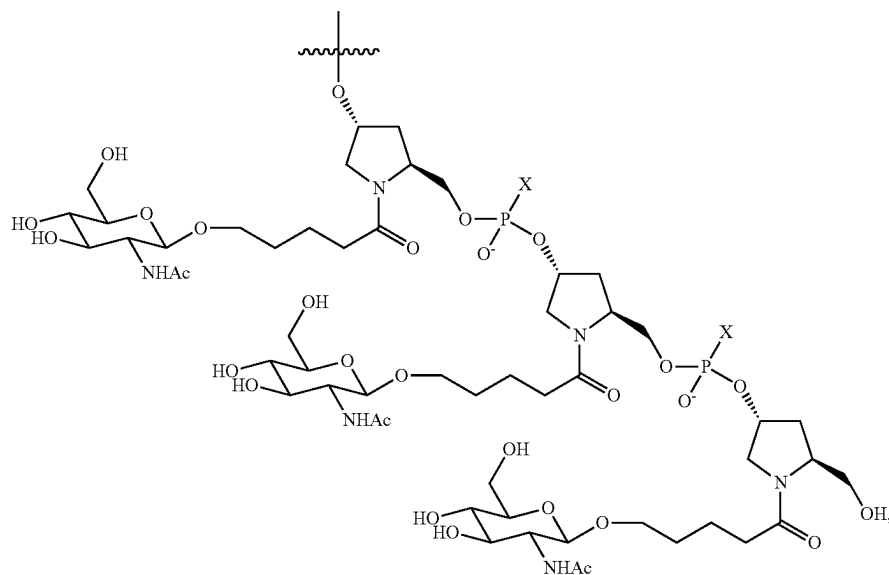
wherein Y is O or S and n is 3-6;

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Formula XXVI

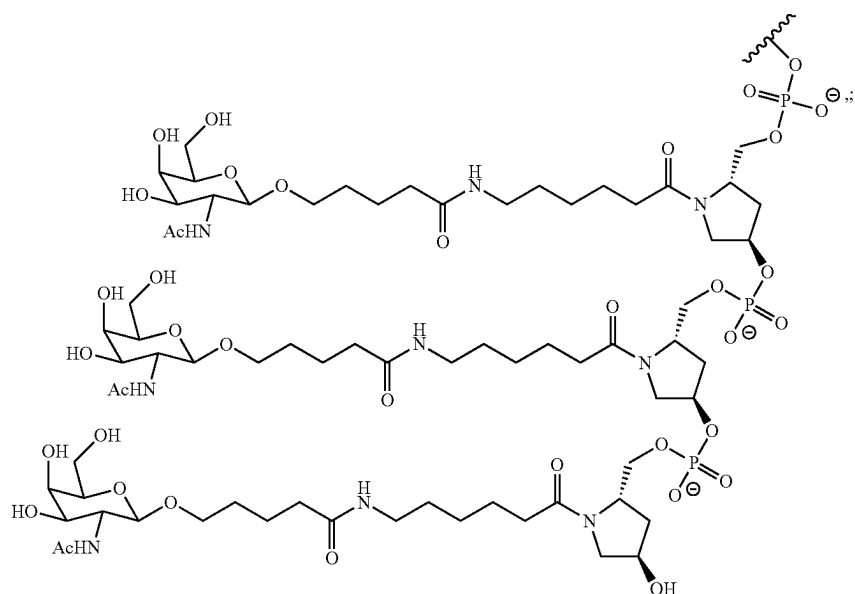


(Formula XXVII)



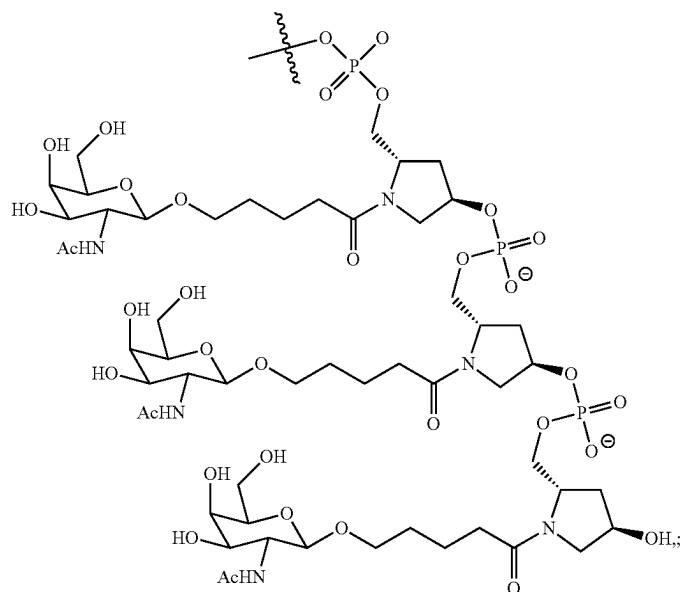
wherein X is O or S;

Formula XXVIII

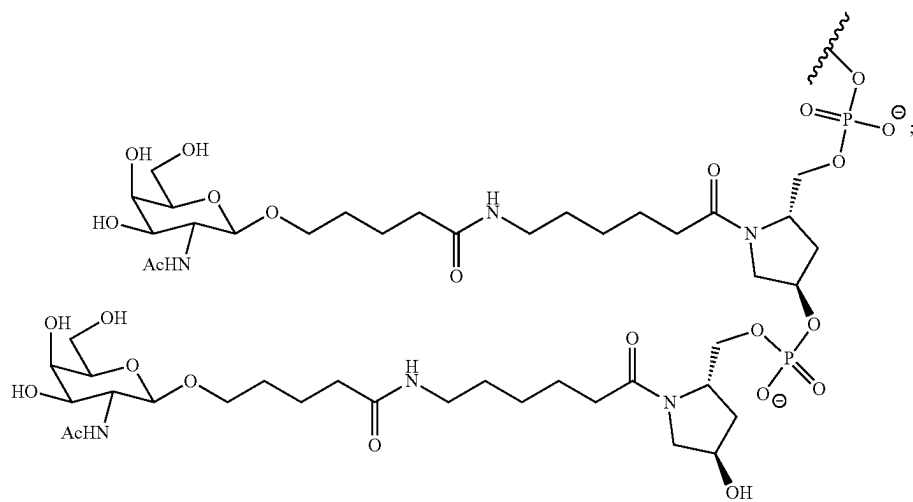


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Formula XXIX



Formula XXX

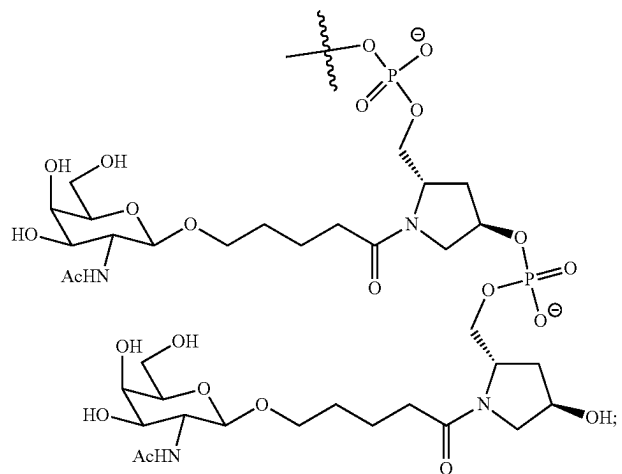


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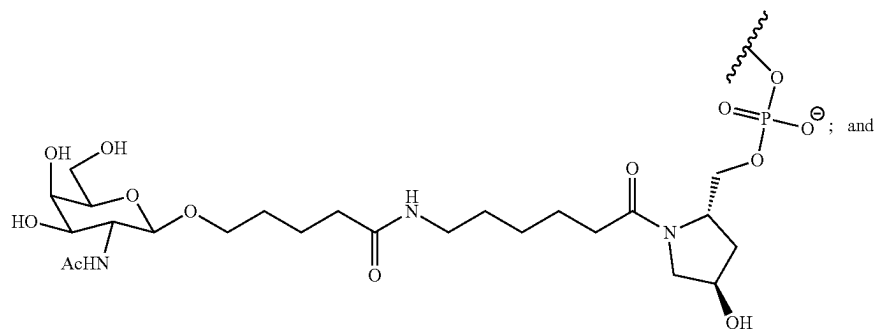
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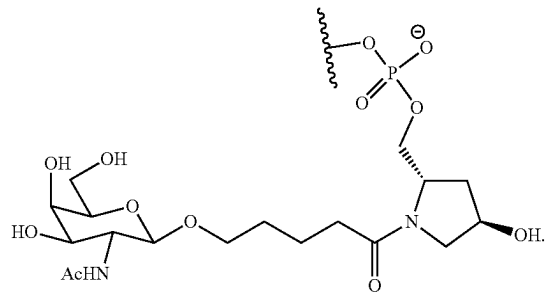
Formula XXXI



Formula XXXII



Formula XXXIII

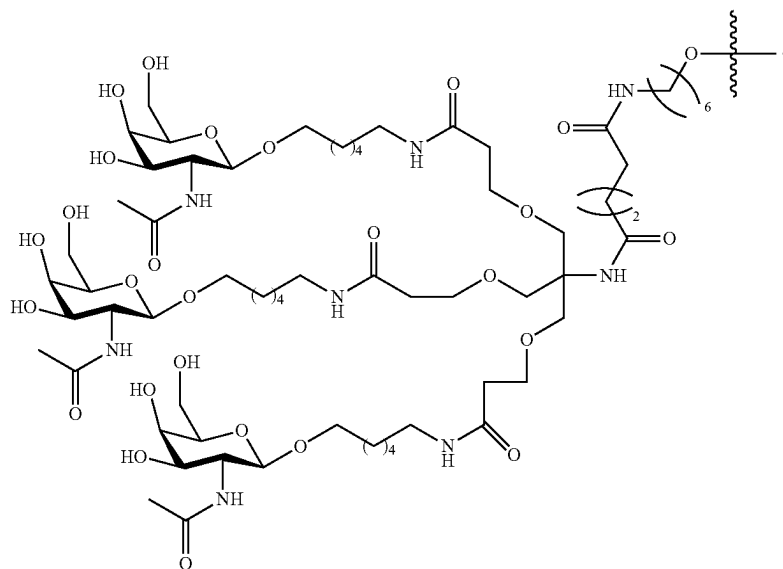


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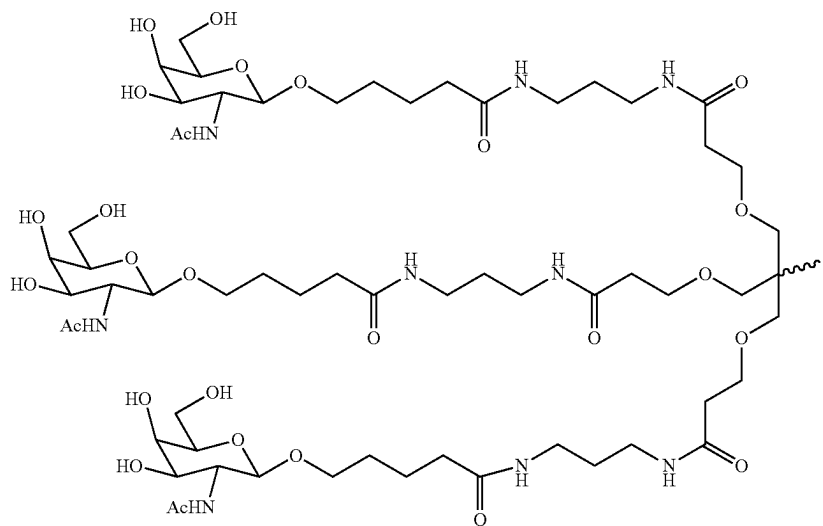
Formula XXXIV



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In another embodiment, a carbohydrate conjugate for use in the compositions and methods of the invention is a monosaccharide. In one embodiment, the monosaccharide is an N-acetylgalactosamine, such as

Formula II

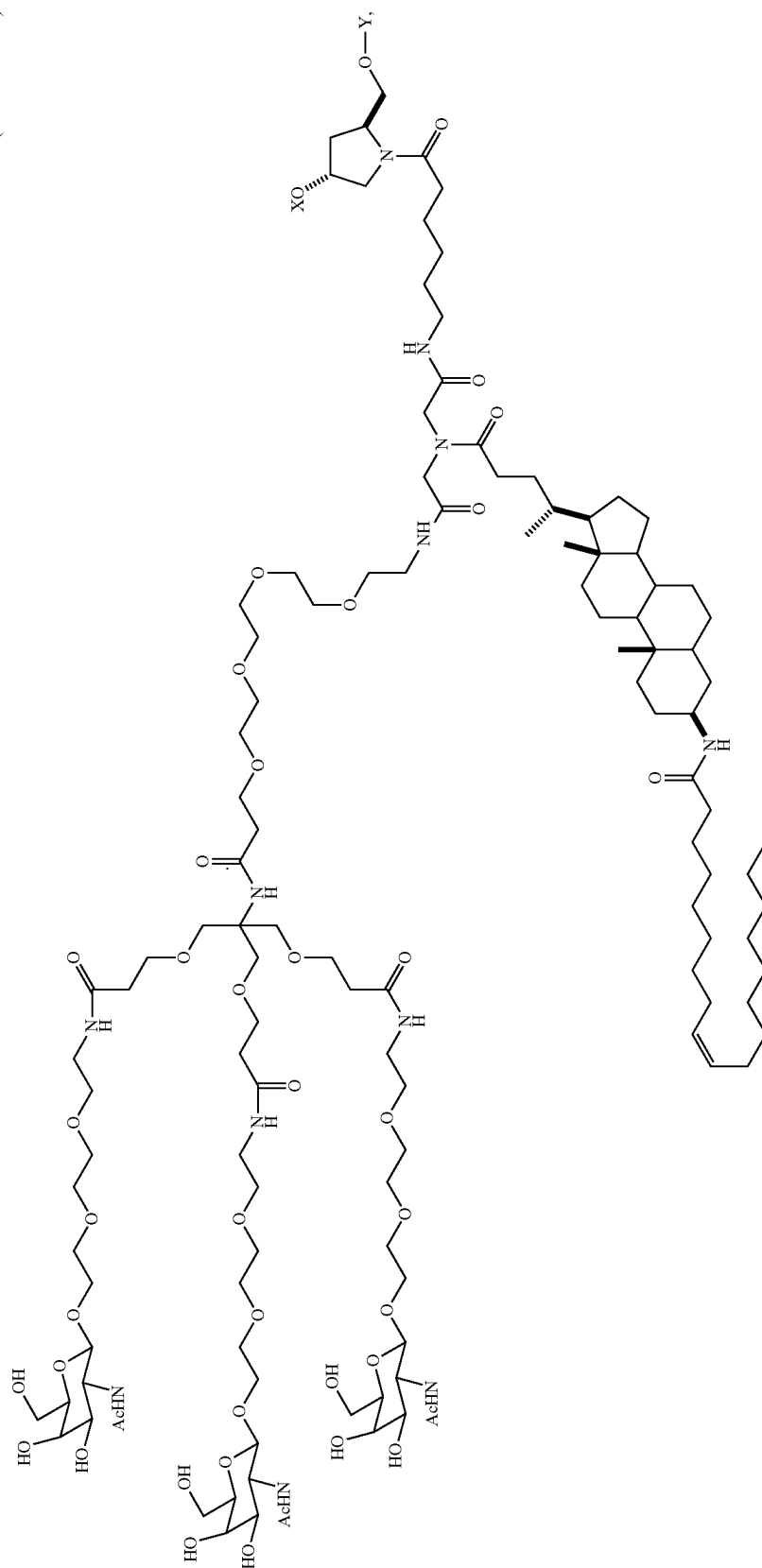


Another representative carbohydrate conjugate for use in the embodiments described herein includes, but is not limited to,

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(Formula XXXV)



when one of X or Y is an oligonucleotide, the other is a hydrogen.

In certain embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a monovalent linker. In some embodiments, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a bivalent linker. In yet other embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a trivalent linker.

In one embodiment, the double stranded RNAi agents of the invention comprise one GalNAc or GalNAc derivative attached to the iRNA agent, e.g., the 3' or 5' end of the sense strand of a dsRNA agent as described herein. In another embodiment, the double stranded RNAi agents of the invention comprise a plurality (e.g., 2, 3, 4, 5, or 6) of GalNAc or GalNAc derivatives, each independently attached to a plurality of nucleotides of the double stranded RNAi agent through a plurality of monovalent linkers.

In some embodiments, for example, when the two strands of an iRNA agent of the invention are part of one larger molecule connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming a hairpin loop comprising, a plurality of unpaired nucleotides, each unpaired nucleotide within the hairpin loop may independently comprise a GalNAc or GalNAc derivative attached via a monovalent linker.

In some embodiments, the carbohydrate conjugate further comprises one or more additional ligands as described above, such as, but not limited to, a PK modulator and/or a cell permeation peptide.

Additional carbohydrate conjugates (and linkers) suitable for use in the present invention include those described in PCT Publication Nos. WO 2014/179620 and WO 2014/179627, the entire contents of each of which are incorporated herein by reference.

D. Linkers

In some embodiments, the conjugate or ligand described herein can be attached to an iRNA oligonucleotide with various linkers that can be cleavable or non-cleavable.

The term "linker" or "linking group" means an organic moiety that connects two parts of a compound, e.g., covalently attaches two parts of a compound. Linkers typically comprise a direct bond or an atom such as oxygen or sulfur, a unit such as NR₈, C(O), C(O)NH, SO, SO₂, SO₂NH or a chain of atoms, such as, but not limited to, substituted or unsubstituted alkyl, substituted or unsubstituted alkenyl, substituted or unsubstituted alkynyl, arylalkyl, arylalkenyl, arylalkynyl, heteroarylalkyl, heteroarylalkenyl, heteroarylalkynyl, heterocyclalkyl, heterocyclalkenyl, heterocyclalkynyl, aryl, heteroaryl, heterocycl, cycloalkyl, cycloalkenyl, alkylarylalkyl, alkylarylalkenyl, alkylarylalkynyl, alkenylarylalkyl, alkenylarylalkenyl, alkenylarylalkynyl, alkynylarylalkyl, alkynylarylalkenyl, alkynylarylalkynyl, alkylheteroarylalkyl, alkylheteroarylalkenyl, alkylheteroarylalkynyl, alkenylheteroarylalkyl, alkenylheteroarylalkenyl, alkenylheteroarylalkynyl, alkynylheteroarylalkyl, alkynylheteroarylalkenyl, alkynylheteroarylalkynyl, alkylheterocyclalkyl, alkylheterocyclalkenyl, alkylheterocyclalkynyl, alkenylheterocyclalkyl, alkenylheterocyclalkenyl, alkenylheterocyclalkynyl, alkynylheterocyclalkyl, alkynylheterocyclalkenyl, alkynylheterocyclalkynyl, alkylarylalkenyl, alkylarylalkynyl, alkylheteroaryl, alkenylheteroaryl, alkynylheteroaryl, which one or more methylenes can be interrupted or terminated by O, S, S(O), SO₂, N(R₈), C(O), substituted

or unsubstituted aryl, substituted or unsubstituted heteroaryl, substituted or unsubstituted heterocyclic; where R₈ is hydrogen, acyl, aliphatic or substituted aliphatic. In one embodiment, the linker is between about 1-24 atoms, 2-24, 3-24, 4-24, 5-24, 6-24, 6-18, 7-18, 8-18 atoms, 7-17, 8-17, 6-16, 7-17, or 8-16 atoms.

A cleavable linking group is one which is sufficiently stable outside the cell, but which upon entry into a target cell is cleaved to release the two parts the linker is holding together. In a preferred embodiment, the cleavable linking group is cleaved at least about 10 times, 20, times, 30 times, 40 times, 50 times, 60 times, 70 times, 80 times, 90 times or more, or at least about 100 times faster in a target cell or under a first reference condition (which can, e.g., be selected to mimic or represent intracellular conditions) than in the blood of a subject, or under a second reference condition (which can, e.g., be selected to mimic or represent conditions found in the blood or serum).

Cleavable linking groups are susceptible to cleavage agents, e.g., pH, redox potential or the presence of degradative molecules. Generally, cleavage agents are more prevalent or found at higher levels or activities inside cells than in serum or blood. Examples of such degradative agents include: redox agents which are selected for particular substrates or which have no substrate specificity, including, e.g., oxidative or reductive enzymes or reductive agents such as mercaptans, present in cells, that can degrade a redox cleavable linking group by reduction; esterases; endosomes or agents that can create an acidic environment, e.g., those that result in a pH of five or lower; enzymes that can hydrolyze or degrade an acid cleavable linking group by acting as a general acid, peptidases (which can be substrate specific), and phosphatases.

A cleavable linkage group, such as a disulfide bond can be susceptible to pH. The pH of human serum is 7.4, while the average intracellular pH is slightly lower, ranging from about 7.1-7.3. Endosomes have a more acidic pH, in the range of 5.5-6.0, and lysosomes have an even more acidic pH at around 5.0. Some linkers will have a cleavable linking group that is cleaved at a preferred pH, thereby releasing a cationic lipid from the ligand inside the cell, or into the desired compartment of the cell.

A linker can include a cleavable linking group that is cleavable by a particular enzyme. The type of cleavable linking group incorporated into a linker can depend on the cell to be targeted. For example, a liver-targeting ligand can be linked to a cationic lipid through a linker that includes an ester group. Liver cells are rich in esterases, and therefore the linker will be cleaved more efficiently in liver cells than in cell types that are not esterase-rich. Other cell-types rich in esterases include cells of the lung, renal cortex, and testis.

Linkers that contain peptide bonds can be used when targeting cell types rich in peptidases, such as liver cells and synovocytes.

In general, the suitability of a candidate cleavable linking group can be evaluated by testing the ability of a degradative agent (or condition) to cleave the candidate linking group. It will also be desirable to also test the candidate cleavable linking group for the ability to resist cleavage in the blood or when in contact with other non-target tissue. Thus, one can determine the relative susceptibility to cleavage between a first and a second condition, where the first is selected to be indicative of cleavage in a target cell and the second is selected to be indicative of cleavage in other tissues or biological fluids, e.g., blood or serum. The evaluations can be carried out in cell free systems, in cells, in cell culture, in organ or tissue culture, or in whole animals. It can be useful

to make initial evaluations in cell-free or culture conditions and to confirm by further evaluations in whole animals. In preferred embodiments, useful candidate compounds are cleaved at least about 2, 4, 10, 20, 30, 40, 50, 60, 70, 80, 90, or about 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood or serum (or under in vitro conditions selected to mimic extracellular conditions).

i. Redox Cleavable Linking Groups

In one embodiment, a cleavable linking group is a redox cleavable linking group that is cleaved upon reduction or oxidation. An example of reductively cleavable linking group is a disulphide linking group (—S—S—). To determine if a candidate cleavable linking group is a suitable “reductively cleavable linking group,” or for example is suitable for use with a particular iRNA moiety and particular targeting agent one can look to methods described herein. For example, a candidate can be evaluated by incubation with dithiothreitol (DTT), or other reducing agent using reagents known in the art, which mimic the rate of cleavage which would be observed in a cell, e.g., a target cell. The candidates can also be evaluated under conditions which are selected to mimic blood or serum conditions. In one, candidate compounds are cleaved by at most about 10% in the blood. In other embodiments, useful candidate compounds are degraded at least about 2, 4, 10, 20, 30, 40, 50, 60, 70, 80, 90, or about 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood (or under in vitro conditions selected to mimic extracellular conditions). The rate of cleavage of candidate compounds can be determined using standard enzyme kinetics assays under conditions chosen to mimic intracellular media and compared to conditions chosen to mimic extracellular media.

ii. Phosphate-Based Cleavable Linking Groups

In another embodiment, a cleavable linker comprises a phosphate-based cleavable linking group. A phosphate-based cleavable linking group is cleaved by agents that degrade or hydrolyze the phosphate group. An example of an agent that cleaves phosphate groups in cells are enzymes such as phosphatases in cells. Examples of phosphate-based linking groups are —O—P(O)(ORk)—O— , —O—P(S)(ORk)—O— , —O—P(S)(SRk)—O— , —S—P(O)(ORk)—O— , —O—P(O)(ORk)—S— , —S—P(O)(ORk)—S— , —O—P(S)(ORk)—S— , —S—P(S)(ORk)—O— , —O—P(O)(Rk)—O— , —O—P(S)(Rk)—O— , —S—P(O)(Rk)—O— , —S—P(S)(Rk)—O— , —S—P(O)(Rk)—S— , —O—P(S)(Rk)—S— . Preferred embodiments are —O—P(O)(OH)—O— , —O—P(S)(OH)—O— , —O—P(S)(SH)—O— , —S—P(O)(OH)—O— , —O—P(O)(OH)—S— , —S—P(O)(OH)—S— , —O—P(S)(OH)—S— , —S—P(S)(OH)—O— , —O—P(O)(H)—O— , —O—P(S)(H)—O— , —S—P(O)(H)—O— , —S—P(S)(H)—O— , —S—P(O)(H)—S— , —O—P(S)(H)—S— . A preferred embodiment is —O—P(O)(OH)—O— . These candidates can be evaluated using methods analogous to those described above.

iii. Acid Cleavable Linking Groups

In another embodiment, a cleavable linker comprises an acid cleavable linking group. An acid cleavable linking group is a linking group that is cleaved under acidic conditions. In preferred embodiments acid cleavable linking groups are cleaved in an acidic environment with a pH of about 6.5 or lower (e.g., about 6.0, 5.75, 5.5, 5.25, 5.0, or lower), or by agents such as enzymes that can act as a general acid. In a cell, specific low pH organelles, such as endosomes and lysosomes can provide a cleaving environment for acid cleavable linking groups. Examples of acid cleavable linking groups include but are not limited to hydrazones, esters, and esters of amino acids. Acid cleavable groups can have the general formula —C=NN— , C(O)O— , or —OC(O)— . A preferred embodiment is when the carbon attached to the oxygen of the ester (the alkoxy group) is an aryl group, substituted alkyl group, or tertiary alkyl group such as dimethyl pentyl or t-butyl. These candidates can be evaluated using methods analogous to those described above.

iv. Ester-Based Linking Groups

In another embodiment, a cleavable linker comprises an ester-based cleavable linking group. An ester-based cleavable linking group is cleaved by enzymes such as esterases and amidases in cells. Examples of ester-based cleavable linking groups include but are not limited to esters of alkylene, alkenylene and alkynylene groups. Ester cleavable linking groups have the general formula —C(O)O— , or —OC(O)— . These candidates can be evaluated using methods analogous to those described above.

v. Peptide-Based Cleaving Groups

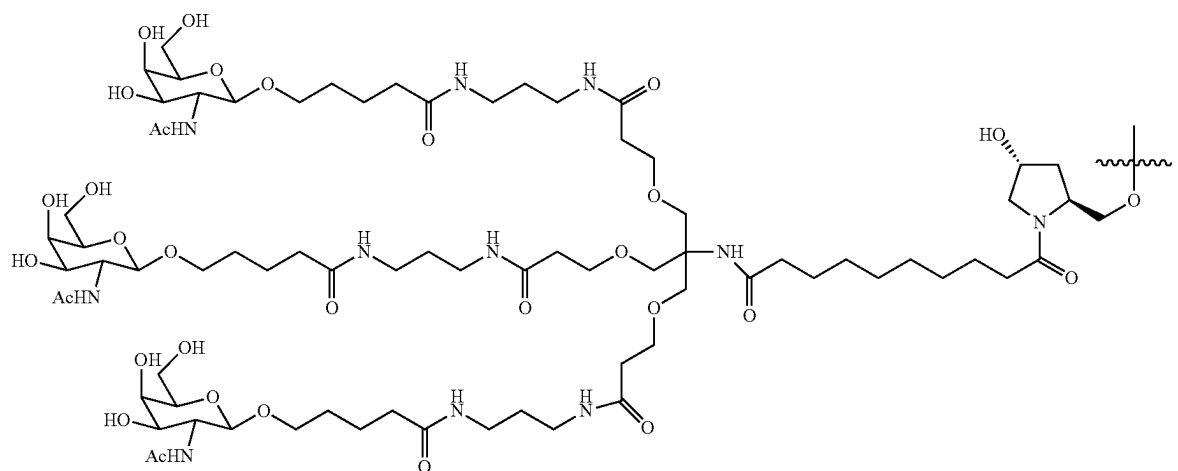
In yet another embodiment, a cleavable linker comprises a peptide-based cleavable linking group. A peptide-based cleavable linking group is cleaved by enzymes such as peptidases and proteases in cells. Peptide-based cleavable linking groups are peptide bonds formed between amino acids to yield oligopeptides (e.g., dipeptides, tripeptides etc.) and polypeptides. Peptide-based cleavable groups do not include the amide group (—C(O)NH—). The amide group can be formed between any alkylene, alkenylene or alkynylene. A peptide bond is a special type of amide bond formed between amino acids to yield peptides and proteins. The peptide based cleavage group is generally limited to the peptide bond (i.e., the amide bond) formed between amino acids yielding peptides and proteins and does not include the entire amide functional group. Peptide-based cleavable linking groups have the general formula $\text{—NHCHRAC(O)NHCHRBC(O)—}$, where RA and RB are the R groups of the two adjacent amino acids. These candidates can be evaluated using methods analogous to those described above.

In one embodiment, an iRNA of the invention is conjugated to a carbohydrate through a linker. Non-limiting examples of iRNA carbohydrate conjugates with linkers of the compositions and methods of the invention include, but are not limited to,

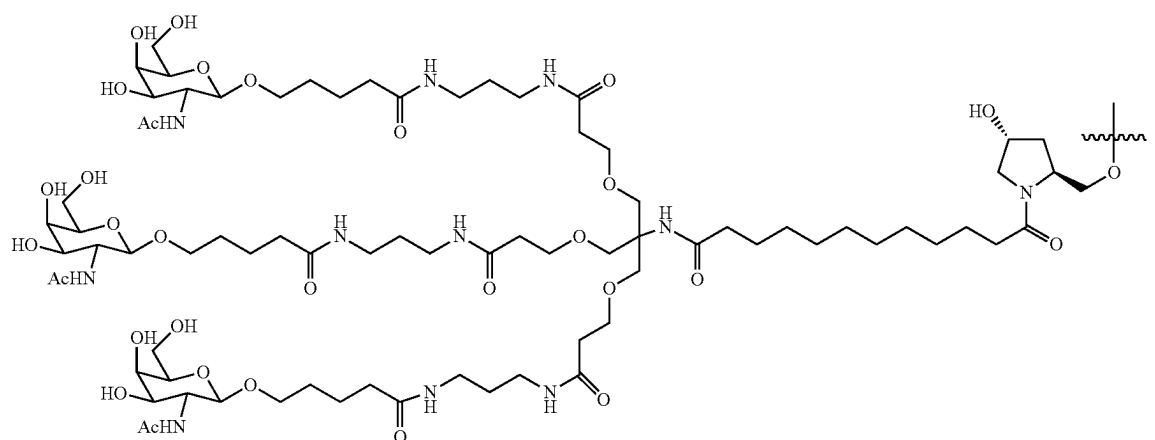
111

112

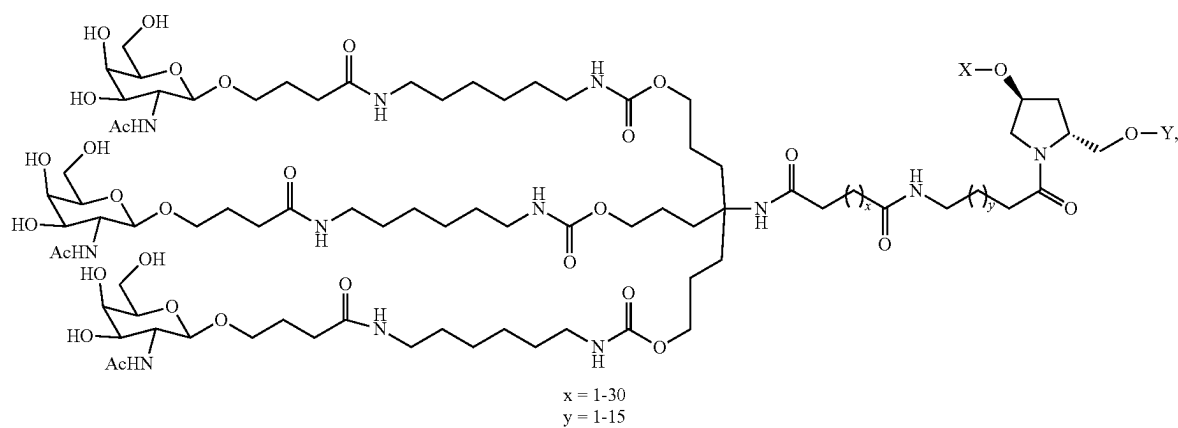
(Formula XXXVI)



(Formula XXXVII)



(Formula XXXVIII)

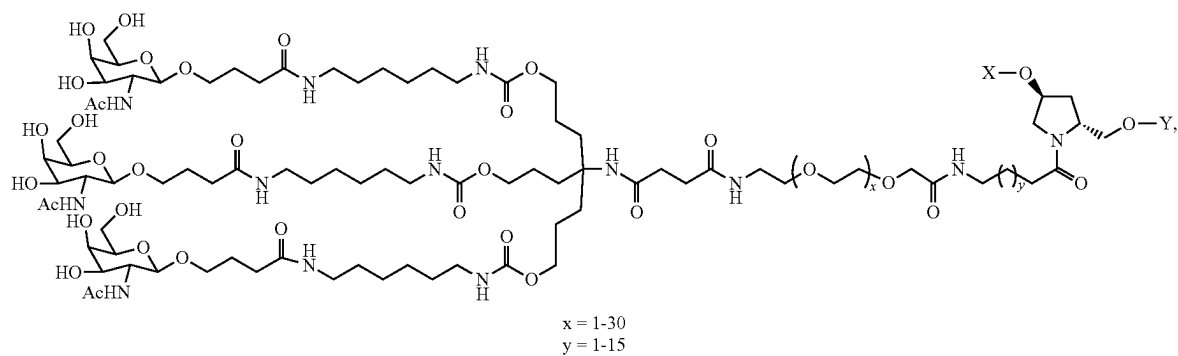


113

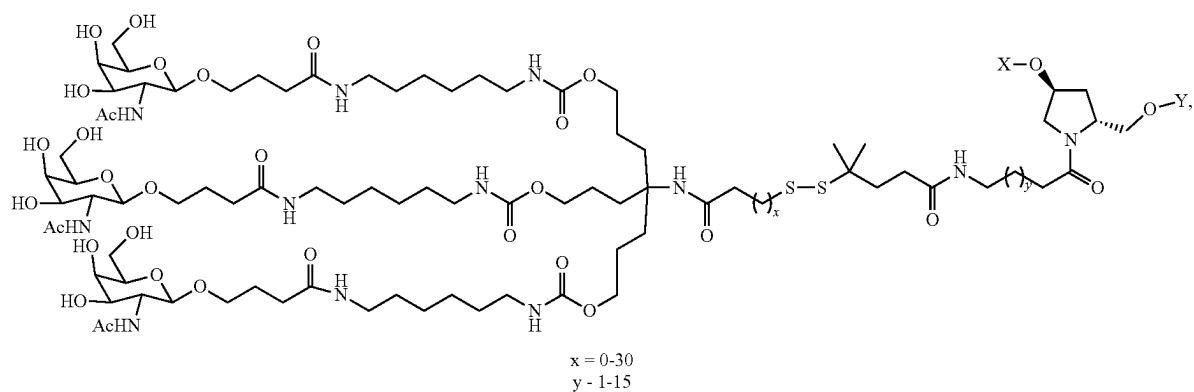
114

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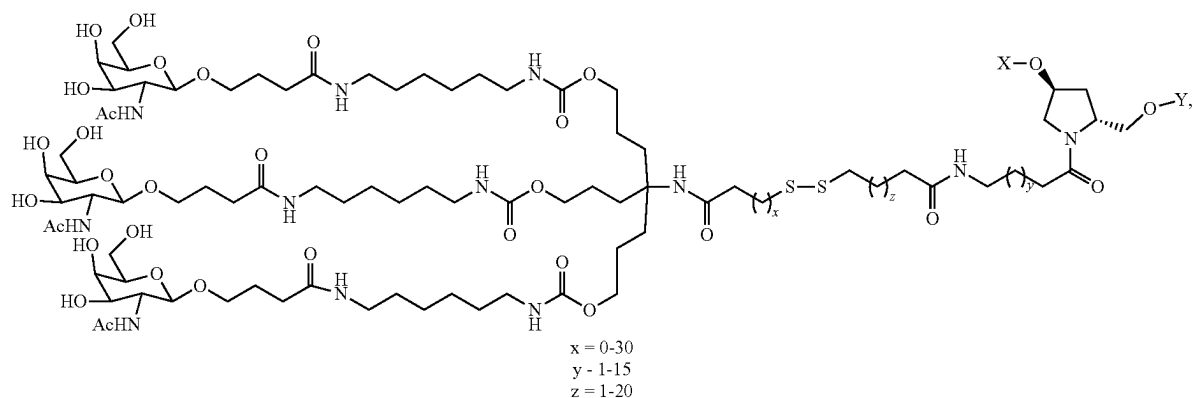
(Formula XXXIX)



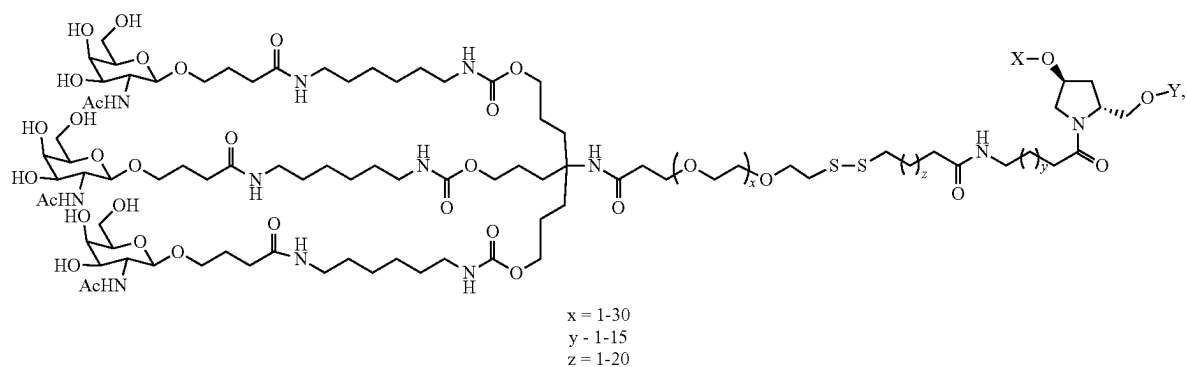
(Formula XL)



(Formula XLI)



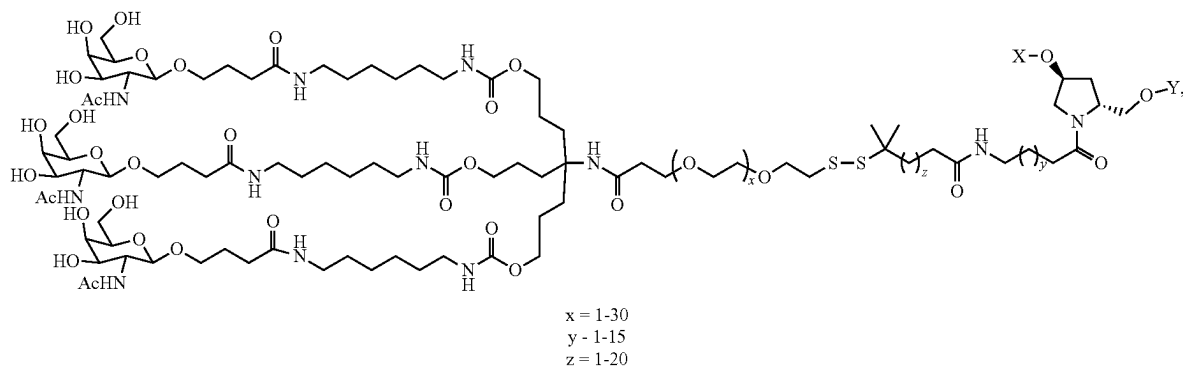
(Formula XLII)



and

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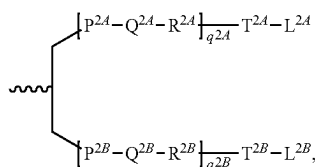
(Formula XLIII)



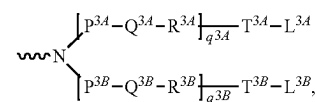
when one of X or Y is an oligonucleotide, the other is a hydrogen.

In certain embodiments of the compositions and methods of the invention, a ligand is one or more GalNAc (N-acetyl-galactosamine) derivatives attached through a bivalent or trivalent branched linker.

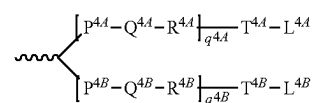
In one embodiment, a dsRNA of the invention is conjugated to a bivalent or trivalent branched linker selected from the group of structures shown in any of Formula XLIV-XLVII:



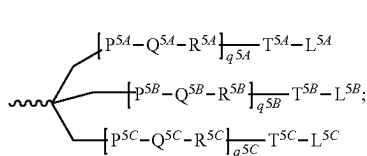
Formula XLIV



Formula XLV



Formula XLVI

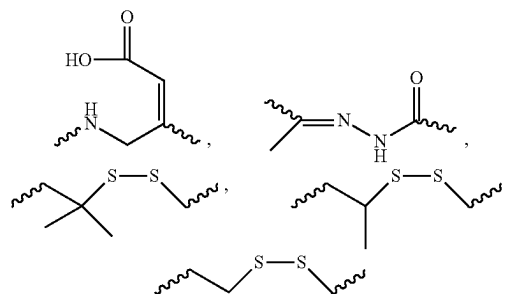


Formula XLVII

wherein:

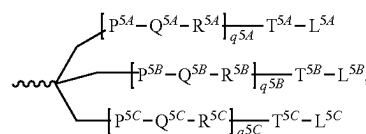
q2A, q2B, q3A, q3B, q4A, q4B, q5A, q5B and q5C represent independently for each occurrence 0-20 and wherein the repeating unit can be the same or different; p^{2A}, p^{2B}, p^{3A}, p^{3B}, p^{4A}, p^{4B}, p^{5A}, p^{5B}, p^{5C}, T^{2A}, T^{2B}, T^{3A}, T^{3B}, T^{4A}, T^{4B}, T^{5A}, T^{5B}, T^{5C} are each independently for each occurrence absent, CO, NH, O, S, OC(O), NHC(O), CH₂, CH₂NH or CH₂O; Q^{2A}, Q^{2B}, Q^{3A}, Q^{3B}, Q^{4A}, Q^{4B}, Q^{5A}, Q^{5B}, Q^{5C} are independently for each occurrence absent, alkylene, substituted alkylene wherein one or more methylenes can be

interrupted or terminated by one or more of O, S, S(O), SO₂, N(R^N), C(R')=C(R''), C≡C or C(O); R^{2A}, R^{2B}, R^{3A}, R^{3B}, R^{4A}, R^{4B}, R^{5A}, R^{5B}, R^{5C} are each independently for each occurrence absent, NH, O, S, CH₂, C(O)O, C(O)NH, NHCH(R^a)C(O), -C(O)-CH(R^a)-NH-, CO, CH=N-O,



or heterocyclyl;

L^{2A}, L^{2B}, L^{3A}, L^{3B}, L^{4A}, L^{4B}, L^{5A}, L^{5B} and L^{5C} represent the ligand; i.e. each independently for each occurrence a monosaccharide (such as GalNAc), disaccharide, trisaccharide, tetrasaccharide, oligosaccharide, or polysaccharide; and R^a is H or amino acid side chain. Trivalent conjugating GalNAc derivatives are particularly useful for use with RNAi agents for inhibiting the expression of a target gene, such as those of formula XLVIII:



Formula XLVIII

wherein L^{5A}, L^{5B} and L^{5C} represent a monosaccharide, such as GalNAc derivative.

Examples of suitable bivalent and trivalent branched linker groups conjugating GalNAc derivatives include, but are not limited to, the structures recited above as formulas II, VII, XI, X, and XIII.

Representative U.S. patents that teach the preparation of RNA conjugates include, but are not limited to, U.S. Pat.

Nos. 4,828,979; 4,948,882; 5,218,105; 5,525,465; 5,541,313; 5,545,730; 5,552,538; 5,578,717; 5,580,731; 5,591,584; 5,109,124; 5,118,802; 5,138,045; 5,414,077; 5,486,603; 5,512,439; 5,578,718; 5,608,046; 4,587,044; 4,605,735; 4,667,025; 4,762,779; 4,789,737; 4,824,941; 4,835,263; 4,876,335; 4,904,582; 4,958,013; 5,082,830; 5,112,963; 5,214,136; 5,082,830; 5,112,963; 5,214,136; 5,245,022; 5,254,469; 5,258,506; 5,262,536; 5,272,250; 5,292,873; 5,317,098; 5,371,241; 5,391,723; 5,416,203; 5,451,463; 5,510,475; 5,512,667; 5,514,785; 5,565,552; 5,567,810; 5,574,142; 5,585,481; 5,587,371; 5,595,726; 5,597,696; 5,599,923; 5,599,928 and 5,688,941; 6,294,664; 6,320,017; 6,576,752; 6,783,931; 6,900,297; 7,037,646; 8,106,022, the entire contents of each of which are hereby incorporated herein by reference.

It is not necessary for all positions in a given compound to be uniformly modified, and in fact more than one of the aforementioned modifications can be incorporated in a single compound or even at a single nucleoside within an iRNA. The present invention also includes iRNA compounds that are chimeric compounds.

"Chimeric" iRNA compounds or "chimeras," in the context of this invention, are iRNA compounds, preferably dsRNAs, which contain two or more chemically distinct regions, each made up of at least one monomer unit, i.e., a nucleotide in the case of a dsRNA compound. These iRNAs typically contain at least one region wherein the RNA is modified so as to confer upon the iRNA increased resistance to nuclease degradation, increased cellular uptake, and/or increased binding affinity for the target nucleic acid. An additional region of the iRNA can serve as a substrate for enzymes capable of cleaving RNA:DNA or RNA:RNA hybrids. By way of example, RNase H is a cellular endonuclease which cleaves the RNA strand of an RNA:DNA duplex. Activation of RNase H, therefore, results in cleavage of the RNA target, thereby greatly enhancing the efficiency of iRNA inhibition of gene expression. Consequently, comparable results can often be obtained with shorter iRNAs when chimeric dsRNAs are used, compared to phosphorothioate deoxy dsRNAs hybridizing to the same target region. Cleavage of the RNA target can be routinely detected by gel electrophoresis and, if necessary, associated nucleic acid hybridization techniques known in the art.

In certain instances, the RNA of an iRNA can be modified by a non-ligand group. A number of non-ligand molecules have been conjugated to iRNAs in order to enhance the activity, cellular distribution or cellular uptake of the iRNA, and procedures for performing such conjugations are available in the scientific literature. Such non-ligand moieties have included lipid moieties, such as cholesterol (Kubo, T. et al., *Biochem. Biophys. Res. Comm.*, 2007, 365(1):54-61; Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86:6553), cholic acid (Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1994, 4:1053), a thioether, e.g., hexyl-S-tritylthiol (Manoharan et al., *Ann. N.Y. Acad. Sci.*, 1992, 660:306; Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1993, 3:2765), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20:533), an aliphatic chain, e.g., dodecandiol or undecyl residues (Saison-Behmoaras et al., *EMBO J.*, 1991, 10:111; Kabanov et al., *FEBS Lett.*, 1990, 259:327; Svinarchuk et al., *Biochimie*, 1993, 75:49), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethylammonium 1,2-di-O-hexadecyl-rac-glycero-3-H-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651; Shea et al., *Nucl. Acids Res.*, 1990, 18:3777), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14:969), or adamantane acetic acid (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651), a palmitoyl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264:229), or an octadecylamine or hexylamino-carbonyl-oxycholesterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277:923). Representative United States patents that teach the preparation of such RNA conjugates have been listed above. Typical conjugation protocols involve the synthesis of an RNAs bearing an aminolinker at one or more positions of the sequence. The amino group is then reacted with the molecule being conjugated using appropriate coupling or activating reagents. The conjugation reaction can be performed either with the RNA still bound to the solid support or following cleavage of the RNA, in solution phase. Purification of the RNA conjugate by HPLC typically affords the pure conjugate.

V. Delivery of an iRNA of the Invention

The delivery of an iRNA of the invention to a cell e.g., a cell within a subject, such as a human subject (e.g., a subject in need thereof, such as a subject having a disorder of lipid metabolism) can be achieved in a number of different ways. For example, delivery may be performed by contacting a cell with an iRNA of the invention either in vitro or in vivo. In vivo delivery may also be performed directly by administering a composition comprising an iRNA, e.g., a dsRNA, to a subject. Alternatively, in vivo delivery may be performed indirectly by administering one or more vectors that encode and direct the expression of the iRNA. These alternatives are discussed further below.

In general, any method of delivering a nucleic acid molecule (in vitro or in vivo) can be adapted for use with an iRNA of the invention (see e.g., Akhtar S. and Julian R L., (1992) *Trends Cell. Biol.* 2(5):139-144 and WO94/02595, which are incorporated herein by reference in their entireties). For in vivo delivery, factors to consider in order to deliver an iRNA molecule include, for example, biological stability of the delivered molecule, prevention of non-specific effects, and accumulation of the delivered molecule in the target tissue. The non-specific effects of an iRNA can be minimized by local administration, for example, by direct injection or implantation into a tissue or topically administering the preparation. Local administration to a treatment site maximizes local concentration of the agent, limits the exposure of the agent to systemic tissues that can otherwise be harmed by the agent or that can degrade the agent, and permits a lower total dose of the iRNA molecule to be administered. Several studies have shown successful knock-down of gene products when an iRNA is administered locally. For example, intraocular delivery of a VEGF dsRNA by intravitreal injection in cynomolgus monkeys (Tolentino, M J. et al., (2004) *Retina* 24:132-138) and subretinal injections in mice (Reich, S J. et al. (2003) *Mol. Vis.* 9:210-216) were both shown to prevent neovascularization in an experimental model of age-related macular degeneration. In addition, direct intratumoral injection of a dsRNA in mice reduces tumor volume (Pille, J. et al. (2005) *Mol. Ther.* 11:267-274) and can prolong survival of tumor-bearing mice (Kim, W J. et al., (2006) *Mol. Ther.* 14:343-350; Li, S. et al., (2007) *Mol. Ther.* 15:515-523). RNA interference has also shown success with local delivery to the CNS by direct injection (Dorn, G. et al., (2004) *Nucleic Acids* 32: e49; Tan, P H. et al. (2005) *Gene Ther.* 12:59-66; Makimura, H. et al. (2002) *BMC Neurosci.* 3:18; Shishkina, G T., et al. (2004) *Neuroscience* 129:521-528; Thakker, E R., et al. (2004) *Proc. Natl. Acad. Sci. U.S.A.* 101:17270-17275; Akaneya, Y., et al. (2005) *J. Neurophysiol.* 93:594-602) and

to the lungs by intranasal administration (Howard, K. A. et al., (2006) *Mol. Ther.* 14:476-484; Zhang, X. et al., (2004) *J. Biol. Chem.* 279:10677-10684; Bitko, V. et al., (2005) *Nat. Med.* 11:50-55). For administering an iRNA systemically for the treatment of a disease, the RNA can be modified or alternatively delivered using a drug delivery system; both methods act to prevent the rapid degradation of the dsRNA by endo- and exo-nucleases in vivo. Modification of the RNA or the pharmaceutical carrier can also permit targeting of the iRNA composition to the target tissue and avoid undesirable off-target effects. iRNA molecules can be modified by chemical conjugation to lipophilic groups such as cholesterol to enhance cellular uptake and prevent degradation. For example, an iRNA directed against ApoB conjugated to a lipophilic cholesterol moiety was injected systemically into mice and resulted in knockdown of apoB mRNA in both the liver and jejunum (Soutschek, J. et al., (2004) *Nature* 432:173-178). Conjugation of an iRNA to an aptamer has been shown to inhibit tumor growth and mediate tumor regression in a mouse model of prostate cancer (McNamara, J. O. et al., (2006) *Nat. Biotechnol.* 24:1005-1015). In an alternative embodiment, the iRNA can be delivered using drug delivery systems such as a nanoparticle, a dendrimer, a polymer, liposomes, or a cationic delivery system. Positively charged cationic delivery systems facilitate binding of an iRNA molecule (negatively charged) and also enhance interactions at the negatively charged cell membrane to permit efficient uptake of an iRNA by the cell. Cationic lipids, dendrimers, or polymers can either be bound to an iRNA, or induced to form a vesicle or micelle (see e.g., Kim S. H. et al., (2008) *Journal of Controlled Release* 129(2):107-116) that encases an iRNA. The formation of vesicles or micelles further prevents degradation of the iRNA when administered systemically. Methods for making and administering cationic-iRNA complexes are well within the abilities of one skilled in the art (see e.g., Sorensen, D. R., et al. (2003) *J. Mol. Biol.* 327:761-766; Verma, U. N. et al., (2003) *Clin. Cancer Res.* 9:1291-1300; Arnold, A. S. et al., (2007) *J. Hypertens.* 25:197-205, which are incorporated herein by reference in their entirety). Some non-limiting examples of drug delivery systems useful for systemic delivery of iRNAs include DOTAP (Sorensen, D. R., et al. (2003), *supra*; Verma, U. N. et al., (2003), *supra*), Oligofectamine, "solid nucleic acid lipid particles" (Zimmermann, T. S. et al., (2006) *Nature* 441:111-114), cardiolipin (Chien, P. Y. et al., (2005) *Cancer Gene Ther.* 12:321-328; Pal, A. et al., (2005) *Int. J. Oncol.* 26:1087-1091), polyethyleneimine (Bonnet M. E. et al., (2008) *Pharm. Res.* August 16 Epub ahead of print; Aigner, A. (2006) *J. Biomed. Biotechnol.* 71659), Arg-Gly-Asp (RGD) peptides (Liu, S. (2006) *Mol. Pharm.* 3:472-487), and polyamidoamines (Tomalia, D. A. et al., (2007) *Biochem. Soc. Trans.* 35:61-67; Yoo, H. et al., (1999) *Pharm. Res.* 16:1799-1804). In some embodiments, an iRNA forms a complex with cyclodextrin for systemic administration. Methods for administration and pharmaceutical compositions of iRNAs and cyclodextrins can be found in U.S. Patent No. 7, 427, 605, which is herein incorporated by reference in its entirety.

A. Vector encoded iRNAs of the Invention

iRNA targeting the CIDEb gene can be expressed from transcription units inserted into DNA or RNA vectors (see, e.g., Couture, A. et al., *ITG*. (1996), 12:5-10; Skillern, A., et al., International PCT Publication No. WO 00/22113, Conrad, International PCT Publication No. WO 00/22114, and Conrad, U.S. Pat. No. 6,054,299). Expression can be transient (on the order of hours to weeks) or sustained (weeks to

months or longer), depending upon the specific construct used and the target tissue or cell type. These transgenes can be introduced as a linear construct, a circular plasmid, or a viral vector, which can be an integrating or non-integrating vector. The transgene can also be constructed to permit it to be inherited as an extrachromosomal plasmid (Gassmann, et al., (1995) *Proc. Natl. Acad. Sci. USA* 92:1292).

The individual strand or strands of an iRNA can be transcribed from a promoter on an expression vector. Where two separate strands are to be expressed to generate, for example, a dsRNA, two separate expression vectors can be co-introduced (e.g., by transfection or infection) into a target cell. Alternatively, each individual strand of a dsRNA can be transcribed by promoters both of which are located on the same expression plasmid. In one embodiment, a dsRNA is expressed as inverted repeat polynucleotides joined by a linker polynucleotide sequence such that the dsRNA has a stem and loop structure.

iRNA expression vectors are generally DNA plasmids or viral vectors. Expression vectors compatible with eukaryotic cells, preferably those compatible with vertebrate cells, can be used to produce recombinant constructs for the expression of an iRNA as described herein. Eukaryotic cell expression vectors are well known in the art and are available from a number of commercial sources. Typically, such vectors are provided containing convenient restriction sites for insertion of the desired nucleic acid segment. Delivery of iRNA expressing vectors can be systemic, such as by intravenous or intramuscular administration, by administration to target cells ex-planted from the patient followed by reintroduction into the patient, or by any other means that allows for introduction into a desired target cell.

Viral vector systems which can be utilized with the methods and compositions described herein include, but are not limited to, (a) adenovirus vectors; (b) retrovirus vectors, including but not limited to lentiviral vectors, moloney murine leukemia virus, etc.; (c) adeno-associated virus vectors; (d) herpes simplex virus vectors; (e) SV 40 vectors; (f) polyoma virus vectors; (g) papilloma virus vectors; (h) picornavirus vectors; (i) pox virus vectors such as an orthopox, e.g., vaccinia virus vectors or avipox, e.g. canary pox or fowl pox; and (j) a helper-dependent or gutless adenovirus. Replication-defective viruses can also be advantageous. Different vectors will or will not become incorporated into the cells' genome. The constructs can include viral sequences for transfection, if desired. Alternatively, the construct can be incorporated into vectors capable of episomal replication, e.g. EPV and EBV vectors. Constructs for the recombinant expression of an iRNA will generally require regulatory elements, e.g., promoters, enhancers, etc., to ensure the expression of the iRNA in target cells. Other aspects to consider for vectors and constructs are known in the art.

VI. Pharmaceutical Compositions of the Invention

The present invention also includes pharmaceutical compositions and formulations which include the iRNAs of the invention. Accordingly, in one embodiment, provided herein are pharmaceutical compositions comprising a double stranded ribonucleic acid (dsRNA) agent that inhibits expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell, such as a liver cell, wherein the dsRNA agent comprises a sense strand and an antisense strand, wherein the sense strand comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO:1, and said antisense

strand comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO: 2; and a pharmaceutically acceptable carrier. In some embodiments, the dsRNA agent comprises a sense strand and an antisense strand, wherein the sense strand comprises at least 15 contiguous nucleotides from the nucleotide sequence of SEQ ID NO:1, and said antisense strand comprises at least 15 contiguous nucleotides from the nucleotide sequence of SEQ ID NO: 2.

In another embodiment, provided herein are pharmaceutical compositions comprising a dsRNA agent that inhibits expression of CIDEB in a cell, such as a liver cell, wherein the dsRNA agent comprises a sense strand and an antisense strand, the antisense strand comprising a region of complementarity which comprises at least 15 contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from any one of the antisense sequences listed in Tables 3-6; and a pharmaceutically acceptable carrier. In some embodiments, the dsRNA agent comprises a sense strand and an antisense strand, the antisense strand comprising a region of complementarity which comprises at least 15 contiguous nucleotides from any one of the antisense sequences listed in Tables 3-6.

The pharmaceutical compositions containing the iRNA of the invention are useful for treating a disease or disorder associated with the expression or activity of a CIDEB gene, e.g., a chronic inflammatory disease.

Such pharmaceutical compositions are formulated based on the mode of delivery. One example is compositions that are formulated for systemic administration via parenteral delivery, e.g., by intravenous (IV), intramuscular (IM) or for subcutaneous delivery. Another example is compositions that are formulated for direct delivery into the liver, e.g., by infusion into the liver, such as by continuous pump infusion. The pharmaceutical compositions of the invention may be administered in dosages sufficient to inhibit expression of a CIDEB gene. In general, a suitable dose of an iRNA of the invention will be in the range of about 0.001 to about 200.0 milligrams per kilogram body weight of the recipient per day, generally in the range of about 1 to 50 mg per kilogram body weight per day. Typically, a suitable dose of an iRNA of the invention will be in the range of about 0.1 mg/kg to about 5.0 mg/kg, preferably about 0.3 mg/kg and about 3.0 mg/kg.

A repeat-dose regimen may include administration of a therapeutic amount of iRNA on a regular basis, such as every other day to once a year. In certain embodiments, the iRNA is administered about once per week, once every 7-10 days, once every 2 weeks, once every 3 weeks, once every 4 weeks, once every 5 weeks, once every 6 weeks, once every 7 weeks, once every 8 weeks, once every 9 weeks, once every 10 weeks, once every 11 weeks, once every 12 weeks, once per month, once every 2 months, once every 3 months (once per quarter), once every 4 months, once every 5 months, or once every 6 months.

After an initial treatment regimen, the treatments can be administered on a less frequent basis.

The skilled artisan will appreciate that certain factors can influence the dosage and timing required to effectively treat a subject, including but not limited to the severity of the disease or disorder, previous treatments, the general health and/or age of the subject, and other diseases present. Moreover, treatment of a subject with a therapeutically effective amount of a composition can include a single treatment or a series of treatments. Estimates of effective dosages and in vivo half-lives for the individual iRNAs encompassed by the invention can be made using conventional methodologies or

on the basis of in vivo testing using an appropriate animal model, as described elsewhere herein.

Advances in mouse genetics have generated a number of mouse models for the study of various human diseases, such as a CIDEB-associated disease, disorder, or condition that would benefit from reduction in the expression of CIDEB. Such models can be used for in vivo testing of iRNA, as well as for determining a therapeutically effective dose. Such models can be used for in vivo testing of iRNA, as well as for determining a therapeutically effective dose. Suitable mouse models are known in the art and include, for example, mice and rats fed a high fat diet (HFD; also referred to as a Western diet), a methionine-choline deficient (MCD) diet, or a high-fat (15%), high-cholesterol (1%) diet (HFHC), an obese (ob/ob) mouse containing a mutation in the obese (ob) gene (Wiegman et al., (2003) *Diabetes*, 52:1081-1089); a mouse containing homozygous knock-out of an LDL receptor (LDLR $-/-$ mouse; Ishibashi et al., (1993) *J Clin Invest* 92(2):883-893); diet-induced atherosclerosis mouse model (Ishida et al., (1991) *J. Lipid. Res.*, 32:559-568); heterozygous lipoprotein lipase knockout mouse model (Weistock et al., (1995) *J. Clin. Invest.* 96(6):2555-2568); mice and rats fed a choline-deficient, L-amino acid-defined, high-fat diet (CDAHFD) (Matsumoto et al. (2013) *Int. J. Exp. Path.* 94:93-103); mice and rats fed a high-trans-fat, cholesterol diet (HTF-C) (Clapper et al. (2013) *Am. J. Physiol. Gastrointest. Liver Physiol.* 305: G483-G495); mice and rats fed a high-fat, high-cholesterol, bile salt diet (HF/HC/BS) (Matsumoto et al. (2007) *Hepatology* 46:1392-1403); and mice and rats fed a high-fat diet+fructose (30%) water (Softic et al. (2018) *J. Clin. Invest.* 128(1)-85-96).

The pharmaceutical compositions of the present invention can be administered in a number of ways depending upon whether local or systemic treatment is desired and upon the area to be treated. Administration can be topical (e.g., by a transdermal patch), pulmonary, e.g., by inhalation or insufflation of powders or aerosols, including by nebulizer; intratracheal, intranasal, epidermal and transdermal, oral or parenteral. Parenteral administration includes intravenous, intraarterial, subcutaneous, intraperitoneal or intramuscular injection or infusion; subdermal, e.g., via an implanted device; or intracranial, e.g., by intraparenchymal, intrathecal or intraventricular, administration.

The iRNA can be delivered in a manner to target a particular cell or tissue, such as the liver (e.g., the hepatocytes of the liver).

In some embodiments, the pharmaceutical compositions of the invention are suitable for intramuscular administration to a subject. In other embodiments, the pharmaceutical compositions of the invention are suitable for intravenous administration to a subject. In some embodiments of the invention, the pharmaceutical compositions of the invention are suitable for subcutaneous administration to a subject, e.g., using a 29 g or 30 g needle.

The pharmaceutical compositions of the invention may include an RNAi agent of the invention in an unbuffered solution, such as saline or water, or in a buffer solution, such as a buffer solution comprising acetate, citrate, prolamine, carbonate, or phosphate or any combination thereof.

In one embodiment, the pharmaceutical compositions of the invention, e.g., such as the compositions suitable for subcutaneous administration, comprise an RNAi agent of the invention in phosphate buffered saline (PBS). Suitable concentrations of PBS include, for example, 1 mM, 1.5 mM, 2 mM, 2.5 mM, 3 mM, 3.5 mM, 4 mM, 4.5 mM, 5 mM, 6.5 mM, 7 mM, 7.5 mM, 9 mM, 8.5 mM, 9 mM, 9.5 mM, or about 10 mM PBS. In one embodiment of the invention, a

pharmaceutical composition of the invention comprises an RNAi agent of the invention dissolved in a solution of about 5 mM PBS (e.g., 0.64 mM NaH_2PO_4 , 4.36 mM Na_2HPO_4 , 85 mM NaCl). Values intermediate to the above recited ranges and values are also intended to be part of this invention. In addition, ranges of values using a combination of any of the above recited values as upper and/or lower limits are intended to be included.

The pH of the pharmaceutical compositions of the invention may be between about 5.0 to about 8.0, about 5.5 to about 8.0, about 6.0 to about 8.0, about 6.5 to about 8.0, about 7.0 to about 8.0, about 5.0 to about 7.5, about 5.5 to about 7.5, about 6.0 to about 7.5, about 6.5 to about 7.5, about 5.0 to about 7.2, about 5.25 to about 7.2, about 5.5 to about 7.2, about 5.75 to about 7.2, about 6.0 to about 7.2, about 6.5 to about 7.2, or about 6.8 to about 7.2. Ranges and values intermediate to the above recited ranges and values are also intended to be part of this invention.

The osmolality of the pharmaceutical compositions of the invention may be suitable for subcutaneous administration, such as no more than about 400 mOsm/kg, e.g., between 50 and 400 mOsm/kg, between 75 and 400 mOsm/kg, between 100 and 400 mOsm/kg, between 125 and 400 mOsm/kg, between 150 and 400 mOsm/kg, between 175 and 400 mOsm/kg, between 200 and 400 mOsm/kg, between 250 and 400 mOsm/kg, between 300 and 400 mOsm/kg, between 50 and 375 mOsm/kg, between 75 and 375 mOsm/kg, between 100 and 375 mOsm/kg, between 125 and 375 mOsm/kg, between 150 and 375 mOsm/kg, between 175 and 375 mOsm/kg, between 200 and 375 mOsm/kg, between 250 and 375 mOsm/kg, between 300 and 375 mOsm/kg, between 50 and 350 mOsm/kg, between 75 and 350 mOsm/kg, between 100 and 350 mOsm/kg, between 125 and 350 mOsm/kg, between 150 and 350 mOsm/kg, between 175 and 350 mOsm/kg, between 200 and 350 mOsm/kg, between 250 and 350 mOsm/kg, between 50 and 325 mOsm/kg, between 75 and 325 mOsm/kg, between 100 and 325 mOsm/kg, between 125 and 325 mOsm/kg, between 150 and 325 mOsm/kg, between 175 and 325 mOsm/kg, between 200 and 325 mOsm/kg, between 250 and 325 mOsm/kg, between 300 and 325 mOsm/kg, between 300 and 350 mOsm/kg, between 50 and 300 mOsm/kg, between 75 and 300 mOsm/kg, between 100 and 300 mOsm/kg, between 125 and 300 mOsm/kg, between 150 and 300 mOsm/kg, between 175 and 300 mOsm/kg, between 200 and 300 mOsm/kg, between 250 and 300 mOsm/kg, between 50 and 250 mOsm/kg, between 75 and 250 mOsm/kg, between 100 and 250 mOsm/kg, between 125 and 250 mOsm/kg, between 150 and 250 mOsm/kg, between 175 and 250 mOsm/kg, between 200 and 250 mOsm/kg, e.g., about 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 275, 280, 285, 295, 300, 305, 310, 320, 325, 330, 335, 340, 345, 350, 355, 360, 365, 370, 375, 380, 385, 390, 395, or about 400 mOsm/kg. Ranges and values intermediate to the above recited ranges and values are also intended to be part of this invention.

The pharmaceutical compositions of the invention comprising the RNAi agents of the invention, may be present in a vial that contains about 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, or about 2.0 mL of the pharmaceutical composition. The concentration of the RNAi agents in the pharmaceutical compositions of the invention may be about 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 130, 125, 130, 135, 140, 145, 150, 175, 180, 185, 190, 195, 200, 205, 210,

215, 230, 225, 230, 235, 240, 245, 250, 275, 280, 285, 290, 295, 300, 305, 310, 315, 330, 325, 330, 335, 340, 345, 350, 375, 380, 385, 390, 395, 400, 405, 410, 415, 430, 425, 430, 435, 440, 445, 450, 475, 480, 485, 490, 495, or about 500 mg/mL. In one embodiment, the concentration of the RNAi agents in the pharmaceutical compositions of the invention is about 100 mg/mL. Values intermediate to the above recited ranges and values are also intended to be part of this invention.

The pharmaceutical compositions of the invention may comprise a dsRNA agent of the invention in a free acid form. In other embodiments of the invention, the pharmaceutical compositions of the invention may comprise a dsRNA agent of the invention in a salt form, such as a sodium salt form. In certain embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for substantially all of the phosphodiester and/or phosphorothioate groups present in the agent. Agents in which substantially all of the phosphodiester and/or phosphorothioate linkages have a sodium counterion include not more than 5, 4, 3, 2, or 1 phosphodiester and/or phosphorothioate linkages without a sodium counterion. In some embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for all of the phosphodiester and/or phosphorothioate groups present in the agent.

Pharmaceutical compositions and formulations for topical administration can include transdermal patches, ointments, lotions, creams, gels, drops, suppositories, sprays, liquids and powders. Conventional pharmaceutical carriers, aqueous, powder or oily bases, thickeners and the like can be necessary or desirable. Coated condoms, gloves and the like can also be useful. Suitable topical formulations include those in which the iRNAs featured in the invention are in admixture with a topical delivery agent such as lipids, liposomes, fatty acids, fatty acid esters, steroids, chelating agents and surfactants. Suitable lipids and liposomes include neutral (e.g., dioleoylphosphatidyl DOPE ethanolamine, dimyristoylphosphatidyl choline DMPC, distearoylphosphatidyl choline) negative (e.g., dimyristoylphosphatidyl glycerol DMPG) and cationic (e.g., dioleoyltetramethylaminopropyl DOTAP and dioleoylphosphatidyl ethanolamine DOTMA). iRNAs featured in the invention can be encapsulated within liposomes or can form complexes thereto, in particular to cationic liposomes. Alternatively, iRNAs can be complexed to lipids, in particular to cationic lipids. Suitable fatty acids and esters include but are not limited to arachidonic acid, oleic acid, eicosanoic acid, lauric acid, caprylic acid, capric acid, myristic acid, palmitic acid, stearic acid, linoleic acid, linolenic acid, dicaprate, tricaprate, monoolein, dilaurin, glyceryl 1-monocaprate, 1-dodecylazacycloheptan-2-one, an acylcarnitine, an acylcholine, or a C_{1-20} alkyl ester (e.g., isopropylmyristate IPM), monoglyceride, diglyceride or pharmaceutically acceptable salt thereof. Topical formulations are described in detail in U.S. Pat. No. 6,747,014, which is incorporated herein by reference.

A. iRNA Formulations Comprising Membranous Molecular Assemblies

An iRNA for use in the compositions and methods of the invention can be formulated for delivery in a membranous molecular assembly, e.g., a liposome or a micelle. There are many organized surfactant structures besides microemulsions that have been studied and used for the formulation of drugs. These include monolayers, micelles, bilayers and vesicles. Vesicles, such as liposomes, have attracted great interest because of their specificity and the duration of action they offer from the standpoint of drug delivery. As used in

the present invention, the term "liposome" means a vesicle composed of amphiphilic lipids arranged in a spherical bilayer or bilayers.

Liposomes include unilamellar or multilamellar vesicles which have a membrane formed from a lipophilic material and an aqueous interior. The aqueous portion contains the composition (e.g., iRNA) to be delivered. The lipophilic material isolates the aqueous interior from an aqueous exterior, which typically does not include the iRNA composition, although in some examples, it may. Cationic liposomes possess the advantage of being able to fuse to the cell wall. Non-cationic liposomes, although not able to fuse as efficiently with the cell wall, are taken up by macrophages in vivo.

In order to traverse intact mammalian skin, lipid vesicles must pass through a series of fine pores, each with a diameter less than 50 nm, under the influence of a suitable transdermal gradient. Therefore, it is desirable to use a liposome which is highly deformable and able to pass through such fine pores.

Liposomes are useful for the transfer and delivery of active ingredients to the site of action. Because the liposomal membrane is structurally similar to biological membranes, when liposomes are applied to a tissue, the liposomes start to merge with the cellular membranes and as the merging of the liposome and cell progresses, the liposomal contents are emptied into the cell where the active agent may act.

Liposomal formulations have been the focus of extensive investigation as the mode of delivery for many drugs. There is growing evidence that for topical administration, liposomes present several advantages over other formulations. Such advantages include reduced side-effects related to high systemic absorption of the administered drug, increased accumulation of the administered drug at the desired target, and the ability to administer a wide variety of drugs, both hydrophilic and hydrophobic, into the skin.

Several reports have detailed the ability of liposomes to deliver agents including high-molecular weight DNA into the skin. Compounds including analgesics, antibodies, hormones and high-molecular weight DNAs have been administered to the skin. The majority of applications resulted in the targeting of the upper epidermis.

A liposome containing an iRNA agent can be prepared by a variety of methods. In one example, the lipid component of a liposome is dissolved in a detergent so that micelles are formed with the lipid component. For example, the lipid component can be an amphipathic cationic lipid or lipid conjugate. The detergent can have a high critical micelle concentration and may be nonionic. Exemplary detergents include cholate, CHAPS, octylglucoside, deoxycholate, and lauroyl sarcosine. The iRNA agent preparation is then added to the micelles that include the lipid component. The cationic groups on the lipid interact with the iRNA agent and condense around the iRNA agent to form a liposome. After condensation, the detergent is removed, e.g., by dialysis, to yield a liposomal preparation of iRNA agent.

If necessary a carrier compound that assists in condensation can be added during the condensation reaction, e.g., by controlled addition. For example, the carrier compound can be a polymer other than a nucleic acid (e.g., spermine or spermidine). pH can also be adjusted to favor condensation.

Methods for producing stable polynucleotide delivery vehicles, which incorporate a polynucleotide/cationic lipid complex as structural components of the delivery vehicle, are further described in, e.g., WO 96/37194, the entire contents of which are incorporated herein by reference.

Liposome formation can also include one or more aspects of exemplary methods described in Felgner, P. L. et al., *Proc. Natl. Acad. Sci., USA* 8:7413-7417, 1987; U.S. Pat. Nos. 4,897,355; 5,171,678; Bangham, et al. *Mol. Biol.* 23:238, 1965; Olson, et al. *Biochim. Biophys. Acta* 557:9, 1979; Szoka, et al. *Proc. Natl. Acad. Sci.* 75: 4194, 1978; Mayhew, et al. *Biochim. Biophys. Acta* 775:169, 1984; Kim, et al. *Biochim. Biophys. Acta* 728:339, 1983; and Fukunaga, et al. *Endocrinol.* 115:757, 1984. Commonly used techniques for preparing lipid aggregates of appropriate size for use as delivery vehicles include sonication and freeze-thaw plus extrusion (see, e.g., Mayer, et al. *Biochim. Biophys. Acta* 858:161, 1986). Microfluidization can be used when consistently small (50 to 200 nm) and relatively uniform aggregates are desired (Mayhew, et al. *Biochim. Biophys. Acta* 775:169, 1984). These methods are readily adapted to packaging iRNA agent preparations into liposomes.

Liposomes fall into two broad classes. Cationic liposomes are positively charged liposomes which interact with the negatively charged DNA molecules to form a stable complex. The positively charged DNA/liposome complex binds to the negatively charged cell surface and is internalized in an endosome. Due to the acidic pH within the endosome, the liposomes are ruptured, releasing their contents into the cell cytoplasm (Wang et al., *Biochem. Biophys. Res. Commun.*, 1987, 147, 980-985).

Liposomes which are pH-sensitive or negatively-charged, entrap DNA rather than complex with it. Since both the DNA and the lipid are similarly charged, repulsion rather than complex formation occurs. Nevertheless, some DNA is entrapped within the aqueous interior of these liposomes. pH-sensitive liposomes have been used to deliver DNA encoding the thymidine kinase gene to cell monolayers in culture. Expression of the exogenous gene was detected in the target cells (Zhou et al., *Journal of Controlled Release*, 1992, 19, 269-274).

One major type of liposomal composition includes phospholipids other than naturally-derived phosphatidylcholine. Neutral liposome compositions, for example, can be formed from dimyristoyl phosphatidylcholine (DMPC) or dipalmitoyl phosphatidylcholine (DPPC). Anionic liposome compositions generally are formed from dimyristoyl phosphatidylglycerol, while anionic fusogenic liposomes are formed primarily from dioleoyl phosphatidylethanolamine (DOPE). Another type of liposomal composition is formed from phosphatidylcholine (PC) such as, for example, soybean PC, and egg PC. Another type is formed from mixtures of phospholipid and/or phosphatidylcholine and/or cholesterol.

Examples of other methods to introduce liposomes into cells in vitro and in vivo include U.S. Pat. Nos. 5,283,185 and 5,171,678; WO 94/00569; WO 93/24640; WO 91/16024; Felgner, J. *Biol. Chem.* 269:2550, 1994; Nabel, *Proc. Natl. Acad. Sci.* 90: 11307, 1993; Nabel, *Human Gene Ther.* 3:649, 1992; Gershon, *Biochem.* 32:7143, 1993; and Strauss *EMBO J.* 11:417, 1992.

Non-ionic liposomal systems have also been examined to determine their utility in the delivery of drugs to the skin, in particular systems comprising non-ionic surfactant and cholesterol. Non-ionic liposomal formulations comprising Novasome™ I (glyceryl dilaurate/cholesterol/polyoxyethylene-10-stearyl ether) and Novasome™ II (glyceryl distearate/cholesterol/polyoxyethylene-10-stearyl ether) were used to deliver cyclosporin-A into the dermis of mouse skin. Results indicated that such non-ionic liposomal systems were effective in facilitating the deposition of cyclosporin-A into different layers of the skin (Hu et al. *S. T. P. Pharma. Sci.*, 1994, 4, 6, 466).

Liposomes also include "sterically stabilized" liposomes, a term which, as used herein, refers to liposomes comprising one or more specialized lipids that, when incorporated into liposomes, result in enhanced circulation lifetimes relative to liposomes lacking such specialized lipids. Examples of sterically stabilized liposomes are those in which part of the vesicle-forming lipid portion of the liposome (A) comprises one or more glycolipids, such as monosialoganglioside G_{M1} , or (B) is derivatized with one or more hydrophilic polymers, such as a polyethylene glycol (PEG) moiety. While not wishing to be bound by any particular theory, it is thought in the art that, at least for sterically stabilized liposomes containing gangliosides, sphingomyelin, or PEG-derivatized lipids, the enhanced circulation half-life of these sterically stabilized liposomes derives from a reduced uptake into cells of the reticuloendothelial system (RES) (Allen et al., *FEBS Letters*, 1987, 223, 42; Wu et al., *Cancer Research*, 1993, 53, 3765).

Various liposomes comprising one or more glycolipids are known in the art. Papahadjopoulos et al. (*Ann. N.Y. Acad. Sci.*, 1987, 507, 64) reported the ability of monosialoganglioside G_{M1} , galactocerebroside sulfate and phosphatidylinositol to improve blood half-lives of liposomes. These findings were expounded upon by Gabizon et al. (*Proc. Natl. Acad. Sci. U.S.A.*, 1988, 85, 6949). U.S. Pat. No. 4,837,028 and WO 88/04924, both to Allen et al., disclose liposomes comprising (1) sphingomyelin and (2) the ganglioside G_{M1} or a galactocerebroside sulfate ester. U.S. Pat. No. 5,543,152 (Webb et al.) discloses liposomes comprising sphingomyelin. Liposomes comprising 1,2-sn-dimyristoylphosphatidylcholine are disclosed in WO 97/13499 (Lim et al.).

In some embodiments, cationic liposomes are used. Cationic liposomes possess the advantage of being able to fuse to the cell membrane. Non-cationic liposomes, although not able to fuse as efficiently with the plasma membrane, are taken up by macrophages in vivo and can be used to deliver iRNA agents to macrophages.

Further advantages of liposomes include: liposomes obtained from natural phospholipids are biocompatible and biodegradable; liposomes can incorporate a wide range of water and lipid soluble drugs; liposomes can protect encapsulated iRNAs in their internal compartments from metabolism and degradation (Rosoff, in "Pharmaceutical Dosage Forms," Lieberman, Rieger and Banker (Eds.), 1988, volume 1, p. 245). Important considerations in the preparation of liposome formulations are the lipid surface charge, vesicle size, and the aqueous volume of the liposomes.

A positively charged synthetic cationic lipid, N-[1-(2,3-dioleoyloxy)propyl]-N,N,N-trimethylammonium chloride (DOTMA) can be used to form small liposomes that interact spontaneously with nucleic acid to form lipid-nucleic acid complexes which are capable of fusing with the negatively charged lipids of the cell membranes of tissue culture cells, resulting in delivery of iRNA agent (see, e.g., Felgner, P. L. et al., *Proc. Natl. Acad. Sci.*, USA 8:7413-7417, 1987 and U.S. Pat. No. 4,897,355 for a description of DOTMA and its use with DNA).

A DOTMA analogue, 1,2-bis(oleoyloxy)-3-(trimethylammonia)propane (DOTAP) can be used in combination with a phospholipid to form DNA-complexing vesicles. LipofectinTM (Bethesda Research Laboratories, Gaithersburg, Md.) is an effective agent for the delivery of highly anionic nucleic acids into living tissue culture cells that comprise positively charged DOTMA liposomes which interact spontaneously with negatively charged polynucleotides to form complexes. When enough positively charged liposomes are used, the net charge on the resulting complexes is also

positive. Positively charged complexes prepared in this way spontaneously attach to negatively charged cell surfaces, fuse with the plasma membrane, and efficiently deliver functional nucleic acids into, for example, tissue culture cells. Another commercially available cationic lipid, 1,2-bis(oleoyloxy)-3,3-(trimethylammonia)propane ("DOTAP") (Boehringer Mannheim, Indianapolis, Indiana) differs from DOTMA in that the oleoyl moieties are linked by ester, rather than ether linkages.

Other reported cationic lipid compounds include those that have been conjugated to a variety of moieties including, for example, carboxyspermine which has been conjugated to one of two types of lipids and includes compounds such as 5-carboxyspermylglycine dioctaoyleamide ("DOGS") (TransfectamTM, Promega, Madison, Wisconsin) and dipalmitoylphosphatidylethanolamine 5-carboxyspermylamide ("DPPES") (see, e.g., U.S. Pat. No. 5,171,678).

Another cationic lipid conjugate includes derivatization of the lipid with cholesterol ("DC-Chol") which has been formulated into liposomes in combination with DOPE (See, Gao, X. and Huang, L., *Biochim. Biophys. Res. Commun.* 179:280, 1991). Lipopolylysine, made by conjugating polylysine to DOPE, has been reported to be effective for transfection in the presence of serum (Zhou, X. et al., *Biochim. Biophys. Acta* 1065:8, 1991). For certain cell lines, these liposomes containing conjugated cationic lipids, are said to exhibit lower toxicity and provide more efficient transfection than the DOTMA-containing compositions. Other commercially available cationic lipid products include DMRIE and DMRIE-HP (Vical, La Jolla, California) and Lipofectamine (DOSPA) (Life Technology, Inc., Gaithersburg, Maryland). Other cationic lipids suitable for the delivery of oligonucleotides are described in WO 98/39359 and WO 96/37194.

Liposomal formulations are particularly suited for topical administration; liposomes present several advantages over other formulations. Such advantages include reduced side effects related to high systemic absorption of the administered drug, increased accumulation of the administered drug at the desired target, and the ability to administer iRNA agent into the skin. In some implementations, liposomes are used for delivering iRNA agent to epidermal cells and also to enhance the penetration of iRNA agent into dermal tissues, e.g., into skin. For example, the liposomes can be applied topically. Topical delivery of drugs formulated as liposomes to the skin has been documented (see, e.g., Weiner et al., *Journal of Drug Targeting*, 1992, vol. 2, 405-410 and du Plessis et al., *Antiviral Research*, 18, 1992, 259-265; Mannino, R. J. and Fould-Fogerite, S., *Biotechniques* 6:682-690, 1988; Itani, T. et al. *Gene* 56:267-276, 1987; Nicolau, C. et al. *Meth. Enz.* 149:157-176, 1987; Straubinger, R. M. and Papahadjopoulos, D. *Meth. Enz.* 101:512-527, 1983; Wang, C. Y. and Huang, L., *Proc. Natl. Acad. Sci. USA* 84:7851-7855, 1987).

Non-ionic liposomal systems have also been examined to determine their utility in the delivery of drugs to the skin, in particular systems comprising non-ionic surfactant and cholesterol. Non-ionic liposomal formulations comprising NovasomeTM I (glyceryl dilaurate/cholesterol/polyoxyethylene-10-stearyl ether) and NovasomeTM II (glyceryl distearate/cholesterol/polyoxyethylene-10-stearyl ether) were used to deliver a drug into the dermis of mouse skin. Such formulations with iRNA agent are useful for treating a dermatological disorder.

Liposomes that include iRNA can be made highly deformable. Such deformability can enable the liposomes to penetrate through pore that are smaller than the average

radius of the liposome. For example, transfersomes are a type of deformable liposomes. Transfersomes can be made by adding surface edge activators, usually surfactants, to a standard liposomal composition. Transfersomes that include iRNAs can be delivered, for example, subcutaneously by infection in order to deliver iRNAs to keratinocytes in the skin. In order to cross intact mammalian skin, lipid vesicles must pass through a series of fine pores, each with a diameter less than 50 nm, under the influence of a suitable transdermal gradient. In addition, due to the lipid properties, these transfersomes can be self-optimizing (adaptive to the shape of pores, e.g., in the skin), self-repairing, and can frequently reach their targets without fragmenting, and often self-loading.

Other formulations amenable to the present invention are described in WO 2008/042973.

Transfersomes are yet another type of liposomes and are highly deformable lipid aggregates which are attractive candidates for drug delivery vehicles. Transfersomes may be described as lipid droplets which are so highly deformable that they are easily able to penetrate through pores which are smaller than the droplet. Transfersomes are adaptable to the environment in which they are used, e.g., they are self-optimizing (adaptive to the shape of pores in the skin), self-repairing, frequently reach their targets without fragmenting, and often self-loading. To make transfersomes it is possible to add surface edge-activators, usually surfactants, to a standard liposomal composition. Transfersomes have been used to deliver serum albumin to the skin. The transfersome-mediated delivery of serum albumin has been shown to be as effective as subcutaneous injection of a solution containing serum albumin.

Surfactants find wide application in formulations such as emulsions (including microemulsions) and liposomes. The most common way of classifying and ranking the properties of the many different types of surfactants, both natural and synthetic, is by the use of the hydrophile/lipophile balance (HLB). The nature of the hydrophilic group (also known as the "head") provides the most useful means for categorizing the different surfactants used in formulations (Rieger, in *Pharmaceutical Dosage Forms*, Marcel Dekker, Inc., New York, N.Y., 1988, p. 285).

If the surfactant molecule is not ionized, it is classified as a nonionic surfactant. Nonionic surfactants find wide application in pharmaceutical and cosmetic products and are usable over a wide range of pH values. In general, their HLB values range from 2 to about 18 depending on their structure. Nonionic surfactants include nonionic esters such as ethylene glycol esters, propylene glycol esters, glyceryl esters, polyglyceryl esters, sorbitan esters, sucrose esters, and ethoxylated esters. Nonionic alkanolamides and ethers such as fatty alcohol ethoxylates, propoxylated alcohols, and ethoxylated/propoxylated block polymers are also included in this class. The polyoxyethylene surfactants are the most popular members of the nonionic surfactant class.

If the surfactant molecule carries a negative charge when it is dissolved or dispersed in water, the surfactant is classified as anionic. Anionic surfactants include carboxylates such as soaps, acyl lactylates, acyl amides of amino acids, esters of sulfuric acid such as alkyl sulfates and ethoxylated alkyl sulfates, sulfonates such as alkyl benzene sulfonates, acyl isethionates, acyl taurates and sulfosuccinates, and phosphates. The most important members of the anionic surfactant class are the alkyl sulfates and the soaps.

If the surfactant molecule carries a positive charge when it is dissolved or dispersed in water, the surfactant is classified as cationic. Cationic surfactants include quater-

nary ammonium salts and ethoxylated amines. The quaternary ammonium salts are the most used members of this class.

If the surfactant molecule has the ability to carry either a positive or negative charge, the surfactant is classified as amphoteric. Amphoteric surfactants include acrylic acid derivatives, substituted alkylamides, N-alkylbetaines and phosphatides.

The use of surfactants in drug products, formulations and in emulsions has been reviewed (Rieger, in *Pharmaceutical Dosage Forms*, Marcel Dekker, Inc., New York, N.Y., 1988, p. 285).

The iRNA for use in the methods of the invention can also be provided as micellar formulations. "Micelles" are defined herein as a particular type of molecular assembly in which amphipathic molecules are arranged in a spherical structure such that all the hydrophobic portions of the molecules are directed inward, leaving the hydrophilic portions in contact with the surrounding aqueous phase. The converse arrangement exists if the environment is hydrophobic.

A mixed micellar formulation suitable for delivery through transdermal membranes may be prepared by mixing an aqueous solution of iRNA, an alkali metal C₈ to C₂₂ alkyl sulphate, and a micelle forming compounds. Exemplary micelle forming compounds include lecithin, hyaluronic acid, pharmaceutically acceptable salts of hyaluronic acid, glycolic acid, lactic acid, chamomile extract, cucumber extract, oleic acid, linoleic acid, linolenic acid, monoolein, monooleates, monolaurates, borage oil, evening of primrose oil, menthol, trihydroxy oxo cholanyl glycine and pharmaceutically acceptable salts thereof, glycerin, polyglycerin, lysine, polylysine, triolein, polyoxyethylene ethers and analogues thereof, polidocanol alkyl ethers and analogues thereof, chenodeoxycholate, deoxycholate, and mixtures thereof. The micelle forming compounds may be added at the same time or after addition of the alkali metal alkyl sulphate. Mixed micelles will form with substantially any kind of mixing of the ingredients but vigorous mixing in order to provide smaller size micelles.

In one method a first micellar composition is prepared which contains the RNAi and at least the alkali metal alkyl sulphate. The first micellar composition is then mixed with at least three micelle forming compounds to form a mixed micellar composition. In another method, the micellar composition is prepared by mixing the RNAi, the alkali metal alkyl sulphate and at least one of the micelle forming compounds, followed by addition of the remaining micelle forming compounds, with vigorous mixing.

Phenol or m-cresol may be added to the mixed micellar composition to stabilize the formulation and protect against bacterial growth. Alternatively, phenol or m-cresol may be added with the micelle forming ingredients. An isotonic agent such as glycerin may also be added after formation of the mixed micellar composition.

For delivery of the micellar formulation as a spray, the formulation can be put into an aerosol dispenser and the dispenser is charged with a propellant. The propellant, which is under pressure, is in liquid form in the dispenser. The ratios of the ingredients are adjusted so that the aqueous and propellant phases become one, i.e., there is one phase. If there are two phases, it is necessary to shake the dispenser prior to dispensing a portion of the contents, e.g., through a metered valve. The dispensed dose of pharmaceutical agent is propelled from the metered valve in a fine spray.

Propellants may include hydrogen-containing chlorofluorocarbons, hydrogen-containing fluorocarbons, dimethyl

ether and diethyl ether. In certain embodiments, HFA 134a (1,1,1,2 tetrafluoroethane) may be used.

The specific concentrations of the essential ingredients can be determined by relatively straightforward experimentation. For absorption through the oral cavities, it is often desirable to increase, e.g., at least double or triple, the dosage for through injection or administration through the gastrointestinal tract.

B. Lipid Particles

iRNAs, e.g., dsRNA agents of in the invention may be fully encapsulated in a lipid formulation, e.g., an LNP, or other nucleic acid-lipid particle.

As used herein, the term "LNP" refers to a stable nucleic acid-lipid particle. LNPs typically contain a cationic lipid, a non-cationic lipid, and a lipid that prevents aggregation of the particle (e.g., a PEG-lipid conjugate). LNPs are extremely useful for systemic applications, as they exhibit extended circulation lifetimes following intravenous (i.v.) injection and accumulate at distal sites (e.g., sites physically separated from the administration site). As used herein, the term "SPLP" refers to a nucleic acid-lipid particle comprising plasmid DNA encapsulated within a lipid vesicle. LNPs include "pSPLP," which include an encapsulated condensing agent-nucleic acid complex as set forth in PCT Publication No. WO 00/03683. The particles of the present invention typically have a mean diameter of about 50 nm to about 150 nm, more typically about 60 nm to about 130 nm, more typically about 70 nm to about 110 nm, most typically about 70 nm to about 90 nm, and are substantially nontoxic. In addition, the nucleic acids when present in the nucleic acid-lipid particles of the present invention are resistant in aqueous solution to degradation with a nuclease. Nucleic acid-lipid particles and their method of preparation are disclosed in, e.g., U.S. Pat. Nos. 5,976,567; 5,981,501; 6,534,484; 6,586,410; 6,815,432; and PCT Publication No. WO 96/40964.

In certain embodiments, the lipid to drug ratio (mass/mass ratio) (e.g., lipid to dsRNA ratio) will be in the range of from about 1:1 to about 50:1, from about 1:1 to about 25:1, from about 3:1 to about 15:1, from about 4:1 to about 10:1, from about 5:1 to about 9:1, or about 6:1 to about 9:1. Ranges intermediate to the above recited ranges are also contemplated to be part of the invention.

The cationic lipid may be, for example, N,N-dioleoyl-N,N-dimethylammonium chloride (DODAC), N,N-distearyl-N,N-dimethylammonium bromide (DDAB), N—[(1-(2,3-dioleoyloxy)propyl)-N,N,N-trimethylammonium chloride (DOTAP), N—[(1-(2,3-dioleoyloxy)propyl)-N,N,N-trimethylammonium chloride (DOTMA), N,N-dimethyl-2,3-dioleoyloxypropylamine (DODMA), 1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLinDMA), 1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLenDMA), 1,2-Dilinoleylcarbamoyloxy-3-dimethylaminopropane (DLin-C-DAP), 1,2-Dilinoleoxy-3-(dimethylamino)acetoxyp propane (DLin-DAC), 1,2-Dilinoleoxy-3-morpholinopropane (DLin-MA), 1,2-Dilinoleoyl-3-dimethylaminopropane (DLinDAP), 1,2-Dilinoleylthio-3-dimethylaminopropane (DLin-S-DMA), 1-Linoleoyl-2-linoleoyloxy-3-dimethylaminopropane (DLin-2-DMA), 1,2-Dilinoleoyloxy-3-trimethylaminopropane chloride salt (DLin-TMA.Cl), 1,2-Dilinoleoyl-3-trimethylaminopropane chloride salt (DLin-TAP.Cl), 1,2-Dilinoleoyloxy-3-(N-methylpiperazin)propane (DLin-MPZ), or 3-(N,N-Dilinolenylamino)-1,2-propanediol (DLinAP), 3-(N,N-Dioleylamino)-1,2-propanedio (DOAP), 1,2-Dilinoleoxylo-3-(2-N,N-dimethylamino)ethoxypropane (DLin-EG-DMA), 1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLinDMA),

2,2-Dilinoleyl-4-dimethylaminomethyl-[1,3]-dioxolane (DLin-K-DMA) or analogs thereof, (3aR,5s,6aS)—N,N-dimethyl-2,2-di((9Z,12Z)-octadeca-9,12-dienyl)tetrahydro-3aH-cyclopenta[d][1,3]dioxol-5-amine (ALN100), (6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetraen-19-yl 4-(dimethylamino)butanoate (MC3), 1,1'-(2-(4-((2-bis(2-hydroxydodecyl)amino)ethyl)(2-hydroxydodecyl)amino)ethyl)piperazin-1-yl)ethylazanediyldidodecan-2-ol (Tech G1), or a mixture thereof. The cationic lipid may comprise from about 20 mol % to about 50 mol % or about 40 mol % of the total lipid present in the particle.

In certain embodiments, the compound 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane can be used to prepare lipid-siRNA nanoparticles. Synthesis of 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane is described in U.S. provisional patent application No. 61/107,998 filed on Oct. 23, 2008, which is herein incorporated by reference.

In certain embodiments, the lipid-siRNA particle includes 40% 2, 2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane: 10% DSPC: 40% Cholesterol: 10% PEG-C-DOMG (mole percent) with a particle size of 63.0±20 nm and a 0.027 siRNA/Lipid Ratio.

The non-cationic lipid may be an anionic lipid or a neutral lipid including, but not limited to, distearoylphosphatidylcholine (DSPC), dioleoylphosphatidylcholine (DOPC), dipalmitoylphosphatidylcholine (DPPC), dioleoylphosphatidylglycerol (DOPG), dipalmitoylphosphatidylglycerol (DPPG), dioleoyl-phosphatidylethanolamine (DOPE), palmitoyloleoylphosphatidylcholine (POPC), palmitoyloleoylphosphatidylethanolamine (POPE), dioleoyl-phosphatidylethanolamine 4-(N-maleimidomethyl)-cyclohexane-1-carboxylate (DOPE-mal), dipalmitoyl phosphatidyl ethanolamine (DPPE), dimyristoylphosphoethanolamine (DMPE), distearoyl-phosphatidylethanolamine (DSPE), 16-O-monomethyl PE, 16-O-dimethyl PE, 18-1-trans PE, 1-stearoyl-2-oleoyl-phosphatidylethanolamine (SOPE), cholesterol, or a mixture thereof. The non-cationic lipid may be from about 5 mol % to about 90 mol %, about 10 mol %, or about 58 mol % if cholesterol is included, of the total lipid present in the particle.

The conjugated lipid that inhibits aggregation of particles may be, for example, a polyethyleneglycol (PEG)-lipid including, without limitation, a PEG-diaclylglycerol (DAG), a PEG-dialkylxypropyl (DAA), a PEG-phospholipid, a PEG-ceramide (Cer), or a mixture thereof. The PEG-DAA conjugate may be, for example, a PEG-dilauryloxypropyl (C₁₂), a PEG-dimyristyloxypropyl (C₁₄), a PEG-dipalmitoxypropyl (C₁₆), or a PEG-distearoxypropyl (C₁₈). The conjugated lipid that prevents aggregation of particles may be from 0 mol % to about 20 mol % or about 2 mol % of the total lipid present in the particle.

In some embodiments, the nucleic acid-lipid particle further includes cholesterol at, e.g., about 10 mol % to about 60 mol % or about 48 mol % of the total lipid present in the particle.

LNP01

In certain embodiments, the lipidoid ND98•4HCl (MW 1487) (see U.S. patent application Ser. No. 12/056,230, filed Mar. 26, 2008, which is herein incorporated by reference), Cholesterol (Sigma-Aldrich), and PEG-Ceramide C16 (Avanti Polar Lipids) can be used to prepare lipid-dsRNA nanoparticles (e.g., LNP01 particles). Stock solutions of each in ethanol can be prepared as follows: ND98, 133 mg/ml; Cholesterol, 25 mg/ml, PEG-Ceramide C16, 100 mg/ml. The ND98, Cholesterol, and PEG-Ceramide C16 stock solutions can then be combined in a, e.g., 42:48:10 molar ratio. The combined lipid solution can be mixed with

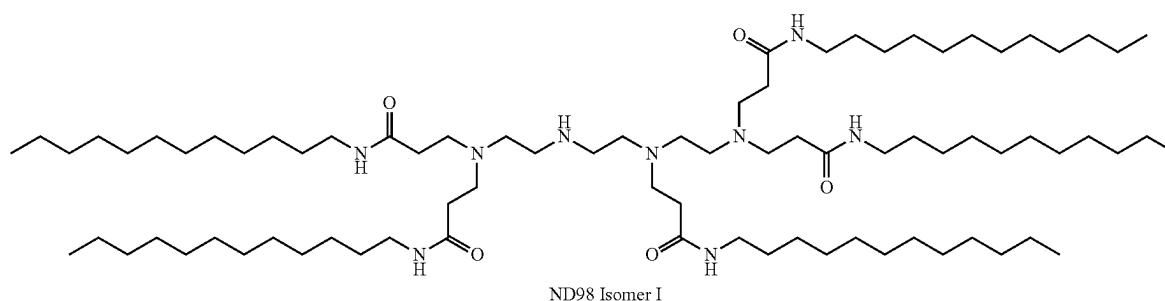
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aqueous dsRNA (e.g., in sodium acetate pH 5) such that the final ethanol concentration is about 35-45% and the final sodium acetate concentration is about 100-300 mM. Lipid-dsRNA nanoparticles typically form spontaneously upon mixing. Depending on the desired particle size distribution, the resultant nanoparticle mixture can be extruded through a polycarbonate membrane (e.g., 100 nm cut-off) using, for example, a thermobarrel extruder, such as Lipex Extruder

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(Northern Lipids, Inc). In some cases, the extrusion step can be omitted. Ethanol removal and simultaneous buffer exchange can be accomplished by, for example, dialysis or tangential flow filtration. Buffer can be exchanged with, for example, phosphate buffered saline (PBS) at about pH 7, e.g., about pH 6.9, about pH 7.0, about pH 7.1, about pH 7.2, about pH 7.3, or about pH 7.4.

Formula I



ND98 Isomer I

LNP01 formulations are described, e.g., in International Application Publication No. WO 2008/042973, which is hereby incorporated by reference.

Additional exemplary lipid-dsRNA formulations are provided in the following Table 1.

TABLE 1

Exemplary lipid formulations		
	Cationic Lipid	cationic lipid/non-cationic lipid/ cholesterol/PEG-lipid conjugate Lipid:siRNA ratio
SNALP	1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLinDMA)	DLinDMA/DPPC/Cholesterol/PEG-cDMA (57.1/7.1/34.4/1.4) lipid:siRNA ~7:1
S-XTC	2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (XTC)	XTC/DPPC/Cholesterol/PEG-cDMA 57.1/7.1/34.4/1.4 lipid:siRNA ~7:1
LNP05	2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (XTC)	XTC/DSPC/Cholesterol/PEG-DMG 57.5/7.5/31.5/3.5 lipid:siRNA ~6:1
LNP06	2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (XTC)	XTC/DSPC/Cholesterol/PEG-DMG 57.5/7.5/31.5/3.5 lipid:siRNA ~11:1
LNP07	2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (XTC)	XTC/DSPC/Cholesterol/PEG-DMG 60/7.5/31/1.5, lipid:siRNA ~6:1
LNP08	2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (XTC)	XTC/DSPC/Cholesterol/PEG-DMG 60/7.5/31/1.5, lipid:siRNA ~11:1
LNP09	2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (XTC)	XTC/DSPC/Cholesterol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA 10:1
LNP10	(3aR,5s,6aS)-N,N-dimethyl-2,2-di((9Z,12Z)-octadeca-9,12-dienyl)tetrahydro-3aH-cyclopenta[d][1,3]dioxol-5-amine (ALN100)	ALN100/DSPC/Cholesterol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA 10:1
LNP11	(6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetraen-19-yl 4-(dimethylamino)butanoate (MC3)	MC-3/DSPC/Cholesterol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA 10:1
LNP12	1,1'-(2-(4-(2-(2-(bis(2-hydroxydodecyl)amino)ethyl)(2-hydroxydodecyl)amino)ethyl)piperazin-1-yl)ethylazanediyldidodecan-2-ol (C12-200)	C12-200/DSPC/Cholesterol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA 10:1
LNP13	XTC	XTC/DSPC/Chol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA: 33:1

TABLE 1-continued

Exemplary lipid formulations		
Cationic Lipid		cationic lipid/non-cationic lipid/ cholesterol/PEG-lipid conjugate Lipid:siRNA ratio
LNP14	MC3	MC3/DSPC/Chol/PEG-DMG 40/15/40/5 Lipid:siRNA: 11:1
LNP15	MC3	MC3/DSPC/Chol/PEG-DSG/GalNAc-PEG-DSG 50/10/35/4.5/0.5 Lipid:siRNA: 11:1
LNP16	MC3	MC3/DSPC/Chol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA: 7:1
LNP17	MC3	MC3/DSPC/Chol/PEG-DSG 50/10/38.5/1.5 Lipid:siRNA: 10:1
LNP18	MC3	MC3/DSPC/Chol/PEG-DMG 50/10/38.5/1.5 Lipid:siRNA: 12:1
LNP19	MC3	MC3/DSPC/Chol/PEG-DMG 50/10/35/5 Lipid:siRNA: 8:1
LNP20	MC3	MC3/DSPC/Chol/PEG-DPG 50/10/38.5/1.5 Lipid:siRNA: 10:1
LNP21	C12-200	C12-200/DSPC/Chol/PEG-DSG 50/10/38.5/1.5 Lipid:siRNA: 7:1
LNP22	XTC	XTC/DSPC/Chol/PEG-DSG 50/10/38.5/1.5 Lipid:siRNA: 10:1

DSPC: distearoylphosphatidylcholine

DPPC: dipalmitoylphosphatidylcholine

PEG-DMG: PEG-didimristoyl glycerol (C14-PEG, or PEG-C14) (PEG with avg mol of 2000)

PEG-DSG: PEG-distyryl glycerol (C18-PEG, or PEG-C18) (PEG with avg mol of 2000)

PEG-cDMA: PEG-carbamoyl-1,2-dimyrityloxypropylamine (PEG with avg mol wt of 2000)

SNALP (1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLinDMA)) comprising formulations are described in International Publication No. WO2009/127060, filed Apr. 15, 2009, which is hereby incorporated by reference.

XTC comprising formulations are described, e.g., in U.S. Provisional Ser. No. 61/148,366, filed Jan. 29, 2009; U.S. Provisional Ser. No. 61/156,851, filed Mar. 2, 2009; U.S. Provisional Ser. No. 61/185,712, filed Jun. 10, 2009; U.S. Provisional Ser. No. 61/228,373, filed Jul. 24, 2009; U.S. Provisional Ser. No. 61/239,686, filed Sep. 3, 2009, and International Application No. PCT/US2010/022614, filed Jan. 29, 2010, which are hereby incorporated by reference.

MC3 comprising formulations are described, e.g., in U.S. Provisional Ser. No. 61/244,834, filed Sep. 22, 2009, U.S. Provisional Ser. No. 61/185,800, filed Jun. 10, 2009, and International Application No. PCT/US 10/2 8224, filed Jun. 10, 2010, which are hereby incorporated by reference.

ALNY-100 comprising formulations are described, e.g., International patent application number PCT/US09/63933, filed on Nov. 10, 2009, which is hereby incorporated by reference.

C12-200 comprising formulations are described in U.S. Provisional Ser. No. 61/175,770, filed May 5, 2009 and International Application No. PCT/US10/33777, filed May 5, 2010, which are hereby incorporated by reference.

Compositions and formulations for oral administration include powders or granules, microparticulates, nanoparticulates, suspensions or solutions in water or non-aqueous media, capsules, gel capsules, sachets, tablets or minitables. Thickeners, flavoring agents, diluents, emulsifiers, dispersing aids or binders can be desirable. In some embodiments, oral formulations are those in which dsRNAs featured in the

invention are administered in conjunction with one or more penetration enhancer surfactants and chelators. Suitable surfactants include fatty acids and/or esters or salts thereof, bile acids and/or salts thereof. Suitable bile acids/salts include chenodeoxycholic acid (CDCA) and ursodeoxycholic acid (UDCA), cholic acid, dehydrocholic acid, deoxycholic acid, glucolic acid, glycolic acid, glycodeoxycholic acid, taurocholic acid, taurodeoxycholic acid, sodium tauro-24,25-dihydro-fusidate and sodium glycodihydrofusidate. Suitable fatty acids include arachidonic acid, undecanoic acid, oleic acid, lauric acid, caprylic acid, capric acid, myristic acid, palmitic acid, stearic acid, linoleic acid, linolenic acid, dicaprinate, tricaprinate, monoolein, dilaurin, glyceryl 1-monocaprinate, 1-dodecylazacycloheptan-2-one, an acylcarnitine, an acylcholine, or a monoglyceride, a diglyceride or a pharmaceutically acceptable salt thereof (e.g., sodium). In some embodiments, combinations of penetration enhancers are used, for example, fatty acids/salts in combination with bile acids/salts. One exemplary combination is the sodium salt of lauric acid, capric acid and UDCA. Further penetration enhancers include polyoxyethylene-9-lauryl ether, polyoxyethylene-20-cetyl ether. DsRNAs featured in the invention can be delivered orally, in granular form including sprayed dried particles, or complexed to form micro or nanoparticles. DsRNA complexing agents include poly-amino acids; polyimines; polyacrylates; polyalkylacrylates, polyoxethanes, polyalkylcyanoacrylates; cationized gelatins, albumins, starches, acrylates, polyethyleneglycols (PEG) and starches; polyalkylcyanoacrylates; DEAE-derivatized polyimines, pollulans, celluloses and starches. Suitable complexing agents include chitosan, N-trimethylchitosan, poly-L-lysine, polyhistidine, polyornithine

thine, polyspermines, protamine, polyvinylpyridine, polythiodiethylaminomethylethylene P(TDAE), polyaminostyrene (e.g., p-amino), poly(methylcyanoacrylate), poly(ethylcyanoacrylate), poly(butylcyanoacrylate), poly(isobutylcyanoacrylate), poly(isohexylcyanoacrylate), DEAE-methacrylate, DEAE-hexylacrylate, DEAE-acrylamide, DEAE-albumin and DEAE-dextran, polymethylacrylate, polyhexylacrylate, poly(D,L-lactic acid), poly(DL-lactic-co-glycolic acid (PLGA), alginate, and polyethyleneglycol (PEG). Oral formulations for dsRNAs and their preparation are described in detail in U.S. Pat. No. 6,887,906, US Publication. No. 20030027780, and U.S. Pat. No. 6,747,014, each of which is incorporated herein by reference.

Compositions and formulations for parenteral, intraparenchymal (into the brain), intrathecal, intraventricular or intrahepatic administration can include sterile aqueous solutions which can also contain buffers, diluents and other suitable additives such as, but not limited to, penetration enhancers, carrier compounds and other pharmaceutically acceptable carriers or excipients.

Pharmaceutical compositions of the present invention include, but are not limited to, solutions, emulsions, and liposome-containing formulations. These compositions can be generated from a variety of components that include, but are not limited to, preformed liquids, self-emulsifying solids and self-emulsifying semisolids. Particularly preferred are formulations that target the liver when treating hepatic disorders such as chronic inflammation of the liver.

The pharmaceutical formulations of the present invention, which can conveniently be presented in unit dosage form, can be prepared according to conventional techniques well known in the pharmaceutical industry. Such techniques include the step of bringing into association the active ingredients with the pharmaceutical carrier(s) or excipient(s). In general, the formulations are prepared by uniformly and intimately bringing into association the active ingredients with liquid carriers or finely divided solid carriers or both, and then, if necessary, shaping the product.

The compositions of the present invention can be formulated into any of many possible dosage forms such as, but not limited to, tablets, capsules, gel capsules, liquid syrups, soft gels, suppositories, and enemas. The compositions of the present invention can also be formulated as suspensions in aqueous, non-aqueous or mixed media. Aqueous suspensions can further contain substances which increase the viscosity of the suspension including, for example, sodium carboxymethylcellulose, sorbitol and/or dextran. The suspension can also contain stabilizers.

Many liposomes comprising lipids derivatized with one or more hydrophilic polymers, and methods of preparation thereof, are known in the art. Sunamoto et al. (*Bull. Chem. Soc. Jpn.*, 1980, 53, 2778) described liposomes comprising a nonionic detergent, 2C_{1215G}, that contains a PEG moiety. Illum et al. (*FEBS Lett.*, 1984, 167, 79) noted that hydrophilic coating of polystyrene particles with polymeric glycols results in significantly enhanced blood half-lives. Synthetic phospholipids modified by the attachment of carboxylic groups of polyalkylene glycols (e.g., PEG) are described by Sears (U.S. Pat. Nos. 4,426,330 and 4,534,899). Klivanov et al. (*FEBS Lett.*, 1990, 268, 235) described experiments demonstrating that liposomes comprising phosphatidylethanolamine (PE) derivatized with PEG or PEG stearate have significant increases in blood circulation half-lives. Blume et al. (*Biochimica et Biophysica Acta*, 1990, 1029, 91) extended such observations to other PEG-derivatized phospholipids, e.g., DSPE-PEG, formed from the

combination of distearoylphosphatidylethanolamine (DSPE) and PEG. Liposomes having covalently bound PEG moieties on their external surface are described in European Patent No. EP 0 445 131 B1 and WO 90/04384 to Fisher. Liposome compositions containing 1-20 mole percent of PE derivatized with PEG, and methods of use thereof, are described by Woodle et al. (U.S. Pat. Nos. 5,013,556 and 5,356,633) and Martin et al. (U.S. Pat. No. 5,213,804 and European Patent No. EP 0 496 813 B1). Liposomes comprising a number of other lipid-polymer conjugates are disclosed in WO 91/05545 and U.S. Pat. No. 5,225,212 (both to Martin et al.) and in WO 94/20073 (Zalipsky et al.). Liposomes comprising PEG-modified ceramide lipids are described in WO 96/10391 (Choi et al.). U.S. Pat. No. 5,540,935 (Miyazaki et al.) and U.S. Pat. No. 5,556,948 (Tagawa et al.) describe PEG-containing liposomes that can be further derivatized with functional moieties on their surfaces.

A number of liposomes comprising nucleic acids are known in the art. WO 96/40062 to Thierry et al. discloses methods for encapsulating high molecular weight nucleic acids in liposomes. U.S. Pat. No. 5,264,221 to Tagawa et al. discloses protein-bonded liposomes and asserts that the contents of such liposomes may include a dsRNA. U.S. Pat. No. 5,665,710 to Rahman et al. describes certain methods of encapsulating oligodeoxynucleotides in liposomes. WO 97/04787 to Love et al. discloses liposomes comprising dsRNAs targeted to the raf gene.

C. Additional Formulations

i. Emulsions

The compositions of the present invention can be prepared and formulated as emulsions. Emulsions are typically heterogeneous systems of one liquid dispersed in another in the form of droplets usually exceeding 0.1 μ m in diameter (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, LV., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199; Rosoff, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., Volume 1, p. 245; Block in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 2, p. 335; Higuchi et al., in *Remington's Pharmaceutical Sciences*, Mack Publishing Co., Easton, Pa., 1985, p. 301). Emulsions are often biphasic systems comprising two immiscible liquid phases intimately mixed and dispersed with each other. In general, emulsions can be of either the water-in-oil (w/o) or the oil-in-water (o/w) variety. When an aqueous phase is finely divided into and dispersed as minute droplets into a bulk oily phase, the resulting composition is called a water-in-oil (w/o) emulsion. Alternatively, when an oily phase is finely divided into and dispersed as minute droplets into a bulk aqueous phase, the resulting composition is called an oil-in-water (o/w) emulsion. Emulsions can contain additional components in addition to the dispersed phases, and the active drug which can be present as a solution in either aqueous phase, oily phase or itself as a separate phase. Pharmaceutical excipients such as emulsifiers, stabilizers, dyes, and anti-oxidants can also be present in emulsions as needed. Pharmaceutical emulsions can also be multiple emulsions that are comprised of more than two phases such as, for example, in the case of oil-in-water-in-oil (o/w/o) and water-in-oil-in-water (w/o/w) emulsions. Such complex formulations often provide certain advantages that simple binary emulsions do not. Multiple

emulsions in which individual oil droplets of an o/w emulsion enclose small water droplets constitute a w/o/w emulsion. Likewise, a system of oil droplets enclosed in globules of water stabilized in an oily continuous phase provides an o/w/o emulsion.

Emulsions are characterized by little or no thermodynamic stability. Often, the dispersed or discontinuous phase of the emulsion is well dispersed into the external or continuous phase and maintained in this form through the means of emulsifiers or the viscosity of the formulation. Either of the phases of the emulsion can be a semisolid or a solid, as is the case of emulsion-style ointment bases and creams. Other means of stabilizing emulsions entail the use of emulsifiers that can be incorporated into either phase of the emulsion. Emulsifiers can broadly be classified into four categories: synthetic surfactants, naturally occurring emulsifiers, absorption bases, and finely dispersed solids (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199).

Synthetic surfactants, also known as surface active agents, have found wide applicability in the formulation of emulsions and have been reviewed in the literature (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rieger, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 285; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), Marcel Dekker, Inc., New York, N.Y., 1988, volume 1, p. 199). Surfactants are typically amphiphilic and comprise a hydrophilic and a hydrophobic portion. The ratio of the hydrophilic to the hydrophobic nature of the surfactant has been termed the hydrophile/lipophile balance (HLB) and is a valuable tool in categorizing and selecting surfactants in the preparation of formulations. Surfactants can be classified into different classes based on the nature of the hydrophilic group: nonionic, anionic, cationic and amphoteric (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rieger, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 285).

Naturally occurring emulsifiers used in emulsion formulations include lanolin, beeswax, phosphatides, lecithin and acacia. Absorption bases possess hydrophilic properties such that they can soak up water to form w/o emulsions yet retain their semisolid consistencies, such as anhydrous lanolin and hydrophilic petrolatum. Finely divided solids have also been used as good emulsifiers especially in combination with surfactants and in viscous preparations. These include polar inorganic solids, such as heavy metal hydroxides, nonswelling clays such as bentonite, attapulgite, hectorite, kaolin, montmorillonite, colloidal aluminum silicate and colloidal magnesium aluminum silicate, pigments and nonpolar solids such as carbon or glyceryl tristearate.

A large variety of non-emulsifying materials are also included in emulsion formulations and contribute to the properties of emulsions. These include fats, oils, waxes, fatty acids, fatty alcohols, fatty esters, humectants, hydrophilic colloids, preservatives and antioxidants (Block, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and

Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 335; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199).

Hydrophilic colloids or hydrocolloids include naturally occurring gums and synthetic polymers such as polysaccharides (for example, acacia, agar, alginic acid, carrageenan, guar gum, karaya gum, and tragacanth), cellulose derivatives (for example, carboxymethylcellulose and carboxypolycellulose), and synthetic polymers (for example, carbomers, cellulose ethers, and carboxyvinyl polymers). These disperse or swell in water to form colloidal solutions that stabilize emulsions by forming strong interfacial films around the dispersed-phase droplets and by increasing the viscosity of the external phase.

Since emulsions often contain a number of ingredients such as carbohydrates, proteins, sterols and phosphatides that can readily support the growth of microbes, these formulations often incorporate preservatives. Commonly used preservatives included in emulsion formulations include methyl paraben, propyl paraben, quaternary ammonium salts, benzalkonium chloride, esters of p-hydroxybenzoic acid, and boric acid. Antioxidants are also commonly added to emulsion formulations to prevent deterioration of the formulation. Antioxidants used can be free radical scavengers such as tocopherols, alkyl gallates, butylated hydroxyanisole, butylated hydroxytoluene, or reducing agents such as ascorbic acid and sodium metabisulfite, and antioxidant synergists such as citric acid, tartaric acid, and lecithin.

The application of emulsion formulations via dermatological, oral and parenteral routes and methods for their manufacture have been reviewed in the literature (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199). Emulsion formulations for oral delivery have been very widely used because of ease of formulation, as well as efficacy from an absorption and bioavailability standpoint (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199). Mineral-oil base laxatives, oil-soluble vitamins and high fat nutritive preparations are among the materials that have commonly been administered orally as o/w emulsions.

ii. Microemulsions

In one embodiment of the present invention, the compositions of iRNAs and nucleic acids are formulated as microemulsions. A microemulsion can be defined as a system of water, oil and amphiphile which is a single optically isotropic and thermodynamically stable liquid solution (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245). Typically, microemulsions are systems that are prepared by first dispersing an oil in an aqueous surfactant solution and then adding a sufficient

amount of a fourth component, generally an intermediate chain-length alcohol to form a transparent system. Therefore, microemulsions have also been described as thermodynamically stable, isotropically clear dispersions of two immiscible liquids that are stabilized by interfacial films of surface-active molecules (Leung and Shah, in: *Controlled Release of Drugs: Polymers and Aggregate Systems*, Rosoff, M., Ed., 1989, VCH Publishers, New York, pages 185-215). Microemulsions commonly are prepared via a combination of three to five components that include oil, water, surfactant, cosurfactant and electrolyte. Whether the microemulsion is of the water-in-oil (w/o) or an oil-in-water (o/w) type is dependent on the properties of the oil and surfactant used and on the structure and geometric packing of the polar heads and hydrocarbon tails of the surfactant molecules (Schott, in *Remington's Pharmaceutical Sciences*, Mack Publishing Co., Easton, Pa., 1985, p. 271).

The phenomenological approach utilizing phase diagrams has been extensively studied and has yielded a comprehensive knowledge, to one skilled in the art, of how to formulate microemulsions (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L.V., Popovich N G., and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245; Block, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 335). Compared to conventional emulsions, microemulsions offer the advantage of solubilizing water-insoluble drugs in a formulation of thermodynamically stable droplets that are formed spontaneously.

Surfactants used in the preparation of microemulsions include, but are not limited to, ionic surfactants, non-ionic surfactants, Brij 96, polyoxyethylene oleyl ethers, polyglycerol fatty acid esters, tetraglycerol monolaurate (ML310), tetraglycerol monooleate (MO310), hexaglycerol monooleate (PO310), hexaglycerol pentaoleate (PO500), decaglycerol monocaprate (MCA750), decaglycerol monooleate (MO750), decaglycerol sequioleate (SO750), decaglycerol decaoleate (DAO750), alone or in combination with cosurfactants. The cosurfactant, usually a short-chain alcohol such as ethanol, 1-propanol, and 1-butanol, serves to increase the interfacial fluidity by penetrating into the surfactant film and consequently creating a disordered film because of the void space generated among surfactant molecules. Microemulsions can, however, be prepared without the use of cosurfactants and alcohol-free self-emulsifying microemulsion systems are known in the art. The aqueous phase can typically be, but is not limited to, water, an aqueous solution of the drug, glycerol, PEG300, PEG400, polyglycerols, propylene glycols, and derivatives of ethylene glycol. The oil phase can include, but is not limited to, materials such as Captex 300, Captex 355, Capmul MCM, fatty acid esters, medium chain (C8-C12) mono, di, and tri-glycerides, polyoxyethylated glyceryl fatty acid esters, fatty alcohols, polyglycolized glycerides, saturated polyglycolized C8-C10 glycerides, vegetable oils and silicone oil.

Microemulsions are particularly of interest from the standpoint of drug solubilization and the enhanced absorption of drugs. Lipid based microemulsions (both o/w and w/o) have been proposed to enhance the oral bioavailability of drugs, including peptides (see e.g., U.S. Pat. Nos. 6,191, 105; 7,063,860; 7,070,802; 7,157,099; Constantinides et al., *Pharmaceutical Research*, 1994, 11, 1385-1390; Ritschel, *Meth. Find. Exp. Clin. Pharmacol.*, 1993, 13, 205). Microemulsions afford advantages of improved drug solubiliza-

tion, protection of drug from enzymatic hydrolysis, possible enhancement of drug absorption due to surfactant-induced alterations in membrane fluidity and permeability, ease of preparation, ease of oral administration over solid dosage forms, improved clinical potency, and decreased toxicity (see e.g., U.S. Pat. Nos. 6,191,105; 7,063,860; 7,070,802; 7,157,099; Constantinides et al., *Pharmaceutical Research*, 1994, 11, 1385; Ho et al., *J. Pharm. Sci.*, 1996, 85, 138-143). Often microemulsions can form spontaneously when their components are brought together at ambient temperature. This can be particularly advantageous when formulating thermolabile drugs, peptides or iRNAs. Microemulsions have also been effective in the transdermal delivery of active components in both cosmetic and pharmaceutical applications. It is expected that the microemulsion compositions and formulations of the present invention will facilitate the increased systemic absorption of iRNAs and nucleic acids from the gastrointestinal tract, as well as improve the local cellular uptake of iRNAs and nucleic acids.

Microemulsions of the present invention can also contain additional components and additives such as sorbitan monostearate (Grill 3), Labrasol, and penetration enhancers to improve the properties of the formulation and to enhance the absorption of the iRNAs and nucleic acids of the present invention. Penetration enhancers used in the microemulsions of the present invention can be classified as belonging to one of five broad categories—surfactants, fatty acids, bile salts, chelating agents, and non-chelating non-surfactants (Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92). Each of these classes has been discussed above.

iii. Microparticles

An RNAi agent of the invention may be incorporated into a particle, e.g., a microparticle. Microparticles can be produced by spray-drying, but may also be produced by other methods including lyophilization, evaporation, fluid bed drying, vacuum drying, or a combination of these techniques.

iv. Penetration Enhancers

In one embodiment, the present invention employs various penetration enhancers to effect the efficient delivery of nucleic acids, particularly iRNAs, to the skin of animals. Most drugs are present in solution in both ionized and nonionized forms. However, usually only lipid soluble or lipophilic drugs readily cross cell membranes. It has been discovered that even non-lipophilic drugs can cross cell membranes if the membrane is to be crossed is treated with a penetration enhancer. In addition to aiding the diffusion of non-lipophilic drugs across cell membranes, penetration enhancers also enhance the permeability of lipophilic drugs.

Penetration enhancers can be classified as belonging to one of five broad categories, i.e., surfactants, fatty acids, bile salts, chelating agents, and non-chelating non-surfactants (see e.g., Malmsten, M. *Surfactants and polymers in drug delivery*, Informa Health Care, New York, NY, 2002; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92). Each of the above mentioned classes of penetration enhancers are described below in greater detail.

Surfactants (or "surface-active agents") are chemical entities which, when dissolved in an aqueous solution, reduce the surface tension of the solution or the interfacial tension between the aqueous solution and another liquid, with the result that absorption of iRNAs through the mucosa is enhanced. In addition to bile salts and fatty acids, these penetration enhancers include, for example, sodium lauryl sulfate, polyoxyethylene-9-lauryl ether and polyoxyethylene-20-cetyl ether (see e.g., Malmsten, M. *Surfactants and polymers in drug delivery*, Informa Health Care, New York,

NY, 2002; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92); and perfluorochemical emulsions, such as FC-43. Takahashi et al., *J. Pharm. Pharmacol.*, 1988, 40, 252).

Various fatty acids and their derivatives which act as penetration enhancers include, for example, oleic acid, lauric acid, capric acid (n-decanoic acid), myristic acid, palmitic acid, stearic acid, linoleic acid, linolenic acid, dicaprate, tricaprate, monoolein (1-monooleoyl-rac-glycerol), dilaurin, caprylic acid, arachidonic acid, glycerol 1-monocaprate, 1-dodecylazacycloheptan-2-one, acylcarnitines, acylcholines, C₁₋₂₀ alkyl esters thereof (e.g., methyl, isopropyl and t-butyl), and mono- and di-glycerides thereof (i.e., oleate, laurate, caprate, myristate, palmitate, stearate, linoleate, etc.) (see e.g., Toutiou, E., et al. *Enhancement in Drug Delivery*, CRC Press, Danvers, MA, 2006; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92; Muranishi, *Critical Reviews in Therapeutic Drug Carrier Systems*, 1990, 7, 1-33; El Hariri et al., *J. Pharm. Pharmacol.*, 1992, 44, 651-654).

The physiological role of bile includes the facilitation of dispersion and absorption of lipids and fat-soluble vitamins (see e.g., Malmsten, M. *Surfactants and polymers in drug delivery*, Informa Health Care, New York, NY, 2002; Brunton, Chapter 38 in: Goodman & Gilman's *The Pharmacological Basis of Therapeutics*, 9th Ed., Hardman et al. Eds., McGraw-Hill, New York, 1996, pp. 934-935). Various natural bile salts, and their synthetic derivatives, act as penetration enhancers. Thus the term "bile salts" includes any of the naturally occurring components of bile as well as any of their synthetic derivatives. Suitable bile salts include, for example, cholic acid (or its pharmaceutically acceptable sodium salt, sodium cholate), dehydrocholic acid (sodium dehydrocholate), deoxycholic acid (sodium deoxycholate), glucolic acid (sodium glucolate), glycholic acid (sodium glycocholate), glycodeoxycholic acid (sodium glycodeoxycholate), taurocholic acid (sodium taurocholate), taurodeoxycholic acid (sodium taurodeoxycholate), chenodeoxycholic acid (sodium chenodeoxycholate), ursodeoxycholic acid (UDCA), sodium tauro-24,25-dihydro-fusidate (STDHF), sodium glycodihydrofusidate and polyoxyethylene-9-lauryl ether (POE) (see e.g., Malmsten, M. *Surfactants and polymers in drug delivery*, Informa Health Care, New York, NY, 2002; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, page 92; Swinyard, Chapter 39 In: Remington's *Pharmaceutical Sciences*, 18th Ed., Gennaro, ed., Mack Publishing Co., Easton, Pa., 1990, pages 782-783; Muranishi, *Critical Reviews in Therapeutic Drug Carrier Systems*, 1990, 7, 1-33; Yamamoto et al., *J. Pharm. Exp. Ther.*, 1992, 263, 25; Yamashita et al., *J. Pharm. Sci.*, 1990, 79, 579-583).

Chelating agents, as used in connection with the present invention, can be defined as compounds that remove metallic ions from solution by forming complexes therewith, with the result that absorption of iRNAs through the mucosa is enhanced. With regards to their use as penetration enhancers in the present invention, chelating agents have the added advantage of also serving as DNase inhibitors, as most characterized DNA nucleases require a divalent metal ion for catalysis and are thus inhibited by chelating agents (Jarrett, *J. Chromatogr.*, 1993, 618, 315-339). Suitable chelating agents include but are not limited to disodium ethylenediaminetetraacetate (EDTA), citric acid, salicylates (e.g., sodium salicylate, 5-methoxysalicylate and homovanilate), N-acyl derivatives of collagen, laureth-9 and N-amino acyl derivatives of beta-diketones (enamines)(see e.g., Katdare, A. et al., *Excipient development for pharma-*

ceutical, biotechnology, and drug delivery, CRC Press, Danvers, MA, 2006; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, page 92; Muranishi, *Critical Reviews in Therapeutic Drug Carrier Systems*, 1990, 7, 1-33; Buur et al., *J. Control Rel.*, 1990, 14, 43-51).

As used herein, non-chelating non-surfactant penetration enhancing compounds can be defined as compounds that demonstrate insignificant activity as chelating agents or as surfactants but that nonetheless enhance absorption of iRNAs through the alimentary mucosa (see e.g., Muranishi, *Critical Reviews in Therapeutic Drug Carrier Systems*, 1990, 7, 1-33). This class of penetration enhancers includes, for example, unsaturated cyclic ureas, 1-alkyl- and 1-alkenylazacyclo-alkanone derivatives (Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, page 92); and non-steroidal anti-inflammatory agents such as diclofenac sodium, indomethacin and phenylbutazone (Yamashita et al., *J. Pharm. Pharmacol.*, 1987, 39, 621-626).

Agents that enhance uptake of iRNAs at the cellular level can also be added to the pharmaceutical and other compositions of the present invention. For example, cationic lipids, such as lipofectin (Junichi et al, U.S. Pat. No. 5,705,188), cationic glycerol derivatives, and polycationic molecules, such as polylysine (Lollo et al., PCT Application WO 97/30731), are also known to enhance the cellular uptake of dsRNAs. Examples of commercially available transfection reagents include, for example Lipofectamine™ (Invitrogen; Carlsbad, CA), Lipofectamine 2000™ (Invitrogen; Carlsbad, CA), 293Fectin™ (Invitrogen; Carlsbad, CA), Cellfectin™ (Invitrogen; Carlsbad, CA), DMRIE-CT™ (Invitrogen; Carlsbad, CA), FreeStyle™ MAX (Invitrogen; Carlsbad, CA), Lipofectamine™ 2000 CD (Invitrogen; Carlsbad, CA), Lipofectamine™ (Invitrogen; Carlsbad, CA), RNAiMAX (Invitrogen; Carlsbad, CA), Oligofectamine™ (Invitrogen; Carlsbad, CA), Optifect™ (Invitrogen; Carlsbad, CA), X-tremeGENE Q2 Transfection Reagent (Roche; Grenzacherstrasse, Switzerland), DOTAP Liposomal Transfection Reagent (Grenzacherstrasse, Switzerland), DOSPER Liposomal Transfection Reagent (Grenzacherstrasse, Switzerland), or Eugene (Grenzacherstrasse, Switzerland), Transfectam® Reagent (Promega; Madison, WI), TransFast™ Transfection Reagent (Promega; Madison, WI), Tfx™-20 Reagent (Promega; Madison, WI), Tfx™-50 Reagent (Promega; Madison, WI), DreamFect™ (OZ Biosciences; Marseille, France), EcoTransfect (OZ Biosciences; Marseille, France), TransPass[®] D1 Transfection Reagent (New England Biolabs; Ipswich, MA, USA), LyoVec™/LipoGen™ (Invitrogen; San Diego, CA, USA), PerFectin Transfection Reagent (Genlantis; San Diego, CA, USA), NeuroPORTER Transfection Reagent (Genlantis; San Diego, CA, USA), GenePORTER Transfection reagent (Genlantis; San Diego, CA, USA), GenePORTER 2 Transfection reagent (Genlantis; San Diego, CA, USA), Cytofectin Transfection Reagent (Genlantis; San Diego, CA, USA), BaculoPORTER Transfection Reagent (Genlantis; San Diego, CA, USA), TroganPORTER™ transfection Reagent (Genlantis; San Diego, CA, USA), RiboFect (Bioline; Taunton, MA, USA), PlasFect (Bioline; Taunton, MA, USA), UniFECTOR (B-Bridge International; Mountain View, CA, USA), SureFECTOR (B-Bridge International; Mountain View, CA, USA), or HiFect™ (B-Bridge International, Mountain View, CA, USA), among others.

Other agents can be utilized to enhance the penetration of the administered nucleic acids, including glycols such as ethylene glycol and propylene glycol, pyrrols such as 2-pyrrol, azones, and terpenes such as limonene and menthone.

v. Carriers

Certain compositions of the present invention also incorporate carrier compounds in the formulation. As used herein, "carrier compound" or "carrier" can refer to a nucleic acid, or analog thereof, which is inert (i.e., does not possess biological activity per se) but is recognized as a nucleic acid by in vivo processes that reduce the bioavailability of a nucleic acid having biological activity by, for example, degrading the biologically active nucleic acid or promoting its removal from circulation. The coadministration of a nucleic acid and a carrier compound, typically with an excess of the latter substance, can result in a substantial reduction of the amount of nucleic acid recovered in the liver, kidney or other extracirculatory reservoirs, presumably due to competition between the carrier compound and the nucleic acid for a common receptor. For example, the recovery of a partially phosphorothioate dsRNA in hepatic tissue can be reduced when it is coadministered with polyinosinic acid, dextran sulfate, polycytidic acid or 4-acetamido-4'-isothiocyano-stilbene-2,2'-disulfonic acid (Miyao et al., *DsRNA Res. Dev.*, 1995, 5, 115-121; Takakura et al., *DsRNA & Nucl. Acid Drug Dev.*, 1996, 6, 177-183).

vi. Excipients

In contrast to a carrier compound, a "pharmaceutical carrier" or "excipient" is a pharmaceutically acceptable solvent, suspending agent or any other pharmacologically inert vehicle for delivering one or more nucleic acids to an animal. The excipient can be liquid or solid and is selected, with the planned manner of administration in mind, so as to provide for the desired bulk, consistency, etc., when combined with a nucleic acid and the other components of a given pharmaceutical composition. Typical pharmaceutical carriers include, but are not limited to, binding agents (e.g., pregelatinized maize starch, polyvinylpyrrolidone or hydroxypropyl methylcellulose, etc.); fillers (e.g., lactose and other sugars, microcrystalline cellulose, pectin, gelatin, calcium sulfate, ethyl cellulose, polyacrylates or calcium hydrogen phosphate, etc.); lubricants (e.g., magnesium stearate, talc, silica, colloidal silicon dioxide, stearic acid, metallic stearates, hydrogenated vegetable oils, corn starch, polyethylene glycols, sodium benzoate, sodium acetate, etc.); disintegrants (e.g., starch, sodium starch glycolate, etc.); and wetting agents (e.g., sodium lauryl sulphate, etc.).

Pharmaceutically acceptable organic or inorganic excipients suitable for non-parenteral administration which do not deleteriously react with nucleic acids can also be used to formulate the compositions of the present invention. Suitable pharmaceutically acceptable carriers include, but are not limited to, water, salt solutions, alcohols, polyethylene glycols, gelatin, lactose, amylose, magnesium stearate, talc, silicic acid, viscous paraffin, hydroxymethylcellulose, polyvinylpyrrolidone and the like.

Formulations for topical administration of nucleic acids can include sterile and non-sterile aqueous solutions, non-aqueous solutions in common solvents such as alcohols, or solutions of the nucleic acids in liquid or solid oil bases. The solutions can also contain buffers, diluents and other suitable additives. Pharmaceutically acceptable organic or inorganic excipients suitable for non-parenteral administration which do not deleteriously react with nucleic acids can be used.

Suitable pharmaceutically acceptable excipients include, but are not limited to, water, salt solutions, alcohol, polyethylene glycols, gelatin, lactose, amylose, magnesium stearate, talc, silicic acid, viscous paraffin, hydroxymethylcellulose, polyvinylpyrrolidone and the like.

vii. Other Components

The compositions of the present invention can additionally contain other adjunct components conventionally found in pharmaceutical compositions, at their art-established usage levels. Thus, for example, the compositions can contain additional, compatible, pharmaceutically-active materials such as, for example, antipruritics, astringents, local anesthetics or anti-inflammatory agents, or can contain additional materials useful in physically formulating various dosage forms of the compositions of the present invention, such as dyes, flavoring agents, preservatives, antioxidants, opacifiers, thickening agents and stabilizers. However, such materials, when added, should not unduly interfere with the biological activities of the components of the compositions of the present invention. The formulations can be sterilized and, if desired, mixed with auxiliary agents, e.g., lubricants, preservatives, stabilizers, wetting agents, emulsifiers, salts for influencing osmotic pressure, buffers, colorings, flavorings and/or aromatic substances and the like which do not deleteriously interact with the nucleic acid(s) of the formulation.

Aqueous suspensions can contain substances which increase the viscosity of the suspension including, for example, sodium carboxymethylcellulose, sorbitol and/or dextran. The suspension can also contain stabilizers.

In some embodiments, pharmaceutical compositions featured in the invention include (a) one or more iRNA compounds and (b) one or more agents which function by a non-RNAi mechanism and which are useful in treating a CIDEA-associated disease, disorder, or condition. Examples of such agents include, but are not limited to pyridoxine, an ACE inhibitor (angiotensin converting enzyme inhibitors), e.g., benazepril (Lotensin); an angiotensin II receptor antagonist (ARB) (e.g., losartan potassium, such as Merck & Co.'s Cozaar®), e.g., Candesartan (Atacand); an HMG-CoA reductase inhibitor (e.g., a statin); calcium binding agents, e.g., Sodium cellulose phosphate (Calcibind); diuretics, e.g., thiazide diuretics, such as hydrochlorothiazide (Microzide); an insulin sensitizer, such as the PPAR γ agonist pioglitazone, a glp-1r agonist, such as liraglutide, vitamin E, an SGLT2 inhibitor, a DPPIV inhibitor, and kidney/liver transplant; or a combination of any of the foregoing.

Toxicity and therapeutic efficacy of such compounds can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD₅₀ (the dose lethal to 50% of the population) and the ED₅₀ (the dose therapeutically effective in 50% of the population). The dose ratio between toxic and therapeutic effects is the therapeutic index and it can be expressed as the ratio LD₅₀/ED₅₀. Compounds that exhibit high therapeutic indices are preferred.

The data obtained from cell culture assays and animal studies can be used in formulating a range of dosage for use in humans. The dosage of compositions featured herein in the invention lies generally within a range of circulating concentrations that include the ED₅₀ with little or no toxicity. The dosage can vary within this range depending upon the dosage form employed and the route of administration utilized. For any compound used in the methods featured in the invention, the therapeutically effective dose can be estimated initially from cell culture assays. A dose can be formulated in animal models to achieve a circulating plasma concentration range of the compound or, when appropriate, of the polypeptide product of a target sequence (e.g., achieving a decreased concentration of the polypeptide) that includes the IC₅₀ (i.e., the concentration of the test compound which achieves a half-maximal inhibition of symptoms) as determined in cell culture. Such information can be

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used to more accurately determine useful doses in humans. Levels in plasma can be measured, for example, by high performance liquid chromatography.

In addition to their administration, as discussed above, the iRNAs featured in the invention can be administered in combination with other known agents effective in treatment of pathological processes mediated by CIDEb expression. In any event, the administering physician can adjust the amount and timing of iRNA administration on the basis of results observed using standard measures of efficacy known in the art or described herein.

Synthesis of Cationic Lipids:

Any of the compounds, e.g., cationic lipids and the like, used in the nucleic acid-lipid particles featured in the invention may be prepared by known organic synthesis techniques. All substituents are as defined below unless indicated otherwise.

"Alkyl" means a straight chain or branched, noncyclic or cyclic, saturated aliphatic hydrocarbon containing from 1 to 24 carbon atoms. Representative saturated straight chain alkyls include methyl, ethyl, n-propyl, n-butyl, n-pentyl, n-hexyl, and the like; while saturated branched alkyls include isopropyl, sec-butyl, isobutyl, tert-butyl, isopentyl, and the like. Representative saturated cyclic alkyls include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, and the like; while unsaturated cyclic alkyls include cyclopentenyl and cyclohexenyl, and the like.

"Alkenyl" means an alkyl, as defined above, containing at least one double bond between adjacent carbon atoms. Alkenyls include both cis and trans isomers. Representative straight chain and branched alkenyls include ethylenyl, propylenyl, 1-butenyl, 2-butenyl, isobutylenyl, 1-pentenyl, 2-pentenyl, 3-methyl-1-butenyl, 2-methyl-2-butenyl, 2,3-dimethyl-2-butenyl, and the like.

"Alkynyl" means any alkyl or alkenyl, as defined above, which additionally contains at least one triple bond between adjacent carbons. Representative straight chain and branched alkynyls include acetylenyl, propynyl, 1-butylnyl, 2-butylnyl, 1-pentylnyl, 2-pentylnyl, 3-methyl-1 butynyl, and the like.

"Acyl" means any alkyl, alkenyl, or alkynyl wherein the carbon at the point of attachment is substituted with an oxo group, as defined below. For example, $-C(=O)$ alkyl, $-C(=O)$ alkenyl, and $-C(=O)$ alkynyl are acyl groups.

"Heterocycle" means a 5- to 7-membered monocyclic, or 7- to 10-membered bicyclic, heterocyclic ring which is either saturated, unsaturated, or aromatic, and which contains from 1 or 2 heteroatoms independently selected from nitrogen, oxygen and sulfur, and wherein the nitrogen and sulfur heteroatoms may be optionally oxidized, and the nitrogen heteroatom may be optionally quaternized, including bicyclic rings in which any of the above heterocycles are fused to a benzene ring. The heterocycle may be attached via any heteroatom or carbon atom. Heterocycles include heteroaryls as defined below. Heterocycles include morpholinyl, pyrrolidinonyl, pyrrolidinyl, piperidinyl, piperizynyl, hydantoinyl, valerolactamyl, oxiranyl, oxetanyl, tetrahydrofuranyl, tetrahydropyranyl, tetrahydropyridinyl, tetrahydroprimidinyl, tetrahydrothiophenyl, tetrahydrothiopyranyl, tetrahydrothiopyrimidinyl, tetrahydrothiopyridinyl, and the like.

The terms "optionally substituted alkyl", "optionally substituted alkenyl", "optionally substituted alkynyl", "option-

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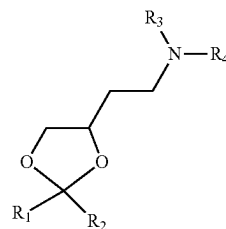
ally substituted acyl", and "optionally substituted heterocycle" means that, when substituted, at least one hydrogen atom is replaced with a substituent. In the case of an oxo substituent ($=O$) two hydrogen atoms are replaced. In this regard, substituents include oxo, halogen, heterocycle, $-CN$, $-OR^x$, $-NR^xR^y$, $-NR^xC(=O)R^y$, $-NR^xSO_2R^y$, $-C(=O)R^x$, $-C(=O)OR^x$, $-C(=O)NR^xR^y$, $-SO_nR^x$ and $-SO_nNR^xR^y$, wherein n is 0, 1 or 2, R^x and R^y are the same or different and independently hydrogen, alkyl or heterocycle, and each of said alkyl and heterocycle substituents may be further substituted with one or more of oxo, halogen, $-OH$, $-CN$, alkyl, $-OR^x$, heterocycle, $-NR^xR^y$, $-NR^xC(=O)R^y$, $-NR^xSO_2R^y$, $-C(=O)R^x$, $-C(=O)OR^x$, $-C(=O)NR^xR^y$, $-SO_nR^x$ and $-SO_nNR^xR^y$.

"Halogen" means fluoro, chloro, bromo and iodo.

In some embodiments, the methods featured in the invention may require the use of protecting groups. Protecting group methodology is well known to those skilled in the art (see, for example, *PROTECTIVE GROUPS IN ORGANIC SYNTHESIS*, Green, T. W. et al., Wiley-Interscience, New York City, 1999). Briefly, protecting groups within the context of this invention are any group that reduces or eliminates unwanted reactivity of a functional group. A protecting group can be added to a functional group to mask its reactivity during certain reactions and then removed to reveal the original functional group. In some embodiments an "alcohol protecting group" is used. An "alcohol protecting group" is any group which decreases or eliminates unwanted reactivity of an alcohol functional group. Protecting groups can be added and removed using techniques well known in the art.

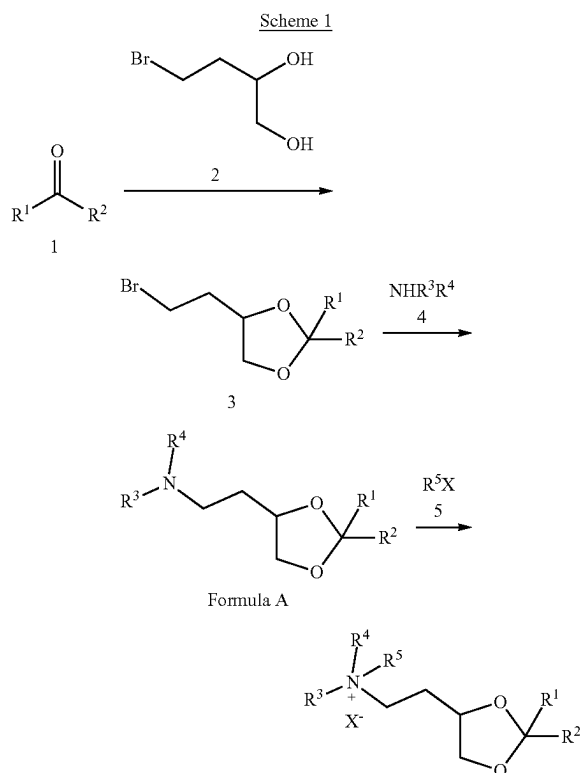
Synthesis of Formula A

In certain embodiments, nucleic acid-lipid particles featured in the invention are formulated using a cationic lipid of formula A:



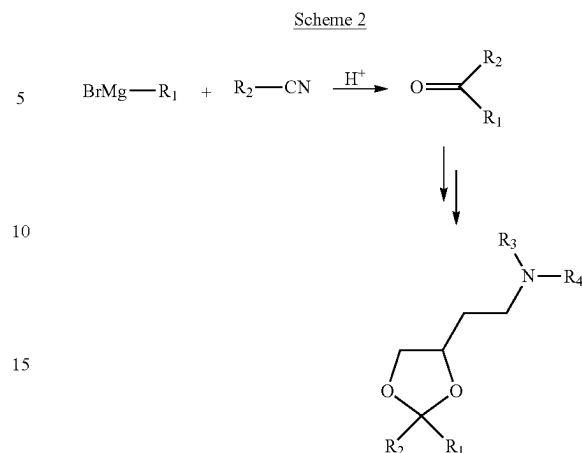
where R1 and R2 are independently alkyl, alkenyl or alkynyl, each can be optionally substituted, and R3 and R4 are independently lower alkyl or R3 and R4 can be taken together to form an optionally substituted heterocyclic ring. In some embodiments, the cationic lipid is XTC (2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane). In general, the lipid of formula A above may be made by the following Reaction Schemes 1 or 2, wherein all substituents are as defined above unless indicated otherwise.

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Lipid A, where R_1 and R_2 are independently alkyl, alkenyl or alkynyl, each can be optionally substituted, and R_3 and R_4 are independently lower alkyl or R_3 and R_4 can be taken together to form an optionally substituted heterocyclic ring, can be prepared according to Scheme 1. Ketone 1 and bromide 2 can be purchased or prepared according to methods known to those of ordinary skill in the art. Reaction of 1 and 2 yields ketal 3. Treatment of ketal 3 with amine 4 yields lipids of formula A. The lipids of formula A can be converted to the corresponding ammonium salt with an organic salt of formula 5, where X is anion counter ion selected from halogen, hydroxide, phosphate, sulfate, or the like.

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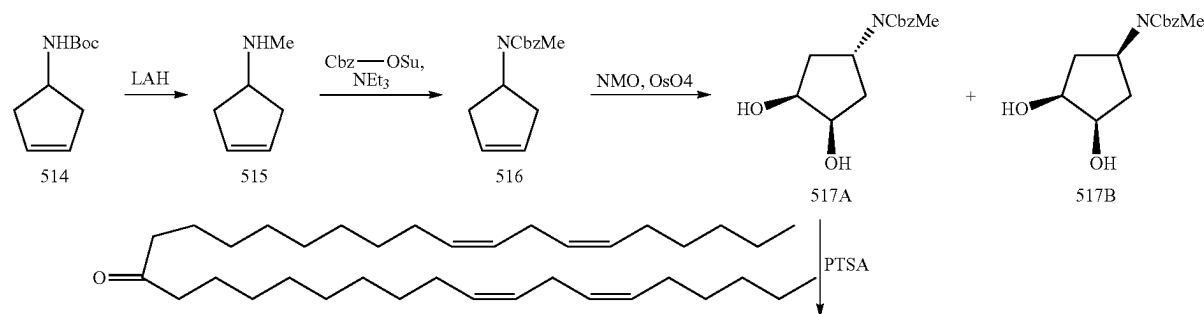
Alternatively, the ketone 1 starting material can be prepared according to Scheme 2. Grignard reagent 6 and cyanide 7 can be purchased or prepared according to methods known to those of ordinary skill in the art. Reaction of 6 and 7 yields ketone 1. Conversion of ketone 1 to the corresponding lipids of formula A is as described in Scheme 1.

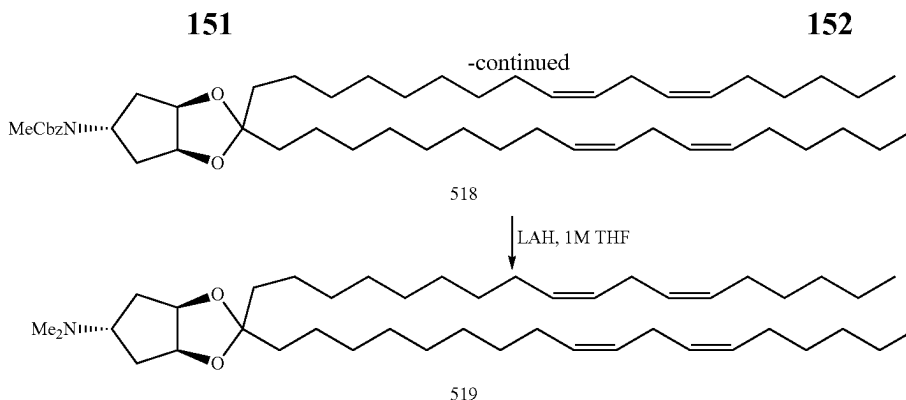
Synthesis of MC3

Preparation of Dlin-M-C3-DMA (i.e., (6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetraen-19-yl 4-(dimethylamino)butanoate) was as follows. A solution of (6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetraen-19-ol (0.53 g), 4-N,N-dimethylaminobutyric acid hydrochloride (0.51 g), 4-N,N-dimethylaminopyridine (0.61 g) and 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride (0.53 g) in dichloromethane (5 mL) was stirred at room temperature overnight. The solution was washed with dilute hydrochloric acid followed by dilute aqueous sodium bicarbonate. The organic fractions were dried over anhydrous magnesium sulphate, filtered and the solvent removed on a rotovap. The residue was passed down a silica gel column (20 g) using a 1-5% methanol/dichloromethane elution gradient. Fractions containing the purified product were combined and the solvent removed, yielding a colorless oil (0.54 g).

Synthesis of ALNY-100

Synthesis of Ketal 519 [ALNY-100] was Performed Using the Following Scheme 3





Synthesis of 515

To a stirred suspension of LiAlH_4 (3.74 g, 0.09852 mol) in 200 ml anhydrous THF in a two neck RBF (1 L), was added a solution of 514 (10 g, 0.04926 mol) in 70 mL of THF slowly at 0°C . under nitrogen atmosphere. After complete addition, reaction mixture was warmed to room temperature and then heated to reflux for 4 h. Progress of the reaction was monitored by TLC. After completion of reaction (by TLC) the mixture was cooled to 0°C . and quenched with careful addition of saturated Na_2SO_4 solution. Reaction mixture was stirred for 4 h at room temperature and filtered off. Residue was washed well with THF. The filtrate and washings were mixed and diluted with 400 mL dioxane and 26 mL conc. HCl and stirred for 20 minutes at room temperature. The volatilities were stripped off under vacuum to furnish the hydrochloride salt of 515 as a white solid. Yield: 7.12 g 1H-NMR (DMSO, 400 MHz): $\delta=9.34$ (broad, 2H), 5.68 (s, 2H), 3.74 (m, 1H), 2.66-2.60 (m, 2H), 2.50-2.45 (m, 5H).

Synthesis of 516

To a stirred solution of compound 515 in 100 mL dry DCM in a 250 mL two neck RBF, was added NEt_3 (37.2 mL, 0.2669 mol) and cooled to 0°C . under nitrogen atmosphere. After a slow addition of N-(benzyloxy-carbonyloxy)-succinimide (20 g, 0.08007 mol) in 50 mL dry DCM, reaction mixture was allowed to warm to room temperature. After completion of the reaction (2-3 h by TLC) mixture was washed successively with 1N HCl solution (1×100 mL) and saturated NaHCO_3 solution (1×50 mL). The organic layer was then dried over anhyd. Na_2SO_4 and the solvent was evaporated to give crude material which was purified by silica gel column chromatography to get 516 as sticky mass. Yield: 11 g (89%). 1H-NMR (CDCl_3 , 400 MHz): $\delta=7.36$ -7.27 (m, 5H), 5.69 (s, 2H), 5.12 (s, 2H), 4.96 (br., 1H) 2.74 (s, 3H), 2.60 (m, 2H), 2.30-2.25 (m, 2H). LC-MS $[\text{M}+\text{H}]^+$ -232.3 (96.94%).

Synthesis of 517A and 517B

The cyclopentene 516 (5 g, 0.02164 mol) was dissolved in a solution of 220 mL acetone and water (10:1) in a single neck 500 mL RBF and to it was added N-methyl morpholine-N-oxide (7.6 g, 0.06492 mol) followed by 4.2 mL of 7.6% solution of 0.0 (0.275 g, 0.00108 mol) in tert-butanol at room temperature. After completion of the reaction (~3 h), the mixture was quenched with addition of solid Na_2SO_3 and resulting mixture was stirred for 1.5 h at room temperature. Reaction mixture was diluted with DCM (300 mL) and

washed with water (2×100 mL) followed by saturated NaHCO_3 (1×50 mL) solution, water (1×30 mL) and finally with brine (1×50 mL). Organic phase was dried over Na_2SO_4 and solvent was removed in vacuum. Silica gel column chromatographic purification of the crude material was afforded a mixture of diastereomers, which were separated by prep HPLC. Yield: ~6 g crude 517A—Peak-1 (white solid), 5.13 g (96%). 1H-NMR (DMSO, 400 MHz): $\delta=7.39$ -7.31 (m, 5H), 5.04 (s, 2H), 4.78-4.73 (m, 1H), 4.48-4.47 (d, 2H), 3.94-3.93 (m, 2H), 2.71 (s, 3H), 1.72-1.67 (m, 4H). LCMS- $[\text{M}+\text{H}]^+$ -266.3, $[\text{M}+\text{NH}_4]^+$ -283.5 present, HPLC-97.86%. Stereochemistry confirmed by X-ray.

Synthesis of 518

Using a procedure analogous to that described for the synthesis of compound 505, compound 518 (1.2 g, 41%) was obtained as a colorless oil. 1H-NMR (CDCl_3 , 400 MHz): $\delta=7.35$ -7.33 (m, 4H), 7.30-7.27 (m, 1H), 5.37-5.27 (m, 8H), 5.12 (s, 2H), 4.75 (m, 1H), 4.58-4.57 (m, 2H), 2.78-2.74 (m, 7H), 2.06-2.00 (m, 8H), 1.96-1.91 (m, 2H), 1.62 (m, 4H), 1.48 (m, 2H), 1.37-1.25 (br m, 36H), 0.87 (m, 6H). HPLC-98.65%.

General Procedure for the Synthesis of Compound 519

A solution of compound 518 (1 eq) in hexane (15 mL) was added in a drop-wise fashion to an ice-cold solution of LAH in THF (1 M, 2 eq). After complete addition, the mixture was heated at 40°C . over 0.5 h then cooled again on an ice bath. The mixture was carefully hydrolyzed with saturated aqueous Na_2SO_4 then filtered through Celite® and reduced to an oil. Column chromatography provided the pure 519 (1.3 g, 68%) which was obtained as a colorless oil. ^{13}C NMR=130.2, 130.1 (×2), 127.9 (×3), 112.3, 79.3, 64.4, 44.7, 38.3, 35.4, 31.5, 29.9 (×2), 29.7, 29.6 (×2), 29.5 (×3), 29.3 (×2), 27.2 (×3), 25.6, 24.5, 23.3, 22.6, 14.1; Electrospray MS (+ve): Molecular weight for $\text{C}_{44}\text{H}_{80}\text{NO}_2$ ($\text{M}+\text{H})^+$ Calc. 654.6, Found 654.6.

Formulations prepared by either the standard or extrusion-free method can be characterized in similar manners. For example, formulations are typically characterized by visual inspection. They should be whitish translucent solutions free from aggregates or sediment. Particle size and particle size distribution of lipid-nanoparticles can be measured by light scattering using, for example, a Malvern Zetasizer Nano ZS (Malvern, USA). Particles should be about 20-300 nm, such as 40-100 nm in size. The particle size distribution should be unimodal. The total dsRNA concentration in the formulation, as well as the entrapped fraction, is estimated using a

dye exclusion assay. A sample of the formulated dsRNA can be incubated with an RNA-binding dye, such as RiboGreen® (Molecular Probes) in the presence or absence of a formulation disrupting surfactant, e.g., 0.5% Triton-X100. The total dsRNA in the formulation can be determined by the signal from the sample containing the surfactant, relative to a standard curve. The entrapped fraction is determined by subtracting the “free” dsRNA content (as measured by the signal in the absence of surfactant) from the total dsRNA content. Percent entrapped dsRNA is typically >85%. For SNALP formulation, the particle size is at least 30 nm, at least 40 nm, at least 50 nm, at least 60 nm, at least 70 nm, at least 80 nm, at least 90 nm, at least 100 nm, at least 110 nm, and at least 120 nm. The suitable range is typically about at least 50 nm to about at least 110 nm, about at least 60 nm to about at least 100 nm, or about at least 80 nm to about at least 90 nm.

VII. Methods of the Invention

The present invention also provides methods of using an iRNA of the invention and/or a composition of the invention to reduce and/or inhibit CIDEB expression in a cell, such as a cell in a subject, e.g., a hepatocyte. The methods include contacting the cell with an RNAi agent or pharmaceutical composition comprising an iRNA agent of the invention. In some embodiments, the cell is maintained for a time sufficient to obtain degradation of the mRNA transcript of a CIDEB gene.

The present invention also provides methods of using an iRNA of the invention and/or a composition of the invention and an iRNA agent targeting a Patatin-like Phospholipase Domain Containing 3 (PNPLA3) gene and/or pharmaceutical composition comprising an iRNA agent targeting PNPLA3 to reduce and/or inhibit PNPLA3 expression in a cell, such as a cell in a subject, e.g., a hepatocyte.

In addition, the present invention provides methods of inhibiting the accumulation and/or expansion of lipid droplets in a cell, such as a cell in a subject, e.g., a hepatocyte. The methods include contacting the cell with an RNAi agent or pharmaceutical composition comprising an iRNA agent of the invention and an iRNA agent targeting a PNPLA3 gene and/or pharmaceutical composition comprising an iRNA agent targeting PNPLA3. In some embodiments, the cell is maintained for a time sufficient to obtain degradation of the mRNA transcript of a CIDEB gene and a PNPLA3 gene.

Reduction in gene expression can be assessed by any methods known in the art. For example, a reduction in the expression of CIDEB may be determined by determining the mRNA expression level of CIDEB using methods routine to one of ordinary skill in the art, e.g., Northern blotting, qRT-PCR; by determining the protein level of CIDEB using methods routine to one of ordinary skill in the art, such as Western blotting, immunological techniques. A reduction in the expression of CIDEB may also be assessed indirectly by measuring a decrease in biological activity of CIDEB, e.g., a decrease in the interaction of CIDEB with ApoB and/or a decrease in lipid maturation in the liver.

Suitable agents targeting a PNPLA3 gene are described in, for example, U.S. Patent Publication No.: 2017/0340661, the entire contents of which are incorporated herein by reference.

In the methods of the invention the cell may be contacted in vitro or in vivo, i.e., the cell may be within a subject.

A cell suitable for treatment using the methods of the invention may be any cell that expresses a CIDEB gene

(and, in some embodiments, a PNPLA3 gene). A cell suitable for use in the methods of the invention may be a mammalian cell, e.g., a primate cell (such as a human cell or a non-human primate cell, e.g., a monkey cell or a chimpanzee cell), a non-primate cell (such as a cow cell, a pig cell, a camel cell, a llama cell, a horse cell, a goat cell, a rabbit cell, a sheep cell, a hamster, a guinea pig cell, a cat cell, a dog cell, a rat cell, a mouse cell, a lion cell, a tiger cell, a bear cell, or a buffalo cell), a bird cell (e.g., a duck cell or a goose cell), or a whale cell. In one embodiment, the cell is a human cell, e.g., a human liver cell.

CIDEB expression is inhibited in the cell by at least about 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or about 100%. In preferred embodiments, CIDEB expression is inhibited by at least 20%.

In some embodiment, PNPLA3 expression is also inhibited in the cell by at least about 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or about 100%. In preferred embodiments, PNPLA3 expression is inhibited by at least 20%.

In one embodiment, the in vivo methods of the invention may include administering to a subject a composition containing an iRNA, where the iRNA includes a nucleotide sequence that is complementary to at least a part of an RNA transcript of the CIDEB gene of the mammal to be treated.

In another embodiment, the in vivo methods of the invention may include administering to a subject a composition containing a first iRNA agent and a second iRNA agent, where the first iRNA includes a nucleotide sequence that is complementary to at least a part of an RNA transcript of the CIDEB gene of the mammal to be treated and the second iRNA includes a nucleotide sequence that is complementary to at least a part of an RNA transcript of the PNPLA3 gene of the mammal to be treated.

When the organism to be treated is a mammal such as a human, the composition can be administered by any means known in the art including, but not limited to oral, intraperitoneal, or parenteral routes, including intracranial (e.g., intraventricular, intraparenchymal and intrathecal), intravenous, intramuscular, subcutaneous, transdermal, airway (aerosol), nasal, rectal, and topical (including buccal and sublingual) administration. In certain embodiments, the compositions are administered by intravenous infusion or injection. In certain embodiments, the compositions are administered by subcutaneous injection.

In some embodiments, the administration is via a depot injection. A depot injection may release the iRNA in a consistent way over a prolonged time period. Thus, a depot injection may reduce the frequency of dosing needed to obtain a desired effect, e.g., a desired inhibition of CIDEB, or a therapeutic or prophylactic effect. A depot injection may also provide more consistent serum concentrations. Depot injections may include subcutaneous injections or intramuscular injections. In preferred embodiments, the depot injection is a subcutaneous injection.

In some embodiments, the administration is via a pump. The pump may be an external pump or a surgically implanted pump. In certain embodiments, the pump is a

subcutaneously implanted osmotic pump. In other embodiments, the pump is an infusion pump. An infusion pump may be used for intravenous, subcutaneous, arterial, or epidural infusions. In preferred embodiments, the infusion pump is a subcutaneous infusion pump. In other embodiments, the pump is a surgically implanted pump that delivers the iRNA to the liver.

An iRNA of the invention may be present in a pharmaceutical composition, such as in a suitable buffer solution. The buffer solution may comprise acetate, citrate, proline, carbonate, or phosphate, or any combination thereof. In one embodiment, the buffer solution is phosphate buffered saline (PBS). The pH and osmolarity of the buffer solution containing the iRNA can be adjusted such that it is suitable for administering to a subject.

Alternatively, an iRNA of the invention may be administered as a pharmaceutical composition, such as a dsRNA liposomal formulation.

The mode of administration may be chosen based upon whether local or systemic treatment is desired and based upon the area to be treated. The route and site of administration may be chosen to enhance targeting.

In one aspect, the present invention also provides methods for inhibiting the expression of a CIDEB gene in a mammal. The methods include administering to the mammal a composition comprising a dsRNA that targets a CIDEB gene in a cell of the mammal, thereby inhibiting expression of the CIDEB gene in the cell.

In some embodiments, the methods include administering to the mammal a composition comprising a dsRNA that targets a CIDEB gene in a cell of the mammal, thereby inhibiting expression of the CIDEB gene in the cell. In another embodiment, the methods include administering to the mammal a pharmaceutical composition comprising a dsRNA agent that targets a CIDEB gene in a cell of the mammal.

In another aspect, the present invention provides use of an iRNA agent or a pharmaceutical composition of the invention for inhibiting the expression of a CIDEB gene in a mammal.

In yet another aspect, the present invention provides use of an iRNA agent of the invention targeting a CIDEB gene or a pharmaceutical composition comprising such an agent in the manufacture of a medicament for inhibiting expression of a CIDEB gene in a mammal.

In another aspect, the present invention also provides methods for inhibiting the expression of a CIDEB gene and a PNPLA3 gene in a mammal. The methods include administering to the mammal a composition comprising a dsRNA that targets a CIDEB gene in a cell of the mammal and a composition comprising a dsRNA that targets a PNPLA3 gene in a cell of the mammal, thereby inhibiting expression of the CIDEB gene and the PNPLA3 gene in the cell. In one embodiment, the methods include administering to the mammal a pharmaceutical composition comprising a dsRNA agent that targets a CIDEB gene and a PNPLA3 gene in a cell of the mammal.

In one aspect, the present invention provides use of an iRNA agent or a pharmaceutical composition of the invention, and a dsRNA that targets a PNPLA3 gene or a pharmaceutical composition comprising such an agent for inhibiting the expression of a CIDEB gene and a PNPLA3 gene in a mammal.

In yet another aspect, the present invention provides use of an iRNA agent of the invention targeting a CIDEB gene or a pharmaceutical composition comprising such an agent, and a dsRNA that targets a PNPLA3 gene or a pharma-

ceutical composition comprising such an agent in the manufacture of a medicament for inhibiting expression of a CIDEB gene and a PNPLA3 gene in a mammal.

Reduction in gene expression can be assessed by any methods known in the art and by methods, e.g. qRT-PCR, described herein. Reduction in protein production can be assessed by any methods known in the art and by methods, e.g. ELISA, enzymatic activity, described herein.

The present invention also provides therapeutic and prophylactic methods which include administering to a subject having, or prone to developing a fatty liver-associated disease, disorder, or condition, the iRNA agents, pharmaceutical compositions comprising an iRNA agent, or vectors comprising an iRNA of the invention.

In one aspect, the present invention provides methods of treating a subject having a disorder that would benefit from reduction in CIDEB expression, e.g., a CIDEB-associated disease.

The treatment methods (and uses) of the invention include administering to the subject, e.g., a human, a therapeutically effective amount of a dsRNA agent that inhibits expression of CIDEB or a pharmaceutical composition comprising a dsRNA that inhibits expression of CIDEB, thereby treating the subject.

In one aspect, the invention provides methods of preventing at least one symptom in a subject having a disorder that would benefit from reduction in CIDEB expression, e.g., a chronic inflammatory disease. The methods include administering to the subject a prophylactically effective amount of dsRNA agent or a pharmaceutical composition comprising a dsRNA, thereby preventing at least one symptom in the subject.

In one embodiment, a CIDEB-associated disease, disorder, or condition is a chronic inflammatory disease. Non-limiting examples of chronic inflammatory diseases include inflammation of the liver and other tissues. Non-limiting examples of chronic inflammatory liver diseases include liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, hepatocellular carcinoma, insulin insensitivity and/or diabetes.

The present invention also provides therapeutic and prophylactic methods which include administering to a subject having, or prone to developing a fatty liver-associated disease, disorder, or condition, the iRNA agents, pharmaceutical compositions comprising an iRNA agent, or vectors comprising an iRNA of the invention and iRNA agent targeting PNPLA3, pharmaceutical compositions comprising such an iRNA agent, or vectors comprising such an iRNA.

The present invention also provides use of a therapeutically effective amount of an iRNA agent of the invention or a pharmaceutical composition comprising a dsRNA that inhibits expression of CIDEB for treating a subject, e.g., a subject that would benefit from a reduction and/or inhibition of CIDEB expression, e.g., a CIDEB-associated disease, e.g., a chronic inflammatory disease. Non-limiting examples of chronic inflammatory liver diseases include liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, hepatocellular carcinoma, insulin insensitivity and/or diabetes.

In another aspect, the present invention provides use of an iRNA agent, e.g., a dsRNA, of the invention targeting a CIDEB for gene or a pharmaceutical composition comprising an iRNA agent targeting a CIDEB for gene in the manufacture of a medicament for treating a subject, e.g., a subject that would benefit from a reduction and/or inhibition of CIDEB for expression, e.g., a CIDEB-associated disease.

The present invention also provides use of a prophylactically effective amount of an iRNA agent of the invention or a pharmaceutical composition comprising a dsRNA that inhibits expression of CIDEB for preventing at least one symptom in a subject having a disorder that would benefit from reduction in CIDEB expression, e.g., a chronic inflammatory disease. Non-limiting examples of chronic inflammatory liver diseases include liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, hepatocellular carcinoma, insulin insensitivity and/or diabetes.

In another aspect, the present invention provides use of an iRNA agent, e.g., a dsRNA, of the invention targeting a CIDEB gene or a pharmaceutical composition comprising an iRNA agent targeting a CIDEB gene in the manufacture of a medicament for preventing at least one symptom in a subject having a disorder that would benefit from reduction in CIDEB expression, e.g., a chronic inflammatory disease. Non-limiting examples of chronic inflammatory liver diseases include liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, hepatocellular carcinoma, insulin insensitivity and/or diabetes.

In one aspect, the present invention also provides use of a therapeutically effective amount of an iRNA agent of the invention or a pharmaceutical composition comprising a dsRNA that inhibits expression of CIDEB in combination with a dsRNA that targets a PNPLA3 gene or a pharmaceutical composition comprising such an agent for treating a subject, e.g., a subject that would benefit from a reduction and/or inhibition of CIDEB expression, e.g., a CIDEB-associated disease, e.g., a chronic inflammatory disease. Non-limiting examples of chronic inflammatory liver diseases include liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, hepatocellular carcinoma, insulin insensitivity and/or diabetes.

In one aspect, the present invention also provides use of an iRNA agent, e.g., a dsRNA, of the invention targeting a CIDEB gene or a pharmaceutical composition comprising an iRNA agent targeting a CIDEB gene in combination with a dsRNA that targets a PNPLA3 gene or a pharmaceutical composition comprising such an agent for preventing at least one symptom in a subject having a disorder that would benefit from reduction in CIDEB expression, e.g., a chronic inflammatory disease. Non-limiting examples of chronic inflammatory liver diseases include liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, hepatocellular carcinoma, insulin insensitivity and/or diabetes.

The combination methods of the invention for treating a subject, e.g., a human subject, having a CIDEB-associated disease, disorder, or condition, such as a chronic inflammatory disease, e.g., chronic inflammatory liver disease, e.g., NASH, are useful for treating such subjects as silencing of PNPLA3 decreases steatosis (i.e. liver fat).

Accordingly, in one aspect, the present invention provides methods of treating a subject having a disorder that would benefit from reduction in CIDEB expression, e.g., a CIDEB-associated disease, such as a chronic inflammatory disease (e.g., inflammation of the liver and other tissues). In one embodiment, the chronic inflammatory disease is chronic inflammatory liver disease (e.g., liver fibrosis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cirrhosis of the liver, alcoholic steatohepatitis (ASH), alcoholic liver diseases (ALD), HCV-associated cirrhosis, drug induced liver injury, hepatocellular necrosis, and/or hepatocellular carcinoma). In one embodiment, the chronic inflammatory liver disease is NASH.

The combination treatment methods (and uses) of the invention include administering to the subject, e.g., a human subject, a therapeutically effective amount of a dsRNA agent that inhibits expression of CIDEB or a pharmaceutical composition comprising a dsRNA that inhibits expression of CIDEB, and a dsRNA agent that inhibits expression of PNPLA3 or a pharmaceutical composition comprising a dsRNA that inhibits expression of PNPLA3, thereby treating the subject.

In one aspect, the invention provides methods of preventing at least one symptom in a subject having a disorder that would benefit from reduction in CIDEB expression, e.g., a chronic inflammatory disease. The methods include administering to the subject a prophylactically effective amount of dsRNA agent or a pharmaceutical composition comprising a dsRNA that inhibits expression of CIDEB, and a dsRNA agent that inhibits expression of PNPLA3 or a pharmaceutical composition comprising a dsRNA that inhibits expression of PNPLA3, thereby preventing at least one symptom in the subject.

In one embodiment, the subject is heterozygous for the gene encoding the patatin like phospholipase domain containing 3 (PNPLA3) I148M variation, that is, the subject may have an allele of the gene encoding a PNPLA3 I148M variant and the other allele encoding a different variant. In another embodiment, the subject is homozygous for the gene encoding the PNPLA3 I148M variation, that is, both alleles of the gene encode PNPLA3 I148M variant. In one embodiment, the subject is heterozygous for the gene encoding the patatin like phospholipase domain containing 3 (PNPLA3) I144M variation, that is, the subject may have an allele of the gene encoding a PNPLA3 I144M variant and the other allele encoding a different variant. In another embodiment, the subject is homozygous for the gene encoding the PNPLA3 I144M variation, that is, both alleles of the gene encode PNPLA3 I143M variant.

In certain embodiments of the invention the methods may include identifying a subject that would benefit from reduction in CIDEB expression. The methods generally include determining whether or not a sample from the subject comprises a nucleic acid encoding a PNPLA3Ile148Met variant or a PNPLA3Ile144Met variant. The methods may also include classifying a subject as a candidate for treating or inhibiting a liver disease by inhibiting the expression of CIDEB, by determining whether or not a sample from the subject comprises a first nucleic acid encoding a PNPLA3 protein comprising an I148M variation and a second nucleic acid encoding a functional CIDEB protein, and/or a

PNPLA3 protein comprising an I144M variation and a functional CIDEA protein, and classifying the subject as a candidate for treating or inhibiting a liver disease by inhibiting CIDEA when both the first and second nucleic acids are detected and/or when both proteins are detected.

The variant PNPLA3 Ile148Met variant or PNPLA3 Ile144Met variant can be any of the PNPLA3 Ile148Met variants and PNPLA3 Ile144Met variants described herein. The PNPLA3 Ile148Met variant or PNPLA3 Ile144Met variant can be detected by any suitable means, such as ELISA assay, RT-PCR, sequencing.

In some embodiments, the methods further comprise determining whether the subject is homozygous or heterozygous for the PNPLA3 Ile148Met variant or the PNPLA3 Ile144Met variant. In some embodiments, the subject is homozygous for the PNPLA3 Ile148Met variant or the PNPLA3 Ile144Met variant. A subject homozygous for the PNPLA3 Ile148Met variant has both alleles of the gene encoding PNPLA3 Ile148Met variant; a subject homozygous for the PNPLA3 Ile144Met variant has both alleles of the gene encoding PNPLA3 Ile144Met variant. In some embodiments, the subject is heterozygous for the PNPLA3 Ile148Met variant or the PNPLA3 Ile144Met variant. A subject heterozygous for the PNPLA3 Ile148Met variant may have an allele for the gene encoding for PNPLA3 Ile148Met variant and the other allele encoding for a different PNPLA3 variant; a subject heterozygous for the PNPLA3 Ile144Met variant may have an allele for the gene encoding for PNPLA3 Ile144Met variant and the other allele encoding for a different PNPLA3 variant. In some embodiments, the subject is homozygous for the PNPLA3 Ile148Met variant. In some embodiments, the subject is heterozygous for the PNPLA3 Ile148Met variant. In some embodiments, the subject is homozygous for the PNPLA3 Ile144Met variant. In some embodiments, the subject is heterozygous for the PNPLA3 Ile144Met variant.

In some embodiments, the methods further comprise determining whether the subject is obese. In some embodiments, a subject is obese if their body mass index (BMI) is over 30 kg/m². Obesity can be a characteristic of a subject having, or at risk of developing, a liver disease. In some embodiments, the methods further comprise determining whether the subject has a fatty liver. A fatty liver can be a characteristic of a subject having, or at risk of developing, a liver disease. In some embodiments, the methods further comprise determining whether the subject is obese and has a fatty liver.

As used herein, “nonalcoholic fatty liver disease,” used interchangeably with the term “NAFLD,” refers to a disease defined by the presence of macrovascular steatosis in the presence of less than 20 gm of alcohol ingestion per day. NAFLD is the most common liver disease in the United States, and is commonly associated with insulin resistance/type 2 diabetes mellitus and obesity. NAFLD is manifested by steatosis, steatohepatitis, cirrhosis, and sometimes hepatocellular carcinoma. For a review of NAFLD, see Tolman and Dalpiaz (2007) *Ther. Clin. Risk. Manag.*, 3(6):1153-1163 the entire contents of which are incorporated herein by reference.

As used herein, the terms “steatosis,” “hepatic steatosis,” and “fatty liver disease” refer to the accumulation of triglycerides and other fats in the liver cells.

As used herein, the term “NAFLD” refers to non-alcoholic fatty liver disease. NAFLD is the commonest form of liver disease in all regions of the world with modern industrialized economies, including Korea and many other Asian countries. Patients usually present without symptoms

or clinical features are non-specific. Instead, liver abnormalities are found incidentally by hepatic imaging, particularly ultrasonography, and/or there are raised liver enzymes (alanine aminotransferase [ALT] and gamma-glutamyltranspeptidase).

The diagnosis of NAFLD requires exclusion of other disorders, particularly viral hepatitis, significant alcohol intake, and exposure to potentially hepatotoxic medications. By agreements such as the Asia-Pacific Guidelines on NAFLD, the term NAFLD is now retained for cases of fatty liver associated with metabolic complications of over-nutrition, usually with central obesity and overweight (Farrell et al., *Gut Liver*, 6(2): 149-171, 2012).

As used herein, the term “Nonalcoholic steatohepatitis” or “NASH” refers to liver inflammation and damage caused by a buildup of fat in the liver. NASH is considered the progressive form of nonalcoholic fatty liver disease (NAFLD) and is characterized by liver steatosis, inflammation, hepatocellular injury and different degrees of fibrosis. Adipose tissue dysfunction and the hepatic inflammatory response have a fundamental role during NASH development. Cellular and molecular response mechanisms also promote liver inflammation in the absence of a fatty liver by inducing a chronic inflammatory response that results in hepatocyte damage.

NASH resembles alcoholic liver disease, but occurs in people who drink little or no alcohol. The major feature in NASH is fat in the liver, along with inflammation and damage. Most people with NASH feel well and are not aware that they have a liver problem. Nevertheless, NASH can be severe and can lead to cirrhosis, in which the liver is permanently damaged and scarred and no longer able to work properly. NASH is usually first suspected in a person who is found to have elevations in liver tests that are included in routine blood test panels, such as alanine aminotransferase (ALT) or aspartate aminotransferase (AST). When further evaluation shows no apparent reason for liver disease (such as medications, viral hepatitis, or excessive use of alcohol) and when x rays or imaging studies of the liver show fat, NASH is suspected. The only means of proving a diagnosis of NASH and separating it from simple fatty liver is a liver biopsy.

As used herein, the term “cirrhosis,” defined histologically, is a diffuse hepatic process characterized by fibrosis and conversion of the normal liver architecture into structurally abnormal nodules.

As used herein, the term “serum lipid” refers to any major lipid present in the blood. Serum lipids may be present in the blood either in free form or as a part of a protein complex, e.g., a lipoprotein complex. Non-limiting examples of serum lipids may include triglycerides (TG), cholesterol, such as total cholesterol (TC), low density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), very low density lipoprotein cholesterol (VLDL-C) and intermediate-density lipoprotein cholesterol (IDL-C).

In one embodiment, a subject that would benefit from the reduction of the expression of CIDEA (and, in some embodiments, PNPLA3) is, for example, a subject that has type 2 diabetes and prediabetes, or obesity; a subject that has high levels of fats in the blood, such as cholesterol, or has high blood pressure; a subject that has certain metabolic disorders, including metabolic syndrome; a subject that has rapid weight loss; a subject that has certain infections, such as hepatitis C infection, or a subject that has been exposed to some toxins. In one embodiment, a subject that would benefit from the reduction of the expression of CIDEA (and, in some embodiments, PNPLA3) is, for example, a subject

that is middle-aged or older; a subject that is Hispanic, non-Hispanic whites, or African Americans; a subject that takes certain drugs, such as corticosteroids and cancer drugs.

In the methods (and uses) of the invention which comprise administering to a subject a first dsRNA agent targeting CIDEb and a second dsRNA agent targeting PNPLA3, the first and second dsRNA agents may be formulated in the same composition or different compositions and may administered to the subject in the same composition or in separate compositions.

In one embodiment, an “iRNA” for use in the methods of the invention is a “dual targeting RNAi agent.” The term “dual targeting RNAi agent” refers to a molecule comprising a first dsRNA agent comprising a complex of ribonucleic acid molecules, having a duplex structure comprising two anti-parallel and substantially complementary nucleic acid strands, referred to as having “sense” and “antisense” orientations with respect to a first target RNA, i.e., a CIDEb gene, covalently attached to a molecule comprising a second dsRNA agent comprising a complex of ribonucleic acid molecules, having a duplex structure comprising two anti-parallel and substantially complementary nucleic acid strands, referred to as having “sense” and “antisense” orientations with respect to a second target RNA, i.e., a PNPLA3 gene. In some embodiments of the invention, a dual targeting RNAi agent triggers the degradation of the first and the second target RNAs, e.g., mRNAs, through a post-transcriptional gene-silencing mechanism referred to herein as RNA interference or RNAi.

The dsRNA agent may be administered to the subject at a dose of about 0.1 mg/kg to about 50 mg/kg. Typically, a suitable dose will be in the range of about 0.1 mg/kg to about 5.0 mg/kg, preferably about 0.3 mg/kg and about 3.0 mg/kg.

The iRNA can be administered by intravenous infusion over a period of time, on a regular basis. In certain embodiments, after an initial treatment regimen, the treatments can be administered on a less frequent basis.

Administration of the iRNA can reduce CIDEb levels, e.g., in a cell, tissue, blood, urine or other compartment of the patient by at least about 5%, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 39, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or at least about 99% or more. In a preferred embodiment, administration of the iRNA can reduce CIDEb levels, e.g., in a cell, tissue, blood, urine or other compartment of the patient by at least 20%.

Administration of the iRNA can reduce PNPLA3 levels, e.g., in a cell, tissue, blood, urine or other compartment of the patient by at least about 5%, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 39, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or at least about 99% or more. In a preferred embodiment, administration of the iRNA can reduce PNPLA3 levels, e.g., in a cell, tissue, blood, urine or other compartment of the patient by at least 20%.

Before administration of a full dose of the iRNA, patients can be administered a smaller dose, such as a 5% infusion reaction, and monitored for adverse effects, such as an allergic reaction. In another example, the patient can be monitored for unwanted immunostimulatory effects, such as increased cytokine (e.g., TNF-alpha or INF-alpha) levels.

Alternatively, the iRNA can be administered subcutaneously, i.e., by subcutaneous injection. One or more injections may be used to deliver the desired daily dose of iRNA to a subject. The injections may be repeated over a period of time. The administration may be repeated on a regular basis. In certain embodiments, after an initial treatment regimen, the treatments can be administered on a less frequent basis. A repeat-dose regimen may include administration of a therapeutic amount of iRNA on a regular basis, such as every other day or to once a year. In certain embodiments, the iRNA is administered about once per week, once every 7-10 days, once every 2 weeks, once every 3 weeks, once every 4 weeks, once every 5 weeks, once every 6 weeks, once every 7 weeks, once every 8 weeks, once every 9 weeks, once every 10 weeks, once every 11 weeks, once every 12 weeks, once per month, once every 2 months, once every 3 months, once per quarter, once every 4 months, once every 5 months, or once every 6 months.

In one embodiment, the method includes administering a composition featured herein such that expression of the target CIDEb gene is decreased, such as for about 1, 2, 3, 4, 5, 6, 7, 8, 12, 16, 18, 24 hours, 28, 32, or about 36 hours. In one embodiment, expression of the target CIDEb gene is decreased for an extended duration, e.g., at least about two, three, four days or more, e.g., about one week, two weeks, three weeks, or four weeks or longer.

In another embodiment, the method includes administering a composition featured herein such that expression of the target PNPLA3 gene is decreased, such as for about 1, 2, 3, 4, 5, 6, 7, 8, 12, 16, 18, 24 hours, 28, 32, or about 36 hours. In one embodiment, expression of the target PNPLA3 gene is decreased for an extended duration, e.g., at least about two, three, four days or more, e.g., about one week, two weeks, three weeks, or four weeks or longer.

Preferably, the iRNAs useful for the methods and compositions featured herein specifically target RNAs (primary or processed) of the target CIDEb gene (and, in some embodiments, a PNPLA3 gene). Compositions and methods for inhibiting the expression of these genes using iRNAs can be prepared and performed as described herein.

Administration of the dsRNA according to the methods of the invention may result in a reduction of the severity, signs, symptoms, and/or markers of such diseases or disorders in a patient with a disorder of lipid metabolism. By “reduction” in this context is meant a statistically significant decrease in such level. The reduction can be, for example, at least about 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or about 100%.

Efficacy of treatment or prevention of disease can be assessed, for example by measuring disease progression, disease remission, symptom severity, reduction in pain, quality of life, dose of a medication required to sustain a treatment effect, level of a disease marker or any other measurable parameter appropriate for a given disease being treated or targeted for prevention. It is well within the ability of one skilled in the art to monitor efficacy of treatment or prevention by measuring any one of such parameters, or any combination of parameters. For example, efficacy of treatment of a disorder of lipid metabolism may be assessed, for example, by periodic monitoring of one or more serum lipid levels, e.g., triglyceride levels. Comparisons of the later readings with the initial readings provide a physician an indication of whether the treatment is effective. It is well within the ability of one skilled in the art to monitor efficacy of treatment or prevention by measuring any one of such parameters, or any combination of parameters. In connection

with the administration of an iRNA or pharmaceutical composition thereof, "effective against" a disorder of lipid metabolism indicates that administration in a clinically appropriate manner results in a beneficial effect for at least a statistically significant fraction of patients, such as an improvement of symptoms, a cure, a reduction in disease, extension of life, improvement in quality of life, or other effect generally recognized as positive by medical doctors familiar with treating disorder of lipid metabolisms and the related causes.

A treatment or preventive effect is evident when there is a statistically significant improvement in one or more parameters of disease status, or by a failure to worsen or to develop symptoms where they would otherwise be anticipated. As an example, a favorable change of at least 10% in a measurable parameter of disease, and preferably at least 20%, 30%, 40%, 50% or more can be indicative of effective treatment. Efficacy for a given iRNA drug or formulation of that drug can also be judged using an experimental animal model for the given disease as known in the art.

The invention further provides methods for the use of a iRNA agent or a pharmaceutical composition of the invention, e.g., for treating a subject that would benefit from reduction and/or inhibition of CIDEB expression or CIDEB, e.g., a subject having a CIDEB-associated disease disorder, or condition, in combination with other pharmaceuticals and/or other therapeutic methods, e.g., with known pharmaceuticals and/or known therapeutic methods, such as, for example, those which are currently employed for treating these disorders. In some embodiments, the invention provides methods for the use of a iRNA agent or a pharmaceutical composition of the invention and an iRNA agent targeting PNPLA3, e.g., for treating a subject that would benefit from reduction and/or inhibition of CIDEB expression and PNPLA3 expression, e.g., a subject having a CIDEB-associated disease disorder, or condition (e.g., chronic inflammatory disease), in combination with other pharmaceuticals and/or other therapeutic methods, e.g., with known pharmaceuticals and/or known therapeutic methods, such as, for example, those which are currently employed for treating these disorders. For example, in certain embodiments, an iRNA agent or pharmaceutical composition of the invention is administered in combination with, e.g., pyridoxine, an ACE inhibitor (angiotensin converting enzyme inhibitors), e.g., benazepril agents to decrease blood pressure, e.g., diuretics, beta-blockers, ACE inhibitors, angiotensin II receptor blockers, calcium channel blockers, alpha blockers, alpha-2 receptor antagonists, combined alpha- and beta-blockers, central agonists, peripheral adrenergic inhibitors, and blood vessel dilators; or agents to decrease cholesterol, e.g., statins, selective cholesterol absorption inhibitors, resins; lipid lowering therapies; insulin sensitizers, such as the PPAR γ agonist pioglitazone; glp-1r agonists, such as liraglutide; vitamin E; SGLT2 inhibitors; or DPPIV inhibitors; or a combination of any of the foregoing. In one embodiment, an iRNA agent or pharmaceutical

composition of the invention is administered in combination with an agent that inhibits the expression and/or activity of a transmembrane 6 superfamily member 2 (TM6SF2) gene, e.g., an RNAi agent that inhibits the expression of a TM6SF2 gene.

The iRNA agent and an additional therapeutic agent and/or treatment may be administered at the same time and/or in the same combination, e.g., subcutaneously, or the additional therapeutic agent can be administered as part of a separate composition or at separate times and/or by another method known in the art or described herein.

VIII. Kits

The present invention also provides kits for performing any of the methods of the invention. Such kits include one or more RNAi agent(s) and instructions for use, e.g., instructions for inhibiting expression of a CIDEB in a cell by contacting the cell with an RNAi agent or pharmaceutical composition of the invention in an amount effective to inhibit expression of the CIDEB. The kits may optionally further comprise means for contacting the cell with the RNAi agent (e.g., an injection device), or means for measuring the inhibition of CIDEB (e.g., means for measuring the inhibition of CIDEB mRNA and/or CIDEB protein). Such means for measuring the inhibition of CIDEB may comprise a means for obtaining a sample from a subject, such as, e.g., a plasma sample. The kits of the invention may optionally further comprise means for administering the RNAi agent(s) to a subject or means for determining the therapeutically effective or prophylactically effective amount.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the iRNAs and methods featured in the invention, suitable methods and materials are described below. All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

EXAMPLES

Example 1. CIDEB iRNA Design, Synthesis, and Selection

Nucleic acid sequences provided herein are represented using standard nomenclature. See the abbreviations of Table 2. It will be understood that these monomers, when present in an oligonucleotide, are mutually linked by 5'-3'-phosphodiester bonds. The abbreviations are understood to omit the 3'-phosphate (i.e. they are 3'-OH) when placed at the 3'-terminal position of an oligonucleotide.

TABLE 2

Abbreviations of nucleotide monomers used in nucleic acid sequence representation.	
Abbreviation	Nucleotide(s)
A	Adenosine-3'-phosphate
Ab	beta-L-adenosine-3'-phosphate
Abs	beta-L-adenosine-3'-phosphorothioate
Af	2'-fluoroadenosine-3'-phosphate

TABLE 2-continued

Abbreviations of nucleotide monomers used in nucleic acid sequence representation.	
Abbreviation	Nucleotide(s)
Afs	2'-fluoroadenosine-3'-phosphorothioate
As	adenosine-3'-phosphorothioate
(A2p)	adenosine-2'-phosphate
(A2ps)	adenosine-2'-phosphorothioate
C	cytidine-3'-phosphate
Cb	beta-L-cytidine-3'-phosphate
Cbs	beta-L-cytidine-3'-phosphorothioate
Cf	2'-fluorocytidine-3'-phosphate
Cfs	2'-fluorocytidine-3'-phosphorothioate
Cs	cytidine-3'-phosphorothioate
(C2p)	cytidine-2'-phosphate
(C2ps)	cytidine-2'-phosphorothioate
G	guanosine-3'-phosphate
Gb	beta-L-guanosine-3'-phosphate
Gbs	beta-L-guanosine-3'-phosphorothioate
Gf	2'-fluoroguanosine-3'-phosphate
Gfs	2'-fluoroguanosine-3'-phosphorothioate
Gs	guanosine-3'-phosphorothioate
(G2p)	guanosine-2'-phosphate
(G2ps)	guanosine-2'-phosphorothioate
T	5'-methyluridine-3'-phosphate
Tf	2'-fluoro-5-methyluridine-3'-phosphate
Tfs	2'-fluoro-5-methyluridine-3'-phosphorothioate
Ts	5'-methyluridine-3'-phosphorothioate
U	Uridine-3'-phosphate
Uf	2'-fluorouridine-3'-phosphate
Ufs	2'-fluorouridine-3'-phosphorothioate
Us	uridine-3'-phosphorothioate
(U2p)	uridine-2'-phosphate
(U2ps)	uridine-2'-phosphorothioate
N	any nucleotide (G, A, C, T or U)
a	2'-O—methyladenosine-3'-phosphate
as	2'-O—methyladenosine-3'-phosphorothioate
c	2'-O—methylcytidine-3'-phosphate
cs	2'-O—methylcytidine-3'-phosphorothioate
g	2'-O—methylguanosine-3'-phosphate
gs	2'-O—methylguanosine-3'-phosphorothioate
t	2'-O—methyl-5-methyluridine-3'-phosphate
ts	2'-O—methyl-5-methyluridine-3'-phosphorothioate
u	2'-O—methyluridine-3'-phosphate
us	2'-O—methyluridine-3'-phosphorothioate
s	phosphorothioate linkage
L96 ¹	N-[tris(GalNAc-alkyl)-amidodecanoyl]-4-hydroxyprolinol; or (2S,4R)-1-[29-[[2-(acetylamino)-2-deoxy-β-D-galactopyranosyl]oxy]-14,14-bis[[3-[[[3-[[5-[[2-(acetylamino)-2-deoxy-β-D-galactopyranosyl]oxy]-1-oxopentyl]amino]propyl]amin 3-oxopropoxy]methyl]-1,12,19,25-tetraoxo-16-oxa-13,20,24-triazanonecos-1-yl]-4-hydroxy-2-hydroxymethylpyrrolidine
uL96 ²	2'-O—methyluridine-3'-phosphate ((2S,4R)-1-[29-[[2-(acetylamino)-2-deoxy-β-D-galactopyranosyl]oxy]-14,14-bis[[3-[[[3-[[5-[[2-(acetylamino)-2-deoxy-β-D-galactopyranosyl]oxy]-1-oxopentyl]amino]propyl]amin 3-oxopropoxy]methyl]-1,12,19,25-tetraoxo-16-oxa-13,20,24-triazanonecos-1-yl]-4-hydroxy-2-pyrrolidinyl)methyl ester
P	Phosphate
VP	Vinyl-phosphate
dA	2'-deoxyadenosine-3'-phosphate
dAs	2'-deoxyadenosine-3'-phosphorothioate
dC	2'-deoxycytidine-3'-phosphate
dCs	2'-deoxycytidine-3'-phosphorothioate
dG	2'-deoxyguanosine-3'-phosphate
dGs	2'-deoxyguanosine-3'-phosphorothioate
dT	2'-deoxythymidine-3'-phosphate
dTs	2'-deoxythymidine-3'-phosphorothioate
dU	2'-deoxyuridine
dUs	2'-deoxyuridine-3'-phosphorothioate
Y34	2-hydroxymethyl-tetrahydrofuran-4-methoxy-3-phosphate (abasic 2'-OMe furanose)
Y44	inverted abasic DNA (2-hydroxymethyl-tetrahydrofuran-5-phosphate)
(Agn)	Adenosine-glycol nucleic acid (GNA)
(Cgn)	Cytidine-glycol nucleic acid (GNA)
(Ggn)	Guanosine-glycol nucleic acid (GNA)
(Tgn)	Thymidine-glycol nucleic acid (GNA) S-Isomer
(Aam)	2'-O—(N-methylacetamide)adenosine-3'-phosphate
(Aams)	2'-O—(N-methylacetamide)adenosine-3'-phosphorothioate
(Gam)	2'-O—(N-methylacetamide)guanosine-3'-phosphate
(Gams)	2'-O—(N-methylacetamide)guanosine-3'-phosphorothioate
(Tam)	2'-O—(N-methylacetamide)thymidine-3'-phosphate
(Tams)	2'-O—(N-methylacetamide)thymidine-3'-phosphorothioate
(Aeo)	2'-O—methoxyethyladenosine-3'-phosphate
(Aeos)	2'-O—methoxyethyladenosine-3'-phosphorothioate

Briefly, siRNA sequences were synthesized at 1 mol scale on a Mermade 192 synthesizer (BioAutomation) using the solid support mediated phosphoramidite chemistry. The solid support was controlled pore glass (500 Å) loaded with custom GalNAc ligand or universal solid support (AM 5 10 biochemical). Ancillary synthesis reagents, 2'-F and 2'-O-Methyl RNA and deoxy phosphoramidites were obtained from Thermo-Fisher (Milwaukee, WI) and Hongene (China). 2'-F 2'-O-Methyl, GNA (glycol nucleic acids), 5'phosphate and other modifications were introduced using the corresponding phosphoramidites. Synthesis of 3' GalNAc conjugated single strands was performed on a GalNAc modified CPG support. Custom CPG universal solid support was used for the synthesis of antisense single strands. Coupling time for all phosphoramidites (100 mM in acetonitrile) was 5 min employing 5-Ethylthio-1H-tetrazole (ETT) as activator (0.6 M in acetonitrile). Phosphorothioate linkages were generated using a 50 mM solution of 3-((Dimethylamino-methylidene)amino)-3H-1,2,4-dithiazole-3-thione (DDTT, obtained from Chemgenes (Wilmington, MA, USA)) in anhydrous acetonitrile/pyridine (1:1 v/v). Oxidation time was 3 minutes. All sequences were synthesized with final removal of the DMT group ("DMT off").

Upon completion of the solid phase synthesis, oligoribonucleotides were cleaved from the solid support and deprotected in sealed 96 deep well plates using 200 µL Aqueous Methylamine reagents at 60° C. for 20 minutes. For sequences containing 2' ribo residues (2'-OH) that are protected with a tert-butyl dimethyl silyl (TBDMS) group, a second step deprotection was performed using TEA.3HF (triethylamine trihydro fluoride) reagent. To the methylamine deprotection solution, 200 µL of dimethyl sulfoxide (DMSO) and 300 µL TEA.3HF reagent was added and the solution was incubated for additional 20 min at 60° C. At the end of cleavage and deprotection step, the synthesis plate 35 was allowed to come to room temperature and was precipi-

tated by addition of 1 mL of acetone: ethanol mixture (9:1). The plates were cooled at -80° C. for 2 hrs, supernatant decanted carefully with the aid of a multi-channel pipette. The oligonucleotide pellet was re-suspended in 20 mM NaOAc buffer and were desalted using a 5 mL HiTrap size exclusion column (GE Healthcare) on an AKTA Purifier System equipped with an A905 autosampler and a Frac 950 fraction collector. Desalted samples were collected in 96-well plates. Samples from each sequence were analyzed by LC-MS to confirm the identity, UV (260 nm) for quantification and a selected set of samples by IEX chromatography to determine purity.

Annealing of single strands was performed on a Tecan liquid handling robot. Equimolar mixture of sense and antisense single strands were combined and annealed in 96 well plates. After combining the complementary single strands, the 96-well plate was sealed tightly and heated in an oven at 100° C. for 10 minutes and allowed to come slowly to room temperature over a period 2-3 hours. The concentration of each duplex was normalized to 10 M in 1xPBS and then submitted for in vitro screening assays.

In some instances, a duplex (dsRNA) was synthesized more than once. Different batches are labeled with different extensions. For example, AD-1686813.1 and AD-1686813.2 are different batches of the same duplex. Duplexes having the same ID but without an extension, or with different extensions, have the same nucleotide sequences of the sense strand and antisense strand, e.g., AD-1686813, AD-1686813.1, and AD-1686813.2 have the same nucleotide sequences of the sense strand and antisense strand.

A detailed list of the unmodified nucleotide sequences of the sense strand and antisense strand sequences is shown in Tables 3 and 5.

A detailed list of the modified nucleotide sequences of the sense strand and antisense strand sequences is shown in Tables 4 and 6.

TABLE 3

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1
AD-1685156	CUGCAGAAGGU UGACUGCGUU	7	31-51	AACGCAGUCAAC CUUCUGCAGGC	142	29-51
AD-1685173	AGCCGAAGGCA AGCACGAUGU	8	69-89	ACAUCGUGCUUG CCUUCGGCUUG	143	67-89
AD-1685198	CCAGCGUCACG CUGUAGCAGU	9	156-176	ACUGCUACAGCG UGACGCUUGCA	144	154-176
AD-1685207	CGCUGUAGCAG CCGAGCAUCU	10	165-185	AGAUGCUCGGCU GCUACAGCGUG	145	163-185
AD-1685217	GCCGAGCAUCA GCCCGAAAGU	11	175-195	ACUUUCGGGCU AUGCUCGGCUG	146	173-195
AD-1685228	GCCCGAAAGGA AGCACGAAAU	12	186-206	AUUUCGUGCUUC CUUUCGGGCU	147	184-206
AD-1685240	GCACGAAAGCG GUCAGAGUCU	13	198-218	AGACUCUGACCG CUUUCGUGCUU	148	196-218
AD-1685250	GGUCAGAGUCU CCAGGCUCAU	14	208-228	AUGAGCCUGGAG ACUCUGACCGC	149	206-228
AD-1685271	GCACAGCUGGC AUACGCGGUU	15	259-279	AACCGCGUAUGC CAGCUGUGCCA	150	257-279

TABLE 3-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1
AD-1685284	ACGCGGUCCCU CCACAGGUGU	16	272-292	ACACCUGUGGAG GGACCGCGUAU	151	270-292
AD-1685366	GCUGGCGUACA UGCUGAGCGU	17	448-468	ACGCUCAGCAUG UACGCCAGCGU	152	446-468
AD-1685379	CUGAGCGCGCA CACGUAGUAU	18	461-481	AUACUACGUGUG CGCGCUCAGCA	153	459-481
AD-1685388	CACACGUAGUA CACCGCCUUU	19	470-490	AAAGGCGGUGUA CUACGUGUGCG	154	468-490
AD-1685406	CCGGGUCAGGA AGGCCACAAU	20	520-540	AUUGUGGCCUUC CUGACCCGGCA	155	518-540
AD-1685418	GGCCACAAAGA GCGGCGUGAU	21	532-552	AUCACGCCGCUC UUUGUGCCUU	156	530-552
AD-1685469	CCACACCACGA AGCCGUUGCU	22	643-663	AGCAACGGCUUC GUGGUGUGGAG	157	641-663
AD-1685491	AGCAGGAAGGC UGUGCCUGUU	23	689-709	AACAGGCACAGC CUUCCUGCUGC	158	687-709
AD-1685503	CCUGUGGCCCG CGAAGUCUUU	24	704-724	AAAGACUUCGCG GGCCACAGGCA	159	702-724
AD-1685512	CGCGAAGUCUU CCAGCUCAGU	25	713-733	ACUGAGCUGGAA GACUUCGCGGG	160	711-733
AD-1685528	UCAGCAGUGUC UCGUUCCCUU	26	729-749	AAGGGAACGAGA CACUGCUGAGC	161	727-749
AD-1685538	GCAGACCGACA UCCUUCUGGU	27	760-780	ACCAGAAGGAUG UCGGUCUGCUA	162	758-780
AD-1685549	UCCUUCUGGGC CUACAGCCUU	28	771-791	AAGGCUGUAGGC CCAGAAGGAUG	163	769-791
AD-1685561	UACAGCCUGCC UCUUUUCUGU	29	783-803	ACAGAAAAGAGG CAGGCUGUAGG	164	781-803
AD-1685570	CCUCUUUUCUG CCUGGGAGUU	30	792-812	AACUCCCAGGCA GAAAAGAGGCA	165	790-812
AD-1685587	AGUCCUGACUU CCACGAGGAU	31	809-829	AUCCUCGUGGAA GUCAGGACUCC	166	807-829
AD-1685599	CUCAAACACAA CUCCUUCUUU	32	841-861	AAAGAAGGAGUU GUGUUUGAGGU	167	839-861
AD-1685610	CUCCUUCUUGG AACCAGAUU	33	852-872	AAUCUGGGUUC AAGAAGGAGUU	168	850-872
AD-1685614	UGCUCACAGUC AGUUGACCUU	34	876-896	AAGGUCAACUGA CUGGGAGCAGG	169	874-896
AD-1685627	CCUCCUUCCCA GAGCUCAGUU	35	909-929	AACUGAGCUCUG GGAAGGAGGCC	170	907-929
AD-1685637	AGAGCUCAGUG GACACAGAAU	36	919-939	AUUCUGUGUCCA CUGAGCUCUGG	171	917-939
AD-1685648	ACCCUACAAGG AUCCUUGGCU	37	960-980	AGCCAAGGAUCC UUGUAGGGUCC	172	958-980
AD-1685664	UGGCAGGAAAG CAGGGAUUGU	38	976-996	ACAAUCCUCGCU UUCCUGCCAAG	173	974-996
AD-1685673	AGCAGGGAUUG UGUUCAUUUU	39	985-1005	AAAAUGAACACA AUCCUGCUUUU	174	983-1005
AD-1685689	AUUUGAGGGUU UCACUGUCAU	40	1001-1021	AUGACAGUGAAA CCCUCAAAGA	175	999-1021

TABLE 3-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1
AD-1685699	UUCACUGUCAG UGAGAGUCUU	41	1011-1031	AAGACUCUCACU GACAGUGAAAC	176	1009-1031
AD-1685708	AGUGAGAGUCU CAGCUUCCAU	42	1020-1040	AUGGAAGCUGAG ACUCUCACUGA	177	1018-1040
AD-1685717	CUCAGCUUCCA UGCAACUGUU	43	1029-1049	AACAGUUGCAUG GAAGCUGAGAC	178	1027-1049
AD-1685726	CAUGCAACUGU CCAUCACGGU	44	1038-1058	ACCGUGAUGGAC AGUUGCAUGGA	179	1036-1058
AD-1685735	GUCCAUCACGG CUGCAACUGU	45	1047-1067	ACAGUUGCAGCC GUGAUGGACAG	180	1045-1067
AD-1685744	GGCUGCAACUG AAAUCAAGAU	46	1056-1076	ACUCUGAUUUCA GUUGCAGCCGU	181	1054-1076
AD-1685770	CACAGCGCACC AGAAGCUAAU	47	1082-1102	AUUAGCUUCUGG UGCUCUGUGUC	182	1080-1102
AD-1685779	CCAGAAGCUAA AGUCUUGAUU	48	1091-1111	AAUCAAGACUUU AGCUUCUGGUG	183	1089-1111
AD-1685788	AAAGUCUUGAU GCCAUCAAAU	49	1100-1120	AUUUGAUGGCAU CAAGACUUUAG	184	1098-1120
AD-1685798	UGCAUCAAAAG GACAUCCCUU	50	1110-1130	AAGGGAUGUCCU UUGAUGGCAUC	185	1108-1130
AD-1685810	UCUCUGUCACG UCCACUAAUU	51	1142-1162	AAUUGAUGGACG UGACAGAGAUG	186	1140-1162
AD-1685826	UAAUCGGCAAA AGGAGAAAUA	52	1158-1178	AUUUUCUCCUUU UGCCGAUUAGU	187	1156-1178
AD-1685838	AAGUGAGAGAA GAUGACCUAU	53	1176-1196	AUAGGUCAUCUU CUCUCACUUUU	188	1174-1196
AD-1685847	AAGAUGACCUA AGUGUGACUU	54	1185-1205	AAGUCACACUUA GGUCAUCUUCU	189	1183-1205
AD-1685856	UAAGUGUGACU GCAGCAGGCU	55	1194-1214	AGCCUGCUGCAG UCACACUUAGG	190	1192-1214
AD-1685869	AGCAGGCAGCU CUGGAAAAUU	56	1207-1227	AAUUUUCAGAG CUGCCUGCUGC	191	1205-1227
AD-1685878	CUCUGGAAAAU GAAGCCAGAU	57	1216-1236	AUCUGGCUUCAU UUUCCAGAGCU	192	1214-1236
AD-1685889	GAAGCCAGAGC AGUGAGCCAU	58	1227-1247	AUGGCUCACUGC UCUGGCUUCAU	193	1225-1247
AD-1685898	CCGACCAAGGA GGAAGGAAAUA	59	1256-1276	AUUUCCUCCUCC CUUGGUCGGAG	194	1254-1276
AD-1685902	GAGGAAGGAAA GAGCAGAUUCU	60	1265-1285	AGAUCUGCUCUU UCCUCCUCCU	195	1263-1285
AD-1685912	AGAGCAGAUCC CAGGUUUGUU	61	1275-1295	AACAAACCUGGG AUCUGCUCUUU	196	1273-1295
AD-1685921	CCCAGGUUUGU AACAGAAAUA	62	1284-1304	AUUUUCUGUUAC AAACCUGGGAU	197	1282-1304
AD-1685931	UAACAGAAAAC ACCACUAAAUA	63	1294-1314	AUUUAGUGGUGU UUUCUGUUACA	198	1292-1314

TABLE 3-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1
AD-1685935	CAGCACAGGAG AGAACCACCU	64	1318-1338	AGGUGGUUCUCU CCUGUGCUGGG	199	1316-1338
AD-1685956	AGCCCAGAAGU UCCAGGGAU	65	1339-1359	AUUCCUGGAAC UUCUGGCUGG	200	1337-1359
AD-1685969	CAGGGAAGGAA CUCUCCGGUU	66	1352-1372	AACCGGAGAGUU CCUCCCCUGGA	201	1350-1372
AD-1685983	UCCGGUCCACC AUGGAGUACU	67	1366-1386	AGUACUCCAUGG UGGACCGGAGA	202	1364-1386
AD-1685994	AUGGAGUACCU CUCAGCUCUU	68	1377-1397	AAGAGCUGAGAG GUACUCCAUGG	203	1375-1397
AD-1686007	UUACUCAGGUC AGUAUCUAAU	69	1410-1430	AUUAGAUACUGA CCUGAGUAAGU	204	1408-1430
AD-1686016	UCAGUAUCUAA UAUAAGCUCU	70	1419-1439	AGAGCUUAUAUU AGAUACUGACC	205	1417-1439
AD-1686028	AUAAGCUCGGA GUUUGGACGU	71	1431-1451	ACGUCCAAACUC CGAGCUUAUAU	206	1429-1451
AD-1686053	UCUGGACCUC GCUCCACCAU	72	1456-1476	AUGGUGGAGCUG AGGUCCAGACC	207	1454-1476
AD-1686057	AGCGACCUUUC CGUGUCUGUU	73	1480-1500	AACAGACACGGA AAGUUCGCUGG	208	1478-1500
AD-1686066	UCCGUGUCUGU GAUCACAAGU	74	1489-1509	ACUUGUGAUCAC AGACACGGA	209	1487-1509
AD-1686075	GUGAUCACAAG CGGACCAUCU	75	1498-1518	AGAUGGUCCGCU UGUGAUCACAG	210	1496-1518
AD-1686086	CGGACCAUCCG GAAAGGCCUU	76	1509-1529	AAGGCCUUUCCG GAUGUCCGCU	211	1507-1529
AD-1686098	AAAGGCCUGAC AGCUGCCACU	77	1521-1541	AGUGGCAGCUGU CAGGCCUUUCC	212	1519-1541
AD-1686116	CCAGGAGCUGC UAGCCAAAGU	78	1544-1564	ACUUUGGCUAGC AGCUCUUGGCG	213	1542-1564
AD-1686126	CUAGCCAAAGC AUUGGAGACU	79	1554-1574	AGUCUCCAAUGC UUUGGCUAGCA	214	1552-1574
AD-1686136	CAUUGGAGACC CUACUGCUGU	80	1564-1584	ACAGCAGUAGGG UCUCCAAUGCU	215	1562-1584
AD-1686147	CUACUGCUGAA UGGAGUGC UU	81	1575-1595	AAGCACUCCA UU CAGCAGUAGGG	216	1573-1595
AD-1686159	GGAGUGCUAAC CCUGGUGC UU	82	1587-1607	AAGCACCAGGGU UAGCACUCCAU	217	1585-1607
AD-1686171	CUGGUGCUGA GGAGGAUGGU	83	1599-1619	ACCAUCCUCCUC UAGCACCAGGG	218	1597-1619
AD-1686181	AGGAGGAUGGA ACUGCAGUGU	84	1609-1629	ACACUGCAGUUC CAUCCUCCUCU	219	1607-1629
AD-1686197	AGUGGACAGUG AGGACUUCUU	85	1625-1645	AAGAAGUCCUCA CUGUCCACUGC	220	1623-1645
AD-1686207	GAGGACUUCUU CCAGCUGC UU	86	1635-1655	AAGCAGCUGGAA GAAGUCCUCAC	221	1633-1655
AD-1686216	UUCCAGCUGCU GGAGGAUGAU	87	1644-1664	AUCAUCCUCCAG CAGCUGGAAGA	222	1642-1664
AD-1686225	CUGGAGGAUGA CACGUGCCUU	88	1653-1673	AAGGCACGUGUC AUCCUCCAGCA	223	1651-1673

TABLE 3-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1
AD-1686242	CCUGAUGGUGU UGCAGUCUGU	89	1670-1690	ACAGACUGCAAC ACCAUCAGGCA	224	1668-1690
AD-1686251	GUUGCAGUCUG GUCAGAGCUU	90	1679-1699	AAGCUCUGACCA GACUGCAACAC	225	1677-1699
AD-1686265	AGAGCUGGAGC CCUACAAGGU	91	1693-1713	ACCUUGUAGGGC UCCAGCUCUGA	226	1691-1713
AD-1686274	GCCCUACAAGG AGUGGAGUGU	92	1702-1722	ACACUCCACUCC UUGUAGGGCUC	227	1700-1722
AD-1686286	GUGGAGUGCUG UCAUAUGGCU	93	1714-1734	AGCCAUUAUGACA GCACUCCACUC	228	1712-1734
AD-1686319	CAAGCACAGCA AGGACAUCGU	94	1751-1771	ACGAUGUCCUUG CUGUGCUUGGG	229	1749-1771
AD-1686334	CAUCGCCCGAU UCACCUUUGU	95	1766-1786	ACAAAGGUGAAU CGGGCGAUGUC	230	1764-1786
AD-1686343	AUUCACCUUUG ACGUGUACAU	96	1775-1795	AUGUACACGUCA AAGGUGAAUCG	231	1773-1795
AD-1686354	ACGUGUACAAG CAAAACCCUU	97	1786-1806	AAGGGUUUUGCU UGUACACGUCA	232	1784-1806
AD-1686366	AAAACCCUCGA GACCUCUUUU	98	1798-1818	AAAAGAGGUCUC GAGGGUUUUGC	233	1796-1818
AD-1686377	GACCUCUUUGG CAGCCUGAAU	99	1809-1829	AUUCAGGCUGCC AAAGAGGUCUC	234	1807-1829
AD-1686394	GAAUGUCAAAG CCACAUUCUU	100	1826-1846	AAGAAUGUGGCU UUGACAUUCAG	235	1824-1846
AD-1686403	AGCCACAUUCU ACGGGCUCUU	101	1835-1855	AAGAGCCCGUAG AAUGUGGCUUU	236	1833-1855
AD-1686417	GGCUCUACUCU AUGAGUUGUU	102	1849-1869	AACAACUCAUAG AGUAGAGCCCG	237	1847-1869
AD-1686426	CUAUGAGUUGU GACUUUCAAU	103	1858-1878	AUUGAAAGUCAC AACUCAUAGAG	238	1856-1878
AD-1686435	GUGACUUUCAA GGACUUGGCU	104	1867-1887	AGCCAAGUCCUU GAAAGUCACAA	239	1865-1887
AD-1686451	UGGCCCAAAGA AAGUACUCAU	105	1883-1903	AUGAGUACUUUC UUUGGGCCAAG	240	1881-1903
AD-1686466	ACUCAGGGAGC UCCUUCGUUU	106	1898-1918	AAACGAAGGAGC UCCUGAGUAC	241	1896-1918
AD-1686475	GCUCCUUCGUU GGACCUCCAU	107	1907-1927	AUGGAGGUCCAA CGAAGGAGCUC	242	1905-1927
AD-1686491	UCCACACUGCU GCAAGGCCUU	108	1923-1943	AAGGCCUUGCAG CAGUGUGGAGG	243	1921-1943
AD-1686508	CCUGGGCCAUA UGUUGCUGGU	109	1940-1960	ACCAGCAACAUA UGGCCCAGGCC	244	1938-1960
AD-1686518	AUGUUGCUGGG AAUUUCUCUCU	110	1950-1970	AGAGGAAAUUC CAGCAACAUAU	245	1948-1970
AD-1686531	UUUCCUCCACC CUUCGUCAUU	111	1963-1983	AAUGACGAAGGG UGGAGGAAAUU	246	1961-1983
AD-1686540	CCCUUCGUCAU GCAGUGGAGU	112	1972-1992	ACUCCACUGCAU GACGAAGGGUG	247	1970-1992

TABLE 3-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1
AD-1686543	CUGAGCAGUGG CAGCAGAAGU	113	1996-2016	ACUUCUGCUGCC ACUGCUCAGCC	248	1994-2016
AD-1686556	CAGAAGGGCCG CCUCCAUUCU	114	2010-2030	AGAAUGGAGGCG GCCCUCUGCU	249	2008-2030
AD-1686565	CGCCUCCAUUC CUACUAAGGU	115	2019-2039	ACCUUAGUAGGA AUGGAGGCGGC	250	2017-2039
AD-1686566	CAGAAUCAUUC CAACCGACCU	116	2058-2078	AGGUCGGUUGGA AUGAUUCUGGG	251	2056-2078
AD-1686576	CCAACCGACCC ACUGCAAAGU	117	2068-2088	ACUUUGCAGUGG GUCGGUUGGAA	252	2066-2088
AD-1686585	CCACUGCAAAG ACUAUGACAU	118	2077-2097	AUGUCAUAGUCU UUGCAGUGGGU	253	2075-2097
AD-1686597	CUAUGACAGCA UCAAAUUCU	119	2089-2109	AGAAAUUUGAUG CUGUCAUAGUC	254	2087-2109
AD-1686606	CAUAAAUUUC AGGACCUGCU	120	2098-2118	AGCAGGUCCUGA AAUUUGAUGCU	255	2096-2118
AD-1686616	CAGGACCUGCA GACAGUACAU	121	2108-2128	AUGUACUGUCUG CAGGUCCUGAA	256	2106-2128
AD-1686626	AGACAGUACAG GCUAGAUAU	122	2118-2138	AUUUUCUAGCCU GUACUGUCUGC	257	2116-2138
AD-1686639	UAGAUAAACCA CCCAAUUCU	123	2131-2151	AGAAAUUGGGUG GGUUAUCUAGC	258	2129-2151
AD-1686649	AGAACCUUUA GCAUAACGCU	124	2178-2198	AGCGUUAUGCUG AAAGGUUCUGU	259	2176-2198
AD-1686658	CAGCAUAACGC CUCACAUCCU	125	2187-2207	AGGAUGUGAGGC GUUAUGCUGAA	260	2185-2207
AD-1686669	CUCACAUCCCA AGUCUAUACU	126	2198-2218	AGUAUAGACUUG GGAUGUGAGGC	261	2196-2218
AD-1686680	AGUCUAUACCC UUACCUGAAU	127	2209-2229	AUUCAGGUAAAG GUUAUAGACUUG	262	2207-2229
AD-1686693	ACCUGAAGAAU GUGUUCUUU	128	2222-2242	AAAGAACAGCAU UCUUCAGGUAA	263	2220-2242
AD-1686709	UCUUUCCUAGC CACCUUUCUU	129	2238-2258	AAGAAAGGUGGC UAGGAAAGAAC	264	2236-2258
AD-1686720	CACCUUUCUGG CCUCCACUU	130	2249-2269	AAGUGGGAGGCC AGAAAGGUGGC	265	2247-2269
AD-1686729	GGCCUCCACU UGCCUGAAU	131	2258-2278	AUUCAGGGCAAG UGGAGGCCAG	266	2256-2278
AD-1686738	CUUGCCUGAA AGGCCAAGAU	132	2267-2287	AUCUUGGCCUUU CAGGGCAAGUG	267	2265-2287
AD-1686747	AAAGCCAAGA UCAAGAUGUU	133	2276-2296	AACAUCUUUGAUC UUGCCUUUCA	268	2274-2296
AD-1686751	AGGCAUCUUGA UCCAGCCUU	134	2301-2321	AAGGCUGGGAUC AAGAUGCUGG	269	2299-2321
AD-1686761	AUCCAGCCUG ACUGCUGCUU	135	2311-2331	AAGCAGCAGUCA GGCUGGGAUCA	270	2309-2331
AD-1686770	UGACUGCUGCU ACAUCUAAUU	136	2320-2340	AAUUAGAUGUAG CAGCAGUCAGG	271	2318-2340
AD-1686781	GCCUCCUGUCC CUAAACUCCU	137	2351-2371	AGGAGUUUAGGG ACAGGAGGCAU	272	2349-2371

TABLE 3-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO:	Range in NM_001393338.1
AD-1686783	AGCAUACUGAU GACAGCCCUU	138	2373-2393	AAGGGCUGUCAU CAGUAUGCUGG	273	2371-2393
AD-1686794	GACAGCCCUU CUGACUUUAU	139	2384-2404	AUAAAGUCAGAG AGGGCUGUCAU	274	2382-2404
AD-1686803	CUCUGACUUUA CCUUGAGAUAU	140	2393-2413	AAUCUCAAGGUA AAGUCAGAGAG	275	2391-2413
AD-1686813	ACCUUGAGAU UGUCUUCAUU	141	2403-2423	AAUGAAGACAGA UCUCAAGGUAA	276	2401-2423

TABLE 4

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents					
Duplex ID	Sense Sequence 5' to 3'	SEQAntisense ID Sequence NO: 5' to 3'	SEQ mRNA Target ID Sequence NO: 5' to 3'	SEQ ID NO:	
AD-1685156	csusgcagAfaGfGfUfuga cugcgauL96	277 asAfscgcAfgUfCfaa ccUfuCfugcagsgsc	412 GCCUGCAGAAGGU UGACUGCGUG	547	
AD-1685173	asgscggaAfgGfCfAfac acgauguL96	278 asCfsaucGfuGfCfu gcCfuUfcggcusug	413 CAAGCCGAAGGCA AGCACGAUGG	548	
AD-1685198	csacagcGfCfAfcGfcug uagcaguL96	279 asCfsugcUfaCfAfgc guGfaCfugcgscsa	414 UGCCAGCGUCACG CUGUAGCAGC	549	
AD-1685207	csugcuguAfgCfAfcGfcg agcaucuL96	280 asGfsaugCfuCfGfgc ugCfuAfcagcgsg	415 CACGUCGUAGCAG CCGAGCAUCA	550	
AD-1685217	gscscgagCfaUfCfAfgcc cgaaaguL96	281 asCfsuuCfGfGfcu gaUfgCfugcgcsug	416 CAGCCGAGCAUCA GCCCCGAAAGG	551	
AD-1685228	gscscggaAfaGfGfAfac acgaaaL96	282 asUfsuucGfuGfCfu ccUfuUfcggcsug	417 CAGCCGAAAGGA AGCACGAAAG	552	
AD-1685240	gsacgaAfaGfGfGfugc agagucuL96	283 asGfsacuCfuGfAfcc gcUfuUfcgugcsusu	418 AAGCACGAAAGCG GUCAGAGUCU	553	
AD-1685250	gsgsucagAfgUfCfUfcc ggcucauL96	284 asUfsgagCfcUfGfga gaCfuCfugaccsgsc	419 GCGUCAGAGUCU CCAGGCUCAG	554	
AD-1685271	gsacagCfuGfGfCfaa cgcgguL96	285 asAfscgcCfGfUfAfu ccAfgCfugugcsa	420 UGGCACAGCUGGC AUACGCGGUC	555	
AD-1685284	ascsccgGfCfCfUfcc cagguL96	286 asCfsaccUfgUfGfga ggGfaCfcgcsusu	421 AUACGCGGUCCCU CCACAGGUGG	556	
AD-1685366	gsccgGfUfCfAfu ugagcuL96	287 asCfsgcuCfaGfCfau guAfcGfccagcgsu	422 ACGCUGGCGUACA UGCUGAGCGC	557	
AD-1685379	csugagcGfcGfCfAfcac guaguauL96	288 asUfsacuAfcGfUfgu gcGfcGfcucagscsa	423 UGCUGAGCGCGCA CACGUAGUAC	558	
AD-1685388	csacacgUfaGfUfAfcac cgccuuL96	289 asAfsaggCfGfUfgu acUfaCfugugscsg	424 CGCACAGUAGUA CACCCTCUGG	559	
AD-1685406	csccggguCfaGfGfAfac ccacaauL96	290 asUfsuguGfGfCfu ccUfgAfcgcsa	425 UGCCGGGUCAGGA AGGCCACAAA	560	
AD-1685418	gsgscacAfaAfcGfAfcg gcgugauL96	291 asUfscacGfcCfGfcu cuUfuGfugccsusu	426 AAGGCCACAAAGA GCGGCGUGAG	561	

TABLE 4-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents				
Duplex ID	Sense Sequence 5' to 3'	SEQAntisense ID Sequence NO: 5' to 3'	SEQ mRNA Target ID Sequence NO: 5' to 3'	SEQ ID NO:
AD-1685469	csacsacacCfaCfGfAfagc cgugcuL96	292 asGfscaaCfGfCfu cgUfgGfuggsasg	427 CUCCACACCACGA AGCCGUUGCC	562
AD-1685491	asgscaggAfaGfGfCfugu gccuguuL96	293 asAfscagGfCfCfag ccUfuCfcugcusgsc	428 GCAGCAGGAAGGC UGUGCCUGUG	563
AD-1685503	csugsugGfCfCfGfcga agucuuL96	294 asAfsagaCfuUfCfGc ggGfCfCfaggsa	429 UGCCUGUGGCCCG CGAAGUCUUC	564
AD-1685512	csagscgaGfCfUfUfcca gcucaguL96	295 asCfsugaGfCfGfGf agAfcUfucgcsagsg	430 CCCGCGAAGUCUU CCAGCUCAGC	565
AD-1685528	uscsagcaGfUfUfCfucg uuccuuL96	296 asAfsaggAfaCfGfag acAfcUfGfcugagsc	431 GCUCAGCAGUGUC UCGUUCCUG	566
AD-1685538	gsagsagacCfGfAfcAfucc uucugguL96	297 asCfscagAfaGfGfau guCfGfGfucgcsusa	432 UAGCAGACCGACA UCCUUCUGGG	567
AD-1685549	uscsuucUfGfGfCfcua cagccuuL96	298 asAfsaggUfGfUfAgg ccCfaGfaaggasusg	433 CAUCCUUCUGGGC CUACAGCCUG	568
AD-1685561	usascagcCfuGfCfCfcu uuucuguL96	299 asCfsagaAfaAfcGfag gcAfgGfcuguasgsg	434 CCUACAGCCUGCC UCUUUUCUGC	569
AD-1685570	csucsucuuUfuCfUfGfcu gggaguL96	300 asAfsucCfCfAfcGfGc agAfaAfcaggsa	435 UGCCUCUUUUCUG CCUGGGAGUC	570
AD-1685587	asgsuccuGfaCfUfUfcca cgaggauL96	301 asUfscuUfGfUfGfGf agUfcAfcgagacscsc	436 GGAGUCCUGACUU CCACAGGAC	571
AD-1685599	csuscaaaCfaCfAfcAfcuc cuucuuL96	302 asAfsagaAfcGfAfcu ugUfGfUfugagsu	437 ACCUCAAACACAA CUCCUUCUG	572
AD-1685610	csusccuuCfuUfGfGfaac ccagauL96	303 asAfsucUfGfGfUfuc caAfcAfcagagsu	438 AACUCCUUCUUGG AACCAGAU	573
AD-1685614	usgscuccCfaGfUfCfagu ugaccuuL96	304 asAfsaggCfaAfcCfug acUfGfGfagcagsg	439 CCUGCUCUCCAGUC AGUUGACCUG	574
AD-1685627	csusuccuUfcCfCfAfcag cucaguL96	305 asAfsucUfGfCfUfcu ggGfaAfcgaggsa	440 GGCCUCCUCCCA GAGCUCAGUG	575
AD-1685637	asgsagcuCfaGfUfGfgac acagaauL96	306 asUfsucUfGfUfUfcc acUfGfAfcgaggsa	441 CCAGAGCUCAGUG GACACAGAA	576
AD-1685648	ascsccuaCfaAfcGfGfauc cuuggcuL96	307 asGfsccaAfcGfAfcu cuUfGfUfaggsa	442 GGACCCUACAAGG AUCCUUGCA	577
AD-1685664	usgscagGfaAfcAfcGfcag ggauuguL96	308 asCfsaaUfCfCfUfGc uuUfcCfugccasag	443 CUUGGCAGGAAAG CAGGGAUUGU	578
AD-1685673	asgscaggGfaUfUfGfugu ucauuuL96	309 asAfsaaUfGfaAfcCfag aaUfcCfugcusu	444 AAAGCAGGGAUUG UGUUCAUUG	579
AD-1685689	asusugaGfGfUfUfuca cugucauL96	310 asUfsgagAfcUfGfGfa acCfCfcaaausgsa	445 UCAUUGAGGGUU UCACUGUCAG	580
AD-1685699	ususcacuGfuCfAfcGfuga gagucuL96	311 asAfsagUfcUfCfCfag ugAfcAfcgagaasasc	446 GUUUCACUGUCAG UGAGAGUCUC	581
AD-1685708	asgsugagAfcUfCfUfcag cuuccauL96	312 asUfsggaAfcCfUfGf gaCfuCfucaggsa	447 UCAGUGAGAGUCU CAGCUUCCA	582
AD-1685717	csuscagcUfuCfCfAfcug aacuguL96	313 asAfsagUfuGfCfCfau ggAfaGfcagagsa	448 GUCUCAGCUCCA UGCAACUGUC	583
AD-1685726	csasugcaAfcUfGfUfcca ucacggL96	314 asCfscguGfaUfGfGf caGfuUfGcaggsa	449 UCCAUGCAACUGU CCAUCACGGC	584

TABLE 4-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents				
Duplex ID	Sense Sequence 5' to 3'	SEQAntisense ID Sequence NO: 5' to 3'	SEQ mRNA Target ID Sequence NO: 5' to 3'	SEQ ID NO:
AD-1685735	gusccauCfaCfGfGfcug caacugul96	315 asCfsaguUfgCfAfgc cgUfgAfuggacsasg	450 CUGUCCAUCACGG CUGCAACUGA	585
AD-1685744	gsgscugcAfaCfUfGfaaa ucagaauL96	316 asCfsucuGfaUfUfuc agUfuGfcagccsgsu	451 ACGGCUGCAACUG AAAUCAGAGC	586
AD-1685770	csascagcGfcAfCfCfaga agcuauL96	317 asUfsuagCfuUfCfug guGfcGfcugugsusc	452 GACACAGCGCACC AGAAGCUAAA	587
AD-1685779	csacsagaaGfcUfAfAfa cuugaauL96	318 asAfsucaAfgAfCfu uaGfcUfucggsusg	453 CACCAGAAGCUAA AGUCUUGAUG	588
AD-1685788	asasagucUfuGfAfUfgcc aucaauL96	319 asUfsuugAfuGfGfca ucAfaGfacuuasasg	454 CUAAGUCUUGAU GCCAUCAAAG	589
AD-1685798	usgsccauCfaAfAfGfgac aucccuL96	320 asAfsgggAfuGfUfcc uuUfgAfuggcasusc	455 GAUGCCAUCAAAG GACAUCCUG	590
AD-1685810	uscsucugUfcAfCfGfucc acuaauL96	321 asAfsuuuGfuGfGfac guGfaCfagagasusc	456 CAUCUCUGUCACG UCCACUAAUC	591
AD-1685826	usasacugGfcAfAfAfa agaaaaL96	322 asUfsuuuCfuCfCfu uuGfcCfgauuasgsu	457 ACUAAUCGGCAA AGGAGAAAAG	592
AD-1685838	asasgugaGfaGfAfAfa gaccuauL96	323 asUfsaggUfcAfUfcu ucUfcUfcacuusus	458 AAAAGUGAGAGAA GAUGACCUAA	593
AD-1685847	asasgaugAfcCfUfAfa gugacuL96	324 asAfsugcAfcAfCfu agGfuCfaucuuuscsu	459 AGAAGAUACCUA AGUGUGACUG	594
AD-1685856	usasagugUfgAfCfUfgca gcaggcuL96	325 asGfscuuGfcUfGfca guCfaCfacuuaasgsu	460 CCUAAGUGUGACU GCAGCAGGCA	595
AD-1685869	asgscaggCfaGfCfUfcug gaaaauL96	326 asAfsuuuUfcCfAfa gcUfgCfcugcusgsc	461 GCAGCAGGCAGCU CUGGAAAUG	596
AD-1685878	csuscuggAfaAfAfUfgaa gccagauL96	327 asUfscugGfcUfUfca uuUfuCfcagagscsu	462 AGCUCUGGAAA GAAGCCAGAG	597
AD-1685889	gsasagccAfgAfGfCfagu gagccauL96	328 asUfsggcUfcAfCfug cuCfuGfgcuucsasu	463 AUGAAGCCAGAGC AGUGAGCCAG	598
AD-1685898	csccgaccAfaGfGfAfgga aggaaauL96	329 asUfsuucCfuUfCfcu ccUfuGfgucggsasg	464 CUCCGACCAAGGA GGAAGGAAAG	599
AD-1685902	gsasggaaGfgAfAfAfa cagaucuL96	330 asGfsaucUfgCfUfcu uuCfcUfuccucscsu	465 AGGAGGAAGGAA AGAGCAGAUCC	600
AD-1685912	asgsagcaGfaUfCfCfcag guuuguL96	331 asAfscaaAfcCfUfgg gaUfcUfgcucususu	466 AAAGAGCAGAUCC CAGGUUUGUA	601
AD-1685921	csccaggUfuUfGfUfaac agaaaaL96	332 asUfsuuuCfuGfUfua caAfaCfcugggsasu	467 AUCCAGGUUUGU AACAGAAAAC	602
AD-1685931	usasacagAfaAfAfCfacc acuaauL96	333 asUfsuuuGfuGfGfug uuUfuCfuguuascsa	468 UGUAAACAGAAAAC ACCACUAAAG	603
AD-1685935	csasgcacAfgGfAfGfaga accaccuL96	334 asGfsgugGfuUfCfuc ucCfuGfugcugsgsg	469 CCCAGCACAGGAG AGAACCACCC	604
AD-1685956	asgscccaGfaAfGfUfucc agggaauL96	335 asUfsuccCfuGfGfaa cuUfcUfgggcusgsg	470 CCAGCCCAGAAGU UCCAGGGAAG	605
AD-1685969	csasgggaAfgGfAfAfcuc uccgguL96	336 asAfsccgGfaGfAfgu ucCfuUfccugsgsa	471 UCCAGGGAAGGAA CUCUCCGGUC	606
AD-1685983	uscsccguCfcAfCfCfaug gaguacuL96	337 asGfsuacUfcCfAfug guGfgAfccggasgsa	472 UCUCCGGUCCACC AUGGAGUACC	607
AD-1685994	asusggagUfaCfCfUfcuc agcucuL96	338 asAfsagagCfuGfAfga ggUfaCfuccausgsu	473 CCAUGGAGUACCU CUCAGCUCUG	608

TABLE 4-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents				
Duplex ID	Sense Sequence 5' to 3'	SEQAntisense ID Sequence NO: 5' to 3'	SEQ mRNA Target ID Sequence NO: 5' to 3'	SEQ ID NO:
AD-1686007	ususcacucAfgGfUfCfagu aucuaauL96	339 asUfsuagAfuAfCfug acCfuGfaguasgsu	474 ACUUAUCAGGUC AGUAUCUAAU	609
AD-1686016	uscsaguaUfcUfAfAfau aagcuculL96	340 asGfsagcUfuAfUfau uaGfaUfacugascs	475 GGUCAGUAUCUAA UAUAAGCUCG	610
AD-1686028	asusaagcUfcGfGfAfguu uggacguL96	341 asCfsgucCfaAfAfcu ccGfaGfcuausasu	476 AUUAAGCUCGGA GUUUGGACGG	611
AD-1686053	uscsuggaCfcUfCfAfgcu ccaccauL96	342 asUfsgguGfgAfGfcu gaGfgUfccagascsc	477 GGUCUGGACCUCA GCUCCACCAC	612
AD-1686057	asgscgacCfuUfUfCfcgu gucuguuL96	343 asAfscagAfcAfCfgy aaAfgGfucgusgs	478 CCAGCGACCUUUC CGUGUCUGUG	613
AD-1686066	uscsugcUfcUfGfUfgau cacaaguL96	344 asCfsuugUfgAfUfca caGfaCfacggasasa	479 UUUCGUGUCUGU GAUCACAAGC	614
AD-1686075	gsusgaucAfcAfAfGfcgg accaucuL96	345 asGfsaugGfuCfCfgy uuGfuGfaucacsas	480 CUGUGAUCACAAG CGGACCAUCC	615
AD-1686086	csagsgaccAfuCfCfGfgaa aggccuuL96	346 asAfsggcCfuUfUfcc ggAfuGfguccgsu	481 AGCGGACCAUCCG GAAAGGCCUG	616
AD-1686098	asasaggcCfuGfAfCfagc ugccacuL96	347 asGfsuggCfaGfCfug ucAfgGfccuuuscs	482 GGAAAGGCCUGAC AGCUGCCACC	617
AD-1686116	csagsgaGfcUfGfCfuag ccaaaguL96	348 asCfsuuGfgCfUfag caGfcUfccuggscs	483 CGCCAGGAGCUGC UAGCCAAAGC	618
AD-1686126	csusagccAfaAfGfCfauu ggagacuL96	349 asGfsucCfAfAfug cuUfuGfgcuagscsa	484 UGCUAGCCAAAGC AUUGGAGACC	619
AD-1686136	csasuggAfgAfCfCfcua cugcuguL96	350 asCfsagcAfgUfAfgg guCfuCfcaugscsu	485 AGCAUUGGAGACC CUACUGCUGA	620
AD-1686147	csusacugCfuGfAfAfugg agugcuuL96	351 asAfsgcCfuCfCfau ucAfgCfaguagsgs	486 CCCUACUGCUGAA UGGAGUGCUA	621
AD-1686159	gsagsagugCfuAfAfCfcu ggugcuuL96	352 asAfsgcCfAfGfgg uuAfgCfacuccsas	487 AUGGAGUGCUAAC CCUGGUGCUA	622
AD-1686171	csusggugCfuAfGfAfgga ggauuguL96	353 asCfscauCfCfCfcu cuAfgCfaccagsgs	488 CCCUGGUGCUAGA GGAGGAUGGA	623
AD-1686181	asgsgaggAfuGfGfAfacu gcaguguL96	354 asCfsacuGfAfGfuu ccAfuCfuccuscsu	489 AGAGGAGGAUGG AACUGCAGUGG	624
AD-1686197	asgsuggaCfaGfUfGfagg acuucuuL96	355 asAfsgaaGfuCfCfuc acUfgUfccacusgs	490 GCAGUGGACAGUG AGGACUUCUU	625
AD-1686207	gsasggacUfuCfUfUfcca gcugcuuL96	356 asAfsgcaGfCfGfga agAfaGfuccusasc	491 GUGAGGACUUCUU CCAGCUGCUG	626
AD-1686216	ususcagCfuGfCfUfgga ggauuguL96	357 asUfscacCfCfCfca gcAfgCfuggaasgsa	492 UCUCAGCUGCU GGAGGAUGAC	627
AD-1686225	csusggagGfaUfGfAfcac gugccuuL96	358 asAfsggcAfcGfUfgu caUfcCfuccagscsa	493 UGCUGGAGGAUGA CACGUGCCUG	628
AD-1686242	csusgaguGfgUfGfUfgc agucuguL96	359 asCfsagaCfuGfCfaa caCfAfucaggsa	494 UGCCUGAUGGUGU UGCAGUCUGG	629
AD-1686251	gsusugcaGfuCfUfGfguc agagcuuL96	360 asAfsgcuCfuGfAfc agAfcUfgcaacsasc	495 GUGUUGCAGUCUG GUCAGAGCUG	630
AD-1686265	asgsagcuGfgAfGfCfcu acaagguL96	361 asCfscuuGfuAfGfgg cuCfAfgcucsgsa	496 UCAGAGCUGGAGC CCUACAAGGA	631
AD-1686274	gscccaCfaAfGfGfagu ggaguguL96	362 asCfsacuCfAfCfuc cuUfgUfaggcsusc	497 GAGCCCUACAAGG AGUGGAGUGC	632

TABLE 4-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents				
Duplex ID	Sense Sequence 5' to 3'	SEQAntisense ID Sequence NO: 5' to 3'	SEQ mRNA Target ID Sequence NO: 5' to 3'	SEQ ID NO:
AD-1686286	gusggagUfgCfUfGfuca uauugcuL96	363 asGfsccaUfaUfGfac agCfaCfuccacsusc	498 GAGUGGAGUGCUG UCAUAUGGCC	633
AD-1686319	csasagcaCfaGfCfAfaagg acaucguL96	364 asCfsgauGfuCfCfu gcUfgUfgcuugsgsg	499 CCCAAGCACAGCA AGGACAUCGC	634
AD-1686334	csasucgcCfcGfAfUfuca ccuuuguL96	365 asCfsaaaGfgUfGfaa ucGfgGfcgaugsusc	500 GACAUCGCCCGAU UCACCUUUGA	635
AD-1686343	asusucacCfuUfUfGfacg uguacauL96	366 asUfsguaCfaCfGfuc laaAfgGfugaauscsg	501 CGAUUACCCUUUG ACGUGUACAA	636
AD-1686354	ascsuguguAfcAfAfGfcaa aaccuuL96	367 asAfsgggUfuUfUfgc uuGfuAfcaguscscsa	502 UGACGUGUACAAG CAAAACCCUC	637
AD-1686366	asasaaccCfuCfGfAfgac cucuuuL96	368 asAfsaagAfgGfUfcu cgAfgGfguuusgsc	503 GCAAAACCCUCGA GACCUCUUUG	638
AD-1686377	gsasccucUfuUfGfGfcag ccugaauL96	369 asUfsucaGfgCfUfgc caAfaGfaggucscsc	504 GAGACCUUUUGG CAGCCUGAAU	639
AD-1686394	gsasauguCfaAfAfGfccca cauucuL96	370 asAfsgaaUfgUfGfgc uuUfgAfcuucscasg	505 CUGAAUGUCAAG CCACAUCUA	640
AD-1686403	asgscacAfUfCfUfacg ggcuuL96	371 asAfsgagCfcCfGfua gaAfUfggucscscsu	506 AAAGCCACAUCU ACGGGCUCUA	641
AD-1686417	gsgscucuAfcUfCfUfaug aguuguL96	372 asAfscaaCfuCfAfua gaGfuAfgagccscscg	507 CGGGCUCUACUCU AUGAGUUGUG	642
AD-1686426	csusaugaGfuUfGfUfgac uuucauL96	373 asUfsugaAfaGfUfca caAfcUfcuagscasg	508 CUCUAUGAGUUGU GACUUUCAAG	643
AD-1686435	gsusgacuUfuCfAfAfgga cuugguL96	374 asGfsccaAfgUfCfcu ugAfaAfgucacsasa	509 UUGUGACUUCAA GGACUUGGCC	644
AD-1686451	usgsgcccAfaAfGfAfaag uacucaL96	375 asUfsgagUfaCfUfu cuUfuGfggcccscasg	510 CUUGGCCCAAAGA AAGUACUCAG	645
AD-1686466	ascsucagGfgAfgCfucc uucguuL96	376 asAfsacgAfaGfGfag cuCfcCfugaguscsc	511 GUACUCAGGGAGC UCCUUCGUUG	646
AD-1686475	gscsuccuUfcUfUfUfgga ccuccauL96	377 asUfsggaGfgUfCfca acGfaAfggagcscusc	512 GAGCUCCUUCGUU GGACCUCAC	647
AD-1686491	uscscacaCfuGfCfUfgca aggccuL96	378 asAfsggcCfuUfGfca gcAfgUfggagsgsg	513 CCUCCACACUGCU GCAAGGCCUG	648
AD-1686508	cscsugggCfcAfUfAfugu ugcugguL96	379 asCfscagCfaAfCfau auGfgCfccaggscsc	514 GGCCUGGGCCAUA UGUUGCUGGG	649
AD-1686518	asusguugCfuGfGfGfaau uucccuL96	380 asGfsaggAfaAfUfuc ccAfgCfaacausasu	515 AUAUGUUGCUGGG AAUUUCCUCC	650
AD-1686531	usussuccCfcAfCfCfcuu cgucauL96	381 asAfsugaCfGfAfAfgg guGfgAfggaaasusu	516 AAUUUCCUCCACC CUUCGUCAUG	651
AD-1686540	csescuucGfuCfAfUfgca guggaguL96	382 asCfsuccAfcUfGfca ugAfcGfaagggscusc	517 CACCUCUUCGUCAU GCAGUGGAGG	652
AD-1686543	csusgagcAfgUfGfGfcag cagaaguL96	383 asCfsuucUfgCfUfgc caCfuGfcucagscsc	518 GGCUGAGCAGUGG CAGAGAAGG	653
AD-1686556	csasgaagGfgCfCfGfccu ccauucuL96	384 asGfsaauGfgAfGfgc ggCfcCfuucugscsu	519 AGCAGAAAGGCCG CCUCCAUUCC	654
AD-1686565	csagccucCfaUfUfCfcua cuaagguL96	385 asCfscuuAfgUfAfgg aaUfgGfaggcgsgsc	520 GCCGCCUCCAUUC CUACUAAGGG	655
AD-1686566	csasgaauCfaUfUfCfcaa ccgaccuL96	386 asGfsgucGfgUfUfgg aaUfgAfuucugsgsg	521 CCCAGAAUCAUUC CAACCGACCC	656

TABLE 4-continued

Modified Sense and Antisense Strand Sequences of Human CIDEb dsRNA Agents				
Duplex ID	Sense Sequence 5' to 3'	SEQAntisense ID Sequence NO: 5' to 3'	SEQ mRNA Target ID Sequence NO: 5' to 3'	SEQ ID NO:
AD-1686576	csesaaccGfaCfCfCfacu gcaaagL96	387asCfsuuuGfcAfGfug ggUfcGfguuggsasa	522 UCCAACCGACCC ACUGCAAAGA	657
AD-1686585	csesacugCfaAfAfGfacu augacauL96	388asUfsgucAfuAfGfuc uuUfgCfaguggsusu	523 ACCCACUGCAAAG ACUUGACAG	658
AD-1686597	csusaugaCfaGfCfAfuca aauuucL96	389asGsaaaUfuUfGfaug cUfgUfcuaggsusc	524 GACUAUGACAGCA UCAAAUUUCA	659
AD-1686606	csasucuaAfuUfUfCfagg accugcuL96	390asGfscagGfuCfCfug aaAfuUfugaugscsu	525 AGCAUCAAUUUC AGGACCUGCA	660
AD-1686616	csasggacCfuGfCfAfgac aguacauL96	391asUfsguaCfuGfUfcu gcAfgGfuccugsasa	526 UUCAGGACCUGCA GACAGUACAG	661
AD-1686626	asgsacagUfaCfAfGfgcu agauaaL96	392asUfsuauCfuAfGfcc ugUfaCfugucugsgc	527 GCAGACAGUACAG GCUAGAUAAC	662
AD-1686639	usasgauaAfcCfCfAfccc aauuucL96	393asGfsaaaUfuGfGfgu ggGfuUfaucuasgsc	528 GCUAGAUACCCA CCCAAUUUCC	663
AD-1686649	asgsaacUfuUfCfAfgca uaacgcuL96	394asGfscguUfaUfGfcu gaAfaGfguucugsu	529 ACAGAACCUUCA GCAUAACGCC	664
AD-1686658	csasgcuaAfaCfGfCfcuc acauccuL96	395asGfsgauGfuGfAfgg cgUfuAfugcugsasa	530 UUCAGCAUACGC CUCACAUCCC	665
AD-1686669	csuscacaUfcCfCfAfa cuauacuL96	396asGfsuauAfgAfCfu ggGfaUfgugagsgsc	531 GCCUCACAUCCCA AGUCUAUACC	666
AD-1686680	asgsucuaUfaCfCfCuua ccugaauL96	397asUfsucaGfgUfAfa ggUfaUfagacusug	532 CAAGUCUAUACCC UUACCUGAAG	667
AD-1686693	ascscugaAfgAfAfUfgcu guucuuL96	398asAfsagaAfcAfGfca uuCfuUfcaggusasa	533 UUACCUGAAGAAU GCUGUUCUUU	668
AD-1686709	uscsuuucCfuAfGfCfcac cuuucuuL96	399asAfsagaaAfgGfUfgg cuAfgGfaaagasasc	534 GUUCUUUCCUAGC CACCUUUCUG	669
AD-1686720	csasccuuUfcUfGfGfccu cccacuuL96	400asAfsugugGfgAfGfgc caGfaAfaugugsgsc	535 GCCACCUUUCUGG CCUCCCACUU	670
AD-1686729	gsgscuccCfcAfCfUfugc ccugaauL96	401asUfsucaGfgGfCfaa guGfgGfaggccsasg	536 CUGGCCUCCACU UGCCUGAAA	671
AD-1686738	csusugccCfuGfAfAfa ccaagauL96	402asUfscuuGfgCfCfu ucAfgGfgcaagsusg	537 CACUUGCCUGAA AGGCCAAGAU	672
AD-1686747	asasaggcCfaAfGfAfuca agauguL96	403asAfscauCfuUfGfau cuUfgGfccuuuscsa	538 UGAAAGGCCAAGA UCAAGAUGUC	673
AD-1686751	asgsgcauCfuUfGfAfucc cagccuuL96	404asAfsggcUfgGfGfau caAfgAfugccusgsg	539 CCAGGCAUCUUGA UCCACGCCUG	674
AD-1686761	asuscccaGfcCfUfGfacu gcugcuL96	405asAfsagcaGfcAfGfuc agGfcUfgggauscsa	540 UGAUCCAGCCUG ACUGCUGCUA	675
AD-1686770	usgsacugCfuGfCfUfaca ucuaauL96	406asAfsuuaGfaUfGfua gcAfgCfagucasgsg	541 CCUGACUGCUGCU ACAUCUAAUC	676
AD-1686781	gscscuccUfgUfCfCfcua aacuccuL96	407asGfsgagUfuUfAfgg gaCfaGfagggcsasu	542 AUGCCUCCUGUCC CUAAACUCCC	677
AD-1686783	asgscauaCfuGfAfUfgac agcccuL96	408asAfsgggCfuGfUfca ucAfgUfaugcusgsg	543 CCAGCAUACUGAU GACAGCCUC	678
AD-1686794	gsascagcCfcUfCfUfcug acuuuauL96	409asUfsaaaGfuCfAfga gaGfgGfcugucsasu	544 AUGACAGCCUCU CUGACUUUAC	679
AD-1686803	csuscugaCfuUfUfAfccu ugagaauL96	410asAfsucuCfaAfGfgu aaAfgUfcagagsasg	545 CUCUCUGACUUUA CCUUGAGAUC	680

TABLE 4-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents			
Sense Sequence	SEQAntisense ID Sequence	SEQ mRNA Target ID Sequence	SEQ ID
Duplex ID 5' to 3'	NO: 5' to 3'	NO: 5' to 3'	NO:
AD-1686813ascscuugAfgAfUfCfugu cuucauL96	411asAfsugaAfgAfCfag auCfuCfaaggusasa	546 UUACCUUGAGAUC UGUCUUCAUA	681

TABLE 5

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1699864.1	GUCCUGGGCUG AGAUGCCAGU	682	1269-1289	ACUGGGAUCUCAG CCCAGGACAA	1130	1267-1289
AD- 1699867.1	CUGGGCUGAGA UCCCAGGUUU	683	1272-1292	AAACCUGGGAUUCU CAGCCAGGA	1131	1270-1292
AD- 1699868.1	UGGGCUGAGAU CCCAGGUUUU	684	1273-1293	AAAACCTGGGATC UCAGCCAGG	1132	1271-1293
AD- 1699869.1	GGGCUGAGAUC CCAGGUUUUGU	685	1274-1294	ACAAACCUGGGAU CUCAGCCAG	1133	1272-1294
AD- 1699870.1	GGCUGAGAUGC CAGGUUUUGU	686	1275-1295	AACAAACCUGGGA UCUCAGCCCA	1134	1273-1295
AD- 1699871.1	GCUGAGAUGCC AGGUUUUGAU	687	1276-1296	ATACAAACCUGGG AUCUCAGCCC	1135	1274-1296
AD- 1699872.1	CUGAGAUGCCA GGUUUGUAAU	688	1277-1297	ATUACAAACCUGG GAUCUCAGCC	1136	1275-1297
AD- 1699873.1	UGAGAUGCCAG GUUUGUAACU	689	1278-1298	AGUUACAAACCTG GGAUCUCAGC	1137	1276-1298
AD- 1699875.1	AGAUGCCAGGU UUGUAACAGU	690	1280-1300	ACUGTUACAAACC UGGGAUCUCA	1138	1278-1300
AD- 1699882.1	AGGUUUUGAAC AGAAAACACU	691	1287-1307	AGUGTUTUCUGTU ACAAACCUGG	1139	1285-1307
AD- 1699891.1	ACAGAAAACAC CACUAAAGCU	692	1296-1316	AGCUTUAGUGGTG UUUUCUGUUA	1140	1294-1316
AD- 1699892.1	CAGAAAACACC ACUAAAGCCU	693	1297-1317	AGGCTUTAGUGGU GUUUUCUGUU	1141	1295-1317
AD- 1699904.1	AGAACCACCCA GCCAGAGAGU	694	1329-1349	ACUUCUGGGCUGG GUGGUUCUCU	1142	1327-1349
AD- 1699907.1	ACCACCCAGCC CAGAAGUUCU	695	1332-1352	AGAACUTCUGGGC UGGGUGGUUC	1143	1330-1352
AD- 1699948.1	CACCAUGGAGU ACCUUCACAGU	696	1373-1393	ACUGAGAGGUACU CCAUGGUGGA	1144	1371-1393
AD- 1699949.1	ACCAUGGAGUA CCUCUCAGCU	697	1374-1394	AGCUGAGAGGUAC UCCAUGGUGG	1145	1372-1394
AD- 1699951.1	CAUGGAGUACC UCUCAGCUCU	698	1376-1396	AGAGCUGAGAGGU ACUCCAUGGU	1146	1374-1396
AD- 1699964.1	CUUACUCAGGU CAGUAUCUAU	699	1409-1429	ATAGAUACUGACC UGAGUAAGUC	1147	1407-1429

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1699967.1	ACUCAGGUCAG UAUCUAAUUAU	700	1412-1432	ATAUTAGAUACTG ACCUGAGUAA	1148	1410-1432
AD- 1699970.1	CAGGUCAGUAU CUAAUAUAAU	701	1415-1435	ATUATATUAGATA CUGACCUGAG	1149	1413-1435
AD- 1699971.1	AGGUCAGUAUC UAAUAUAAGU	702	1416-1436	ACUUAUAUUAGAU ACUGACCUGA	1150	1414-1436
AD- 1699972.1	GGUCAGUAUCU AAUAUAAGCU	703	1417-1437	AGCUTATAUUAGA UACUGACCUG	1151	1415-1437
AD- 1699973.1	GUCAGUAUCUA AUAAUAGCUU	704	1418-1438	AAGCTUAUAUUAG AUACUGACCU	1152	1416-1438
AD- 1699976.1	AGUAUCUAAUA UAAGCUCGGU	705	1421-1441	ACCGAGCUUUAU UAGAUACUGA	1153	1419-1441
AD- 1699977.1	GUAUCUAAUAU AAGCUCGGAU	706	1422-1442	ATCCGAGCUUATA UUAGAUACUG	1154	1420-1442
AD- 1699978.1	UAUCUAAUAUA AGCUCGGAGU	707	1423-1443	ACUCCGAGCUUUAU AUUAGAUACU	1155	1421-1443
AD- 1699979.1	AUCUAAUAUAA GCUCGGAGUU	708	1424-1444	AACUCCGAGCUTA UAUUAGAUAC	1156	1422-1444
AD- 1699982.1	UAAUAUAAGCU CGGAGUUUGU	709	1427-1447	ACAACTCCGAGC UUUAUAUAGA	1157	1425-1447
AD- 1699983.1	AAUAUAAGCUC GGAGUUUGGU	710	1428-1448	ACCAAACUCCGAG CUUAUAUUAG	1158	1426-1448
AD- 1699984.1	AUAUAAGCUCG GAGUUUGGAU	711	1429-1449	ATCCAAACUCCGA GCUUAUAUUA	1159	1427-1449
AD- 1699985.1	UAUAAGCUCGG AGUUUGGACU	712	1430-1450	AGUCCAAACUCCG AGCUUAUAUU	1160	1428-1450
AD- 1699986.1	AUAAGCUCGGA GUUUGGACGU	713	1431-1451	ACGUCCAAACUCC GAGCUUAUAU	1161	1429-1451
AD- 1699987.1	UAAGCUCGGAG UUUGGACGGU	714	1432-1452	ACCGTCCAACTC CGAGCUUAUA	1162	1430-1452
AD- 1699988.1	AAGCUCGGAGU UUGGACGGAU	715	1433-1453	ATCCGUCCAAACU CCGAGCUUAU	1163	1431-1453
AD- 1699989.1	AGCUCGGAGUU UGGACGGAGU	716	1434-1454	ACUCCGTCCAAAC UCCGAGCUUA	1164	1432-1454
AD- 1699990.1	GCUCGGAGUUU GGACGGAGGU	717	1435-1455	ACCUCGUCUCCAA CUCCGAGCUU	1165	1433-1455
AD- 1700015.1	AGCGACCUUUC CGUGUCUGUU	718	1480-1500	AACAGACACGGAA AGGUCGUCUGG	1166	1478-1500
AD- 1700035.1	GAUCACAAGCG GACCAUCCGU	719	1500-1520	ACGGAUGGUCCGC UUGUGAUCAC	1167	1498-1520
AD- 1700037.1	UCACAAGCGGA CCAUCCGGAU	720	1502-1522	ATCCGGAUGGUCC GCUUGUGAUC	1168	1500-1522
AD- 1700038.1	CACAAGCGGAC CAUCCGGAU	721	1503-1523	ATUCCGGAUGGTC CGCUUGUGAU	1169	1501-1523
AD- 1700039.1	ACAAGCGGACC AUCCGAAAU	722	1504-1524	ATUCCGGAUGGU CCGCUUGUGA	1170	1502-1524

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700040.1	CAAGCGGACCA UCCGGAAGGU	723	1505-1525	ACUUTCCGGAUGG UCCGCUUGUG	1171	1503-1525
AD- 1700041.1	AAGCGGACCAU CCGGAAGGU	724	1506-1526	ACCUTUCCGGATG GUCCGCUUGU	1172	1504-1526
AD- 1700077.1	GGAGCUCGUAG CCAAAGCAUU	725	1547-1567	AAUGCUTUGGCTA GCAGCUCCUG	1173	1545-1567
AD- 1700080.1	GCUGCUAGCCA AAGCAUUGGU	726	1550-1570	ACCAAUGC UUUGG CUAGCAGCUC	1174	1548-1570
AD- 1700081.1	CUGCUAGCCAA AGCAUUGGAU	727	1551-1571	ATCCAATGCUUTG GCUAGCAGCU	1175	1549-1571
AD- 1700082.1	UGCUGCCAAA GCAUUGGAGU	728	1552-1572	ACUCCAAUGCUTU GGCUAGCAGC	1176	1550-1572
AD- 1700083.1	GCUGCCAAAG CAUUGGAGAU	729	1553-1573	ATCUCCAAUGC TU UGGCUAGCAG	1177	1551-1573
AD- 1700091.1	AAGCAUUGGAG ACCCUACUGU	730	1561-1581	ACAGTAGGGUCTC CAAUGCUUUG	1178	1559-1581
AD- 1700092.1	AGCAUUGGAGA CCCUCACUGU	731	1562-1582	AGCAGUAGGGUCU CCAAUGCUUU	1179	1560-1582
AD- 1700094.1	CAUUGGAGACC CUACUGCUGU	732	1564-1584	ACAGCAGUAGGGU CUCCAAUGCU	1180	1562-1584
AD- 1700097.1	UGGAGACCCUA CUGCUGAAUU	733	1567-1587	AAUUCAGCAGUAG GGUCUCCAAU	1181	1565-1587
AD- 1700099.1	GAGACCCUACU GCUGAAUGGU	734	1569-1589	ACCATUCAGCAGU AGGGUCUCCA	1182	1567-1589
AD- 1700100.1	AGACCCUACUG CUGAAUGGAU	735	1570-1590	ATCCAUTCAGCAG UAGGGUCUCC	1183	1568-1590
AD- 1700101.1	GACCCUACUGC UGAAUGGAGU	736	1571-1591	ACUCCATUCAGCA GUAGGGUCUC	1184	1569-1591
AD- 1700104.1	CCUACUGCUGA AUGGAGUGCU	737	1574-1594	AGCACUCCAUAU GCAGUAGGGU	1185	1572-1594
AD- 1700109.1	UGCUGAAUGGA GUGCUAACCU	738	1579-1599	AGGUTAGCACUCC AUUCAGCAGU	1186	1577-1599
AD- 1700112.1	UGAAUGGAGUG CUAACCUGU	739	1582-1602	ACAGGGTUAGCAC UCCAUUCAGC	1187	1580-1602
AD- 1700113.1	GAAUGGAGUGC UAACCUGGU	740	1583-1603	ACCAGGGUUAGCA CUCCAUUCAG	1188	1581-1603
AD- 1700114.1	AAUGGAGUGCU AACCUGGUU	741	1584-1604	AACCAGGGUUAGC ACUCCAUAUA	1189	1582-1604
AD- 1700115.1	AUGGAGUGCUA ACCCUGGUGU	742	1585-1605	ACACCAGGGUUAG CACUCCAUAU	1190	1583-1605
AD- 1700116.1	UGGAGUGCUAA CCCUGGUGCU	743	1586-1606	AGCACCAGGGUTA GCACUCCAUA	1191	1584-1606
AD- 1700121.1	UGCUAACCCUG GUGCUAGAGU	744	1591-1611	ACUCTAGCACCAG GGUUAGCACU	1192	1589-1611
AD- 1700122.1	GCUAACCCUGG UGCUGAGAGU	745	1592-1612	ACCUCUAGCACCA GGGUUAGCAC	1193	1590-1612

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700125.1	AACCCUGGUGC UAGAGGAGGU	746	1595-1615	ACCUCCTCUAGCA CCAGGGUUG	1194	1593-1615
AD- 1700148.1	GAACUGCAGUG GACAGUGAGU	747	1618-1638	ACUACTGUCCAC UGCAGUCCA	1195	1616-1638
AD- 1700149.1	AACUGCAGUGG ACAGUGAGGU	748	1619-1639	ACCUCACUGUCCA CUGCAGUUC	1196	1617-1639
AD- 1700156.1	GUGGACAGUGA GGACUUCUUU	749	1626-1646	AAAGAAGUCCUCA CUGUCCACUG	1197	1624-1646
AD- 1700158.1	GGACAGUGAGG ACUUCUCCU	750	1628-1648	AGGAAGAAGUCCU CACUGUCCAC	1198	1626-1648
AD- 1700159.1	GACAGUGAGGA CUUCUCCAU	751	1629-1649	ATGGAAGAAGUCC UCACUGUCCA	1199	1627-1649
AD- 1700160.1	ACAGUGAGGAC UUCUCCAGU	752	1630-1650	ACUGGAAGAAGTC CUCACUGUCC	1200	1628-1650
AD- 1700166.1	AGGACUUCUUC CAGCUGCUGU	753	1636-1656	ACAGCAGCUGGAA GAAGUCCUCA	1201	1634-1656
AD- 1700167.1	GGACUUCUCC AGCUGCUGGU	754	1637-1657	ACCAGCAGCUGGA AGAAGUCCUC	1202	1635-1657
AD- 1700180.1	CUGCUGGAGGA UGACACGUGU	755	1650-1670	ACACGUGUCAUCC UCCAGCAGCU	1203	1648-1670
AD- 1700187.1	AGGAUGACACG UGCCUGAGUGU	756	1657-1677	ACAUCAGGCACGU GUCAUCCUCC	1204	1655-1677
AD- 1700188.1	GGAUGACACGU GCCUGAUGGU	757	1658-1678	ACCATCAGGCACG UGUCAUCCUC	1205	1656-1678
AD- 1700189.1	GAUGACACGUG CCUGAUGGUU	758	1659-1679	AACCAUCAGGCAC GUGUCAUCCU	1206	1657-1679
AD- 1700190.1	AUGACACGUGC CUGAUGGUGU	759	1660-1680	ACACCATCAGGCA CGUGUCAUCC	1207	1658-1680
AD- 1700191.1	UGACACGUGCC UGAUGGUGUU	760	1661-1681	AACACCAUCAGGC ACGUGUCAUC	1208	1659-1681
AD- 1700193.1	ACACGUGCCUG AUGGUGUUGU	761	1663-1683	ACAACCAUCAGAG GCACGUGUCA	1209	1661-1683
AD- 1700213.1	CAGUCUGGUCA GAGCUGGAGU	762	1683-1703	ACUCCAGCUCUGA CCAGACUGCA	1210	1681-1703
AD- 1700242.1	GAGUGGAGUGC UGUCAUAUGU	763	1712-1732	ACAU AUGACAGCA CUCCACUCCU	1211	1710-1732
AD- 1700243.1	AGUGGAGUGCU GUCAUAUGGU	764	1713-1733	ACCATATGACAGC ACUCCACUCC	1212	1711-1733
AD- 1700244.1	GUGGAGUGCUG UCAUAUGGCU	765	1714-1734	AGCCAUAUGACAG CACUCCACUC	1213	1712-1734
AD- 1700245.1	UGGAGUGCUGU CAUAUGGCCU	766	1715-1735	AGGCCATAUGACA GCACUCCACU	1214	1713-1735
AD- 1700248.1	AGUGCUGUCAU AUGGCCUGGU	767	1718-1738	ACCAGGCCAUATG ACAGCACUCC	1215	1716-1738
AD- 1700249.1	GUGCUGUCAUA UGGCCUGGGU	768	1719-1739	ACCCAGGCCAUAU GACAGCACUC	1216	1717-1739

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEb dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700250.1	UGCUGUCAUUAU GGCCUGGGGAU	769	1720-1740	ATCCCAGGCCATA UGACAGCACU	1217	1718-1740
AD- 1700252.1	CUGUCAUAUGG CCUGGGACGCU	770	1722-1742	ACGUCCCAGGCCA UAUGACAGCA	1218	1720-1742
AD- 1700272.1	AGGCCCCAAGCA CAGCAAGGAU	771	1746-1766	ATCCTUGCUGUGC UUGGGCCUCU	1219	1744-1766
AD- 1700279.1	AGCACAGCAAG GACAUCGCCU	772	1753-1773	AGGCGATGUCCTU GCUGUGCUUG	1220	1751-1773
AD- 1700280.1	GCACAGCAAGG ACAUCGCCCU	773	1754-1774	AGGGCGAUGUCCU UGCUGUGCUU	1221	1752-1774
AD- 1700303.1	UCACCUUUGAC GUGUACAAGU	774	1777-1797	ACUUGUACACGTC AAAGGUGAAU	1222	1775-1797
AD- 1700309.1	UUGACGUGUAC AAGCAAAACU	775	1783-1803	AGUUTUGCUUGTA CACGUCAAAG	1223	1781-1803
AD- 1700312.1	ACGUGUACAAG CAAAACCCUU	776	1786-1806	AAGGGTUUGCTU GUACACGUCA	1224	1784-1806
AD- 1700314.1	GUGUACAAGCA AAACCCUCGU	777	1788-1808	ACGAGGGUUUUGC UUGUACACGU	1225	1786-1808
AD- 1700315.1	UGUACAAGCAA AACCUCGGAU	778	1789-1809	ATCGAGGGUUUTG CUUGUACACG	1226	1787-1809
AD- 1700316.1	GUACAAGCAAA ACCCUCGAGU	779	1790-1810	ACUCGAGGGUUTU GCUUGUACAC	1227	1788-1810
AD- 1700317.1	UACAAGCAAAA CCCUCGAGAU	780	1791-1811	ATCUCGAGGGUTU UGCUGUACA	1228	1789-1811
AD- 1700318.1	ACAAGCAAAAC CCUCGAGACU	781	1792-1812	AGUCTCGAGGGTU UUGCUUGUAC	1229	1790-1812
AD- 1700323.1	CAAAACCCUCG AGACCUCUUU	782	1797-1817	AAAGAGGUCUCGA GGGUUUUGCU	1230	1795-1817
AD- 1700324.1	AAAACCCUCGA GACCUCUUUU	783	1798-1818	AAAAGAGGUCUCG AGGGUUUUGC	1231	1796-1818
AD- 1700325.1	AAACCCUCGAG ACCUCUUUGU	784	1799-1819	ACAAAGAGGUCTC GAGGGUUUUG	1232	1797-1819
AD- 1700327.1	ACCCUCGAGAC CUCUUUGGCU	785	1801-1821	AGCCAAAGAGGTC UCGAGGGUUU	1233	1799-1821
AD- 1700328.1	CCCUCGAGACC UCUUUGGCAU	786	1802-1822	ATGCCAAAGAGGU CUCGAGGGUU	1234	1800-1822
AD- 1700329.1	CCUCGAGACCU CUUUGGCAGU	787	1803-1823	ACUGCCAAAGAGG UCUCGAGGGU	1235	1801-1823
AD- 1700336.1	ACCUCUUUGGC AGCCUGAAUU	788	1810-1830	AAUUCAGGCUGCC AAAGAGGUCU	1236	1808-1830
AD- 1700339.1	UCUUUGGCAGC CUGAAUGUCU	789	1813-1833	AGACAUTCAGGCU GCCAAAGAGG	1237	1811-1833
AD- 1700344.1	GGCAGCCUGAA UGUCAAGCU	790	1818-1838	AGCUTUGACAUTC AGGCUGCCAA	1238	1816-1838
AD- 1700350.1	CUGAAUGUCAA AGCCACAUUU	791	1824-1844	AAAUGUGGCUUTG ACAUUCAGGC	1239	1822-1844

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD-1700352.1	GAAUGUCAAG CCACAUUCUU	792	1826-1846	AAGAAUGUGGCTU UGACAUUCAG	1240	1824-1846
AD-1700353.1	AAUGUCAAGC CACAUUCUUA	793	1827-1847	ATAGAATGUGGCU UUGACAUUCA	1241	1825-1847
AD-1700354.1	AUGUCAAGCC ACAUCUACU	794	1828-1848	AGUAGAAUGUGGC UUUGACAUUC	1242	1826-1848
AD-1700355.1	UGUCAAGCCA CAUUCUACGU	795	1829-1849	ACGUAGAAUGUGG CUUUGACAUI	1243	1827-1849
AD-1700356.1	GUCAAAGCCAC AUUCUACGGU	796	1830-1850	ACCGTAGAAUGTG GCUUUGACAUI	1244	1828-1850
AD-1700357.1	UCAAAGCCACA UUCUACGGGU	797	1831-1851	ACCCGUAGAAUGU GCUUUGACA	1245	1829-1851
AD-1700358.1	CAAAGCCACAU UCUACGGGCU	798	1832-1852	AGCCCGTAGAATG UGGCUUUGAC	1246	1830-1852
AD-1700359.1	AAAGCCACAUU CUACGGGCUU	799	1833-1853	AAGCCCGUAGAAU GUGGCUUUGA	1247	1831-1853
AD-1700365.1	ACAUCUACGG GCUCUACUCU	800	1839-1859	AGAGTAGAGCCCG UAGAAUGUGG	1248	1837-1859
AD-1700366.1	CAUUCUACGGG CUCUACUCUU	801	1840-1860	AAGAGUAGAGCCC GUAGAAUGUG	1249	1838-1860
AD-1700368.1	UUCUACGGGCU CUACUCUAUU	802	1842-1862	AAUAGAGUAGAGC CCGUAGAAUG	1250	1840-1862
AD-1700369.1	UCUACGGGCUC UACUCUAUGU	803	1843-1863	ACAUAGAGUAGAG CCCGUAGAAU	1251	1841-1863
AD-1700370.1	CUACGGGCUCU ACUCUAUGAU	804	1844-1864	ATCATAGAGUAGA GCCCCUAGAA	1252	1842-1864
AD-1700371.1	UACGGGCUCUA CUCUAUGAGU	805	1845-1865	ACUCAUAGAGUAG AGCCCCUAGA	1253	1843-1865
AD-1700372.1	ACGGGCUCUAC UCUAUGAGUU	806	1846-1866	AACUCATAGAGTA GAGCCCCUAG	1254	1844-1866
AD-1700374.1	GGGCUCUACUC UAUGAGUUGU	807	1848-1868	ACAACUCAUAGAG UAGAGCCCCU	1255	1846-1868
AD-1700375.1	GGCUCUACUCU AUGAGUUGUU	808	1849-1869	AACAACUCAUAGA GUAGAGCCCCG	1256	1847-1869
AD-1700376.1	GCUCUACUCUA UGAGUUGUGU	809	1850-1870	ACACAACUCAUAG AGUAGAGCCC	1257	1848-1870
AD-1700378.1	UCUACUCUAUG AGUUGUGACU	810	1852-1872	AGUCACAACUCAU AGAGUAGAGC	1258	1850-1872
AD-1700383.1	UCUAUGAGUUG UGACUUUCAU	811	1857-1877	ATGAAAGUCACAA CUCAUAGAGU	1259	1855-1877
AD-1700384.1	CUAUGAGUUGU GACUUUCAAU	812	1858-1878	ATUGAAAGUCACA ACUCAUAGAG	1260	1856-1878
AD-1700385.1	UAUGAGUUGUG ACUUUCAAGU	813	1859-1879	ACUUGAAAGUCAC AACUCAUAGA	1261	1857-1879
AD-1700386.1	AUGAGUUGUGA CUUCAAGGU	814	1860-1880	ACCUTGAAAGUCA CAACUCAUAG	1262	1858-1880

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700387.1	UGAGUUGUGAC UUUCAAGGAU	815	1861-1881	ATCCTUGAAAGTC ACAACUCAUA	1263	1859-1881
AD- 1700388.1	GAGUUGUGACU UUCAAGGACU	816	1862-1882	AGUCCUTGAAAGU CACACUCAU	1264	1860-1882
AD- 1700408.1	UUGGCCCAAAG AAAGUACUCU	817	1882-1902	AGAGTACUUUCTU UGGGCCAAAGU	1265	1880-1902
AD- 1700410.1	GGCCCAAAGAA AGUACUCAGU	818	1884-1904	ACUGAGTACUUTC UUUGGGCCAA	1266	1882-1904
AD- 1700411.1	GCCCAAAGAAA GUACUCAGGU	819	1885-1905	ACCUGAGUACUTU CUUUGGGCCA	1267	1883-1905
AD- 1700413.1	CCAAAGAAAGU ACUCAGGGAU	820	1887-1907	ATCCCUGAGUACU UUCUUUGGGC	1268	1885-1907
AD- 1700414.1	CAAAGAAAGUA CUCAGGGAGU	821	1888-1908	ACUCCCTGAGUAC UUUCUUUGGG	1269	1886-1908
AD- 1700422.1	GUACUCAGGGA GCUCCUUCGU	822	1896-1916	ACGAAGGAGCUCC CUGAGUACUU	1270	1894-1916
AD- 1700423.1	UACUCAGGGAG CUCCUUCGUU	823	1897-1917	AACGAAGGAGCTC CCUGAGUACU	1271	1895-1917
AD- 1700424.1	ACUCAGGGAGC UCCUUCGUUU	824	1898-1918	AAACGAAGGAGCU CCCUGAGUAC	1272	1896-1918
AD- 1700425.1	CUCAGGGAGCU CCUUCGUUGU	825	1899-1919	ACAACGAAGGAGC UCCUGAGUA	1273	1897-1919
AD- 1700426.1	UCAGGGAGCUC CUUCGUUGGU	826	1900-1920	ACCAACGAAGGAG CUCCCUGAGU	1274	1898-1920
AD- 1700427.1	CAGGGAGCUCC UUCGUUGGAU	827	1901-1921	ATCCAACGAAGGA GCUCCCUGAG	1275	1899-1921
AD- 1700428.1	AGGGAGCUCCU UCGUUGGACU	828	1902-1922	AGUCCAACGAAGG AGCUCCCUGA	1276	1900-1922
AD- 1700439.1	UCGUUGGACCU CCACACUGCU	829	1913-1933	AGCAGUGUGGAGG UCCAACGAAG	1277	1911-1933
AD- 1700461.1	CAAGGCCUGGG CCAUAUGUUU	830	1935-1955	AAACAUAUGGCC AGGCCUUGCA	1278	1933-1955
AD- 1700462.1	AAGGCCUGGGC CAUAUGUUGU	831	1936-1956	ACAACATAUGGCC CAGGCCUUGC	1279	1934-1956
AD- 1700464.1	GGCCUGGGCCA UAUGUUGCUU	832	1938-1958	AAGCAACAUUUGG CCCAGGCCUU	1280	1936-1958
AD- 1700465.1	GCCUGGGCCAU AUGUUGCUGU	833	1939-1959	ACAGCAACAUATG GCCCAGGCCU	1281	1937-1959
AD- 1700468.1	UGGGCCAUAUG UUGCUGGGAU	834	1942-1962	ATCCCAGCAACAU AUGGCCCAGG	1282	1940-1962
AD- 1700473.1	CAUAUGUUGCU GGGAAUUCU	835	1947-1967	AGAAAUTCCCAGC AACAUUUGGC	1283	1945-1967
AD- 1700474.1	AUAUGUUGCUG GGAAUUCU	836	1948-1968	AGGAAATUCCAG CAACAUAUGG	1284	1946-1968
AD- 1700476.1	AUGUUGCUGGG AAUUUCCUCU	837	1950-1970	AGAGGAAAUUCCC AGCAACAUAU	1285	1948-1970

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700477.1	UGUUGCUGGGA AUUUCUCCU	838	1951-1971	AGGAGGAAAUUCC CAGCAACAUA	1286	1949-1971
AD- 1700479.1	UUGCUGGGAU UUCCUCCACU	839	1953-1973	AGUGGAGGAAATU CCCAGCAACA	1287	1951-1973
AD- 1700482.1	CUGGGAUUUC CUCCACCCUU	840	1956-1976	AAGGGUGGAGGAA AUUCCAGCA	1288	1954-1976
AD- 1700485.1	GGAAUUUCCUC CACCCUUCGU	841	1959-1979	ACGAAGGGUGGAG GAAAUUCCCA	1289	1957-1979
AD- 1700486.1	GAAUUUCCUCC ACCCUUCGUU	842	1960-1980	AACGAAGGGUGGA GGAAAUUCCC	1290	1958-1980
AD- 1700487.1	AAUUUCCUCCA CCCUUCGUU	843	1961-1981	AGACGAAGGGUGG AGGAAAUUCC	1291	1959-1981
AD- 1700488.1	AUUUCCUCCAC CCUUCGUCAU	844	1962-1982	ATGACGAAGGGTG GAGGAAAUUC	1292	1960-1982
AD- 1700489.1	UUUCCUCCACC CUUCGUCAUU	845	1963-1983	AAUGACGAAGGGU GGAGGAAAUU	1293	1961-1983
AD- 1700490.1	UUCCUCCACCC UUCGUCAUGU	846	1964-1984	ACAUGACGAAGGG UGGAGGAAAU	1294	1962-1984
AD- 1700517.1	AAGGCCGCCCU CCAUCCUUAU	847	2013-2033	ATAGGAAUGGAGG CGGCCCUUCU	1295	2011-2033
AD- 1700519.1	GGGCCGCCCUCC AUUCCUACUU	848	2015-2035	AAGUAGGAAUGGA GGCGGCCCUU	1296	2013-2035
AD- 1700520.1	GGCCGCCCUCCA UUCCUACUUAU	849	2016-2036	ATAGTAGGAAUGG AGGCGGCCCU	1297	2014-2036
AD- 1700522.1	CCGCCUCCAUAU CCUACUAAGU	850	2018-2038	ACUUAGTAGGAAU GGAGGCGGCC	1298	2016-2038
AD- 1700542.1	CCCACUGCAAA GACUAUGACU	851	2076-2096	AGUCAUAGUCUTU GCAGUGGGUC	1299	2074-2096
AD- 1700544.1	CACUGCAAAGA CUAUGACAGU	852	2078-2098	ACUGTCAUAGUCU UUGCAGUGGG	1300	2076-2098
AD- 1700550.1	AAAGACUAUGA CAGCAUCAAU	853	2084-2104	ATUGAUGCUGUCA UAGUCUUUGC	1301	2082-2104
AD- 1700553.1	GACUAUGACAG CAUCAAUUUU	854	2087-2107	AAAUTUGAUGCTG UCAUAGUCUU	1302	2085-2107
AD- 1700554.1	ACUAUGACAGC AUCAAUUUUU	855	2088-2108	AAAATUTGAUGCU GUCAUAGUCU	1303	2086-2108
AD- 1700555.1	CUAUGACAGCA UCAAAUUUCU	856	2089-2109	AGAAAUTUGAUGC UGUCAUAGUC	1304	2087-2109
AD- 1700556.1	UAUGACAGCAU CAAAUUUCAU	857	2090-2110	ATGAAATUUGATG CUGUCAUAGU	1305	2088-2110
AD- 1700557.1	AUGACAGCAUC AAAUUUCAGU	858	2091-2111	ACUGAAAUUUGAU GCUGUCAUAG	1306	2089-2111
AD- 1700558.1	UGACAGCAUCA AAUUUCAGGU	859	2092-2112	ACCUGAAAUUGA UGCUGUCAUA	1307	2090-2112
AD- 1700560.1	ACAGCAUCAA UUUCAGGACU	860	2094-2114	AGUCCUGAAAUTU GAUGCUGUCA	1308	2092-2114

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700563.1	GCAUCAAUUU CAGGACCUGU	861	2097-2117	ACAGGUCCUGAAA UUUGAUGCUG	1309	2095-2117
AD- 1700566.1	UCAAUUUCAG UGACCUGCAGU	862	2100-2120	ACUGCAGGUCCUG AAAUUUUGAUG	1310	2098-2120
AD- 1700573.1	UCAGGACCUGC AGACAGUACU	863	2107-2127	AGUACUGUCUGCA GGUCCUGAAA	1311	2105-2127
AD- 1700575.1	AGGACCUGCAG ACAGUACAGU	864	2109-2129	ACUGTACUGUCTG CAGGUCCUGA	1312	2107-2129
AD- 1700576.1	GGACCUGCAGA CAGUACAGGU	865	2110-2130	ACCUGUACUGUCU GCAGGUCCUG	1313	2108-2130
AD- 1700578.1	ACCUGCAGACA GUACAGGCUU	866	2112-2132	AAGCCUGUACUGU CUGCAGGUCC	1314	2110-2132
AD- 1700580.1	CUGCAGACAGU ACAGGCUAGU	867	2114-2134	ACUAGCCUGUACU GUCUGCAGGU	1315	2112-2134
AD- 1700582.1	GCAGACAGUAC AGGCUAGAUU	868	2116-2136	AAUCTAGCCUGTA CUGUCUGCAG	1316	2114-2136
AD- 1700623.1	ACGCCUCACAU CCCAAGUCUU	869	2194-2214	AAGACUTGGGATG UGAGGCGUUA	1317	2192-2214
AD- 1700670.1	UCCUAGCCAC CUUUCUGGCU	870	2241-2261	AGCCAGAAAGGTG GCUAGGAAAG	1318	2239-2261
AD- 1700671.1	UCCUAGCCACC UUUCUGGCCU	871	2242-2262	AGGCCAGAAAGGU GGCUAGGAAA	1319	2240-2262
AD- 1700680.1	CCUUUCUGGCC UCCACUUGU	872	2251-2271	ACAAGUGGGAGGC CAGAAAGGUG	1320	2249-2271
AD- 1700681.1	CUUUCUGGCCU CCCACUUGCU	873	2252-2272	AGCAAGTGGGAGG CCAGAAAGGU	1321	2250-2272
AD- 1700684.1	UCUGGCCUCCC ACUUGCCCUU	874	2255-2275	AAGGGCAAGUGGG AGGCCAGAAA	1322	2253-2275
AD- 1700710.1	GGCAUCUUGAU CCCAGCCUGU	875	2302-2322	ACAGGCTGGGATC AAGAUGCCUG	1323	2300-2322
AD- 1700736.1	AAUGCCUCCUG UCCUAAACU	876	2348-2368	AGUUTAGGGACAG GAGGCAUUGG	1324	2346-2368
AD- 1700737.1	AUGCCUCCUGU CCCUAAACUU	877	2349-2369	AAGUTUAGGGACA GGAGGCAUUG	1325	2347-2369
AD- 1700738.1	UGCCUCCUGUC CCUAAACUCU	878	2350-2370	AGAGTUTAGGGAC AGGAGGCAUU	1326	2348-2370
AD- 1700780.1	CAAACUAACAA AAACAUUUCU	879	2434-2454	AGAAAUGUUUUTG UUAGUUUGAG	1327	2432-2454
AD- 1700781.1	AAACUAACAAA AACAUUCCU	880	2435-2455	AGGAAATGUUUUTU GUUAGUUUGA	1328	2433-2455
AD- 1700782.1	AACUAACAAAA ACAUUCCAU	881	2436-2456	ATGGAAAUGUUUTU UGUUAGUUUG	1329	2434-2456
AD- 1700791.1	GUCCUGGGCUG AGAUGCCAGU	882	1269-1289	ACUGGGAUCUCAG CCCAGGACAA	1330	1267-1289
AD- 1700793.1	UGAGAUCCAG GUUUGUAACU	883	1278-1298	AGUUACAAACCUG GGAUCUCAGC	1331	1276-1298

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700794.1	GAGAUGCCAGG UUUGUAACAU	884	1279-1299	AUGUTACAAACCU GGGAUCUCAG	1332	1277-1299
AD- 1700795.1	GAUCCAGGUU UGUAACAGAU	885	1281-1301	AUCUGUTACAAAC CUGGGAUUCU	1333	1279-1301
AD- 1700796.1	CCAGGUUGUA ACAGAAAACU	886	1285-1305	AGUUTUCUGUUAC AAACCUGGGA	1334	1283-1305
AD- 1700797.1	CAGGUUGUAA CAGAAAACAU	887	1286-1306	AUGUTUTUCGUUA CAAACCUGGG	1335	1284-1306
AD- 1700798.1	AGGUUUGUAA AGAAAACACU	888	1287-1307	AGUGTUTUCUGUU ACAAACCUGG	1336	1285-1307
AD- 1700799.1	GGUUGUAAAC GAAAACACCU	889	1288-1308	AGGUGUTUUCUGU UACAAACCUG	1337	1286-1308
AD- 1700800.1	UACAGAAAAC ACCACUAAAU	890	1294-1314	AUUUAGTGGUGUU UUCUGUUACA	1338	1292-1314
AD- 1700801.1	CAGAAAACACC ACUAAAGCCU	891	1297-1317	AGGCTUTAGUGGU GUUUUCUGUU	1339	1295-1317
AD- 1700802.1	CACAGGAGAGA ACCACCCAGU	892	1321-1341	ACUGGGTGGUUCU CUCCUGUGCU	1340	1319-1341
AD- 1700804.1	GAACCAACCAG CCCAGAAGUU	893	1330-1350	AACUTCTGGGCUG GGUGGUUCUC	1341	1328-1350
AD- 1700805.1	AACCACCCAGC CCAGAAGUUU	894	1331-1351	AAACTUCUGGGCU GGGUGGUUCU	1342	1329-1351
AD- 1700806.1	ACCACCCAGCC CAGAAGUUCU	895	1332-1352	AGAACUTCUGGGC UGGGUGGUUC	1343	1330-1352
AD- 1700807.1	CCACCCAGCCC AGAAGUCCU	896	1333-1353	AGGAACUUCUGGG CUGGGUGGUU	1344	1331-1353
AD- 1700808.1	CACCCAGCCCA GAAGUCCAU	897	1334-1354	AUGGAACUUCUGG GCUGGGUGGU	1345	1332-1354
AD- 1700809.1	CCAGAAGUCC AGGGAAGGAU	898	1342-1362	AUCCTUCCUGGA ACUUCUGGGC	1346	1340-1362
AD- 1700810.1	CAGAAGUCCA GGGAAGGAAU	899	1343-1363	AUUCCTCCCUGG AACUUCUGGG	1347	1341-1363
AD- 1700811.1	AGAAGUCCAG GGAAGGAACU	900	1344-1364	AGUUCCTUCCUG GAACUUCUGG	1348	1342-1364
AD- 1700812.1	GAAGUCCAGG GAAGGAACUU	901	1345-1365	AAGUTCCUCCCU GGAACUUCUG	1349	1343-1365
AD- 1700813.1	AAGUCCAGGG AAGGAACUCU	902	1346-1366	AGAGTUCCUCCC UGGAACUUCU	1350	1344-1366
AD- 1700814.1	AGUCCAGGGA AGGAACUCUU	903	1347-1367	AAGAGUTCCUCC CUGGAACUUC	1351	1345-1367
AD- 1700815.1	GUUCCAGGGAA GGAACUCUCU	904	1348-1368	AGAGAGTUCCUUC CCUGGAACUU	1352	1346-1368
AD- 1700816.1	GUCCACCAUGG AGUACCUCUU	905	1370-1390	AAGAGGTACUCCA UGGUGGACCG	1353	1368-1390
AD- 1700817.1	CACCAUGGAGU ACCUCUCAGU	906	1373-1393	ACUGAGAGGUACU CCAUGGUGGA	1354	1371-1393

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700819.1	CCAUGGAGUAC CUCUCAGCUU	907	1375-1395	AAGCTGAGAGGUA CUCCAUGGUG	1355	1373-1395
AD- 1700821.1	AUGGAGUACCU CUCAGCUCUU	908	1377-1397	AAGAGCTGAGAGG UACUCCAUGG	1356	1375-1397
AD- 1700822.1	UUACUCAGGUC AGUAUCUAAU	909	1410-1430	AUUAGATACUGAC CUGAGUAAGU	1357	1408-1430
AD- 1700824.1	UCAGUAUCUAA UAUAAGCUCU	910	1419-1439	AGAGCUTAUAUUA GAUACUGACC	1358	1417-1439
AD- 1700825.1	CAGUAUCUAAU AUAAGCUCGU	911	1420-1440	ACGAGCTUAUAUU AGAUACUGAC	1359	1418-1440
AD- 1700826.1	AGUAUCUAAUA UAAGCUCGGU	912	1421-1441	ACCGAGCUUAUAU UAGAUACUGA	1360	1419-1441
AD- 1700828.1	UCUAAUAUAAG CUCGGAGUUU	913	1425-1445	AAACTCCGAGCUU AUAUUAGAU	1361	1423-1445
AD- 1700829.1	CUAAUAUAAGC UCGGAGUUUU	914	1426-1446	AAAACUCCGAGCU UAUAUUAGAU	1362	1424-1446
AD- 1700830.1	UAAGCUCGGAG UUUGGACGGU	915	1432-1452	ACCGTCCAAACUC CGAGCUUAUA	1363	1430-1452
AD- 1700831.1	AAGCUCGGAGU UUGGACGGAU	916	1433-1453	AUCCGUCCAAACU CCGAGCUUAU	1364	1431-1453
AD- 1700832.1	GGAGUUUGGAC GGAGGGUCUU	917	1439-1459	AAGACCCUCCGUC CAAACUCCGA	1365	1437-1459
AD- 1700833.1	UGGACGGAGGG UCUGGACCUU	918	1445-1465	AAGGTCCAGACCC UCCGUCCAAA	1366	1443-1465
AD- 1700834.1	AGCGACCUUUC CGUGUCUGUU	919	1480-1500	AACAGACACGGAA AGGUCGCUGG	1367	1478-1500
AD- 1700837.1	CAAGCGGACCA UCCGAAAGU	920	1505-1525	ACUUTC CGGAUGG UCCGCUUGUG	1368	1503-1525
AD- 1700838.1	CAUCCGAAAG GCCUGACAGU	921	1514-1534	ACUGTCAGGCCUU UCCGGAUGGU	1369	1512-1534
AD- 1700839.1	AGGAGCUGCUA GCCAAAGCAU	922	1546-1566	AUGCTUTGGCUAG CAGCUCCUGG	1370	1544-1566
AD- 1700840.1	GGAGCUGCUAG CCAAAGCAUU	923	1547-1567	AAUGCUTUGGCUA GCAGCUCCUG	1371	1545-1567
AD- 1700841.1	GAGCUGCUAGC CAAAGCAUUU	924	1548-1568	AAAUGCTUUGGCU AGCAGCUCCU	1372	1546-1568
AD- 1700842.1	AGCUGCUAGCC AAAGCAUUGU	925	1549-1569	ACAATGCUUUGGC UAGCAGCUCC	1373	1547-1569
AD- 1700843.1	CUAGCCAAAGC AUUGGAGACU	926	1554-1574	AGUCTCCAAUGCU UUGGCUAGCA	1374	1552-1574
AD- 1700844.1	UAGCCAAAGCA UUGGAGACCU	927	1555-1575	AGGUCUCCAAUGC UUUGGCUAGC	1375	1553-1575
AD- 1700845.1	AGCAUUGGAGA CCCUACUGCU	928	1562-1582	AGCAGUAGGGUCU CCAAUGCUUU	1376	1560-1582
AD- 1700846.1	GCAUUGGAGAC CCUACUGCUU	929	1563-1583	AAGCAGTAGGGUC UCCAAUGCUU	1377	1561-1583

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700848.1	AUUGGAGACCC UACUGCUGAU	930	1565-1585	AUCAGCAGUAGGG UCUCCA AUGC	1378	1563-1585
AD- 1700850.1	GGAGACCCUAC UGCUGAAUGU	931	1568-1588	ACAUTCAGCAGUA GGGUCUCCAA	1379	1566-1588
AD- 1700851.1	GAGACCCUACU GCUGAAUGGU	932	1569-1589	ACCATUCAGCAGU AGGGUCUCCA	1380	1567-1589
AD- 1700852.1	ACCCUACUGCU GAAUGGAGUU	933	1572-1592	AACUCCA UUCAGC AGUAGGGUCU	1381	1570-1592
AD- 1700853.1	CCCUACUGCUG AAUGGAGUGU	934	1573-1593	ACACTCCA UUCAG CAGUAGGGUC	1382	1571-1593
AD- 1700854.1	CCUACUGCUGA AUGGAGUGCU	935	1574-1594	AGCACUCCA UUCA GCAGUAGGGU	1383	1572-1594
AD- 1700855.1	CUACUGCUGAA UGGAGUGCUU	936	1575-1595	AAGCACTCCA UUC AGCAGUAGGG	1384	1573-1595
AD- 1700856.1	UACUGCUGAAU GGAGUGCUAU	937	1576-1596	AUAGCACUCCA UUC CAGCAGUAGG	1385	1574-1596
AD- 1700858.1	GCUGAAUGGAG UGCUAACCCU	938	1580-1600	AGGGTUAGCACUC CAUUCAGCAG	1386	1578-1600
AD- 1700859.1	CUGAAUGGAGU GCUAACCCU	939	1581-1601	AAGGGUTAGCACU CCAUCAGCA	1387	1579-1601
AD- 1700860.1	UGAAUGGAGUG CUAACCCUGU	940	1582-1602	ACAGGGTUAGCAC UCCAUCAGC	1388	1580-1602
AD- 1700863.1	UGGAGUGC UAA CCCUGGUGCU	941	1586-1606	AGCACCAGGGUUA GCACUCCA U	1389	1584-1606
AD- 1700864.1	GGAGUGC UAA CCUGGUGCUU	942	1587-1607	AAGCACCAGGGUU AGCACUCCA U	1390	1585-1607
AD- 1700865.1	GAGUGC UAA CUGGUGCUAU	943	1588-1608	AUAGCACCAGGGU UAGCACUCCA	1391	1586-1608
AD- 1700867.1	AACCCUGGUGC UAGAGGAGGU	944	1595-1615	ACCUCCTCUAGCA CCAGGGUUA G	1392	1593-1615
AD- 1700868.1	CCCUGGUGC UA GAGGAGGAUU	945	1597-1617	AAUCCUCCUCUAG CACCAGGGUU	1393	1595-1617
AD- 1700869.1	GGUGC UAGAGG AGGAUGGAAU	946	1601-1621	AUUCCATCCUCCU CUAGCACCAG	1394	1599-1621
AD- 1700870.1	GUGC UAGAGGA GGAUGGAACU	947	1602-1622	AGUCCA UCCUCC UCUAGCACCA	1395	1600-1622
AD- 1700871.1	CUAGAGGAGGA UGGAACUGCU	948	1605-1625	AGCAGUTC CAUCC UCCUCUAGCA	1396	1603-1625
AD- 1700872.1	GAGGAUGGAAC UGCAGUGGAU	949	1611-1631	AUCCACTGCAGUU CCAUCCUCCU	1397	1609-1631
AD- 1700873.1	GGAUGGAACUG CAGUGGACAU	950	1613-1633	AUGUCCACUGCAG UCCA UCCUC	1398	1611-1633
AD- 1700874.1	GAUGGAACUGC AGUGGACAGU	951	1614-1634	ACUGTCCACUGCA GUCCA UCCU	1399	1612-1634
AD- 1700875.1	AUGGAACUGCA GUGGACAGUU	952	1615-1635	AACUGUCCACUGC AGUCCA UCC	1400	1613-1635
AD- 1700876.1	UGGAACUGCAG UGGACAGUGU	953	1616-1636	ACACTGTCCACUG CAGUCCAUC	1401	1614-1636

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700877.1	GAACUGCAGUG GACAGUGAGU	954	1618-1638	ACUCCACTGUCCAC UGCAGUUGCA	1402	1616-1638
AD- 1700878.1	ACUGCAGUGGA CAGUGAGGAGU	955	1620-1640	AUCCTCACUGUCC ACUGCAGUUC	1403	1618-1640
AD- 1700879.1	CUGCAGUGGAC AGUGAGGACU	956	1621-1641	AGUCCUCACUGUC CACUGCAGUU	1404	1619-1641
AD- 1700880.1	UGCAGUGGACA GUGAGGACUU	957	1622-1642	AAGUCCTCACUGU CCACUGCAGU	1405	1620-1642
AD- 1700881.1	GCAGUGGACAG UGAGGACUUU	958	1623-1643	AAAGTCCUCACUG UCCACUGCAG	1406	1621-1643
AD- 1700882.1	CAGUGGACAGU GAGGACUUCU	959	1624-1644	AGAAGUCCUCACU GUCCACUGCA	1407	1622-1644
AD- 1700883.1	AGUGGACAGUG AGGACUUCUU	960	1625-1645	AAGAAGTCCUCAC UGUCCACUGC	1408	1623-1645
AD- 1700885.1	UGGACAGUGAG GACUUCUUCU	961	1627-1647	AGAAGAAGUCCUC ACUGUCCACU	1409	1625-1647
AD- 1700887.1	CAGUGAGGACU UCUCCAGCU	962	1631-1651	AGCUGGAAGAAGU CCUCACUGUC	1410	1629-1651
AD- 1700888.1	UGAGGACUUCU UCCAGCUGCU	963	1634-1654	AGCAGCTGGAAGA AGUCCUCACU	1411	1632-1654
AD- 1700889.1	GAGGACUUCUU CCAGCUGCUU	964	1635-1655	AAGCAGCUGGAAG AAGUCCUCAC	1412	1633-1655
AD- 1700891.1	GGACUUCUUCC AGCUGCUGGU	965	1637-1657	ACCAGCAGCUGGA AGAAGUCCUC	1413	1635-1657
AD- 1700892.1	UCUCCAGCUG CUGGAGGAUU	966	1642-1662	AAUCCUCCAGCAG CUGGAAGAAG	1414	1640-1662
AD- 1700893.1	CAGCUGCUGGA GGAUGACACU	967	1647-1667	AGUGTCAUCCUCC AGCAGCUGGA	1415	1645-1667
AD- 1700894.1	GCUGCUGGAGG AUGACACGUU	968	1649-1669	AACGTGTCAUCCU CCAGCAGCUG	1416	1647-1669
AD- 1700895.1	CUGGAGGAUGA CACGUGCCUU	969	1653-1673	AAGGCACGUGUCA UCCUCCAGCA	1417	1651-1673
AD- 1700897.1	GGAUGACACGU GCCUGAUGGU	970	1658-1678	ACCATCAGGCACG UGUCAUCCUC	1418	1656-1678
AD- 1700898.1	GAUGACACGUG CCUGAUGGUU	971	1659-1679	AACCAUCAGGCAC GUGUCAUCCU	1419	1657-1679
AD- 1700899.1	GACACGUGCCU GAUGGUGUUU	972	1662-1682	AAACACCAUCAGG CACGUGUCAU	1420	1660-1682
AD- 1700900.1	GCAGUCUGGUC AGAGCUGGAU	973	1682-1702	AUCCAGCUCUGAC CAGACUGCAA	1421	1680-1702
AD- 1700902.1	UACAAGGAGUG GAGUGCUGUU	974	1706-1726	AACAGCACUCCAC UCCUUGUAGG	1422	1704-1726
AD- 1700903.1	ACAAGGAGUGG AGUGCUGUCU	975	1707-1727	AGACAGCACUCCA CUCCUUGUAG	1423	1705-1727
AD- 1700904.1	AAGGAGUGGAG UGCUGUCAUU	976	1709-1729	AAUGACAGCACUC CACUCCUUGU	1424	1707-1729

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700905.1	GGAGUGGAGUG CUGUCAUAUU	977	1711-1731	AAUATGACAGCAC UCCACUCCUU	1425	1709-1731
AD- 1700906.1	UGGAGUGGUGU CAUAUGGCCU	978	1715-1735	AGGCCATAUGACA GCACUCCACU	1426	1713-1735
AD- 1700907.1	GGAGUGGUGUC AUAUGGCCUU	979	1716-1736	AAGGCCAUUAUGAC AGCACUCCAC	1427	1714-1736
AD- 1700908.1	GAGUGGUGUCA UAUGGCCUGU	980	1717-1737	ACAGGCCAUUAUGA CAGCACUCCA	1428	1715-1737
AD- 1700909.1	AGUGGUGUCAU AUGGCCUGGU	981	1718-1738	ACCAGGCCAUUAUG ACAGCACUCC	1429	1716-1738
AD- 1700912.1	GCUGUCAUAUG GCCUGGGACU	982	1721-1741	AGUCCCAGGCCAU AUGACAGCAC	1430	1719-1741
AD- 1700913.1	CUGUCAUAUGG CCUGGGACGU	983	1722-1742	ACGUCCCAGGCCA UAUGACAGCA	1431	1720-1742
AD- 1700915.1	CAAGCACAGCA AGGACAUCGU	984	1751-1771	ACGATGTCCUUGC UGUGCUUGGG	1432	1749-1771
AD- 1700916.1	AUUCACCUUUG ACGUGUACAU	985	1775-1795	AUGUACACGUCAA AGGUGAAUCG	1433	1773-1795
AD- 1700917.1	UUCACCUUUGA CGUGUACAAU	986	1776-1796	AUUGTACACGUCA AAGGUGAAUC	1434	1774-1796
AD- 1700918.1	UCACCUUUGAC GUGUACAAGU	987	1777-1797	ACUUGUACACGUC AAAGGUGAAU	1435	1775-1797
AD- 1700919.1	CACCUUUGACG UGUACAAGCU	988	1778-1798	AGCUTGTACACGU CAAAGGUGAA	1436	1776-1798
AD- 1700920.1	CCUUUGACGUG UACAAGCAAU	989	1780-1800	AUUGCUTGUACAC GUCAAAGGUG	1437	1778-1800
AD- 1700922.1	UGACGUGUACA AGCAAAACCU	990	1784-1804	AGGUTUTGCUUGU ACACGUCAAA	1438	1782-1804
AD- 1700923.1	GACGUGUACAA GCAAAACCCU	991	1785-1805	AGGGTUTUGCUUG UACACGUCAA	1439	1783-1805
AD- 1700924.1	ACGUGUACAAG CAAAACCCUU	992	1786-1806	AAGGGTUTUGCUU GUACACGUCA	1440	1784-1806
AD- 1700925.1	CGUGUACAAGC AAAACCCUCU	993	1787-1807	AGAGGGTUUUGCU UGUACACGUC	1441	1785-1807
AD- 1700929.1	CAAGCAAAACC CUCGAGACCU	994	1793-1813	AGGUCUCGAGGGU UUUGCUUGUA	1442	1791-1813
AD- 1700930.1	AAGCAAAACCC UCGAGACCUU	995	1794-1814	AAGGTCTCGAGGG UUUUGCUUGU	1443	1792-1814
AD- 1700931.1	AGCAAAACCCU CGAGACCUCU	996	1795-1815	AGAGGUCUCGAGG GUUUUGCUUG	1444	1793-1815
AD- 1700934.1	GAGACCUCUUU GGCAGCCUGU	997	1807-1827	ACAGGCTGCCAAA GAGGUCUCGA	1445	1805-1827
AD- 1700936.1	CCUCUUUGGCA GCCUGAAUGU	998	1811-1831	ACAUTCAGGCUGC CAAAGAGGUC	1446	1809-1831
AD- 1700937.1	CUCUUUGGCAG CCUGAAUGUU	999	1812-1832	AACATUCAGGCUG CCAAAGAGGU	1447	1810-1832

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700938.1	CUUUGGCAGCC UGAAUGUCAU	1000	1814-1834	AUGACATUCAGGC UGCCAAAGAG	1448	1812-1834
AD- 1700939.1	UUUGGCAGCCU GAAUGUCAAU	1001	1815-1835	AUUGACAUUCAGG CUGCCAAAGA	1449	1813-1835
AD- 1700940.1	UGGCAGCCUGA AUGUCAAGAU	1002	1817-1837	ACUUTGACAUUCA GGCUGCCAAA	1450	1815-1837
AD- 1700942.1	GCAGCCUGAAU GUCAAAGCCU	1003	1819-1839	AGGCTUTGACAUU CAGGCUGCCA	1451	1817-1839
AD- 1700943.1	CAGCCUGAAUG UCAAGCCAU	1004	1820-1840	AUGGCUTUGACAU UCAGGCUGCC	1452	1818-1840
AD- 1700944.1	AGCCUGAAUGU CAAAGCCACU	1005	1821-1841	AGUGGCTUUGACA UUCAGGCUGC	1453	1819-1841
AD- 1700947.1	AUGUCAAGGCC ACAUCUACU	1006	1828-1848	AGUAGAAUGUGGC UUUGACAUUC	1454	1826-1848
AD- 1700948.1	AGCCACAUCU ACGGGCUCUU	1007	1835-1855	AAGAGCCCGUAGA AUGUGGCUUU	1455	1833-1855
AD- 1700949.1	GCCACAUCUA CGGGCUCUAU	1008	1836-1856	AUAGAGCCCGUAG AAUGUGGCUU	1456	1834-1856
AD- 1700950.1	CACAUCUACG GGCUCUACUU	1009	1838-1858	AAGUAGAGCCCGU AGAAUGUGGC	1457	1836-1858
AD- 1700952.1	CAUUCUACGGG CUCUACUCUU	1010	1840-1860	AAGAGUAGAGCCC GUAGAAUGUG	1458	1838-1860
AD- 1700953.1	AUUCUACGGGC UCUACUCUAU	1011	1841-1861	AUAGAGTAGAGCC CGUAGAAUGU	1459	1839-1861
AD- 1700955.1	UCUACGGGCUC UACUCUAUGU	1012	1843-1863	ACAUAGAGUAGAG CCCGUAGAAU	1460	1841-1863
AD- 1700957.1	ACGGGCUCUAC UCUAUGAGUU	1013	1846-1866	AACUCATAGAGUA GAGCCCGUAG	1461	1844-1866
AD- 1700958.1	GGGCUCUACUC UAUGAGUUGU	1014	1848-1868	ACAACUCAUAGAG UAGAGCCCGU	1462	1846-1868
AD- 1700959.1	GGCUCUACUCU AUGAGUUGUU	1015	1849-1869	AACAACTCAUAGA GUAGAGCCCG	1463	1847-1869
AD- 1700960.1	CUACUCUAUGA GUUGUGACUU	1016	1853-1873	AAGUCACAACUCA UAGAGUAGAG	1464	1851-1873
AD- 1700961.1	UACUCUAUGAG UUGUGACUUU	1017	1854-1874	AAAGTCACAACUC AUAGAGUAGA	1465	1852-1874
AD- 1700962.1	ACUCUAUGAGU UGUGACUUUU	1018	1855-1875	AAAAGUCACAACU CAUAGAGUAG	1466	1853-1875
AD- 1700963.1	CUCUAUGAGUU GUGACUUUCU	1019	1856-1876	AGAAAGTCACAAC UCAUAGAGUA	1467	1854-1876
AD- 1700965.1	CUAUGAGUUGU GACUUUCAAU	1020	1858-1878	AUUGAAAGUCACA ACUCAUAGAG	1468	1856-1878
AD- 1700967.1	GAGUUGUGACU UUCAAGGACU	1021	1862-1882	AGUCCUTGAAAGU CACAAACUACU	1469	1860-1882
AD- 1700968.1	AGUUGUGACUU UCAAGGACUU	1022	1863-1883	AAGUCCTUGAAAG UCACAACUCA	1470	1861-1883

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEb dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1700969.1	UUGGCCCAAAG AAAGUACUCU	1023	1882-1902	AGAGTACUUUCUU UGGGCCAAAGU	1471	1880-1902
AD- 1700970.1	UGGCCCAAAGA AAGUACUCAU	1024	1883-1903	AUGAGUACUUUCU UUGGGCCAAG	1472	1881-1903
AD- 1700971.1	GGCCCAAAGAA AGUACUCAGU	1025	1884-1904	ACUGAGTACUUUC UUUGGGCCAA	1473	1882-1904
AD- 1700973.1	CCCAAAGAAAG UACUCAGGGU	1026	1886-1906	ACCCTGAGUACUU UCUUUGGGCC	1474	1884-1906
AD- 1700975.1	AAAGAAAGUAC UCAGGGAGCU	1027	1889-1909	AGCUCUCCUGAGUA CUUUCUUUGG	1475	1887-1909
AD- 1700976.1	AAGAAAGUACU CAGGGAGCUU	1028	1890-1910	AAGCTCCCUGAGU ACUUUCUUUG	1476	1888-1910
AD- 1700977.1	AGAAAGUACUC AGGGAGCUCU	1029	1891-1911	AGAGCUCCUGAG UACUUUCUUU	1477	1889-1911
AD- 1700978.1	AGUACUCAGGG AGCUCCUUCU	1030	1895-1915	AGAAGGAGCUCCC UGAGUACUUU	1478	1893-1915
AD- 1700980.1	AGGGAGCUCCU UCGUUGGACU	1031	1902-1922	AGUCCAACGAAGG AGCUCUCCUGA	1479	1900-1922
AD- 1700981.1	GGAGCUCCUUC GUUGGACCUU	1032	1904-1924	AAGGTCCAACGAA GGAGCUCCCU	1480	1902-1924
AD- 1700982.1	UUCGUUGGACC UCCACACUGU	1033	1912-1932	ACAGTGTGGAGGU CCAACGAAGG	1481	1910-1932
AD- 1700984.1	CGUUGGACCUC CACACUGCUU	1034	1914-1934	AAGCAGTGUGGAG GUCCAACGAA	1482	1912-1934
AD- 1700985.1	GACCUCCACAC UGCUGCAAGU	1035	1919-1939	ACUUGCAGCAGUG UGGAGGUCCA	1483	1917-1939
AD- 1700986.1	GGCCUGGGCCA UAUGUUGCUU	1036	1938-1958	AAGCAACUAUUGG CCCAGGCCUU	1484	1936-1958
AD- 1700988.1	GGGCCAU AUGU UGCUGGAAU	1037	1943-1963	AUUCCAGCAACA UAUGGCCAG	1485	1941-1963
AD- 1700989.1	GGCCAU AUGUU GCUUGGAAUU	1038	1944-1964	AAUCCCAGCAAC AUAUGGCCA	1486	1942-1964
AD- 1700990.1	CCAU AUGUUGC UGGAAUUUU	1039	1946-1966	AAAATUCCAGCA ACAU AUGGCC	1487	1944-1966
AD- 1700991.1	CAU AUGUUGC GGGAAUUUCU	1040	1947-1967	AGAAAUTCCCAGC AACAU AUGGC	1488	1945-1967
AD- 1700992.1	UGUUGCUGGGA AUUUCUCCU	1041	1951-1971	AGGAGGAAAUUCC CAGCAACAU	1489	1949-1971
AD- 1700995.1	UGGGAUUUCC UCCACUUUU	1042	1957-1977	AAAGGTTGGAGGA AAUUCUCCAGC	1490	1955-1977
AD- 1700997.1	UUCUCCACCC UUCGUCAUGU	1043	1964-1984	ACAUGACGAAGGG UGGAGGAAAU	1491	1962-1984
AD- 1700998.1	AAGGGCCGCCU CCAUUCUUAU	1044	2013-2033	AUAGGAAUGGAGG CGGCCUUUCU	1492	2011-2033
AD- 1701001.1	GCCGCCUCCAU UCCUACUAAU	1045	2017-2037	AUUAGUAGGAAUG GAGGCGGCC	1493	2015-2037

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1701002.1	CCGCCUCCA U CCUACUAAGU	1046	2018-2038	ACUUGAGTAGGAAU GGAGGCGGGC	1494	2016-2038
AD- 1701003.1	CCCACUGCAAA GACUAUGACU	1047	2076-2096	AGUCAUAGUCUUU GCAGUGGGUC	1495	2074-2096
AD- 1701004.1	CCACUGCAAAG ACUAUGACAU	1048	2077-2097	AUGUCATAGUCUU UGCAGUGGGU	1496	2075-2097
AD- 1701005.1	CACUGCAAAGA CUAUGACAGU	1049	2078-2098	ACUGTCAUAGUCU UUGCAGUGGG	1497	2076-2098
AD- 1701007.1	AAGACUAUGAC AGCAUCAAU	1050	2085-2105	AUUUGATGCUGUC AUAGUCUUUG	1498	2083-2105
AD- 1701009.1	GACAGCAUCAA AUUUCAGGAU	1051	2093-2113	AUCCTGAAAUUUG AUGCUGUCAU	1499	2091-2113
AD- 1701011.1	CAGCAUCAAU UUCAGGACCU	1052	2095-2115	AGGUCCTGAAAUU UGAUGCUGUC	1500	2093-2115
AD- 1701012.1	AGCAUCAAUU UCAGGACCUU	1053	2096-2116	AAGGTCCUGAAAU UUGAUGCUGU	1501	2094-2116
AD- 1701013.1	GCAUCAAUUU CAGGACCUGU	1054	2097-2117	ACAGGUCCUGAAA UUUGAUGCUG	1502	2095-2117
AD- 1701014.1	CAUCAAUUUC AGGACCUGCU	1055	2098-2118	AGCAGGTCCUGAA AUUUGAUGCU	1503	2096-2118
AD- 1701016.1	AUUUCAGGACC UGCAGACAGU	1056	2104-2124	ACUGTCTGCAGGU CCUGAAAUUU	1504	2102-2124
AD- 1701017.1	UUUCAGGACCU GCAGACAGUU	1057	2105-2125	AACUGUCUGCAGG UCCUGAAAUU	1505	2103-2125
AD- 1701019.1	CAGGACCUGCA GACAGUACAU	1058	2108-2128	AUGUACTGUCUGC AGGUCCUGAA	1506	2106-2128
AD- 1701020.1	GACCUGCAGAC AGUACAGGCU	1059	2111-2131	AGCCTGTACUGUC UGCAGGUCCU	1507	2109-2131
AD- 1701022.1	CCUGCAGACAG UACAGGCUAU	1060	2113-2133	AUAGCCTGUACUG UCUGCAGGUC	1508	2111-2133
AD- 1701023.1	CUGCAGACAGU ACAGGCUAGU	1061	2114-2134	ACUAGCCUGUACU GUCUGCAGGU	1509	2112-2134
AD- 1701024.1	UGCAGACAGUA CAGGCUAGAU	1062	2115-2135	AUCUAGCCUGUAC UGUCUGCAGG	1510	2113-2135
AD- 1701026.1	CAGACAGUACA GGCUAGAUU	1063	2117-2137	AUAUCUAGCCUGU ACUGUCUGCA	1511	2115-2137
AD- 1701027.1	GACAGUACAGG CUAGAUAAU	1064	2119-2139	AGUUUUCUAGCCU GUACUGUCUG	1512	2117-2139
AD- 1701029.1	CCUAGCCACCU UUCUGGCCUU	1065	2243-2263	AAGGCCAGAAAGG UGGCUAGGAA	1513	2241-2263
AD- 1701031.1	CUUUCUGGCCU CCCACUUGCU	1066	2252-2272	AGCAAGTGGGAGG CCAGAAAGGU	1514	2250-2272
AD- 1701032.1	UCUGGCCUCCC ACUUGCCUUU	1067	2255-2275	AAGGGCAAGUGGG AGGCCAGAAA	1515	2253-2275
AD- 1701033.1	GGCAUCUUGAU CCCAGCCUGU	1068	2302-2322	ACAGGCTGGGAUC AAGAUGCCUG	1516	2300-2322

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1701034.1	GCAUCUUGAUC CCAGCCUGAU	1069	2303-2323	AUCAGGCUGGGAU CAAGAUGCCU	1517	2301-2323
AD- 1701035.1	ACCAAUGCCUC CUGUCCCUAU	1070	2345-2365	AUAGGGACAGGAG GCAUUGGUAG	1518	2343-2365
AD- 1701037.1	AUGCCUCCUGU CCCUAAACUU	1071	2349-2369	AAGUTUAGGGACA GGAGGCAUUG	1519	2347-2369
AD- 1701038.1	GCCUCCUGUCC CUAAACUCCU	1072	2351-2371	AGGAGUTUAGGGA CAGGAGGCAU	1520	2349-2371
AD- 1720280.1	AAAACCCUCGA GACCUCUUUU	1073	1798-1818	AAAAGAGGUCUCG AGGGUUUUGC	1521	1796-1818
AD- 1720281.1	CUGGGCUGAGA UCCCGAGUUU	1074	1272-1292	AAACCTGGGAUCU CAGCCAGGA	1522	1270-1292
AD- 1720282.1	GUGGACAGUGA GGACUUCUUU	1075	1626-1646	AAAGAAGUCCUCA CUGUCCACUG	1523	1624-1646
AD- 1720283.1	CAAAACCCUCG AGACCUCUUU	1076	1797-1817	AAAGAGGTCUCGA GGGUUUUGCU	1524	1795-1817
AD- 1720284.1	CUGAAUGUCAA AGCCACAUUU	1077	1824-1844	AAATGTGGCUUTG ACAUUCAGGC	1525	1822-1844
AD- 1720285.1	GGCUCUACUCU AUGAGUUGUU	1078	1849-1869	AACAACTCAUAGA GUAGAGCCCG	1526	1847-1869
AD- 1720286.1	AAUGGAGUGCU AACCUGGUUU	1079	1584-1604	AACCAGGGUUAGC ACUCCAUUCA	1527	1582-1604
AD- 1720287.1	AAUGGAGUGCU AACCUGGUUU	1080	1584-1604	AACCAAGGUUAGC ACUCCAUUCA	1528	1582-1604
AD- 1720288.1	AAUGGAGUGCU AACUCUGGUUU	1081	1584-1604	AACCAGAGUUAGC ACUCCAUUCA	1529	1582-1604
AD- 1720289.1	ACCCUACUGCU GAAUGGAGUU	1082	1572-1592	AACUCCAUUCAGC AGUAGGGUCU	1530	1570-1592
AD- 1720290.1	ACCUGCAGACA GUACAGGCUU	1083	2112-2132	AAGCCTGUACUGU CUGCAGGUCC	1531	2110-2132
AD- 1720291.1	CUGGGAAUUUC CUCCACCCUU	1084	1956-1976	AAGGGTGGAGGAA AUUCCAGCA	1532	1954-1976
AD- 1720292.1	GGGCCGCCUCC AUUCCUACUU	1085	2015-2035	AAGUAGGAAUGGA GGCGGCCCUU	1533	2013-2035
AD- 1720293.1	UUCUACGGGCU CUACUCUAUU	1086	1842-1862	AAUAGAGUAGAGC CCGUAGAAUG	1534	1840-1862
AD- 1720294.1	GCAGACAGUAC AGGCUAGAUU	1087	2116-2136	AAUCUAGCCUGTA CUGUCUGCAG	1535	2114-2136
AD- 1720295.1	UGGAGACCCUA CUGCUGAAUU	1088	1567-1587	AAUTCAGCAGUAG GGUCUCCAAU	1536	1565-1587
AD- 1720296.1	ACCUCUUUGGC AGCCUGAAUU	1089	1810-1830	AAUTCAGGCUGCC AAAGAGGUCU	1537	1808-1830
AD- 1720297.1	CCUUUCUGGCC UCCACUUGU	1090	2251-2271	ACAAGTGGGAGGC CAGAAAGGUG	1538	2249-2271
AD- 1720298.1	UGGAACUGCAG UGGACAGUGU	1091	1616-1636	ACACTGTCCACTG CAGUUCCAUC	1539	1614-1636

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1720299.1	AGGAUGACACG UGCCUGAUGU	1092	1657-1677	ACATCAGGCACGU GUCAUCCUCC	1540	1655-1677
AD- 1720300.1	GAAUGGAGUGC UAACCCUGGU	1093	1583-1603	ACCAGGGTUAGCA CUCCAUUCAG	1541	1581-1603
AD- 1720301.1	GUGCUGUCAUA UGGCCUGGGU	1094	1719-1739	ACCCAGGCCAUAU GACAGCACUC	1542	1717-1739
AD- 1720302.1	GUGCUGUCAUA UGGCUUGGGU	1095	1719-1739	ACCCAAGCCAUAU GACAGCACUC	1543	1717-1739
AD- 1720303.1	GUGCUGUCAUA UGGUCUGGGU	1096	1719-1739	ACCCAGACCAUUA GACAGCACUC	1544	1717-1739
AD- 1720304.1	GCCCAAAGAAA GUACUCAGGU	1097	1885-1905	ACCTGAGUACUTU CUUUGGGCCA	1545	1883-1905
AD- 1720305.1	GUACUCAGGGA GCUCCUUCGU	1098	1896-1916	ACGAAGGAGCUCC CUGAGUACUU	1546	1894-1916
AD- 1720306.1	GGAAUUUCCUC CACCCUUCGU	1099	1959-1979	ACGAAGGGUGGAG GAAAUUCCCA	1547	1957-1979
AD- 1720307.1	GGAAUUUCCUC CACCUUUCGU	1100	1959-1979	ACGAAAGGUGGAG GAAAUUCCCA	1548	1957-1979
AD- 1720308.1	GGAAUUUCCUC CACUUCUUCGU	1101	1959-1979	ACGAAGAGUGGAG GAAAUUCCCA	1549	1957-1979
AD- 1720309.1	GUGUACAAGCA AAACCCUCGU	1102	1788-1808	ACGAGGGTUUUGC UUGUACACGU	1550	1786-1808
AD- 1720310.1	CAGUCUGGUCA GAGCUGGAGU	1103	1683-1703	ACUCCAGCUCUGA CCAGACUGCA	1551	1681-1703
AD- 1720311.1	UGCUAACCCUG GUGCUAGAGU	1104	1591-1611	ACUCUAGCACCAG GGUUAGCACU	1552	1589-1611
AD- 1720312.1	UCAAUUUUCAG GACCUGCAGU	1105	2100-2120	ACUGCAGGUCCTG AAAUUUGAUG	1553	2098-2120
AD- 1720313.1	AGAACCACCCA GCCCAAGAAGU	1106	1329-1349	ACUTCTGGGCUGG GUGGUUCUCU	1554	1327-1349
AD- 1720314.1	CAAAACUAACAA AAACAUUUCU	1107	2434-2454	AGAAATGUUUUTG UUAGUUUGAG	1555	2432-2454
AD- 1720315.1	CAUGGAGUACC UCUCAGCUCU	1108	1376-1396	AGAGCTGAGAGGU ACUCCAUGGU	1556	1374-1396
AD- 1720316.1	UCGUUGGACCU CCACACUGCU	1109	1913-1933	AGCAGTGUGGAGG UCCAACGAAG	1557	1911-1933
AD- 1720317.1	ACCAUGGAGUA CCUCUCAGCU	1110	1374-1394	AGCTGAGAGGUAC UCCAUGGUGG	1558	1372-1394
AD- 1720368.1	GGCAGCCUGAA UGUCAAGCU	1111	1818-1838	AGCTUTGACAUTC AGGCUGCCAA	1559	1816-1838
AD- 1720369.1	UCCUAGCCACC UUUCUGGCCU	1112	2242-2262	AGGCCAGAAAGGU GGCUAGGAAA	1560	2240-2262
AD- 1720370.1	UGCUGAAUGGA GUGCUAACCU	1113	1579-1599	AGGTUAGCACUCC AUUCAGCAGU	1561	1577-1599
AD- 1720371.1	UCAGGACCUGC AGACAGUACU	1114	2107-2127	AGUACTGUCUGCA GGUCCUGAAA	1562	2105-2127

TABLE 5-continued

Unmodified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1	Antisense Sequence 5' to 3'	SEQ ID NO	Range in NM_001393338.1
AD- 1720372.1	ACAGCAUCAA UUUCAGGACU	1115	2094-2114	AGUCCTGAAUTU GAUGCUGUCA	1563	2092-2114
AD- 1720373.1	UUGCUGGGAAU UUCCUCCACU	1116	1953-1973	AGUGGAGGAAATU CCCAGCAACA	1564	1951-1973
AD- 1720374.1	AAUGCCUCCUG UCCCUAAACU	1117	2348-2368	AGUTUAGGGACAG GAGGCAUUGG	1565	2346-2368
AD- 1720375.1	UUGACGUGUAC AAGCAAAACU	1118	1783-1803	AGUTUTGCUUGTA CACGUCAAAG	1566	1781-1803
AD- 1720376.1	GGCCGCCUCCA UUCCUACUUAU	1119	2016-2036	ATAGUAGGAAUGG AGGCGGCCCU	1567	2014-2036
AD- 1720377.1	ACUCAGGUCAG UAUCUAAUUAU	1120	1412-1432	ATATUAGAUACTG ACCUGAGUAA	1568	1410-1432
AD- 1720378.1	UGGGCCAUUAG UUGCUGGGAU	1121	1942-1962	ATCCCAGCAACAU AUGGCCCAGG	1569	1940-1962
AD- 1720379.1	UGCUGUCAUUAU GGCCUGGGAU	1122	1720-1740	ATCCCAGGCCATA UGACAGCACU	1570	1718-1740
AD- 1720380.1	UGAGUUGUGAC UUUCAAGGAU	1123	1861-1881	ATCCUTGAAAGTC ACAACUCAUA	1571	1859-1881
AD- 1720381.1	AGGCCCAAGCA CAGCAAGGAU	1124	1746-1766	ATCCUTGCUGUGC UUGGGCCUCU	1572	1744-1766
AD- 1720382.1	UGUACAAGCAA AACCUCUGAU	1125	1789-1809	ATCGAGGGUUUTG CUUGUACACG	1573	1787-1809
AD- 1720383.1	UGUACAAGCAA AACCUCUGAU	1126	1789-1809	ATCGAAGGUUUTG CUUGUACACG	1574	1787-1809
AD- 1720384.1	UGUACAAGCAA AACUCUCGAU	1127	1789-1809	ATCGAGAGUUUTG CUUGUACACG	1575	1787-1809
AD- 1720385.1	UCUAUGAGUUG UGACUUUCAU	1128	1857-1877	ATGAAAGUCACAA CUCAUAGAGU	1576	1855-1877
AD- 1720386.1	AAAGACUAUGA CAGCAUCAAU	1129	2084-2104	ATUGATGCUGUCA UAGUCUUUGC	1577	2082-2104

TABLE 6

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO	Antisense Sequence 5' to 3'	SEQ ID NO	mRNA target sequence	SEQ ID NO
AD- 1699864.1	gsusccugggCfUfGf agaucccagL96	1578	asdCsugdGgdAucucdAgCf ccaggacsasa	2026	GGAAGGAAAGAGCA GAUCCAGG	2474
AD- 1699867.1	csusgggcugAfGfAf ucccagguuL96	1579	asdAsacdCudGggaudCuCf agcccagsgsa	2027	AGGAAAGAGCAGAU CCCAGGUUU	2475
AD- 1699868.1	usgsggcugaGfAfUf cccagguuuL96	1580	asdAsaadCcdTgggadTcUf cagcccagsgg	2028	GGAAAGAGCAGAU CCAGGUUUU	2476
AD- 1699869.1	gsgsgcugagAfUfCf ccagguuuguL96	1581	asdCsaadAcdCugggdAuCf ucagcccsasg	2029	GAAAGAGCAGAUCC CAGGUUUUGU	2477

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1699870.1	gsgscugagaUfCfCf cagguuuguuL96	1582	asdAscAdAadCcuggdGaUf cucagccscsa	2030	AAAGAGCAGAUCCC AGGUUUGUA	2478
AD- 1699871.1	gscsugagauCfCfCf agguuuguauL96	1583	asdTsacdAadAccugdGgAf ucucagcscsc	2031	AAGAGCAGAUCCCA GGUUUGUAA	2479
AD- 1699872.1	csusgagauCfCfAf gguuuguaauL96	1584	asdTsuaCdAadAccudGgGf aucucagcscsc	2032	AGAGCAGAUCCCAG GUUUGUAA	2480
AD- 1699873.1	usgsagauccCfAfGf guuuguaacuL96	1585	asdGsuaCdAadAaaccdTgGf gaucucagsgsc	2033	GAGCAGAUCCCAGG UUUGUAACA	2481
AD- 1699875.1	asgsaucccaGfGfUf uuguaacaguL96	1586	asdCsugdTuDaadAadCcUf gggaucuscscsa	2034	GCAGAUCCCAGGUU UGUAACAGA	2482
AD- 1699882.1	asgsguuuguAfAfCf agaaaaacacuL96	1587	asdGsugdTuDtucugdTuAf caaaccusgsg	2035	CCAGGUUUUGUAACA GAAAACACC	2483
AD- 1699891.1	ascsagaaaaCfAfCf cacuaaagcuL96	1588	asdGscudTuDaugugdTuUf uuucugususa	2036	UAACAGAAAACACC ACUAAGGCC	2484
AD- 1699892.1	csasgaaacAfCfCf acuaaagccuL96	1589	asdGsgcdTuDtugugdGuGf uuuucugsusu	2037	AACAGAAAACACCA CUAAGCCC	2485
AD- 1699904.1	asgsaaccacCfCfAf gcccaagaauL96	1590	asdCsuaCdCudGggcdGgGf ugguucuscscu	2038	AGAGAACACCCAG CCCAGAAGU	2486
AD- 1699907.1	ascsaccccaGfCfCf cagaaguucuL96	1591	asdGsaadCudTcuggdGcUf ggguggususc	2039	GAACCACCAGCCC AGAAGUCC	2487
AD- 1699948.1	csasccauggAfGfUf accucucaguL96	1592	asdCsugdAgdAgguadCuCf cauggugsgsa	2040	UCCACCAUGGAGUA CCUCUCAGC	2488
AD- 1699949.1	ascscauggaGfUfAf ccucucagcuL96	1593	asdGscudGadGaggudAcUf ccauggusgsg	2041	CCACCAUGGAGUAC CUCUCAGCU	2489
AD- 1699951.1	csasuggaguAfCfCf ucucagcucuL96	1594	asdGsagdCudGagagdGuAf cuccaugsgsu	2042	ACCAUGGAGUACCU CUCAGCUCU	2490
AD- 1699964.1	csusuacucaGfGfUf caguaucuaL96	1595	asdTsagdAudAcugadCcUf gaguaagsusc	2043	GACUUACUCAGGUC AGUAUCUAA	2491
AD- 1699967.1	ascsucagguCfAfGf uaucuaauauL96	1596	asdTsaudTadGauacdTgAf ccugagusasa	2044	UUACUCAGGUCAGU AUCUAUAU	2492
AD- 1699970.1	csasggucagUfAfUf cuauauauauL96	1597	asdTsuaCdTadTuagadTaCf ugaccugsasg	2045	CUCAGGUCAGUAUC UAAUAUAAG	2493
AD- 1699971.1	asgsgucaguAfUfCf uaauauaaguL96	1598	asdCsuaCdAudAuagadAuAf cugaccusgsa	2046	UCAGGUCAGUAUCU AAUAUAAGC	2494
AD- 1699972.1	gsgsucaguaUfCfUf aauuaaagcuL96	1599	asdGscudTadTauuadGaUf acugaccsusg	2047	CAGGUCAGUAUCUA AUUAUAAGCU	2495
AD- 1699973.1	gsuscagauCfUfAf auauaagcuL96	1600	asdAsgcdTuDaauuadAgAf uacugacscsu	2048	AGGUCAGUAUCUAA UAUAAGCUC	2496
AD- 1699976.1	asgsuaucuaAfUfAf uaagcucggauL96	1601	asdCsagdAgdCuuaudAuUf agauacugsa	2049	UCAGUAUCUAUAU AAGCUCGGA	2497
AD- 1699977.1	gsusaucuaAfUfAf aagcucggauL96	1602	asdTsacdGadGcuuadTaUf uagauacsug	2050	CAGUAUCUAUAUA AGCUCGGAG	2498
AD- 1699978.1	usasucuaauAfUfAf agcucggaguL96	1603	asdCsucdCgdAgcuuadAuAf uuagauascsu	2051	AGUAUCUAUAUA GCUCGGAGU	2499
AD- 1699979.1	asuscuauaAfUfAf gcucggaguL96	1604	asdAscudCcdGagcuuadTaUf auuagausasc	2052	GUAUCUAUAUAAG CUCGGAGUU	2500

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1699982.1	usasaauuaaGfCfUf cggaguuguL96	1605	asdCsaadAcdTccgadGcUf uauauuasgsa	2053	UCUAAUUAAGCUC GGAGUUUGG	2501
AD- 1699983.1	asasaauaagCfUfCf ggaguugguL96	1606	asdCscadAadCuccgdAgCf uuauauusag	2054	CUAAUUAAGCUCG GAGUUUGGA	2502
AD- 1699984.1	asusaauaagCfCfGf gaguuggauL96	1607	asdTscddAadAcuccdGaGf cuuauaususa	2055	UAAUUAAGCUCGG AGUUUGGAC	2503
AD- 1699985.1	usasaagcuCfGfGf aguugggacuL96	1608	asdGsucdCadAacudCgAf gcuuauasusu	2056	AAUUAAGCUCGGA GUUUGGACG	2504
AD- 1699986.1	asusaagcuCfGfAf guuugggacguL96	1609	asdCsgudCcdAaacudCcGf agcuuauasusu	2057	AUAUAAGCUCGGAG UUUGGACGG	2505
AD- 1699987.1	usasagcuCfGfAfGf uuuggcggguL96	1610	asdCscgdTcdCaaacdTcCf gagcuuasusa	2058	UAUAAGCUCGGAGU UUGGACGGA	2506
AD- 1699988.1	asasgcucggAfGfUf uuggcggauL96	1611	asdTscddGudCcaaadCuCf cgagcuuasusu	2059	AUAAGCUCGGAGUU UGGACGGAG	2507
AD- 1699989.1	asgsucgggaGfUfUf uggcggaguL96	1612	asdCsucdCgdTccaadAcUf ccgagcususa	2060	UAAGCUCGGAGUUU GGACGGAGG	2508
AD- 1699990.1	gsacsucggagUfUfUf ggacggaggguL96	1613	asdCscudCcdGuccadAaCf uccgagcsusu	2061	AAGCUCGGAGUUUG GACGGAGGG	2509
AD- 1700015.1	asgsccgaccuUfUfCf cgugucuguuL96	1614	asdAscadGadCacggdAaAf ggucgcusgsg	2062	CCAGCGACCUUUC GUGUCUGUG	2510
AD- 1700035.1	gsasucacaaGfCfGf gaccauccguL96	1615	asdCsaggdAudGguccdGcUf ugugaucsasc	2063	GUGAUCACAAGCGG ACCAUCCGG	2511
AD- 1700037.1	uscsacaagcGfGfAf ccauccggauL96	1616	asdTscddGgdAuggudCcGf cuugugasusc	2064	GAUCACAAGCGGAC CAUCCGGAA	2512
AD- 1700038.1	csascaagcgGfAfCf cauccggaauL96	1617	asdTsucdCgdGauggdTcCf gcuugugsasu	2065	AUCACAAGCGGACC AUCCGAAA	2513
AD- 1700039.1	ascsaagcgGfAfCfCf auccggaaauL96	1618	asdTsuudCcdGgaugdGuCf cgcuugusgsa	2066	UCACAAGCGGACCA UCCGGAAAG	2514
AD- 1700040.1	csasagcggaCfCfAf uccggaagguL96	1619	asdCsuidTcdCggauGgUf ccgcuugsusg	2067	CACAAGCGGACCAU CCGGAAGG	2515
AD- 1700041.1	asasgcggacCfAfUf ccggaagguL96	1620	asdCsudTudCcggaTgGf uccgcuusgsu	2068	ACAAGCGGACCAUC CGGAAAGGC	2516
AD- 1700077.1	gsagsagcuCfAfGf ccaaagcauuL96	1621	asdAsugdCudTuggcdTaGf cagcuccsusg	2069	CAGGAGCUGCUAGC CAAAGCAUU	2517
AD- 1700080.1	gscsugcuagCfCfAf aagcauugguL96	1622	asdCscadAudGcuuudGgCf uagcagcsusc	2070	GAGCUGCUAGCCAA AGCAUUGGA	2518
AD- 1700081.1	csusgcuagCfAfAf agcauuggauL96	1623	asdTscddAadTgcuidTgGf cuagcagcsu	2071	AGCUGCUAGCCAAA GCAUUGGAG	2519
AD- 1700082.1	usgsuagccAfAfAf gcauuggaguL96	1624	asdCsudCadAugcudTuGf gcuagcasgsc	2072	GCUGCUAGCCAAAG CAUUGGAGA	2520
AD- 1700083.1	gsusuagccaAfAfGf cauuggagauL96	1625	asdTsudCcdAaugcdTuUf ggcuagcsasg	2073	CUGCUAGCCAAAGC AUUGGAGAC	2521
AD- 1700091.1	asasgcauugGfAfGf accuacuguL96	1626	asdCsagdTadGggucdTcCf aaugcuusug	2074	CAAAGCAUUGGAGA CCCUACUGC	2522
AD- 1700092.1	asgscauugGfAfAf ccuacugcuL96	1627	asdGscadGudAgggudCuCf caaugcususu	2075	AAAGCAUUGGAGAC CCUACUGCU	2523

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700094.1	csasuuggagAfCfCf cuacugcuguL96	1628	asdCsagdCadGuaggdGuCf uccaugsctu	2076	AGCAUUGGAGACCC UACUGCUGA	2524
AD- 1700097.1	usgsgagaccCfUfAf cugcugaauL96	1629	asdAsuudCadGcagudAgGf gucuccasasu	2077	AUUGGAGACCCUAC UGCUGAAUG	2525
AD- 1700099.1	gsasgaccuAfCfUf gcugaauL96	1630	asdCscadTudCagcadGuAf gggucucscsa	2078	UGGAGACCCUACUG CUGAAUGGA	2526
AD- 1700100.1	asgsaccuAfCfUfGf cugaauL96	1631	asdTscddAudTcagcdAgUf aggucucscsc	2079	GGAGACCCUACUGC UGAAUGGAG	2527
AD- 1700101.1	gsasccuacUfGfCf ugaauL96	1632	asdCsucdCadTucagdCaGf uaggucscusc	2080	GAGACCCUACUGCU GAAUGGAGU	2528
AD- 1700104.1	csesuacugcUfGfAf auggagugcuL96	1633	asdGscadCudCcauudCaGf caguaggsgsu	2081	ACCCUACUGCUGAA UGGAGUGCU	2529
AD- 1700109.1	usgscugaauGfGfAf gugcuaaccuL96	1634	asdGsgudTadGcacudCcAf uucagcasgsu	2082	ACUGCUGAAUGGAG UGCUAACCC	2530
AD- 1700112.1	usgsaauggaGfUfGf cuaaccuL96	1635	asdCsagdGgdTuagcdAcUf ccaucacsgsc	2083	GCUGAAUGGAGUGC UAACCCUGG	2531
AD- 1700113.1	gsasauggaUfGfCf uaaccuL96	1636	asdCscadGgdGuagcdCaCf uccauucscasg	2084	CUGAAUGGAGUGCU AACCCUGGU	2532
AD- 1700114.1	asasuggaguGfCfUf aaccuL96	1637	asdAsccdAgdGguuadGcAf cuccauucscsa	2085	UGAAUGGAGUGCUA ACCCUGGUG	2533
AD- 1700115.1	asusggagugCfUfAf accuL96	1638	asdCsacdCadGggudAgCf acuccaususc	2086	GAAUGGAGUGCUAA CCCUGGUGC	2534
AD- 1700116.1	usgsgagugcUfAfAf cccuL96	1639	asdGscadCcdAgggudTaGf cacuccasusu	2087	AAUGGAGUGCUAAC CCUGGUGCU	2535
AD- 1700121.1	usgscuaaccCfUfGf gugcuagaguL96	1640	asdCsucdTadGcaccdAgGf guuagcascsu	2088	AGUGCUAACCCUGG UCUAGAGG	2536
AD- 1700122.1	gsesuaccuUfGfGf ugcuagagguL96	1641	asdCscudCudAgcadcCaGf gguuagcsasc	2089	GUGCUAACCCUGGU GCUAGAGGA	2537
AD- 1700125.1	asasccuL96	1642	asdCscudCcdTcuagcdCaCf caggguusasg	2090	CUAACCCUGGUGCU AGAGGAGGA	2538
AD- 1700148.1	gsasacugcaGfUfGf gacagugaguL96	1643	asdCsucdAcdTgucccdAcUf gcaguucscsa	2091	UGGAACUGCAGUGG ACAGUGAGG	2539
AD- 1700149.1	asascugcagUfGfGf acagugagguL96	1644	asdCscudCadCugucdCaCf ugcaguucscsc	2092	GGAAUCGACAGUGGA CAGUGAGGA	2540
AD- 1700156.1	gsusggacagUfGfAf ggacuucuuL96	1645	asdAsagdAadGuccudCaCf uguccacsusg	2093	CAGUGGACAGUGAG GACUUCUUC	2541
AD- 1700158.1	gsagsacagugAfGfGf acuucuuL96	1646	asdGsgadAgdAagucdCuCf acuguccsasc	2094	GUGGACAGUGAGGA CUUCUCCA	2542
AD- 1700159.1	gsascagugaGfGfAf cuucuuL96	1647	asdTsggdAadGaagudCcUf cacugucscsa	2095	UGGACAGUGAGGAC UUCUCCAG	2543
AD- 1700160.1	ascsagugagGfAfCf uucuuL96	1648	asdCsugdGadAgaagdTcCf ucacugucscsc	2096	GGACAGUGAGGACU UCUCCAGC	2544
AD- 1700166.1	asgsagacuUfUfCf cagcugcuguL96	1649	asdCsagdCadGcuggdAaGf aaguccscsa	2097	UGAGGACUUCUCC AGCUGCUGG	2545
AD- 1700167.1	gsagsacuUfCfCf agcugcugguL96	1650	asdCscadGcdAgcugdGaAf gaaguccsusc	2098	GAGGACUUCUCCA GCUUGGGA	2546

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700180.1	csusgcuggaGfGfAf ugacacguguL96	1651	asdCsacdGudGucaudCcUf ccagcagscsu	2099	AGCUGCUGGAGGAU GACACGUGC	2547
AD- 1700187.1	asgsaugacAfCfGf ugccgauguL96	1652	asdCsaudCadGgcacdGuGf ucauccuscsc	2100	GGAGGAUGACACGU GCCUGAUGG	2548
AD- 1700188.1	gsgsaugacaCfGfUf gccugaugguL96	1653	asdCscadTcdAggacdCgUf gucauccsusc	2101	GAGGAUGACACGUG CCUGAUGGU	2549
AD- 1700189.1	gsasugacacGfUfGf ccugaugguuL96	1654	asdAscddAudCaggcdAcGf ugucaucscsu	2102	AGGAUGACACGUGC CUGAUGGUG	2550
AD- 1700190.1	asusgacacgUfGfCf cugaugguguL96	1655	asdCsacdCadTcaggdCaCf gugucauscsc	2103	GGAUGACACGUGCC UGAUGGUGU	2551
AD- 1700191.1	usgsacacguGfCfCf ugaugguguL96	1656	asdAscddCcdAucagdGcAf cgugucasusc	2104	GAUGACACGUGCCU GAUGGUGUU	2552
AD- 1700193.1	ascsacgugcCfUfGf augguguuguL96	1657	asdCsaadCadCcaudAgGf cacguguscsa	2105	UGACACGUGCCUGA UGGUGUUGC	2553
AD- 1700213.1	csasgucuggUfCfAf gagcuggaguL96	1658	asdCsucdCadGcucudGaCf cagacugscsa	2106	UGCAGUCUGGUCAG AGCUGGAGC	2554
AD- 1700242.1	gsasguggagUfGfCf ugucauauguL96	1659	asdCsaudAudGacagdCaCf uccacuscscsu	2107	AGGAGUGGAGUGCU GUCAUAUGG	2555
AD- 1700243.1	asgsuggaguGfCfUf gucauauguL96	1660	asdCscadTadTgacadGcAf cuccacuscsc	2108	GGAGUGGAGUGCUG UCAUAUGGC	2556
AD- 1700244.1	gsusggagugCfUfGf ucauauggcuL96	1661	asdGsgcdAudAugacdAgCf acuccacsusc	2109	GAGUGGAGUGCUGU CAUAUGGCC	2557
AD- 1700245.1	usgsagugcUfGfUf cauauggccuL96	1662	asdGsgcdCadTaugadCaGf cacuccascsu	2110	AGUGGAGUGCUGUC AUAUGGCCU	2558
AD- 1700248.1	asgsugcuguCfAfUf auggccugguL96	1663	asdCscadGgdCcaudTgAf cagcacuscsc	2111	GGAGUGCUGUCAUA UGGCCUGGG	2559
AD- 1700249.1	gsusgcugucAfUfAf uggccuggguL96	1664	asdCsccdAgdGccaudAuGf acagcacusc	2112	GAGUGCUGUCAUAU GGCCUGGGA	2560
AD- 1700250.1	usgsucuguaUfAfUf ggccugggauL96	1665	asdTsccdCadGgccadTaUf gacagcascsu	2113	AGUGCUGUCAUAUG GCCUGGGAC	2561
AD- 1700252.1	csusgucuaUfGfGf ccugggacguL96	1666	asdCsudCcdCaggcdCaUf augacagscsa	2114	UGCUGUCAUAUGGC CUGGGACGG	2562
AD- 1700272.1	asgsgcccaaGfCfAf cagcaaggauL96	1667	asdTsccdTudGcugudGcUf ugggccuscsc	2115	AGAGGCCCAAGCAC AGCAAGGAC	2563
AD- 1700279.1	asgscacagcAfAfGf gacaucgccuL96	1668	asdGsgcdGadTguccdTuGf cugugcusug	2116	CAAGCACAGCAAGG ACAUCGCC	2564
AD- 1700280.1	gsacsacagcaAfGfGf acaucgccccuL96	1669	asdGsggdCgdAugudCuUf gcugugcsusu	2117	AAGCACAGCAAGGA CAUCGCCCG	2565
AD- 1700303.1	uscsaccuuUfAfCf guguacaaguL96	1670	asdCsudGudAcacgdTcAf aaggugasasu	2118	AUUCACCUUUGACG UGUACAAGC	2566
AD- 1700309.1	ususgacgugUfAfCf aagcaaaacuL96	1671	asdGsudTudGcuugdTaCf acgucaasag	2119	CUUUGACGUGUACA AGCAAAACC	2567
AD- 1700312.1	ascsuguguacAfAfGf caaaaccuuL96	1672	asdAsggdGudTuugcdTuGf uacacguscsc	2120	UGACGUGUACAAGC AAAACCCUC	2568
AD- 1700314.1	gsusguacaaGfCfAf aaaccucguL96	1673	asdCsgadGgdGuuuudGcUf uguacacsgsu	2121	ACGUGUACAAGCAA AACCCUCGA	2569

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD-1700315.1	usgsuacaagCfAfAf aaccucgaul96	1674	asdTscgdAgdGguudTgCf uuguacascsg	2122	CGUGUACAAGCAAA ACCCUCGAG	2570
AD-1700316.1	gsusacaagcAfAfAf accucgagauL96	1675	asdCsucdGadGggauTuGf cuuguacscasc	2123	GUGUACAAGCAAAA CCCUCGAGA	2571
AD-1700317.1	usascaagcaAfAfAf cccucgagauL96	1676	asdTscudCgdAgggudTuUf gcuuguascsa	2124	UGUACAAGCAAAAC CCUCGAGAC	2572
AD-1700318.1	ascsaagcaaAfAfCf ccucgagacuL96	1677	asdGsucdTcdGagggdTuUf ugcuugusasc	2125	GUACAAGCAAAACC CUCGAGACC	2573
AD-1700323.1	csasaaacccUfCfGf agaccucuuuL96	1678	asdAsagdAgdGucudGaGf gguuuugscsu	2126	AGCAAAACCCUCGA GACCUCUUU	2574
AD-1700324.1	asasaacccuCfGfAf gaccucuuuL96	1679	asdAsaadGadGgucudCgAf ggguuuusgsc	2127	GCAAAACCCUCGAG ACCUCUUUG	2575
AD-1700325.1	asasacccucGfAfGf accucuuuguL96	1680	asdCsaadAgdAgguudTcGf aggguuusug	2128	CAAAACCCUCGAGA CCUCUUUGG	2576
AD-1700327.1	ascscucgaGfAfCf cucuuugcuL96	1681	asdGsccdAadAgaggdTcUf cgagggususu	2129	AAACCCUCGAGACC UCUUUGGCA	2577
AD-1700328.1	cscscucgagAfCfCf ucuuugcauL96	1682	asdTsgcdCadAagagdGuCf ucgagggususu	2130	AACCCUCGAGACCU CUUUUGCAG	2578
AD-1700329.1	cscsucgagaCfCfUf cuuuggcaguL96	1683	asdCsugdCcdAaagadGgUf cucgagggsgsu	2131	ACCCUCGAGACCUC UUUGGCAGC	2579
AD-1700336.1	ascscucuuuGfGfCf agccugaauuL96	1684	asdAsuudCadGgcugdCcAf aagagguscsu	2132	AGACCUCUUUGGCA GCCUGAUG	2580
AD-1700339.1	uscscuuuggcAfGfCf cugaauugcuL96	1685	asdGsacdAudTcaggdCuGf ccaaagasgsg	2133	CCUCUUUGGCAGCC UGAAUGUCA	2581
AD-1700344.1	ggsgcagccuGfAfAf ugucaaagcuL96	1686	asdGscudTudGacaudTcAf ggcugccsasa	2134	UUGGCAGCCUGAAU GUCAAAGCC	2582
AD-1700350.1	csusgaauguCfAfAf agccacauuuL96	1687	asdAsaudGudGgcudTgAf cauucagsgsc	2135	GCCUGAAUGUCAAA GCCACAUUC	2583
AD-1700352.1	gsasaugucaAfAfGf ccacauucuuL96	1688	asdAsgadAudGuggcdTuUf gacauucsasg	2136	CUGAAUGUCAAGC CACAUUCUA	2584
AD-1700353.1	asasugucaaAfGfCf cacauucuaL96	1689	asdTsagdAadTguggdCuUf ugacauuscsc	2137	UGAAUGUCAAGCC ACAUUCUAC	2585
AD-1700354.1	asusgucaaaGfCfCf acauucuacuL96	1690	asdGsudGadAugugdGcUf uugacaususc	2138	GAAUGUCAAGCCA CAUUCUACG	2586
AD-1700355.1	usgsucaaagCfCfAf cauucuacguL96	1691	asdCsgudAgdAaugudGgCf uuugacasusu	2139	AAUGUCAAGCCAC AUUCUACGG	2587
AD-1700356.1	gsuscaaagcCfAfCf auucuacggauL96	1692	asdCscgdTadGaaugdTgGf cuuugacsasu	2140	AUGUCAAGCCACA UUCUACGGG	2588
AD-1700357.1	uscscaaagccAfCfAf uucuacggguL96	1693	asdCsccdGudAgaudGuGf gcuuugascsa	2141	UGUCAAGCCACAU UCUACGGGC	2589
AD-1700358.1	csasaagccaCfAfUf ucucgggcuL96	1694	asdGsccdCgdTagaadTgUf ggcuuugsasc	2142	GUCAAAGCCACAUU CUACGGGCU	2590
AD-1700359.1	asasagccacAfUfUf cuacgggcuuL96	1695	asdAsgcdCcdGuagadAuGf uggcuuugsa	2143	UCAAGCCACAUUC UACGGGCUC	2591

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700365.1	ascsaucuaCfGfGf gcucuaucuuL96	1696	asdGsagdTadGagccdCgUf agaaugusgsg	2144	CCACAUUCUACGGG CUCUACUCU	2592
AD- 1700366.1	csasuucuaCfGfGf cucuaucuuL96	1697	asdAsgadGudAgagcdCcGf uagaugsusg	2145	CACAUUCUACGGGC UCUACUCUA	2593
AD- 1700368.1	ususcuaCgGfCfUf cuacucuaauL96	1698	asdAsuadGadGuagadGcCf cguagaasusg	2146	CAUUCUACGGGCUC UACUCUAUG	2594
AD- 1700369.1	uscsuacggGfUfCf uacucuauguL96	1699	asdCsaudAgdAguagdAgCf ccguagasasu	2147	AUUCUACGGGCUCU ACUCUAUGA	2595
AD- 1700370.1	csusacggGfUfCfUf acucuaugauL96	1700	asdTscadTadGaguadGaGf cccguagsasa	2148	UUCUACGGGCUCUA CUCUAUGAG	2596
AD- 1700371.1	usascggGcuCfUfAf cucuaugaguL96	1701	asdCsudAudAgagudAgAf gccguasgsa	2149	UCUACGGGCUCUAC UCUAUGAGU	2597
AD- 1700372.1	ascsgggcucUfAfCf ucuaugaguL96	1702	asdAscudCadTagagdTaGf agcccgusag	2150	CUACGGGCUCUACU CUAUGAGUU	2598
AD- 1700374.1	gsgsgcucuaCfUfCf uauagaguL96	1703	asdCsaadCudCauagdAgUf agagcccsesu	2151	ACGGGCUCUACUCU AUGAGUUGU	2599
AD- 1700375.1	gsgscucuaCfUfUf augaguL96	1704	asdAscadAcdTcauadGaGf uagagccscsg	2152	CGGGCUCUACUCUA UGAGUUGUG	2600
AD- 1700376.1	gscsucuacuCfUfAf ugaguL96	1705	asdCsacdAadCucaudAgAf guagagccsc	2153	GGGCUCUACUCUAU GAGUUGUGA	2601
AD- 1700378.1	uscsuacucUfUfGf aguugugacuL96	1706	asdGsudAcdAacudAuAf gaguagasgsc	2154	GCUCUACUCUAUGA GUUGUGACU	2602
AD- 1700383.1	uscsuagagUfUfGf ugacuL96	1707	asdTsgadAadGucacdAaCf ucauagasgsu	2155	ACUCUAGAGUUGU GACUUUCA	2603
AD- 1700384.1	csusaugaguUfGfUf gacuL96	1708	asdTsugdAadAgucadCaAf cucuaagsag	2156	CUCUAUGAGUUGUG ACUUUCAAG	2604
AD- 1700385.1	usasugaguGfUfGf acuL96	1709	asdCsudGadAagudAcAf acucauasgsa	2157	UCUAUGAGUUGUGA CUUUCAAGG	2605
AD- 1700386.1	asusgaguUfGfAf cuuL96	1710	asdCsudTgdAagudCaCf aacucausag	2158	CUAUGAGUUGUGAC UUUCAAGGA	2606
AD- 1700387.1	usgsaguuguGfAfCf uuuL96	1711	asdTscddTudGaaagdTcAf caacucasusa	2159	UAUGAGUUGUGACU UUCAAGGAC	2607
AD- 1700388.1	gsasguuguAfCfUf uuL96	1712	asdGsudCudTgaaadGuCf acaacucsasu	2160	AUGAGUUGUGACUU UCAAGGACU	2608
AD- 1700408.1	usugggcccaAfAfGf aaaguL96	1713	asdGsagdTadCuucdTuUf gggccaasgsu	2161	ACUUGGCCCAAAGA AAGUACUCA	2609
AD- 1700410.1	gsgscccaaaGfAfAf aguL96	1714	asdCsugdAgdTacuudTcUf uugggccsasa	2162	UUGGCCCAAAGAAA GUACUCAGG	2610
AD- 1700411.1	gscsccaaagAfAfAf guacuL96	1715	asdCsudGadGuacudTuCf uuugggcscsa	2163	UGGCCCAAAGAAAG UACUCAGGG	2611
AD- 1700413.1	cscsaaagaaAfGfUf acuL96	1716	asdTscddCudGaguadCuUf ucuuuggsgsc	2164	GCCCAAAGAAAGUA CUCAGGGAG	2612
AD- 1700414.1	csasaagaaaGfUfAf cucaggguL96	1717	asdCsudCcdTgagudAcUf uucuuugsgsg	2165	CCCAAAGAAAGUAC UCAGGGAGC	2613

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700422.1	gsusacucagGfGfAf gcuccuucguL96	1718	asdCsgadAgdGagcudCcCf ugaguacsusu	2166	AAGUACUCAGGGAG CUCCUUCGU	2614
AD- 1700423.1	usascucaggGfAfGf cuccuucguuL96	1719	asdAscgdAadGgagcdTcCf cugaguascsu	2167	AGUACUCAGGGAGC UCCUUCGUU	2615
AD- 1700424.1	ascsucaggGfGfCf uccuucguuL96	1720	asdAsacdGadAggagdCuCf ccugagusasc	2168	GUACUCAGGGAGCU CCUUCGUUG	2616
AD- 1700425.1	csuscaggGfCfUf ccuucguuL96	1721	asdCsaadCgdAaggadGcUf cccugagsusa	2169	UACUCAGGGAGCUC CUUCGUUGG	2617
AD- 1700426.1	uscsaggGfCfUfCf cuucguugguL96	1722	asdCscadAcidGaaggdAgCf ucccugasgsu	2170	ACUCAGGGAGCUCC UUCGUUGGA	2618
AD- 1700427.1	csasgggagCfCfCf uucguuggauL96	1723	asdTscddAadCgaagdGaGf cucccugsasg	2171	CUCAGGGAGCUCCU UCGUUGGAC	2619
AD- 1700428.1	asgsgggagCfCfUf ucguuggacuL96	1724	asdGsucdCadAcgaadGgAf gcucccusgsa	2172	UCAGGGAGCUCCUU CGUUGGACC	2620
AD- 1700439.1	uscsuuggaCfCfUf ccacacugcuL96	1725	asdGscadGudGuggadGgUf ccaacgasasg	2173	CUUCGUUGGACCUC CACACUGCU	2621
AD- 1700461.1	csasaggccuGfGfGf ccauauguuL96	1726	asdAsacdAudAuggcdCcAf ggccuugscsa	2174	UGCAAGGCCUGGGC CAUAUGUUG	2622
AD- 1700462.1	asasggccuGfGfCf cauauuguuL96	1727	asdCsaadCadTauggdCcCf aggccuusgsc	2175	GCAAGGCCUGGGCC AUAUGUUGC	2623
AD- 1700464.1	gsgscuuggGfCfAf uauugucuuL96	1728	asdAsgcdAadCauaudGgCf ccaggccsusu	2176	AAGGCCUGGGCCAU AUGUUGCUG	2624
AD- 1700465.1	gscscuuggGfCfUf auguugcguL96	1729	asdCsagdCadAcaudTgGf cccaggcsusu	2177	AGGCCUGGGCCAU UGUUGCUGG	2625
AD- 1700468.1	usgsgggcAuAfUfGf uugcugggauL96	1730	asdTscddCadGcaacdAuAf uggcccasgsg	2178	CCUGGGCCAUUGU UGCUGGGAA	2626
AD- 1700473.1	csasuauuguGfCfUf gggaauuucuL96	1731	asdGsaadAudTcccadGcAf acauaugsgsc	2179	GCCAUUGUUGCUG GGAAUUUC	2627
AD- 1700474.1	asusauguGfCfUfGf ggaaauuccuL96	1732	asdGsgadAadTucccdAgCf aacauausgsg	2180	CCAUUGUUGCUGG GAAUUCUCC	2628
AD- 1700476.1	asusguugCfGfGf aaauuccucuL96	1733	asdGsagdGadAauudCcAf gcaacausasu	2181	AUAUGUUGCUGGGA AUUUCUCC	2629
AD- 1700477.1	usgsuugcGfGfAf auuuccuccuL96	1734	asdGsgadGgdAaaudCcCf agcaacasusa	2182	UAUGUUGCUGGGAA UUUCCUCCA	2630
AD- 1700479.1	usugcuggGfAfUf uuccccacuL96	1735	asdGsugdGadGgaadTuCf ccagcaascsa	2183	UGUUGCUGGGAAU UCCUCCACC	2631
AD- 1700482.1	csusgggaaUfUfCf cuccaccuuL96	1736	asdAsggdGudGgaggdAaAf uucccagscsa	2184	UGCUGGGAAUUUCC UCCACCCUU	2632
AD- 1700485.1	gsgsaauuucCfUfCf caccuucguL96	1737	asdCsgadAgdGguggdAgGf aaauuccscsa	2185	UGGGAAUUUCCUCC ACCCUUCGU	2633
AD- 1700486.1	gsasaauuucUfCfCf acccuucguuL96	1738	asdAscgdAadGggugdGaGf gaaauucscsc	2186	GGGAAUUUCCUCCA CCCUUCGUC	2634
AD- 1700487.1	asasaauuucCfCfAf cccuucguuL96	1739	asdGsacdGadAgggudGgAf ggaaauuscsc	2187	GGAAUUUCCUCCAC CCUUCGUCA	2635

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD-1700488.1	asusuccucCfAfCf ccuucgucauL96	1740	asdTsgadCgdAagggdTgGf agggaaususc	2188	GAAUUUCCUCCACC CUUCGUCAU	2636
AD-1700489.1	ususuccuccAfCfCf cuucgucauL96	1741	asdAsugdAcGgaaggdGuGf gaggaasusu	2189	AAUUUCCUCCACCC UUCGUCAUG	2637
AD-1700490.1	ususcuccaCfCfCf uucgucauguL96	1742	asdCsaudGadCgaagdGgUf ggaggaasasu	2190	AUUUCCUCCACCCU UCGUCAUGC	2638
AD-1700517.1	asasgggcccCfCfUf ccuuccuauL96	1743	asdTsagdGadAuggadGgCf ggcccuuscsu	2191	AGAAGGGCCGCCUC CAUCCUAC	2639
AD-1700519.1	gsgsgccgccUfCfCf auuccuacuuL96	1744	asdAsgudAgdGaaugdGaGf gcggcccsusu	2192	AAGGGCCGCCUCCA UUCCUACUA	2640
AD-1700520.1	gsgscgcccuCfCfAf uuccuacuaL96	1745	asdTsagdTadGgaaudGgAf ggcggccscsu	2193	AGGGCCGCCUCCAU UCCUACUAA	2641
AD-1700522.1	cscsgccuccAfUfUf ccuacuaaguL96	1746	asdCsudAgdTaggadAuGf gaggcggscsc	2194	GGCCGCCUCCAUUC CUACUAAGG	2642
AD-1700542.1	cscscacugcAfAfAf gacuaugacuL96	1747	asdGsudAudAgucudTuGf cagugggsusc	2195	GACCCACUGCAAAG ACUAUGACA	2643
AD-1700544.1	csascugcaaAfGfAf cuaugacaguL96	1748	asdCsugdTcdAuagudCuUf ugcagugsgsg	2196	CCCACUGCAAAGAC UAUGACAGC	2644
AD-1700550.1	asasagacuaUfGfAf cagcaucauL96	1749	asdTsugdAudGcugudCaUf agucuuusgsc	2197	GCAAAGACUAUGAC AGCAUCAA	2645
AD-1700553.1	gsascuaugaCfAfGf caucaaaauuL96	1750	asdAsaudTudGaugcdTgUf cauagucsusu	2198	AAGACUAUGACAGC AUCAAAUUU	2646
AD-1700554.1	ascsuauagacAfGfCf aucaaaaauuL96	1751	asdAsaadTudTgaugdCuGf ucauaguscsc	2199	AGACUAUGACAGCA UCAAAUUUC	2647
AD-1700555.1	csusaugacaGfCfAf ucaaaaauuL96	1752	asdGsaudAudTugaudGcUf gucauagsusc	2200	GACUAUGACAGCAU CAAAUUUCA	2648
AD-1700556.1	usasugacagCfAfUf caaaaaucauL96	1753	asdTsgadAadTuugadTgCf ugucauasgsu	2201	ACUAUGACAGCAUC AAAUUUCAG	2649
AD-1700557.1	asusgacagcAfUfCf aaaaauucaguL96	1754	asdCsugdAadAuuugdAuGf cugucausasg	2202	CUAUGACAGCAUCA AAUUUCAGG	2650
AD-1700558.1	usgsacagcaUfCfAf aauuucaggguL96	1755	asdCscudGadAauuudGaUf gcugucasusa	2203	UAUGACAGCAUCAA AUUUCAGGA	2651
AD-1700560.1	ascsgagcauAfAfAf uuucaggacuL96	1756	asdGsudCudGaaudTuGf augcuguscsc	2204	UGACAGCAUCAAU UUCAGGACC	2652
AD-1700563.1	gscsaucaaaUfUfUf caggaccuguL96	1757	asdCsagdGudCcugadAaUf uugaugcsusg	2205	CAGCAUCAAUUUC AGGACCGC	2653
AD-1700566.1	uscsgaaaauCfAfGf gaccugcaguL96	1758	asdCsugdCadGguccdTgAf aauuugasusg	2206	CAUCAAUUUCAGG ACCUGCAGA	2654
AD-1700573.1	uscsgaggaccUfGfCf agacaguacuL96	1759	asdGsudCudGucugdCaGf guccugasasa	2207	UUUCAGGACCUGCA GACAGUACA	2655
AD-1700575.1	asgsgaccugCfAfGf acaguacaguL96	1760	asdCsugdTadCuguedTgCf agguccusgsa	2208	UCAGGACCUGCAGA CAGUACAGG	2656
AD-1700576.1	gsgsaccugcAfGfAf caguacaggguL96	1761	asdCscudGudAcugudCuGf cagguccsusg	2209	CAGGACCUGCAGAC AGUACAGGC	2657

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700578.1	ascscugcagAfCfAf guacaggcuuL96	1762	asdAsgcdCudGuacudGuCf ugcagguscsc	2210	GGACCUGCAGACAG UACAGGCUA	2658
AD- 1700580.1	csusgcagacAfGfUf acaggcuaguL96	1763	asdCsuaadGcdCuguadCuGf ucugcagsgsu	2211	ACCUGCAGACAGUA CAGGCUAGA	2659
AD- 1700582.1	gscsagacagUfAfCf aggcuagauuL96	1764	asdAsucdTadGccugdTcCf ugucugcsasg	2212	CUGCAGACAGUACA GGCUAGAU	2660
AD- 1700623.1	ascsgccucaCfAfUf cccaagucuuL96	1765	asdAsgadCudTgggadTgUf gaggcgususa	2213	UAACGCCUCACAUC CCAAGUCUA	2661
AD- 1700670.1	ususcucagcCfAfCf cuuucugguL96	1766	asdGsgcdAgdAaaggdTgGf cuaggaasasg	2214	CUUUCUAGCCACC UUUCUGGCC	2662
AD- 1700671.1	uscscuagccAfCfCf uuucugccuL96	1767	asdGsgcdCadGaaagdGuGf gcuaggasasa	2215	UUUCCUAGCCACCU UUCUGGCCU	2663
AD- 1700680.1	csesuucugGfCfCf ucccacuuguL96	1768	asdCsaadGudGggagdGcCf agaaaggsusg	2216	CACCUUUCUGGCCU CCCACUUGC	2664
AD- 1700681.1	csusuucuggCfCfUf cccacuugcuL96	1769	asdGscadAgdTgggadGgCf cagaaagsgsu	2217	ACCUUUCUGGCCUC CCACUUGCC	2665
AD- 1700684.1	uscsgggccuCfCfCf acuugcccuL96	1770	asdAsggdGcdAagugdGgAf ggccagasasa	2218	UUUCUGGCCUCCCA CUUGCCUG	2666
AD- 1700710.1	gsgscaucuuGfAfUf cccagccuguL96	1771	asdCsagdTcdTgggadTcAf agaugccsug	2219	CAGGCAUCUUGAUC CCAGCCUGA	2667
AD- 1700736.1	asasugccucCfUfGf ucccuaaacuL96	1772	asdGsuudTadGggacdAgGf aggcauusgsg	2220	CCAAUGCCUCCUGU CCCUAACU	2668
AD- 1700737.1	asusgcccucUfGfUf cccuaaacuL96	1773	asdAsgdTudAgggadCaGf gaggcausug	2221	CAAUGCCUCCUGUC CCUAACUC	2669
AD- 1700738.1	usgsgccuccGfUfCf ccuaaacucuL96	1774	asdGsagdTudTagggdAcAf ggaggcasusu	2222	AAUGCCUCCUGUCC CUAACUCC	2670
AD- 1700780.1	csasaacuaaCfAfAf aaacauuuuL96	1775	asdGsaadAudGuuuudTgUf uaguugsgasg	2223	CUCAAACUAACAAA AACAUUUC	2671
AD- 1700781.1	asasacuaacAfAfAf aacauuuuL96	1776	asdGsgadAadTguuudTuGf uuaguuusgsa	2224	UCAAAACUAACAAA ACAUUUCCA	2672
AD- 1700782.1	asascuaacaAfAfAf acauuuuL96	1777	asdTsggdAadAuguudTuUf guaguusug	2225	CAAACUAACAAAA CAUUUCCA	2673
AD- 1700791.1	gsusccugGfGfUfG fagauccaguL96	1778	asCfsugdGg (Agn) ucucag CfcCfaggacsasa	2226	GGAAGGAAAGAGCA GAUCCAGG	2674
AD- 1700793.1	usgsagauCfcCfAfG fguuuguacuL96	1779	asGfsuudAc (Agn) aaccug GfgAfucucasgsc	2227	GAGCAGAUCCAGG UUUGUAACA	2675
AD- 1700794.1	gsasgaucCfcAfGfG fuuuuuaacuL96	1780	asUfsgudTa (Cgn) aaaccu GfgGfaucucsasg	2228	AGCAGAUCCAGGU UUGUAACAG	2676
AD- 1700795.1	gsasuccAfGfUfU fuguuacagauL96	1781	asUfscudGu (Tgn) acaaac CfuGfggaucsusc	2229	CAGAUCCAGGUUU GUAACAGAA	2677
AD- 1700796.1	csasagguUfGfUfA facagaaaacuL96	1782	asGfsuudTu (Cgn) uguuac AfaAfccuggsgsa	2230	UCCAGGUUUGUAA CAGAAAACA	2678
AD- 1700797.1	csasgguuUfGfUfA fcagaaaacuL96	1783	asUfsgudTu (Tgn) cuguua CfaAfaccugsgsg	2231	CCCAGGUUUGUAA AGAAAACAC	2679

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEb dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD-1700798.1	asgsguuuGfuAfAfC fagaaaacacul96	1784	asGfsugdTu (Tgn)ucuguu AfcAfaaccusgsg	2232	CCAGGUUUGUAACA GAAAACACC	2680
AD-1700799.1	gsgsuuugUfaAfCfA fgaaaacaccul96	1785	asGfsgudGu (Tgn)uucugu UfaCfaaaccsug	2233	CAGGUUUGUAACAG AAAACACCA	2681
AD-1700800.1	usasacagAfaAfAfC faccacuaauL96	1786	asUfsuudAg (Tgn)gguguu UfuCfuguuascsa	2234	UGUAACAGAAAACA CCACUAAAG	2682
AD-1700801.1	csasgaaaAfcAfCfC facuaaagccul96	1787	asGfsgcdTu (Tgn)aguggu GfuUfuucugsusu	2235	AACAGAAAACACCA CUAAGCCC	2683
AD-1700802.1	csascaggAfgAfGfA faccaccagul96	1788	asCfsugdGg (Tgn)gguuu CfuCfcugugscsu	2236	AGCACAGGAGAGAA CCACCCAGC	2684
AD-1700804.1	gsasaccaCfcCfAfG fcccagaaguuL96	1789	asAfsudTc (Tgn)gggcug GfgUfgguuscusc	2237	GAGAACCACCCAGC CCAGAAGUU	2685
AD-1700805.1	asasccacCfcAfGfC fccagaaguuL96	1790	asAfsacdTu (Cgn)ugggcu GfgGfugguuscscu	2238	AGAACCACCCAGCC CAGAAGUUC	2686
AD-1700806.1	ascscaccCfaGfCfC fcagaaguucul96	1791	asGfsaadCu (Tgn)cugggc UfgGfguggususc	2239	GAACCACCCAGCCC AGAAGUUC	2687
AD-1700807.1	cscsacccAfgCfCfC fagaaguuccul96	1792	asGfsgadAc (Tgn)ucuggg CfuGfggugsusu	2240	AACCACCCAGCCCA GAAGUCCA	2688
AD-1700808.1	csascccaGfcCfCfA fgaaguuccauL96	1793	asUfsggdAa (Cgn)uucugg GfcUfgggugsusu	2241	ACCACCCAGCCCAG AAGUCCAG	2689
AD-1700809.1	cscsagaaGfuUfCfC fagggaggauL96	1794	asUfscddTu (Cgn)ccugga AfcUfucggsgsc	2242	GCCCAGAGUCCA GGGAAGGAA	2690
AD-1700810.1	csasgaagUfuCfCfA fgggaaggauL96	1795	asUfsuudCu (Tgn)cccugg AfaCfuucugsgsg	2243	CCCAGAAGUCCAG GGAAGGAA	2691
AD-1700811.1	asgsaaguUfcCfAfG fggaaggaacul96	1796	asGfsuudCc (Tgn)ucccug GfaAfcuucugsgsg	2244	CCAGAAGUCCAGG GAAGGAACU	2692
AD-1700812.1	gsasaguUfcAfGfG fgaaggaacul96	1797	asAfsudTc (Cgn)uucccu GfgAfacuucsu	2245	CAGAAGUCCAGGG AAGGAACUC	2693
AD-1700813.1	asasguucCfaGfGfG faaggaacucul96	1798	asGfsagdTu (Cgn)cuuccc UfgGfaacuuscscu	2246	AGAAGUCCAGGGA AGGAACUCU	2694
AD-1700814.1	asgsuuccAfgGfGfA faggaacucul96	1799	asAfsagdGu (Tgn)ccuucc CfuGfgaacuscusc	2247	GAAGUCCAGGGAA GGAACUCUC	2695
AD-1700815.1	gsusuccaGfgGfAfA fggaacucucul96	1800	asGfsagdAg (Tgn)uccuuc CfcUfggaacsusu	2248	AAGUCCAGGGAA GAACUCUC	2696
AD-1700816.1	gsusccacCfaUfGfG faguaccucul96	1801	asAfsagdGg (Tgn)acucca UfgGfggacscsg	2249	CGGUCCACCAUGGA GUACUCUC	2697
AD-1700817.1	csasccauGfgAfGfU faccucacul96	1802	asCfsugdAg (Agn)gguauc CfcAfuggugsusa	2250	UCCACCAUGGAGUA CCUCUCAGC	2698
AD-1700819.1	cscsauggAfgUfAfC fcucucacul96	1803	asAfsagdTg (Agn)gaggua CfuCfcauggsug	2251	CACCAUGGAGUACC UCUCAGCUC	2699
AD-1700821.1	asusggagUfaCfCfU fcucagcucul96	1804	asAfsagdGc (Tgn)gagagg UfaCfucausgsg	2252	CCAUGGAGUACCUC UCAGCUCUG	2700
AD-1700822.1	ususacucAfgGfUfC faguaucuaul96	1805	asUfsuudGa (Tgn)acugac CfuGfaguaasgsu	2253	ACUUACUCAGGUCA GUAUCUAU	2701

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD-1700824.1	uscsaguaUfcUfAfA fuauaagcucuL96	1806	asGfsagdCu (Tgn) auauua GfaUfacugascsc	2254	GGUCAGUAUCUAAU AUAAGCUCG	2702
AD-1700825.1	csasguauCfuAfAfU fauaagcucguL96	1807	asCfsgadGc (Tgn) uauauu AfgAfuacugsasc	2255	GUCAGUUAUCUAAUA UAAGCUCGG	2703
AD-1700826.1	asgsuauCfuAfAfU fuaagcucgguL96	1808	asCfscgdAg (Cgn) uuauau UfaGfauacusgsa	2256	UCAGUAUCUAAUAU AAGCUCGGA	2704
AD-1700828.1	uscsuauAfUfAfAfG fcucggaguuuL96	1809	asAfsacdTc (Cgn) gagcuu AfuAfuuagasusa	2257	UAUCUAAUAUAAGC UCGGAGUUU	2705
AD-1700829.1	csusaauUfaAfGfC fucggaguuuuL96	1810	asAfsaadCu (Cgn) cgagcu UfaUfauuagsasu	2258	AUCUAAUAUAAGCU CGGAGUUUG	2706
AD-1700830.1	usasagcuCfGfAfGf fuuuggacgguL96	1811	asCfscgdTc (Cgn) aaacuc CfGfGcuuasusa	2259	UAUAAGCUCGGAGU UUGGACGGA	2707
AD-1700831.1	asasgcucGfGfAfGfU fuuggacggauL96	1812	asUfscddGu (Cgn) caaacu CfcGfagcuusasu	2260	AUAAGCUCGGAGUU UGGACGAG	2708
AD-1700832.1	gsgsaguuUfGfAfCf fggagggucuuL96	1813	asAfsagdCc (Cgn) uccguc CfaAfacuccsgsa	2261	UCGGAGUUUGGACG GAGGGUCUG	2709
AD-1700833.1	usgsgacGfGfGfGfG fucuggaccuuL96	1814	asAfsaggdTc (Cgn) agaccc UfcCfguccasasa	2262	UUUGGACGGAGGGU CUGGACCUC	2710
AD-1700834.1	asgscgacCfuUfUfCf fcgugucuguuL96	1815	asAfsacdGa (Cgn) acggaa AfgGfucgcsugsg	2263	CCAGCGACCUUCC GUGUCUGUG	2711
AD-1700837.1	csasagcgGfAcfCfA fuccggaaaguL96	1816	asCfsuudTc (Cgn) ggaugg UfcCfgeuugsusg	2264	CACAAGCGGACCAU CCGGAAAGG	2712
AD-1700838.1	csasuccGfAcfAfGfC fgccugacaguL96	1817	asCfsugdTc (Agn) ggccuu UfcCfggauugsusu	2265	ACCAUCCGGAAGG CCUGACAGC	2713
AD-1700839.1	asgsgagcUfGfCfUfA fgccaaagcauL96	1818	asUfsgcdTu (Tgn) ggcuag CfaGfcuccugsgg	2266	CCAGGAGCUGCUAG CCAAAGCAU	2714
AD-1700840.1	gsgsagcuGfcUfAfGf fccaaagcauuL96	1819	asAfsugdCu (Tgn) uggcuu GfcAfgcuccusug	2267	CAGGAGCUGCUAGC CAAAGCAU	2715
AD-1700841.1	gsasgcugCfuAfGfCf fccaaagcauuL96	1820	asAfsaudGc (Tgn) uuggcu AfgCfagcucscsu	2268	AGGAGCUGCUAGCC AAAGCAUUG	2716
AD-1700842.1	asgscugcUfaGfCfCf faaagcauuguL96	1821	asCfsaadTg (Cgn) uuuggc UfaGfcagcuscsc	2269	GGAGCUGCUAGCCA AAGCAUUGG	2717
AD-1700843.1	csusagccAfaAfGfCf fauuggagacuL96	1822	asGfsucdTc (Cgn) aaugcu UfuGfgeuagscsa	2270	UGCUAGCCAAAGCA UUGGAGACC	2718
AD-1700844.1	usasgccaAfaGfCfA fuuggagaccuL96	1823	asGfsgudCu (Cgn) caaugc UfuUfggcuasgsc	2271	GCUAGCCAAAGCAU UGGAGACCC	2719
AD-1700845.1	asgscuuGfGfAfGfA fcccuacugcuL96	1824	asGfscadGu (Agn) gggucu CfcAfaugcususu	2272	AAAGCAUUGGAGAC CCUACUGCU	2720
AD-1700846.1	gscsaaugGfAcfCfCf fccuacugcuuL96	1825	asAfsagdAg (Tgn) aggguc UfcCfaaugcsusu	2273	AAGCAUUGGAGACC CUACUGCUG	2721
AD-1700848.1	asusuggaGfAcfCfCf fuacugcugauL96	1826	asUfscadGc (Agn) guaggg UfcUfccaaugsgsc	2274	GCAUUGGAGACCCU ACUGCUGAA	2722
AD-1700850.1	gsgsagacCfcUfAfCf fugcugaauuL96	1827	asCfsaudTc (Agn) gcagua GfgGfucuccsasa	2275	UUGGAGACCCUACU GCUGAAUGG	2723

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700851.1	gsasgaccCfuAfCfU fgcugaauuguL96	1828	asCfscadTu (Cgn) agcagu AfgGfgucucscsa	2276	UGGAGACCCUACUG CUGAAUGGA	2724
AD- 1700852.1	ascsccuaCfuGfCfU fgaauaggaguL96	1829	asAfsudCc (Agn) uucagc AfgUfagguscsu	2277	AGACCCUACUGCUG AAUGGAGUG	2725
AD- 1700853.1	cscscuacUfgCfUfG faauaggaguL96	1830	asCfsacdTc (Cgn) auucag CfaGfuagggsusc	2278	GACCCUACUGCUGA AUGGAGUGC	2726
AD- 1700854.1	cscsuacuGfCfGfA fauggagugcuL96	1831	asGfscadCu (Cgn) cauua GfcAfguaggsgsu	2279	ACCCUACUGCUGAA UGGAGUGCU	2727
AD- 1700855.1	csusacugCfuGfAfA fuggagugcuL96	1832	asAfsgcdAc (Tgn) ccauuc AfgCfaguaggsgg	2280	CCCUACUGCUGAAU GGAGUGCUA	2728
AD- 1700856.1	usascugcUfgAfAfU fggagugcuL96	1833	asUfsagdCa (Cgn) uccauu CfaGfcaguasggg	2281	CCUACUGCUGAAUG GAGUGCUAA	2729
AD- 1700858.1	gsacsugaaUfgGfAfG fugcuaacccuL96	1834	asGfsggdTu (Agn) gcacuc CfaUfucagcsasg	2282	CUGCUGAAUGGAGU GCUAACCCU	2730
AD- 1700859.1	csusgaauGfGfAfGfU fgcuaacccuL96	1835	asAfsggdGu (Tgn) agcacu CfcAfuucagscsa	2283	UGCUGAAUGGAGUG CUAACCCUG	2731
AD- 1700860.1	usgsaaugGfAfGfUfG fcuaacccuL96	1836	asCfsagdGg (Tgn) uagcac UfcCfauucasgsc	2284	GCUGAAUGGAGUGC UAACCCUGG	2732
AD- 1700863.1	usgsagugGfCfAfAfA fccuggugcuL96	1837	asGfscadCc (Agn) ggguaa GfcAfcuccasusu	2285	AAUGGAGUGCUAAC CCUGGUGCU	2733
AD- 1700864.1	gsagsagugCfuAfAfC fccuggugcuL96	1838	asAfsgcdAc (Cgn) aggguu AfgCfcuccsasus	2286	AUGGAGUGCUAAC CUGGUGCUA	2734
AD- 1700865.1	gsasgugCfUfaAfCfC fcuggugcuL96	1839	asUfsagdCa (Cgn) cagggu UfaGfcacucscsa	2287	UGGAGUGCUAACCC UGGUGCUAG	2735
AD- 1700867.1	asascccuGfGfUfGfC fuaggaggguL96	1840	asCfscudCc (Tgn) cuagca CfcAfggguusasg	2288	CUAACCCUGGUGCU AGAGGAGGA	2736
AD- 1700868.1	cscscuggUfgCfUfA fgaggaggauL96	1841	asAfsudCu (Cgn) cucuag CfaCfcagggsusu	2289	AACCCUGGUGCUAG AGGAGGAUG	2737
AD- 1700869.1	gsugsugcuAfgAfGfG faggauggaauL96	1842	asUfsudCa (Tgn) ccuccu CfuAfgcaccsasg	2290	CUGGUGCUAGAGGA GGAUGGAAC	2738
AD- 1700870.1	gsusgcuGfAfGfGfA fggauggaacuL96	1843	asGfsuudCc (Agn) uccucc UfcUfagcacscsa	2291	UGGUGCUAGAGGAG GAUGGAACU	2739
AD- 1700871.1	csusagagGfAfGfGfA fuggaacugcuL96	1844	asGfscadGu (Tgn) ccaucc UfcCfucagscsa	2292	UGCUAGAGGAGGAU GGAACUGCA	2740
AD- 1700872.1	gsasggauGfGfAfAfC fugcaguggauL96	1845	asUfscddAc (Tgn) gcaguu CfcAfuccucscsu	2293	AGGAGGAUGGAACU GCAGUGGAC	2741
AD- 1700873.1	gsagsaugAfaCfUfG fcaguggacauL96	1846	asUfsgudCc (Agn) cugcag UfuCfcauccsusc	2294	GAGGAUGGAACUGC AGUGGACAG	2742
AD- 1700874.1	gsasuggaAfCfUfGfC faguggacaguL96	1847	asCfsugdTc (Cgn) acugca GfuUfccaucscsu	2295	AGGAUGGAACUGCA GUGGACAGU	2743
AD- 1700875.1	asusggaaCfuGfCfA fuggacagauL96	1848	asAfsudGu (Cgn) cacugc AfgUfuccaucsc	2296	GGAUGGAACUGCAG UGGACAGUG	2744
AD- 1700876.1	usgsaacUfgCfAfG fuggacagauL96	1849	asCfsacdTg (Tgn) ccacug CfaGfuuccasusc	2297	GAUGGAACUGCAGU GGACAGUGA	2745

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700877.1	gsasacugCfaGfUfG fgacagugaguL96	1850	asCfsucdAc (Tgn) guccac UfgCfaguucscsa	2298	UGGAACUGCAGUGG ACAGUGAGG	2746
AD- 1700878.1	ascsgucaGfuGfGfA fcagugaggauL96	1851	asUfscddTc (Agn) cugucc AfcUfgcagusc	2299	GAACUGCAGUGGAC AGUGAGGAC	2747
AD 1700879.1	csusgcagUfgGfAfC fagugaggacuL96	1852	asGfsucdCu (Cgn) acuguc CfaCfugcagsusu	2300	AACUGCAGUGGACA GUGAGGACU	2748
AD- 1700880.1	usgscaguGfgAfCfA fgugaggacuL96	1853	asAfsugdCc (Tgn) cacugu CfcAfcugcagsu	2301	ACUGCAGUGGACAG UGAGGACUU	2749
AD- 1700881.1	gscsagugGfaCfAfG fugaggacuuL96	1854	asAfsagdTc (Cgn) ucacug UfcCfacugcsasg	2302	CUGCAGUGGACAGU GAGGACUUC	2750
AD- 1700882.1	csasgugGfCfAfGfU fgaggacuucL96	1855	asGfsaadGu (Cgn) cucacu GfuCfcacugscsa	2303	UGCAGUGGACAGUG AGGACUUCU	2751
AD- 1700883.1	asgsuggaCfaGfUfG faggacuucL96	1856	asAfsgadAg (Tgn) ccucac UfgUfccacusc	2304	GCAGUGGACAGUGA GGACUUCU	2752
AD- 1700885.1	usgsgacaGfuGfAfG fgacuucuuL96	1857	asGfsaadGa (Agn) guccuc AfcUfguccascsu	2305	AGUGGACAGUGAGG ACUUCUUC	2753
AD- 1700887.1	csasgugaGfgAfCfU fucuuccagcuL96	1858	asGfscudGg (Agn) agaagu CfcUfcacugsusc	2306	GACAGUGAGGACUU CUUCCAGCU	2754
AD- 1700888.1	usgsaggaCfuUfCfU fuccagcugcuL96	1859	asGfscadGc (Tgn) ggaaga AfgUfccucascsu	2307	AGUGAGGACUUCUU CCAGCUGCU	2755
AD- 1700889.1	gsasggacUfuCfUfU fccagcugcuL96	1860	asAfsgcdAg (Cgn) uggaag AfaGfuccucsc	2308	GUGAGGACUUCUUC CAGCUGCUG	2756
AD- 1700891.1	gsagsacuCfuUfCfC fagcugcugguL96	1861	asCfscadGc (Agn) gcugga AfgAfcagucsc	2309	GAGGACUUCUCCA GCUUGGGA	2757
AD- 1700892.1	uscsuuccAfgCfUfG fcuggaggaauL96	1862	asAfsucdCu (Cgn) cagcag CfuGfgaagasasg	2310	CUUCUCCAGCUGC UGGAGGAUG	2758
AD- 1700893.1	csasgcugCfuGfGfA fggaugacacuL96	1863	asGfsugdTc (Agn) uccucc AfgCfagcugsgsa	2311	UCCAGCUGCUGGAG GAUGACACG	2759
AD- 1700894.1	gscsugcuGfgAfGfG faugacacguL96	1864	asAfsagdTg (Tgn) cauccu CfcAfcgagcsusg	2312	CAGCUGCUGGAGGA UGACACGUG	2760
AD- 1700895.1	csusggagGfaUfGfA fcacugccuL96	1865	asAfsaggCa (Cgn) guguca UfcCfuccagscsa	2313	UGCUGGAGGAUGAC ACGUGCCUG	2761
AD- 1700897.1	gsgsaugaCfaCfGfU fgccugaugguL96	1866	asCfscadTc (Agn) ggcacg UfgUfcaucscsu	2314	GAGGAUGACACGUG CCUGAUGGU	2762
AD- 1700898.1	gsasugacAfcGfUfG fccugaugguL96	1867	asAfsccdAu (Cgn) aggcac GfuGfcaucscsu	2315	AGGAUGACACGUGC CUGAUGGUG	2763
AD- 1700899.1	gsascacgUfgCfCfU fgaugguuuL96	1868	asAfsacdAc (Cgn) aucagg CfaCfugucsas	2316	AUGACACGUGCCUG AUGGUGUUG	2764
AD- 1700900.1	gscsagucUfgGfUfC fagagcugguL96	1869	asUfscddAg (Cgn) ucugac CfaGfacugcsasa	2317	UUGCAGUCUGGUCA GAGCUGGAG	2765
AD- 1700902.1	usascaagGfaGfUfG fgagugcuguL96	1870	asAfsacdGc (Agn) cuccac UfcCfuuguasgsg	2318	CCUACAAGGAGUGG AGUGCUGUC	2766
AD- 1700903.1	ascsaaggAfgUfGfG fagugcugcuL96	1871	asGfsacdAg (Cgn) acucca CfuCfcuugusasg	2319	CUACAAGGAGUGGA GUGCUGUCA	2767

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD-1700904.1	asasggagUfgGfAfG fugcugucauuL96	1872	asAfsugdAc (Agn)gcacuc CfaCfuccuusgsu	2320	ACAAGGAGUGGAGU GCUGUCAUA	2768
AD-1700905.1	gsgsagugGfaGfUfG fcugucauuL96	1873	asAfsuadTg (Agn) cagcac UfcCfacuccsusu	2321	AAGGAGUGGAGUGC UGUCAUAUG	2769
AD-1700906.1	usgsagugGfcUfGfU fcuaauggccuL96	1874	asGfsgcdCa (Tgn) augaca GfcAfcuccascsu	2322	AGUGGAGUGCUGUC AUAUGGCCU	2770
AD-1700907.1	gsgsagugCfuGfUfC fauauggccuL96	1875	asAfsaggdCc (Agn) uaugac AfgCfacuccsasc	2323	GUGGAGUGCUGUCA UAUGGCCUG	2771
AD-1700908.1	gsasgugcUfgUfCfA fuauggccuguL96	1876	asCfsagdgGc (Cgn) auauga CfaGfcacucscsa	2324	UGGAGUGCUGUCAU AUGGCCUGG	2772
AD-1700909.1	asgsugcuGfuCfAfU fauggccuguL96	1877	asCfscadGg (Cgn) cauaug AfcAfgcacuscsc	2325	GGAGUGCUGUCAUA UGGCCUGGG	2773
AD-1700912.1	gsesugucAfuAfUfG fgccugggacuL96	1878	asGfsucdCc (Agn) ggccau AfuGfacagcsasc	2326	GUGCUGUCAUAUGG CCUGGGACG	2774
AD-1700913.1	csusgucaUfaUfGfG fccugggacguL96	1879	asCfsgudCc (Cgn) aggcca UfaUfgacagscsa	2327	UGCUGUCAUAUGGC CUGGGACGG	2775
AD-1700915.1	csasagcaCfaGfCfA faggacaucguL96	1880	asCfsgadTg (Tgn) ccuugc UfgUfgcuugsgsg	2328	CCCAAGCACAGCAA GGACAUCGC	2776
AD-1700916.1	asusucacCfuUfUfG facguguacauL96	1881	asUfsgudAc (Agn) cgucua AfgGfugaauscsg	2329	CGAUUACCUUUGA CGUGUACAA	2777
AD-1700917.1	ususcaccUfuUfGfA fcguguacaauL96	1882	asUfsugdTa (Cgn) acguca AfaGfgugaasusc	2330	GAUUCACCUUUGAC GUGUACAAG	2778
AD-1700918.1	uscsaccuUfuGfAfC fguguacaaguL96	1883	asCfsuudGu (Agn) cacguc AfaAfggugasasu	2331	AUUCACCUUUGACG UGUACAAGC	2779
AD-1700919.1	csasccuuUfgAfCfG fuguacaagcuL96	1884	asGfscudTg (Tgn) acacgu CfaAfaaggugsasa	2332	UUCACCUUUGACGU GUACAAGCA	2780
AD-1700920.1	csesuuugAfcGfUfG fuacaagcaauL96	1885	asUfsugdCu (Tgn) guacac GfuCfaaaggsusg	2333	CACCUUUGACGUGU ACAAGCAA	2781
AD-1700922.1	usgsacguGfuAfCfA fagcaaaaccuL96	1886	asGfsgudTu (Tgn) gcuugu AfcAfcgucasasa	2334	UUUGACGUGUACAA GCAAAACCC	2782
AD-1700923.1	gsascgugUfaCfAfA fgcaaaaccuL96	1887	asGfsggdTu (Tgn) ugcuug UfaCfacgucsasa	2335	UUGACGUGUACAAG CAAAACCCU	2783
AD-1700924.1	ascsguguAfcAfAfG fcaaaaccuL96	1888	asAfsaggdGu (Tgn) uugcuu GfuAfcacguscsc	2336	UGACGUGUACAAGC AAAACCCUC	2784
AD-1700925.1	csgsuguaCfaAfGfC faaaaccuL96	1889	asGfsagdgGg (Tgn) uuugcu UfgUfacacgsusc	2337	GACGUGUACAAGCA AAACCCUCG	2785
AD-1700929.1	csasagcaAfaAfCfC fcucgagaccuL96	1890	asGfsgudCu (Cgn) gagggg UfuUfgcuugsusa	2338	UACAAGCAAAACCC UCGAGACCU	2786
AD-1700930.1	asasgcaaAfaCfCfC fucgagaccuL96	1891	asAfsaggdTc (Tgn) cgaggg UfuUfgcuusgsu	2339	ACAAGCAAAACCCU CGAGACCU	2787
AD-1700931.1	asgscaaaAfcCfCfU fcgagaccuL96	1892	asGfsagdgGu (Cgn) ucgagg GfuUfuugcusug	2340	CAAGCAAAACCCUC GAGACCUCU	2788
AD-1700934.1	gsasgaccUfcUfUfU fggcagccuL96	1893	asCfsagdgGc (Tgn) gccaaa GfaGfgucucsgsa	2341	UCGAGACCUCUUUG GCAGCCUGA	2789

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEb dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700936.1	csccsucuUfgGfCfA fgccugaauL96	1894	asCfsaudTc (Agn)ggcugc CfaAfaagaggsusc	2342	GACCUCUUUGGCAG CCUGAAUGU	2790
AD- 1700937.1	csuscuuUfgGfCfA fccugaauL96	1895	asAfsadTu (Cgn) aggcug CfcAfaagagsgsu	2343	ACCUCUUUGGCAGC CUGAAUGUC	2791
AD- 1700938.1	csusuuggCfaGfCfC fugaauL96	1896	asUfsgadCa (Tgn)ucaggc UfgCfcaagsasg	2344	CUCUUUGGCAGCCU GAAUGUCAA	2792
AD- 1700939.1	ususuggCfGfCfU fgaauL96	1897	asUfsgdAc (Agn)uucagg CfuGfcaagsa	2345	UCUUUGGCAGCCUG AAUGUCAA	2793
AD- 1700940.1	usgsgcagCfCfGfA fauguaaL96	1898	asCfsuudTg (Agn)cauuc GfgCfugccasasa	2346	UUUGGCAGCCUGAA UGUCAAAGC	2794
AD- 1700942.1	gscsagccUfgAfaFfU fguaaL96	1899	asGfsgcdTu (Tgn)gacauu CfaGfgcugcscsa	2347	UGGCAGCCUGAAUG UCAAGCCA	2795
AD- 1700943.1	csasgcccUfaFfUfG fuaaL96	1900	asUfsggdCu (Tgn)ugacau UfcAfggcugscsc	2348	GGCAGCCUGAAUGU CAAAGCCAC	2796
AD- 1700944.1	asgsgccUfaFfGfU fcaaaL96	1901	asGfsugdGc (Tgn)uugaca UfuCfaggcugsgc	2349	GCAGCCUGAAUGUC AAAGCCACA	2797
AD- 1700947.1	asusgucaAfaGfCfC facauL96	1902	asGfsuadGa (Agn)uguggc UfuUfgacaususc	2350	GAAUGUCAAAGCCA CAUUCUACG	2798
AD- 1700948.1	asgsgccAfuUfCfU facggL96	1903	asAfsagadGc (Cgn)cgua AfuGfggcususu	2351	AAAGCCACAUUCUA CGGGCUCUA	2799
AD- 1700949.1	gscscacaUfuCfUfA fcggL96	1904	asUfsagadAg (Cgn)ccguag AfaUfggcususu	2352	AAGCCACAUUCUAC GGGUCUAC	2800
AD- 1700950.1	csascauUfuAfcfG fggcL96	1905	asAfsagdAg (Agn)gcccgu AfgAfaugsgsgc	2353	GCCACAUUCUACGG GCUCUACUC	2801
AD- 1700952.1	csasuucUfcGfGfG fcucuL96	1906	asAfsagadGu (Agn)gagccc GfuAfgaugsusg	2354	CACAUUCUACGGGC UCUACUCUA	2802
AD- 1700953.1	asusucUfcGfGfC fucuL96	1907	asUfsagadAg (Tgn)agagcc CfGfagaugsu	2355	ACAUCUACGGGCU CUACUCUAU	2803
AD- 1700955.1	uscscacGfGfCfUfC fuacuL96	1908	asCfsaudAg (Agn)guagag CfcCfaguasasu	2356	AUUCUACGGGCU ACUCUAUGA	2804
AD- 1700957.1	ascsgggUfcUfAfc fucuaL96	1909	asAfsadCa (Tgn)agagua GfaGfcccusag	2357	CUACGGGCUACUCU CUAUGAGUU	2805
AD- 1700958.1	gsgsgcUfaCfUfC fuaL96	1910	asCfsaadCu (Cgn)auagag UfaGfagccsgsu	2358	ACGGGCUACUCU AUGAGUUGU	2806
AD- 1700959.1	gsgscuAfcUfCfU faugaL96	1911	asAfsadAc (Tgn)cauaga GfuAfgagccscg	2359	CGGGCUACUCUA UGAGUUGU	2807
AD- 1700960.1	csusacuUfaUfGfA fguL96	1912	asAfsagdCa (Cgn)aacuca UfaGfaguagsag	2360	CUCUACUCUAUGAG UUGUGACUU	2808
AD- 1700961.1	usasucuAfuGfAfc fuugL96	1913	asAfsagdTc (Agn)caacuc AfuAfgaguasga	2361	UCUACUCUAUGAGU UGUGACUUU	2809
AD- 1700962.1	ascscuUfgAfcGfU fugL96	1914	asAfsaadGu (Cgn)acaacu CfaUfagagusag	2362	CUACUCUAUGAGUU GUGACUUUC	2810
AD- 1700963.1	csuscuaUfaGfUfU fguL96	1915	asGfsaadAg (Tgn)cacaac UfcAfuagagsusa	2363	UACUCUAUGAGUUG UGACUUUCA	2811

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700965.1	csusaugaGfuUfGfU fgacuuccaauL96	1916	asUfsugdAa (Agn) gucaCa AfcUfcuagsgasg	2364	CUCUAUGAGUUGUG ACUUUCAAG	2812
AD- 1700967.1	gsasguugUfgAfCfU fuucaaggacuL96	1917	asGfsucdCu (Tgn) gaaagu CfaCfaacuscasa	2365	AUGAGUUGUGACUU UCAAGGACU	2813
AD- 1700968.1	asgsuuguGfaCfUfU fucaaggacuL96	1918	asAfsugdCc (Tgn) ugaaag UfcAfcaacuscasa	2366	UGAGUUGUGACUUU CAAGGACUU	2814
AD- 1700969.1	ususggcccCfaAfAfG faaaguacucuL96	1919	asGfsagdTa (Cgn) uuucuu UfgGfgccaaasgsu	2367	ACUUGGCCCAAAGA AAGUACUCA	2815
AD- 1700970.1	usgsgccCfaAfGfA faaguacucuL96	1920	asUfsgadGu (Agn) cuuucu UfuGfggccasasg	2368	CUUGGCCCAAAGAA AGUACUCAG	2816
AD- 1700971.1	gsgscccaAfaGfAfA faguacucaguL96	1921	asCfsugdAg (Tgn) acuuuc UfuUfgggccsasa	2369	UUGGCCCAAAGAAA GUACUCAGG	2817
AD- 1700973.1	cscscaaaGfaAfAfG fuacucaggguL96	1922	asCfscddTg (Agn) guacuu UfcUfuugggscsc	2370	GGCCCAAAGAAAGU ACUCAGGGA	2818
AD- 1700975.1	asasagaaAfgUfAfC fucaggagacuL96	1923	asGfscudCc (Cgn) ugagua CfuUfucuuusgsu	2371	CCAAAGAAAGUACU CAGGGAGCU	2819
AD- 1700976.1	asasgaaaGfuAfCfU fcaggagacuL96	1924	asAfsgcdTc (Cgn) cugagu AfcUfuucuuusug	2372	CAAAGAAAGUACUC AGGGAGCUC	2820
AD- 1700977.1	asgsaaaGfUfAfCfU faggagacuL96	1925	asGfsagdCu (Cgn) ccugag UfaCfuucucusu	2373	AAAGAAAGUACUCA GGGAGCUC	2821
AD- 1700978.1	asgsuacuCfaGfGfG fagcuccuucuL96	1926	asGfsaadGg (Agn) gcuccc UfgAfguacucusu	2374	AAAGUACUCAGGGA GCUCUUCG	2822
AD- 1700980.1	asgsaggagCfuCfCfU fucguaggacuL96	1927	asGfsucdCa (Agn) cgaagg AfgCfuccucgsa	2375	UCAGGGAGCUCUUU CGUUGGACC	2823
AD- 1700981.1	gsagsagcuCfCfUfC fguaggaccuuL96	1928	asAfsaggdTc (Cgn) aacgaa GfgAfgcuccscsu	2376	AGGGAGCUCUUUCG UUGGACCUC	2824
AD- 1700982.1	ususcguuGfgAfCfC fuccacacuguL96	1929	asCfsagdTg (Tgn) ggaggu CfcAfacgaasgsu	2377	CCUUCGUUGGACCU CCACACUGC	2825
AD- 1700984.1	asgsuaggAfcCfUfC fcacacugcuL96	1930	asAfsgcdAg (Tgn) guggag GfuCfcaacgsasa	2378	UUCGUUGGACCUC ACACUGCUG	2826
AD- 1700985.1	gsasccucCfaCfAfC fugcugcaaguL96	1931	asCfsuudGc (Agn) gcagug UfgGfaggucscsa	2379	UGGACCUCACACU GCUGCAAGG	2827
AD- 1700986.1	gsgsccugGfgCfCfA fuauguugcuL96	1932	asAfsgcdAa (Cgn) auaugg CfcCfaggccsusu	2380	AAGGCCUGGGCCAU AUGUUGCUG	2828
AD- 1700988.1	gsgscccaUfaUfGfU fugcugggaauL96	1933	asUfsucdCc (Agn) gcaaca UfaUfgggccsasa	2381	CUGGGCCAUUGUU GCUGGGAU	2829
AD- 1700989.1	gsgsccauAfuGfUfU fgcugggaauL96	1934	asAfsuudCc (Cgn) agcaac AfuAfuaggccscsa	2382	UGGGCCAUUGUUG CUGGGAAU	2830
AD- 1700990.1	cscsauauGfuUfGfC fugggaauuuL96	1935	asAfsaadTu (Cgn) ccagca AfcAfuauaggscsc	2383	GGCCAUUGUUGCU GGGAUUUC	2831
AD- 1700991.1	csasuaugUfuGfCfU fgggaauuuL96	1936	asGfsaadAu (Tgn) cccagc AfaCfauaugsgsc	2384	GCCAUUGUUGCUG GGAAUUUC	2832
AD- 1700992.1	usgsuugcUfgGfGfA fauuuccuccuL96	1937	asGfsagdGg (Agn) aaaucc CfaGfcaacasusa	2385	UAUGUUGCUGGGAA UUUCCUCCA	2833

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD- 1700995.1	usgsggaaUfuUfCfC fuccacccuuuL96	1938	asAfsagdGg (Tgn) ggagga AfaUfucccasgsc	2386	GCUGGGAAUUUCCU CCACCCUUC	2834
AD- 1700997.1	ususccucCfaCfCfC fuucgucauguL96	1939	asCfsaudGa (Cgn) gaaggg UfgGfaggaasasu	2387	AUUUCCUCCACCCU UCGUCAUGC	2835
AD- 1700998.1	asasgggcCfGfCfCfU fccauuccuauL96	1940	asUfsagdGa (Agn) uggagg CfGfGfccuuscscu	2388	AGAAGGGCCGCCUC CAUCCUAC	2836
AD- 1701001.1	gscscgccUfcCfAfU fuccuacuaauL96	1941	asUfsuadGu (Agn) ggaaug GfaGfgcgscscsc	2389	GGCCGCCUCCAUU CCUACUAG	2837
AD- 1701002.1	cscsgccuCfcAfUfU fccuacuaaguL96	1942	asCfsuudAg (Tgn) aggaau GfgAfgggcgscsc	2390	GGCCGCCUCCAUUC CUACUAGG	2838
AD- 1701003.1	cscscacuGfcAfAfA fgacuauagacuL96	1943	asGfsuadAu (Agn) gucuuu GfcAfgugggsusc	2391	GACCCACUGCAAAG ACUAGACA	2839
AD- 1701004.1	cscscacugCfaAfAfG facuagacauL96	1944	asUfsgudCa (Tgn) agucuu UfgCfaguggsgsu	2392	ACCCACUGCAAAGA CUAUGACAG	2840
AD- 1701005.1	csascugcAfaAfGfA fcuauagacaguL96	1945	asCfsugdTc (Agn) uagucu UfuGfcaguggsgsg	2393	CCCACUGCAAAGAC UAUGACAGC	2841
AD- 1701007.1	asasgacuAfuGfAfC fagcaucaaauL96	1946	asUfsuudGa (Tgn) gcuguc AfuAfgucuuusug	2394	CAAAGACUAUGACA GCAUCAAAU	2842
AD- 1701009.1	gsascagcAfuCfAfA fauuucaggauL96	1947	asUfscddTg (Agn) aauuug AfuGfcugucsasu	2395	AUGACAGCAUCAA UUUCAGGAC	2843
AD- 1701011.1	csasgcauCfaAfAfU fuucaggaccuL96	1948	asGfsgudCc (Tgn) gaaaau UfgAfuGfcugusc	2396	GACAGCAUCAAUU UCAGGACCU	2844
AD- 1701012.1	asgscaucAfaAfUfU fucaggaccuuL96	1949	asAfsaggdTc (Cgn) ugaaa UfuGfaugcusgsu	2397	ACAGCAUCAAUUU CAGGACCU	2845
AD- 1701013.1	gscscauAfaUfUfU fcaggaccuguL96	1950	asCfsagdGu (Cgn) cugaaa UfuUfgaugcsusg	2398	CAGCAUCAAUUUUC AGGACCU	2846
AD- 1701014.1	csasucaaAfuUfUfC faggaccugcuL96	1951	asGfscadGg (Tgn) ccugaa AfuUfugaugscsu	2399	AGCAUCAAUUUCA GGACCU	2847
AD- 1701016.1	asusucaGfGfAfCfC fugcagacaguL96	1952	asCfsugdTc (Tgn) gcaggu CfcUfgaaususu	2400	AAAUUUCAGGACCU GCAGACAGU	2848
AD- 1701017.1	ususucagGfaCfCfU fgcagacaguL96	1953	asAfsuadGu (Cgn) ugcagg UfcCfugaasusu	2401	AAUUUCAGGACCU CAGACAGUA	2849
AD- 1701019.1	csasggacCfuGfCfA fgacaguacauL96	1954	asUfsgudAc (Tgn) gucugc AfgGfuccgsasa	2402	UUCAGGACCU	2850
AD- 1701020.1	gsasccugCfaGfAfC faguacaggcuL96	1955	asGfscddTg (Tgn) acuguc UfgCfaggucscsu	2403	AGGACCU	2851
AD- 1701022.1	cscsugcaGfaCfAfG fuacaggcuauL96	1956	asUfsagdCc (Tgn) guacug UfcUfgcaggusc	2404	GACCU	2852
AD- 1701023.1	csusgcagAfcAfGfU facaggcuaguL96	1957	asCfsuadGc (Cgn) uguacu GfuCfugcagsgsu	2405	ACCUGCAGACAGUA CAGGCU	2853
AD- 1701024.1	usgscagaCfaGfUfA fcaggcuagauL96	1958	asUfscudAg (Cgn) cuguac UfgUfcugcagsgg	2406	CCUGCAGACAGUAC AGGCUAGAU	2854
AD- 1701026.1	csasgacaGfuAfCfA fggcuagauauL96	1959	asUfsaudCu (Agn) gccugu AfcUfgucugscsa	2407	UGCAGACAGUACAG GCUAGAUAA	2855

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD-1701027.1	gsascaguAfcAfGfG fcuagauaacuL96	1960	asGfsuudAu (Cgn) uagccu GfuAfcugucsusg	2408	CAGACAGUACAGGC UAGAUAAACC	2856
AD-1701029.1	csesuagcCfaCfCfU fuucuggccuuL96	1961	asAfsaggdCc (Agn) gaaagg UfgGfcuaggsasa	2409	UUCCUAGCCACCUU UCUGGCCUC	2857
AD-1701031.1	csusuucuGfgCfCfU fcccacuugcuL96	1962	asGfscadAg (Tgn) gggagg CfcAfgaaagsgsu	2410	ACCUUUUCUGGCCUC CCACUUGCC	2858
AD-1701032.1	uscsuggcCfuCfCfC facuugccuuL96	1963	asAfsaggdGc (Agn) aguggg AfgGfccagasa	2411	UUUCUGGCCUCCCA CUUGCCUCG	2859
AD-1701033.1	gsgscaucUfuGfAfU fcccagccuL96	1964	asCfsagdGc (Tgn) gggauC AfaGfaugccsusg	2412	CAGGCAUCUUGAUC CCAGCCUGA	2860
AD-1701034.1	gscsaucUfgAfUfC fccagccuL96	1965	asUfscadGg (Cgn) ugggau CfaAfgaugcscsu	2413	AGGCAUCUUGAUCC CAGCCUGAC	2861
AD-1701035.1	ascscaauGfcCfUfC fcuguccuL96	1966	asUfsagdGg (Agn) caggag GfcAfuuggusasg	2414	CUACCAUAGCCUCC UGUCCUAA	2862
AD-1701037.1	asusgccuCfcUfGfU fcccuaaacuuL96	1967	asAfsagdTu (Agn) gggaca GfgAfggcausug	2415	CAAUGCCUCCUGUC CCUAAACUC	2863
AD-1701038.1	gscscuccUfgUfCfC fcuaaacuccuL96	1968	asGfsgadGu (Tgn) uaggga CfaGfgaggcsasu	2416	AUGCCUCCUGUCCC UAAACUCCC	2864
AD-1720280.1	asasaacccuCfGfAf gaccucuL96	1969	asdAsadAg (Agn) ggucudC gAfgGfguuusgsc	2417	GCAAAACCCUCGAG ACCUCUUUG	2865
AD-1720281.1	csusgggcuGfGfAf ucccagguuL96	1970	asdAsadCc (Tgn) gggaudC uCfaGfccagsgsa	2418	AGGAAAGAGCAGAU CCCAGGUUU	2866
AD-1720282.1	gsusgggagCfGfAf ggacuL96	1971	asdAsadGa (Agn) guccudC aCfuGfuccacsusg	2419	CAGUGGACAGUGAG GACUUCUUC	2867
AD-1720283.1	csasaaacccUfCfGf agaccucuL96	1972	asdAsadGg (Tgn) cuCdG aGfgGfuuuugscsu	2420	AGCAAAACCCUCGA GACCUCUUU	2868
AD-1720284.1	csusggauguCfAfAf agccacauuL96	1973	asdAsadTg (Tgn) ggcuudT gAfcAfuucagsgsc	2421	GCCUGAAUGUCAAA GCCACAUUC	2869
AD-1720285.1	gsgscucuAfcUfCfU faugaguL96	1974	asdAscadAc (Tgn) cauadG aGfuAfgagccscsg	2422	CGGGCCUACUCUA UGAGUUGUG	2870
AD-1720286.1	asasuggaguGfCfUf aaccucguuL96	1975	asdAscc (Agn) gdGguadG cAfcUfccauuscsa	2423	UGAAUGGAGUGCUA ACCCUGGUG	2871
AD-1720287.1	asasuggaguGfCfUf aaccucguuL96	1976	asdAsccdAadGguadGcAf cuccauuscsa	2424	UGAAUGGAGUGCUA ACCCUGGUG	2872
AD-1720288.1	asasuggaguGfCfUf aacucugguuL96	1977	asdAsccdAgdAguadGcAf cuccauuscsa	2425	UGAAUGGAGUGCUA ACCCUGGUG	2873
AD-1720289.1	ascscuccaCfuGfCfU fgauggaguL96	1978	asdAscudCc (Agn) uucadG cAfgUfagguscsu	2426	AGACCUACUGCUG AAUGGAGUG	2874
AD-1720290.1	ascscugcagAfcCfAf guacaggcuL96	1979	asdAsgdCc (Tgn) guacudG uCfuGfcagguscsu	2427	GGACCUAGACAG UACAGGCUA	2875
AD-1720291.1	csusgggaaUfUfCf cuccaccuuL96	1980	asdAsgdGg (Tgn) ggaggdA aAfuUfccagscsa	2428	UGCUGGGAUUUCC UCCACCUU	2876
AD-1720292.1	gsgsgccgccUfCfCf auuccuacuL96	1981	asdAsguadGg (Agn) augdG aGfgCfggcccsusu	2429	AAGGGCCGCCUCCA UUCCUACUA	2877

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEA dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD-1720293.1	ususcucacggGfCfUf cuacucuaauL96	1982	asdAsudAg (Agn) guagadG cCfcGfuagaasusg	2430	CAUUCUACGGGCUC UACUCUAUG	2878
AD-1720294.1	gscsagacagUfAfCf aggcuaagauL96	1983	asdAsudCu (Agn) gccugdT aCfuGfucgcsasg	2431	CUGCAGACAGUACA GGCUAAGUA	2879
AD-1720295.1	usgsgagaccCfUfAf cugcugaauL96	1984	asdAsudTc (Agn) gcagudA gGfgUfcuccasasu	2432	AUUGGAGACCCUAC UGCUGAUG	2880
AD-1720296.1	ascscucuuUfGfCf agccugaauL96	1985	asdAsudTc (Agn) gccugdC cAfaAfgagguscsu	2433	AGACCUCUUUGGCA GCCUGAUG	2881
AD-1720297.1	cscsuuucugGfCfCf uccacacuugL96	1986	asdCsadAg (Tgn) gggagdG cCfaGfaaaggsusg	2434	CACCUUUCUGGCCU CCCACUUGC	2882
AD-1720298.1	usgsgaacUfGfCfAf fuggacagugL96	1987	asdCsacdTg (Tgn) ccacdT gCfaGfuuccasusc	2435	GAUGGAACUGCAGU GGACAGUGA	2883
AD-1720299.1	asgsgaugacAfCfGf ugccugaugL96	1988	asdCsadTc (Agn) ggcacdG uGfuCFauccuscsc	2436	GGAGGAUGACACGU GCCUGAUGG	2884
AD-1720300.1	gsasauggagUfGfCf uaaccucggL96	1989	asdCsacdGg (Tgn) uagdC aCfuCFcauucsasg	2437	CUGAAUGGAGUGCU AACCCUGGU	2885
AD-1720301.1	gsusgcugucAfUfAf uggcucggguL96	1990	asdCscc (Agn) gdGccaudA uGfaCfagcacsusc	2438	GAGUGCUGUCAUAU GGCCUGGGA	2886
AD-1720302.1	gsusgcugucAfUfAf uggcucggguL96	1991	asdCsccdAadGccaudAuGf acagcacsusc	2439	GAGUGCUGUCAUAU GGCCUGGGA	2887
AD-1720303.1	gsusgcugucAfUfAf uggcucggguL96	1992	asdCsccdAgdAccaudAuGf acagcacsusc	2440	GAGUGCUGUCAUAU GGCCUGGGA	2888
AD-1720304.1	gscsccaaagAfAfAf guacucagguL96	1993	asdCscdTg (Agn) guacudT uCfuUfugggcsca	2441	UGGCCAAAGAAAG UACUCAGGG	2889
AD-1720305.1	gsusacucagGfGfAf gcuccuucguL96	1994	asdCsagaadGg (Agn) gcudC cCfuGfaguacsusu	2442	AAGUACUCAGGGAG CUCCUUCGU	2890
AD-1720306.1	gsgsaauuucCfUfCf caccuucguL96	1995	asdCsga (Agn) gdGguggdA gGfaAfauccscsa	2443	UGGGAAUUUCCUCC ACCCUUCGU	2891
AD-1720307.1	gsgsaauuucCfUfCf caccuucguL96	1996	asdCsgadAadGguggdAgGf aaauccscsa	2444	UGGGAAUUUCCUCC ACCCUUCGU	2892
AD-1720308.1	gsgsaauuucCfUfCf cacucucguL96	1997	asdCsgadAgdAguggdAgGf aaauccscsa	2445	UGGGAAUUUCCUCC ACCCUUCGU	2893
AD-1720309.1	gsusguacaaGfCfAf aaaccucguL96	1998	asdCsagadGg (Tgn) uuudG cUfuGfuacacsusu	2446	ACGUGUACAAGCAA AACCCUCGA	2894
AD-1720310.1	csasgucuggUfCfAf gagcuggaguL96	1999	asdCsudCc (Agn) gcucudG aCfcAfgacugscsa	2447	UGCAGUCUGGUCAG AGCUGGAGC	2895
AD-1720311.1	usgscuaaccCfUfGf gugcuagaguL96	2000	asdCsudCu (Agn) gcaccdA gGfgUfuagcascsu	2448	AGUGCUAACCCUGG UGCUAGAGG	2896
AD-1720312.1	uscasaauuucfAfGf gaccugcaguL96	2001	asdCsudGc (Agn) gguccdT gAfaAfuuugasusg	2449	CAUCAAUUUCAGG ACCUGCAGA	2897
AD-1720313.1	asgsaaccacCfCfAf gccagaaguL96	2002	asdCsudTc (Tgn) gggcudG gGfuGfguucuscscu	2450	AGAGAACCCACCCAG CCCAGAAGU	2898
AD-1720314.1	csasaacuaaCfAfAf aaacauuucL96	2003	asdGsadAa (Tgn) guuuudT gUfuAfguuugsasg	2451	CUCAAACUAACAAA AACAUUCC	2899

TABLE 6-continued

Modified Sense and Antisense Strand Sequences of Human CIDEB dsRNA Agents						
Duplex ID	Sense Sequence 5' to 3'	SEQ ID NO:	Antisense Sequence 5' to 3'	SEQ ID NO:	mRNA target sequence	SEQ ID NO:
AD-1720315.1	csasuggaguAfCfCf ucucagcucuL96	2004	asdGsadGc (Tgn) gagagdG uAfcUfccaugsgsu	2452	ACCAUGGAGUACCU CUCAGCUCU	2900
AD-1720316.1	uscsguuggaCfCfUf ccacacugcuL96	2005	asdGsdcAg (Tgn) guggadG gUfcCfaacgasasg	2453	CUUCGUUGGACCUC CACACUGCU	2901
AD-1720317.1	ascscuaggagfUfAf ccucucagcuL96	2006	asdGsdcTg (Agn) gaggudA cUfcCfauggusgsg	2454	CCACCAUGGAGUAC CUCUCAGCU	2902
AD-1720368.1	gsgscagccuGfAfAf ugucaagcuL96	2007	asdGsdcTu (Tgn) gacaudT cAfgGfcugccsasa	2455	UUGGCAGCCUGAAU GUCAAAGCC	2903
AD-1720369.1	uscscuagccAfCfCf uuucuggccuL96	2008	asdGsdcCc (Agn) gaaagdG uGfgCfuaggasasa	2456	UUUCCUAGCCACCU UUCUGGCCU	2904
AD-1720370.1	usgscugaauGfGfAf gugcuaccuL96	2009	asdGsdcTu (Agn) gcacudC cAfuUfcagcasgsu	2457	ACUGCUGAAUGGAG UGCUAACCC	2905
AD-1720371.1	uscscaggaccUfGfCf agacaguacuL96	2010	asdGsudAc (Tgn) gucugdC aGfgUfccugasasa	2458	UUUCAGGACCUGCA GACAGUACA	2906
AD-1720372.1	ascscagcauAfAfAf uuucaggacuL96	2011	asdGsudCc (Tgn) gaaudT uGfaUfgcuguscsa	2459	UGACAGCAUCAAU UUCAGGACC	2907
AD-1720373.1	ususcugggAfAfUf uuucccacuL96	2012	asdGsudGg (Agn) ggaaadT uCfcCfagcaascsa	2460	UGUUGCUGGGAAU UCCUCCACC	2908
AD-1720374.1	asasugccuUfGfGf uccuuaacuL96	2013	asdGsudTu (Agn) gggacdA gGfaGfgcauugsgsg	2461	CCAAUGCCUCCUGU CCCUAAACU	2909
AD-1720375.1	ususcagcugUfAfCf aagcaaacuL96	2014	asdGsudTu (Tgn) gcuugdT aCfaCfgucaasasg	2462	CUUUGACGUGUACA AGCAAAACC	2910
AD-1720376.1	gsgscgccuUfCfAf uuuccuacuL96	2015	asdTsadGu (Agn) ggaaudG gAfgGfcgcccscsu	2463	AGGGCCGCCUCCAU UCCUACUAA	2911
AD-1720377.1	ascsucagguCfAfGf uaucuaauL96	2016	asdTsadTu (Agn) gauacdT gAfcCfugagusasa	2464	UUACUCAGGUCAGU AUCUAAUUAU	2912
AD-1720378.1	usgsggccauAfUfGf uugcugggauL96	2017	asdTscdCc (Agn) gcaacdA uAfuGfgcccasgsg	2465	CCUGGGCCAUUGU UGCUGGGAA	2913
AD-1720379.1	usgscuguaUfAfUf ggccugggauL96	2018	asdTscdCc (Agn) ggccadT aUfgAfcagcascsu	2466	AGUGCUGUCAUAUG GCCUGGGAC	2914
AD-1720380.1	usgsaguuguGfAfCf uuucaaggauL96	2019	asdTscdCu (Tgn) gaaagdT cAfcAfacucasusa	2467	UAUGAGUUGUGACU UUCAAGGAC	2915
AD-1720381.1	asgsgcccaGfCfAf cagcaaggauL96	2020	asdTscdCu (Tgn) gcugudG cUfuGfggccuscscu	2468	AGAGGCCCAAGCAC AGCAAGGAC	2916
AD-1720382.1	usgsuacaagCfAfAf aaccucgauL96	2021	asdTscg (Agn) gdGguudT gCfuUfguacascsg	2469	CGUGUACAAGCAAA ACCCUCGAG	2917
AD-1720383.1	usgsuacaagCfAfAf aaccuucgauL96	2022	asdTscgdAadGguudTgCf uuguacascsg	2470	CGUGUACAAGCAAA ACCCUCGAG	2918
AD-1720384.1	usgsuacaagCfAfAf aacucucgauL96	2023	asdTscgdAgdAguudTgCf uuguacascsg	2471	CGUGUACAAGCAAA ACCCUCGAG	2919
AD-1720385.1	uscscuagagUfUfGf ugacuuuL96	2024	asdTsgdAa (Agn) gucacdA aCfuCfauagasgsu	2472	ACUCUAUGAGUUGU GACUUUCAA	2920
AD-1720386.1	asasagacuUfGfAf cagcaucauL96	2025	asdTsudGa (Tgn) gcugudC aUfaGfucuuusgsc	2473	GCAAAGACUAUGAC AGCAUCAA	2921

Example 2. In Vitro Screening of CIDEB siRNA

Experimental Methods

Cell Culture and Transfections:

Hepa1c7 cells are grown to near confluence at 37° C. in an atmosphere of 5% CO₂ in Minimum Essential Medium Alpha (Gibco) supplemented with 10% FBS (ATCC) before being released from the plate by trypsinization. Transfection is carried out by adding 14.8 µl of Opti-MEM plus 0.2 µl of Lipofectamine RNAiMax per well (Invitrogen, Carlsbad CA. cat #13778-150) to 5 µl of each siRNA duplex to an individual well in a 96-well plate. The mixture is then incubated at room temperature for 15 minutes. Eighty µl of complete growth media without antibiotic containing ~2×10⁴ cells are then added to the siRNA mixture. Cells are incubated for 24 hours prior to RNA purification. Single dose experiments are performed at 10 nM, 1 nM and/or 0.1 nM final duplex concentration.

Panc-1 cells are grown to near confluence at 37° C. in an atmosphere of 5% CO₂ in Minimum Essential Medium Alpha (Gibco) supplemented with 10% FBS (ATCC) before being released from the plate by trypsinization. Transfection is carried out by adding 14.6 µl of Opti-MEM plus 0.4 µl of Lipofectamine 2000 per well to 5 µl of each siRNA duplex to an individual well in a 96-well plate. The mixture is then incubated at room temperature for 15 minutes. Eighty µl of complete growth media without antibiotic containing ~1.5×10⁴ cells are then added to the siRNA mixture. Cells are incubated for 24 hours prior to RNA purification. Dose experiments are performed at 10 nM and 0.1 nM final duplex concentration.

Hep3B cells were grown to near confluence at 37° C. in an atmosphere of 5% CO₂ in Minimum Essential Medium Alpha (Gibco) supplemented with 10% FBS (ATCC) before being released from the plate by trypsinization. Transfection was carried out by adding 14.6 µl of Opti-MEM plus 0.4 µl of Lipofectamine RNAiMax per well to 5 µl of each siRNA duplex to an individual well in a 96-well plate. The mixture was then incubated at room temperature for 15 minutes. Eighty µl of complete growth media without antibiotic containing ~1.5×10⁴ cells were then added to the siRNA mixture. Cells were incubated for 24 hours prior to RNA purification. Dose experiments were performed at 10 nM, 1 nM and 0.1 nM final duplex concentration.

Total RNA Isolation Using DYNABEADS mRNA Isolation Kit:
RNA was isolated using an automated protocol on a BioTek-EL406 platform using DYNABEADS (Invitrogen, cat #61012). Briefly, 70 µl of Lysis/Binding Buffer and 10 µl of lysis buffer containing 3 µl of magnetic beads were added to the plate with cells. Plates were incubated on an electromagnetic shaker for 10 minutes at room temperature and then magnetic beads were captured and the supernatant was removed. Bead-bound RNA was then washed 2 times with 150 µl Wash Buffer A and once with Wash Buffer B. Beads were then washed with 150 µl Elution Buffer, re-captured and supernatant removed.

Synthesis Using ABI High Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Foster City, CA, Cat #4368813):

Ten µl of a master mix containing 1 µl 10× Buffer, 0.4 µl 25× dNTPs, 1 µl 1 Ox Random primers, 0.5 µl Reverse Transcriptase, 0.5 µl RNase inhibitor and 6.6 µl of H₂O per reaction was added to RNA isolated above. Plates were sealed, mixed, and incubated on an electromagnetic shaker for 10 minutes at room temperature, followed by 2 h 37° C.

Real Time PCR:

Two µl of cDNA and 5 µl Lightcycler 480 probe master mix (Roche Cat #04887301001) were added to either 0.5 µl of human GAPDH TaqMan Probe and 0.5 µl human CIDEB probe per well in a 384 well plates (Roche cat #04887301001). Real time PCR was done in a LightCycler480 Real Time PCR system (Roche). Each duplex was tested at least two times and data were normalized to cells transfected with a non-targeting control siRNA. To calculate relative fold change, real time data were analyzed using the ΔΔCt method and normalized to assays performed with cells transfected with a non-targeting control siRNA.

The results of the screening of the dsRNA agents listed in Table 6 in Hep3B cells are shown in Table 7. An additional single dose (10 nM) screening of the dsRNA agents listed in Table 4 in Hep3B cells was performed. CIDEB signal was normalized to GAPDH. The results are shown in Table 8.

TABLE 7

CIDEB Single Dose Screens in Hep3B Cells						
Duplex ID	10 nM Dose		1 nM Dose		0.1 nM Dose	
	Avg % CIDEB mRNA Remaining	SD	Avg % CIDEB mRNA Remaining	SD	Avg % CIDEB mRNA Remaining	SD
AD-1700791.1	88.90	10.50	99.00	15.60	151.20	9.60
AD-1699864.1	92.80	20.20	107.70	22.30	140.30	17.20
AD-1699867.1	27.40	8.30	41.80	6.50	72.30	3.30
AD-1720281.1	77.20	10.40	91.20	14.40	92.70	19.30
AD-1699868.1	41.80	9.30	40.30	12.80	73.30	4.20
AD-1699869.1	63.10	8.70	93.40	1.80	94.50	9.70
AD-1699870.1	54.70	10.50	77.00	15.50	95.70	6.10
AD-1699871.1	32.60	6.70	55.70	7.40	62.00	7.70
AD-1699872.1	49.30	7.50	79.70	18.10	87.40	24.30
AD-1700793.1	44.90	7.40	60.30	6.80	67.40	8.50
AD-1699873.1	42.30	8.10	60.70	5.70	74.90	14.20
AD-1700794.1	78.00	14.40	87.00	8.10	82.70	10.40
AD-1699875.1	47.50	13.10	94.00	11.70	104.60	12.80
AD-1700795.1	73.30	11.30	29.70	9.70	73.80	10.10
AD-1700796.1	58.60	15.10	61.10	10.30	67.40	3.60
AD-1700797.1	62.20	7.50	65.00	15.00	61.30	11.70
AD-1700798.1	29.90	2.50	38.50	13.10	72.50	10.40
AD-1699882.1	42.40	7.40	47.40	11.50	68.00	13.80
AD-1700799.1	40.40	3.40	66.10	11.30	95.40	11.80
AD-1700800.1	46.90	8.60	40.00	9.40	75.00	7.50
AD-1699891.1	55.50	7.50	88.80	7.80	115.80	19.50
AD-1700801.1	54.10	3.90	81.00	15.40	96.80	11.20
AD-1699892.1	66.20	2.70	103.90	15.40	110.00	8.60
AD-1700802.1	19.30	4.60	57.70	8.30	110.90	16.20
AD-1699904.1	36.80	9.60	64.00	11.40	98.10	13.00
AD-1720313.1	80.30	14.70	94.40	20.60	91.30	8.30
AD-1700804.1	43.00	6.20	70.40	21.00	102.60	13.70
AD-1700805.1	36.30	6.40	49.90	4.20	83.20	14.30
AD-1700806.1	7.00	2.20	39.90	7.30	79.20	15.20
AD-1699907.1	18.00	4.20	48.80	12.30	94.00	11.30
AD-1700807.1	10.80	1.90	51.50	5.70	98.40	3.80
AD-1700808.1	31.30	3.20	70.10	15.30	94.90	16.40
AD-1700809.1	34.00	3.90	92.20	13.00	113.70	4.20
AD-1700810.1	15.60	1.10	25.70	3.00	99.20	7.70
AD-1700811.1	17.30	2.80	37.20	7.40	78.00	10.10
AD-1700812.1	35.90	5.40	38.10	10.60	33.40	0.30
AD-1700813.1	2.20	0.70	7.90	1.40	59.20	12.30
AD-1700814.1	11.70	2.00	32.60	6.90	110.80	15.90
AD-1700815.1	21.00	4.60	27.10	5.80	58.50	12.30
AD-1700816.1	5.10	1.10	20.70	2.60	87.20	13.70
AD-1700817.1	7.70	0.60	38.60	6.40	106.50	8.40
AD-1699948.1	7.30	0.50	13.90	2.90	99.50	12.00
AD-1699949.1	5.30	0.60	24.20	3.10	89.60	8.60
AD-1720317.1	3.70	1.10	12.20	2.30	98.60	7.60
AD-1700819.1	3.40	0.80	27.70	6.70	62.90	9.10
AD-1699951.1	7.80	1.40	12.50	3.90	37.60	2.60
AD-1720315.1	3.60	1.60	7.70	2.60	40.10	6.20
AD-1700821.1	4.70	1.10	10.80	1.00	71.90	10.20

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TABLE 7-continued

CIDEB Single Dose Screens in Hep3B Cells						
Duplex ID	10 nM Dose		1 nM Dose		0.1 nM Dose	
	Avg % CIDEB mRNA	SD	Avg % CIDEB mRNA	SD	Avg % CIDEB mRNA	SD
	Remaining	SD	Remaining	SD	Remaining	SD
AD-1699964.1	3.50	0.70	6.50	2.30	45.90	11.40
AD-1700822.1	59.70	12.40	58.70	12.40	78.20	14.00
AD-1699967.1	3.60	1.20	14.80	2.10	100.00	12.10
AD-1720377.1	4.00	1.50	38.70	2.60	82.10	12.20
AD-1699970.1	3.90	1.00	7.70	0.80	54.70	10.30
AD-1699971.1	3.50	1.80	7.60	2.00	51.80	1.60
AD-1699972.1	8.00	1.20	44.80	5.80	106.00	7.20
AD-1699973.1	6.00	1.70	16.20	4.20	102.10	14.00
AD-1700824.1	56.40	12.10	40.80	13.00	75.10	14.70
AD-1700825.1	5.00	1.90	17.40	3.70	41.50	23.90
AD-1700826.1	11.70	3.60	22.20	4.60	30.80	8.80
AD-1699976.1	6.20	1.20	7.70	1.10	23.60	4.70
AD-1699977.1	10.80	1.80	37.10	3.60	86.20	20.00
AD-1699978.1	9.60	1.60	31.00	6.60	105.90	13.10
AD-1699979.1	14.40	3.30	37.10	3.20	116.60	19.40
AD-1700828.1	16.10	2.70	27.50	2.90	129.30	10.40
AD-1700829.1	8.90	2.80	15.80	3.40	68.10	4.80
AD-1699982.1	5.50	0.70	15.20	4.50	84.10	10.60
AD-1699983.1	25.10	7.80	85.60	8.80	99.40	15.50
AD-1699984.1	29.70	1.60	37.00	15.20	95.70	14.50
AD-1699985.1	4.60	0.40	14.20	2.80	68.10	11.90
AD-1699986.1	34.00	4.20	48.70	11.30	111.80	15.60
AD-1700830.1	51.30	8.70	62.90	7.00	NA	NA
AD-1699987.1	30.60	6.50	52.60	8.90	44.50	16.50
AD-1699988.1	14.30	1.80	53.90	5.90	95.90	2.20
AD-1700831.1	57.20	8.20	96.80	5.40	124.70	15.30
AD-1699989.1	5.30	0.70	47.50	6.00	101.10	15.20
AD-1699990.1	39.80	11.40	41.50	10.10	88.30	9.30
AD-1700832.1	64.90	16.10	47.00	19.10	83.30	2.90
AD-1700833.1	19.90	3.70	27.60	1.10	81.00	8.50
AD-1700834.1	27.80	3.90	52.90	8.40	51.00	12.80
AD-1700015.1	7.50	2.40	19.60	1.70	93.20	28.00
AD-1700035.1	17.30	2.20	35.20	7.10	63.30	27.50
AD-1700037.1	79.80	21.40	79.30	10.30	121.70	24.40
AD-1700038.1	14.60	1.90	34.80	5.50	71.00	2.50
AD-1700039.1	13.40	2.80	21.20	1.00	66.60	12.70
AD-1700837.1	27.10	8.60	44.20	6.10	96.10	12.00
AD-1700040.1	18.30	3.90	45.90	8.20	80.60	17.60
AD-1700041.1	85.20	12.40	57.70	9.70	93.80	4.80
AD-1700838.1	20.60	6.60	53.80	6.90	114.40	34.60
AD-1700839.1	35.10	2.40	61.20	5.90	97.90	10.40
AD-1700840.1	29.50	1.40	57.10	12.40	88.60	8.60
AD-1700077.1	14.00	1.70	29.60	8.30	87.70	16.40
AD-1700841.1	7.10	1.00	21.90	5.20	105.40	13.10
AD-1700842.1	9.30	1.50	42.90	10.00	117.30	6.70
AD-1700080.1	17.10	1.30	72.40	6.50	62.60	17.90
AD-1700081.1	16.20	6.70	63.30	8.60	104.20	18.00
AD-1700082.1	6.50	2.20	16.60	3.60	67.10	6.00
AD-1700083.1	9.70	1.60	30.30	6.10	91.20	13.40
AD-1700843.1	38.00	11.60	61.70	6.20	54.70	16.90
AD-1700844.1	3.10	1.60	14.20	3.40	75.40	3.70
AD-1700091.1	13.40	0.60	31.60	4.60	68.90	5.00
AD-1700845.1	16.10	5.30	54.50	11.00	49.70	11.80
AD-1700092.1	5.80	0.60	24.80	1.10	93.80	18.00
AD-1700846.1	3.90	0.30	36.70	14.00	118.60	34.70
AD-1700094.1	5.00	1.00	22.30	2.60	88.70	29.10
AD-1700848.1	3.60	1.10	22.10	3.20	99.30	8.80
AD-1700097.1	10.10	2.30	34.30	10.90	85.90	12.50
AD-1720295.1	65.40	5.40	86.20	6.30	102.10	15.80
AD-1700850.1	21.90	7.20	75.90	7.50	86.30	14.20
AD-1700851.1	35.20	6.20	81.10	18.50	78.60	29.30
AD-1700099.1	17.40	3.90	29.40	12.60	42.00	18.00
AD-1700100.1	14.00	2.40	60.00	1.50	110.20	4.40
AD-1700101.1	28.70	8.00	98.10	12.00	123.70	13.20
AD-1700852.1	19.70	2.30	71.30	7.30	106.20	14.50
AD-1720289.1	14.40	3.90	46.30	3.50	41.40	2.30
AD-1700853.1	14.40	1.70	90.70	7.00	116.20	15.10
AD-1700854.1	27.20	6.90	62.20	8.80	81.90	10.10
AD-1700104.1	31.50	28.20	49.30	11.40	98.00	9.90
AD-1700855.1	3.30	1.10	11.70	2.20	84.40	19.20
AD-1700856.1	7.80	1.10	29.30	5.00	69.60	6.90

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TABLE 7-continued

CIDEB Single Dose Screens in Hep3B Cells						
Duplex ID	10 nM Dose		1 nM Dose		0.1 nM Dose	
	Avg % CIDEB mRNA	SD	Avg % CIDEB mRNA	SD	Avg % CIDEB mRNA	SD
	Remaining	SD	Remaining	SD	Remaining	SD
AD-1700109.1	13.60	3.70	37.10	7.90	82.80	15.90
AD-1720370.1	4.00	0.60	13.70	4.40	80.70	15.10
AD-1700858.1	14.70	1.30	74.20	6.60	106.00	3.90
AD-1700859.1	7.70	3.00	32.10	12.30	99.40	24.30
AD-1700860.1	12.70	1.60	64.00	2.20	100.90	1.40
AD-1700112.1	8.70	1.30	20.30	2.90	107.30	11.10
AD-1700113.1	16.60	4.20	63.60	4.10	77.70	10.50
AD-1720300.1	27.30	7.70	81.20	12.70	82.30	21.90
AD-1700114.1	15.10	3.30	37.90	6.00	101.70	8.00
AD-1720286.1	9.10	2.80	25.40	5.00	103.60	10.80
AD-1720287.1	31.10	3.70	74.40	20.90	116.20	21.00
AD-1720288.1	6.30	1.10	34.40	1.60	119.50	5.00
AD-1700115.1	24.50	2.30	70.80	9.20	90.30	29.30
AD-1700863.1	23.20	9.00	56.20	7.00	106.00	10.00
AD-1700116.1	15.70	2.40	47.40	6.60	71.10	11.70
AD-1700864.1	33.10	1.50	79.20	28.00	69.70	15.30
AD-1700865.1	12.00	1.90	41.00	7.60	89.90	6.80
AD-1700121.1	35.50	6.10	92.50	16.90	113.60	16.00
AD-1720311.1	38.70	2.40	61.80	3.00	104.60	11.90
AD-1700122.1	46.30	6.30	40.00	11.20	83.10	9.50
AD-1700867.1	23.70	4.00	38.40	11.00	80.80	12.10
AD-1700125.1	10.90	1.60	42.70	10.50	92.70	7.30
AD-1700868.1	43.00	3.60	85.80	13.80	80.60	17.10
AD-1700869.1	14.20	3.20	47.50	11.00	67.80	6.70
AD-1700870.1	4.00	1.10	28.00	7.20	75.90	14.00
AD-1700871.1	2.00	0.20	13.80	3.70	77.60	6.90
AD-1700872.1	5.30	1.30	29.50	7.60	68.50	14.00
AD-1700873.1	24.00	5.40	49.20	4.60	80.50	12.60
AD-1700874.1	67.90	11.70	95.50	18.20	115.20	7.10
AD-1700875.1	11.20	3.10	45.80	29.40	100.30	5.40
AD-1700876.1	35.20	11.40	83.30	8.90	108.00	17.90
AD-1720298.1	13.70	2.30	47.80	4.50	129.30	28.70
AD-1700877.1	17.70	2.60	78.30	6.40	95.40	12.00
AD-1700148.1	5.40	1.10	22.50	5.00	85.10	8.40
AD-1700149.1	23.20	9.10	51.70	9.30	45.80	15.30
AD-1700878.1	8.90	1.60	57.70	9.30	86.10	11.20
AD-1700879.1	72.70	12.80	74.60	26.30	90.30	15.10
AD-1700880.1	7.70	1.90	27.60	7.70	67.50	6.50
AD-1700881.1	2.60	1.10	9.70	2.30	97.60	13.30
AD-1700882.1	6.50	1.20	23.50	7.20	88.60	19.70
AD-1700883.1	4.30	0.40	20.30	4.80	97.00	15.50
AD-1700156.1	5.20	1.00	18.50	4.10	66.10	8.80
AD-1720282.1	7.20	1.30	12.80	1.00	96.00	5.10
AD-1700885.1	14.20	2.20	45.70	5.20	95.30	8.10
AD-1700158.1	6.20	0.90	29.90	3.50	99.10	7.20
AD-1700159.1	5.20	1.50	18.30	2.70	84.30	11.90
AD-1700160.1	5.40	1.50	19.20	4.20	119.70	13.70
AD-1700887.1	29.30	6.00	51.90	5.70	78.50	11.50
AD-1700888.1	6.70	1.10	37.80	6.30	94.20	11.10
AD-1700889.1	26.40	4.20	77.90	26.00	83.60	11.10
AD-1700166.1	35.00	5.80	109.20	19.10	114.50	16.10
AD-1700167.1	41.80	12.40	78.70	11.50	75.20	9.70
AD-1700891.1	34.90	2.00	97.10	6.20	90.10	15.60
AD-1700892.1	74.40	8.70	84.00	22.10	84.70	9.20
AD-1700893.1	6.40	2.50	29.60	4.80	82.20	3.80
AD-1700894.1	30.60	1.60	84.70	25.00	99.90	6.10
AD-1700180.1	6.00	0.90	27.30	1.70	59.90	8.50
AD-1700895.1	4.10	0.90	26.20	5.80	98.40	21.30
AD-1700187.1	7.70	1.30	30.70	6.80	90.60	11.90
AD-1720299.1	13.70	2.10	54.40	3.90	85.40	12.50
AD-1700188.1	17.30	4.60	47.70	14.40	66.40	16.60
AD-1700897.1	55.90	10.80	99.50	8.20	74.20	4.10
AD-1700898.1	93.60	13.90	110.20	24.00	109.30	12.20
AD-1700189.1	52.60	7.80	77.40	13.30	93.30	7.40
AD-1700190.1	8.50	2.20	51.00	3.20	72.50	10.90
AD-1700191.1	4.70	0.40	16.30	2.80	102.90	5.30
AD-1700899.1	29.20	1.80	36.80	4.80	97.30	7.20
AD-1700193.1	16.60	2.20	57.60	10.60	106.70	5.90
AD-1700900.1	46.40	4.40	73.10	9.30	86.10	9.80
AD-1700213.1	8.50	1.10	71.30	4.20	102.50	17.40
AD-1720310.1	8.60	1.60	39.20	7.10	130.30	5.30

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TABLE 7-continued

CIDEB Single Dose Screens in Hep3B Cells						
Duplex ID	10 nM Dose		1 nM Dose		0.1 nM Dose	
	Avg % CIDEB mRNA	SD	Avg % CIDEB mRNA	SD	Avg % CIDEB mRNA	SD
	Remaining	SD	Remaining	SD	Remaining	SD
AD-1700902.1	8.10	4.30	45.70	4.30	111.90	15.50
AD-1700903.1	16.90	3.30	68.70	15.00	96.10	6.40
AD-1700904.1	15.60	2.30	36.60	6.90	79.30	9.10
AD-1700905.1	7.70	1.70	19.50	2.30	50.80	17.60
AD-1700242.1	8.80	1.00	30.90	4.40	109.10	9.10
AD-1700243.1	24.70	5.30	66.40	7.90	65.50	19.60
AD-1700244.1	11.50	0.90	46.60	6.80	84.90	4.00
AD-1700906.1	86.90	6.40	97.90	10.00	98.40	5.00
AD-1700245.1	49.20	3.70	89.60	8.20	116.20	5.10
AD-1700907.1	34.70	12.20	87.10	23.80	96.70	15.50
AD-1700908.1	63.10	11.60	91.30	13.60	99.30	20.20
AD-1700909.1	59.00	9.20	96.30	8.70	69.00	11.00
AD-1700248.1	26.70	7.90	59.50	10.10	89.30	13.50
AD-1700249.1	45.10	4.50	90.70	4.50	NA	NA
AD-1720301.1	52.70	8.10	85.10	11.10	72.90	16.20
AD-1720302.1	58.90	15.30	68.90	9.00	71.30	21.50
AD-1720303.1	40.30	7.70	79.90	12.30	79.40	21.50
AD-1700250.1	26.10	4.80	75.80	10.10	102.40	16.20
AD-1720379.1	47.20	10.00	97.90	19.90	117.60	12.20
AD-1700912.1	84.60	10.20	86.70	10.50	86.40	20.90
AD-1700913.1	82.20	14.30	105.20	16.40	120.70	10.30
AD-1700252.1	17.90	1.50	60.80	8.90	116.20	14.10
AD-1720381.1	45.80	7.20	70.20	6.30	108.00	15.70
AD-1700272.1	34.00	4.60	86.00	9.90	115.40	20.90
AD-1700915.1	5.00	0.70	14.10	3.80	68.60	8.30
AD-1700279.1	16.50	2.50	60.80	5.10	90.10	13.30
AD-1700280.1	8.00	3.70	25.60	5.90	89.00	10.80
AD-1700916.1	7.10	2.70	23.20	5.90	69.30	14.50
AD-1700917.1	17.60	3.70	25.30	7.90	65.70	8.40
AD-1700918.1	27.80	8.10	51.10	11.90	102.80	5.10
AD-1700303.1	4.70	1.70	13.60	4.60	79.00	15.00
AD-1700919.1	8.50	1.30	47.50	6.90	104.70	9.70
AD-1700920.1	14.10	2.30	24.40	4.60	66.30	18.10
AD-1700309.1	4.30	0.70	17.50	5.70	43.00	6.40
AD-1720375.1	5.90	1.60	14.80	7.20	29.50	5.80
AD-1700922.1	8.60	0.60	29.70	6.10	80.70	12.00
AD-1700923.1	6.10	1.30	23.60	6.30	46.90	25.40
AD-1700312.1	2.70	0.50	6.90	2.50	54.70	5.50
AD-1700924.1	32.60	13.00	57.40	10.30	101.80	15.30
AD-1700925.1	3.10	0.80	9.90	1.50	62.30	8.20
AD-1700314.1	3.40	1.00	3.80	1.40	37.20	11.50
AD-1720309.1	5.10	0.50	12.70	1.40	37.50	21.80
AD-1700315.1	5.30	1.40	11.80	1.00	62.10	11.00
AD-1720382.1	6.30	1.40	14.50	1.80	74.60	7.90
AD-1720383.1	8.70	5.10	28.60	3.20	92.20	3.20
AD-1720384.1	9.10	4.00	11.30	2.30	45.70	4.60
AD-1700316.1	8.80	1.20	38.50	6.20	118.30	5.00
AD-1700317.1	6.10	1.00	17.60	1.70	87.40	5.60
AD-1700318.1	2.20	0.50	10.80	6.10	55.00	12.00
AD-1700929.1	10.10	2.10	41.20	4.60	80.00	7.60
AD-1700930.1	5.00	1.80	9.50	3.20	91.80	8.60
AD-1700931.1	26.40	3.10	44.80	4.30	76.50	5.80
AD-1700323.1	3.60	0.40	9.10	2.20	85.70	8.30
AD-1720283.1	4.40	1.00	10.40	2.90	66.70	3.50
AD-1700324.1	5.50	2.50	10.10	3.00	71.20	9.60
AD-1720280.1	19.30	3.30	14.40	6.30	81.60	7.10
AD-1700325.1	18.40	2.00	66.70	13.20	113.10	9.10
AD-1700327.1	4.00	0.90	40.20	9.10	98.90	18.00
AD-1700328.1	15.60	1.90	64.90	11.20	92.40	14.80
AD-1700329.1	3.00	1.60	29.10	1.50	114.60	3.90
AD-1700934.1	51.80	3.60	101.90	18.00	104.70	7.20
AD-1700336.1	12.10	2.80	41.70	7.80	92.70	10.50
AD-1720296.1	62.10	11.20	96.90	14.60	84.90	19.00
AD-1700936.1	51.10	7.00	112.80	5.70	94.20	11.60
AD-1700937.1	72.00	17.20	108.40	6.20	134.20	13.90
AD-1700339.1	11.40	3.40	32.70	7.90	49.00	16.70
AD-1700938.1	6.90	1.60	39.40	4.40	82.60	7.60
AD-1700939.1	85.80	10.00	81.50	20.20	103.20	26.50
AD-1700940.1	65.80	14.30	99.20	19.70	90.40	15.30
AD-1700344.1	9.30	1.20	40.60	4.10	99.40	3.90
AD-1720368.1	9.20	2.10	36.20	2.60	101.10	15.10

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TABLE 7-continued

CIDEB Single Dose Screens in Hep3B Cells						
Duplex ID	10 nM Dose		1 nM Dose		0.1 nM Dose	
	Avg % CIDEB mRNA	SD	Avg % CIDEB mRNA	SD	Avg % CIDEB mRNA	SD
	Remaining	SD	Remaining	SD	Remaining	SD
AD-1700942.1	35.00	8.20	89.10	21.90	100.00	26.10
AD-1700943.1	5.30	1.40	23.30	7.20	53.90	1.30
AD-1700944.1	5.70	1.20	34.60	2.70	70.60	6.30
AD-1720284.1	4.90	1.20	20.30	1.50	104.40	12.90
AD-1700350.1	31.40	4.90	70.00	4.80	117.60	13.90
AD-1700352.1	3.30	0.80	8.90	1.20	116.20	12.30
AD-1700353.1	12.70	5.10	19.60	3.30	57.00	7.80
AD-1700947.1	5.60	0.30	17.50	5.40	86.00	14.10
AD-1700354.1	2.30	0.30	4.00	1.50	69.50	6.70
AD-1700355.1	5.00	1.10	8.80	1.00	69.30	12.30
AD-1700356.1	16.80	2.40	53.10	8.20	NA	NA
AD-1700357.1	22.00	2.00	72.30	7.70	79.60	14.10
AD-1700358.1	2.90	0.30	19.90	2.60	75.20	3.00
AD-1700359.1	16.70	2.60	50.60	19.70	112.30	19.00
AD-1700948.1	9.70	3.00	31.70	9.50	79.90	13.70
AD-1700949.1	16.30	1.30	40.90	1.30	60.30	6.70
AD-1700950.1	16.90	7.00	35.70	10.10	82.60	18.60
AD-1700365.1	3.30	0.40	17.10	3.40	72.60	11.90
AD-1700952.1	53.20	9.20	87.00	25.10	83.30	16.80
AD-1700366.1	4.20	0.40	41.80	12.00	91.30	18.00
AD-1700953.1	24.50	6.40	64.70	12.30	71.10	8.60
AD-1720293.1	34.70	7.10	13.20	0.40	61.70	25.00
AD-1700368.1	8.90	2.00	32.80	4.00	86.50	5.70
AD-1700955.1	22.40	0.80	68.50	11.50	77.60	15.50
AD-1700369.1	3.90	1.80	25.60	3.70	80.40	10.10
AD-1700370.1	37.10	3.80	43.90	19.10	94.50	10.70
AD-1700371.1	6.70	2.30	21.00	2.90	114.80	4.20
AD-1700957.1	49.80	8.40	84.60	18.90	92.50	26.30
AD-1700372.1	6.60	0.60	27.70	5.80	95.70	17.40
AD-1700374.1	3.50	0.40	11.50	0.60	38.20	8.50
AD-1700958.1	53.10	2.30	100.80	10.40	89.10	11.80
AD-1700959.1	38.60	0.50	71.60	3.90	92.20	6.30
AD-1720285.1	23.80	7.00	47.20	9.00	85.30	23.20
AD-1700375.1	4.60	1.10	30.30	6.40	93.00	16.90
AD-1700376.1	6.10	1.10	19.30	7.60	76.10	6.00
AD-1700378.1	3.10	1.20	27.10	8.40	79.90	13.10
AD-1700960.1	26.70	5.10	38.50	11.60	61.00	3.70
AD-1700961.1	4.30	0.80	11.90	1.30	116.60	6.70
AD-1700962.1	51.00	10.60	30.90	3.10	71.80	22.90
AD-1700963.1	3.40	0.20	28.70	4.40	99.00	8.10
AD-1700383.1	4.60	1.10	11.10	6.40	69.10	10.20
AD-1720385.1	4.00	0.70	16.50	4.40	89.10	4.50
AD-1700965.1	12.20	4.20	16.50	0.60	44.30	7.90
AD-1700384.1	5.30	1.60	11.60	5.40	53.40	20.40
AD-1700385.1	3.10	0.60	11.40	4.40	88.40	10.70
AD-1700386.1	6.10	0.80	10.70	5.10	47.80	19.00
AD-1700387.1	11.70	2.40	37.20	6.20	135.50	4.40
AD-1720380.1	5.50	1.30	12.80	1.80	100.60	18.50
AD-1700967.1	4.40	0.80	20.60	5.10	68.30	10.90
AD-1700388.1	3.80	0.30	17.60	4.30	71.10	20.40
AD-1700968.1	6.10	2.20	9.30	3.70	57.80	14.10
AD-1700969.1	5.40	1.80	20.60	1.00	94.80	12.70
AD-1700408.1	1.80	0.50	5.80	2.00	54.20	10.80
AD-1700970.1	16.10	3.10	54.70	10.10	105.40	13.00
AD-1700971.1	34.30	8.70	82.40	15.30	107.40	15.10
AD-1700410.1	2.50	0.20	12.20	3.80	95.60	18.30
AD-1700411.1	8.20	2.20	21.70	7.50	63.60	14.90
AD-1720304.1	6.30	2.60	14.30	7.70	76.30	12.80
AD-1700973.1	16.10	6.40	64.00	3.00	45.20	12.20
AD-1700413.1	25.70	6.40	83.40	5.30	129.00	8.50
AD-1700414.1	7.20	1.30	45.20	9.60	125.30	17.80
AD-1700975.1	17.50	5.80	50.20	7.50	101.50	16.30
AD-1700976.1	8.80	1.60	32.00	8.10	97.10	6.90
AD-1700977.1	10.70	1.70	26.60	5.00	48.90	4.90
AD-1700978.1	15.30	8.80	21.50	5.00	80.20	13.10
AD-1700422.1	31.50	6.10	62.90	9.00	96.10	4.30
AD-1720305.1	57.00	8.60	73.70	10.10	120.40	6.50
AD-1700423.1	27.10	6.00	27.10	8.50	71.10	26.30
AD-1700424.1	4.00	0.40	15.90	1.80	106.10	16.00
AD-1700425.1	6.50	0.80	43.50	9.50	110.30	8.90
AD-1700426.1	29.00	5.00	80.60	15.50	84.50	21.00

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TABLE 7-continued

CIDEB Single Dose Screens in Hep3B Cells						
Duplex ID	10 nM Dose		1 nM Dose		0.1 nM Dose	
	Avg % CIDEB mRNA		Avg % CIDEB mRNA		Avg % CIDEB mRNA	
	Remaining	SD	Remaining	SD	Remaining	SD
AD-1700427.1	30.30	6.20	49.40	28.80	103.70	7.20
AD-1700980.1	28.90	5.80	57.20	5.30	68.90	13.30
AD-1700428.1	10.00	1.30	37.70	8.40	66.20	18.80
AD-1700981.1	70.90	17.00	89.10	27.90	89.60	12.40
AD-1700982.1	11.70	5.90	36.50	3.40	93.70	21.90
AD-1700439.1	5.30	2.60	38.90	10.50	112.10	8.30
AD-1720316.1	33.30	5.70	66.20	7.50	90.90	11.00
AD-1700984.1	56.50	12.40	98.20	31.40	114.50	13.90
AD-1700985.1	9.20	3.50	41.30	9.80	92.40	9.20
AD-1700461.1	2.90	0.30	6.50	1.50	72.20	18.30
AD-1700462.1	40.30	6.00	92.30	11.20	59.00	6.30
AD-1700986.1	93.90	18.50	105.10	29.10	103.00	24.00
AD-1700464.1	18.80	3.60	48.90	9.50	109.00	10.90
AD-1700465.1	34.70	5.80	84.10	8.60	106.50	18.90
AD-1700468.1	10.40	0.90	54.10	9.10	64.80	13.00
AD-1720378.1	11.50	1.30	72.70	3.80	90.40	13.60
AD-1700988.1	10.70	1.50	34.50	7.80	56.30	20.30
AD-1700989.1	73.60	18.20	92.30	9.60	93.60	12.60
AD-1700990.1	2.40	0.60	7.50	1.00	72.10	11.90
AD-1700473.1	3.30	0.90	7.40	0.90	50.10	5.40
AD-1700991.1	3.20	1.10	9.20	1.60	100.50	8.00
AD-1700474.1	4.90	0.50	23.90	4.30	91.60	5.30
AD-1700476.1	2.30	0.80	5.60	0.60	43.50	5.60
AD-1700992.1	92.10	8.00	107.60	10.00	85.50	17.40
AD-1700477.1	12.70	1.50	65.60	9.30	83.00	9.30
AD-1700479.1	3.50	0.80	17.40	0.80	68.80	11.00
AD-1720373.1	3.10	1.00	11.10	3.30	58.70	12.70
AD-1700482.1	7.50	1.50	38.40	5.20	91.20	6.30
AD-1720291.1	25.70	6.90	24.20	1.80	80.90	24.50
AD-1700995.1	3.10	0.90	5.20	1.80	67.00	15.40
AD-1700485.1	16.40	1.20	36.80	4.10	103.00	14.40
AD-1720306.1	30.40	5.50	34.90	7.10	87.90	10.40
AD-1720307.1	55.10	15.00	71.40	4.10	102.90	9.10
AD-1720308.1	4.70	0.50	10.10	1.40	66.10	5.00
AD-1700486.1	75.60	8.60	66.70	28.30	78.20	19.50
AD-1700487.1	6.80	1.70	10.30	3.20	58.70	8.50
AD-1700488.1	15.60	2.00	51.30	15.20	111.30	23.70
AD-1700489.1	51.30	2.50	83.20	12.30	73.60	5.50
AD-1700997.1	13.30	2.40	93.60	9.10	75.40	19.50
AD-1700490.1	10.50	2.00	70.70	9.00	89.50	11.40
AD-1700998.1	66.00	11.40	64.00	8.40	70.00	10.70
AD-1700517.1	25.80	5.30	61.30	5.90	84.70	11.60
AD-1700519.1	48.60	13.10	84.20	39.10	111.20	10.60
AD-1720292.1	76.80	8.50	70.20	2.80	74.50	12.10
AD-1700520.1	12.80	3.00	46.20	8.80	85.00	10.20
AD-1720376.1	50.40	10.70	79.60	4.60	93.70	7.40
AD-1701001.1	53.90	6.70	88.90	7.10	115.50	27.10
AD-1701002.1	79.10	10.40	110.00	19.60	111.90	8.60
AD-1700522.1	71.90	7.70	94.40	22.70	115.00	17.00
AD-1701003.1	6.60	1.20	67.80	9.00	90.70	14.20
AD-1700542.1	6.30	0.90	30.00	7.00	101.50	13.60
AD-1701004.1	8.00	3.50	17.30	3.40	76.60	10.50
AD-1701005.1	3.50	1.50	16.00	3.90	90.10	10.00
AD-1700544.1	3.60	0.70	11.60	3.80	84.10	9.30
AD-1700550.1	4.10	1.90	4.80	2.20	46.60	6.90
AD-1720386.1	6.20	1.70	11.60	5.40	64.50	9.00
AD-1701007.1	11.30	2.10	11.00	3.70	42.10	5.90
AD-1700553.1	3.80	0.70	6.00	2.30	77.80	6.40
AD-1700554.1	4.80	0.90	6.60	1.20	73.50	17.60
AD-1700555.1	5.10	1.20	6.40	1.00	63.80	9.50
AD-1700556.1	4.40	1.50	10.70	1.80	61.00	10.00
AD-1700557.1	5.10	2.60	23.10	7.50	95.90	9.40
AD-1700558.1	5.90	0.80	6.60	1.40	63.40	12.50
AD-1701009.1	18.30	3.10	46.80	5.00	99.80	6.40
AD-1700560.1	6.20	4.60	17.50	4.90	63.90	3.30
AD-1720372.1	3.20	1.00	7.30	1.30	56.50	14.90
AD-1701011.1	3.10	0.90	18.50	2.70	75.10	7.90
AD-1701012.1	7.50	2.60	15.00	2.80	78.00	13.90
AD-1701013.1	27.60	5.60	68.10	10.00	115.30	30.40
AD-1700563.1	8.20	1.50	29.30	4.50	114.10	11.00
AD-1701014.1	8.40	0.60	29.90	2.60	83.40	10.10

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TABLE 7-continued

CIDEB Single Dose Screens in Hep3B Cells						
Duplex ID	10 nM Dose		1 nM Dose		0.1 nM Dose	
	Avg % CIDEB mRNA		Avg % CIDEB mRNA		Avg % CIDEB mRNA	
	Remaining	SD	Remaining	SD	Remaining	SD
AD-1700566.1	3.00	1.30	8.10	1.60	82.30	8.40
AD-1720312.1	5.80	3.20	17.10	2.10	108.80	7.00
AD-1701016.1	4.70	0.20	14.90	2.40	83.40	12.00
AD-1701017.1	9.30	1.80	26.60	3.50	115.80	5.30
AD-1700573.1	2.80	1.70	12.70	2.20	67.70	8.00
AD-1720371.1	7.90	0.80	12.40	2.00	74.20	10.60
AD-1701019.1	7.50	2.40	18.30	2.90	86.50	14.90
AD-1700575.1	9.70	1.80	18.90	2.60	106.30	9.00
AD-1700576.1	31.80	6.50	61.00	5.90	84.80	8.00
AD-1701020.1	7.00	1.50	41.80	5.50	105.50	10.20
AD-1700578.1	9.20	2.00	18.80	3.70	96.70	3.20
AD-1720290.1	5.40	1.50	19.00	3.20	95.50	1.80
AD-1701022.1	5.00	0.30	14.60	2.00	68.70	11.30
AD-1701023.1	10.20	1.40	36.00	9.10	107.40	12.00
AD-1700580.1	7.60	2.10	33.10	6.10	107.50	11.70
AD-1701024.1	12.10	5.90	39.70	10.90	91.00	8.00
AD-1700582.1	8.20	1.30	14.80	3.80	56.90	7.40
AD-1720294.1	5.40	1.20	14.10	2.80	63.40	14.00
AD-1701026.1	3.20	0.50	28.60	3.40	84.40	8.70
AD-1701027.1	5.20	1.00	27.10	3.30	82.10	8.50
AD-1700623.1	5.50	1.20	13.10	5.90	95.90	10.40
AD-1700670.1	42.70	7.00	93.20	12.10	96.70	6.20
AD-1700671.1	62.40	12.30	81.80	11.60	82.40	19.80
AD-1720369.1	21.80	3.80	67.10	11.30	98.80	11.70
AD-1701029.1	13.80	5.60	52.00	16.10	96.50	25.40
AD-1700680.1	20.10	1.00	82.90	10.20	99.20	14.30
AD-1720297.1	79.80	3.70	105.80	12.80	101.50	17.30
AD-1701031.1	16.10	3.90	19.90	2.10	93.50	2.90
AD-1700681.1	8.10	1.70	35.00	5.60	76.20	15.60
AD-1701032.1	22.40	8.30	65.10	19.30	80.30	1.10
AD-1700684.1	10.10	3.80	29.70	5.60	96.90	8.50
AD-1701033.1	98.60	7.10	117.50	19.10	123.90	3.90
AD-1700710.1	93.80	16.60	112.90	16.20	122.60	15.20
AD-1701034.1	93.50	11.40	105.10	3.80	110.30	8.20
AD-1701035.1	18.80	1.00	65.90	10.50	89.30	7.70
AD-1700736.1	9.00	3.10	18.80	5.60	72.10	5.00
AD-1720374.1	45.60	14.20	55.40	12.00	61.70	10.40
AD-1701037.1	37.60	12.00	66.20	11.70	69.90	13.10
AD-1700737.1	17.30	3.70	58.00	8.00	71.60	19.10
AD-1700738.1	11.70	1.60	37.80	8.60	92.00	10.90
AD-1701038.1	23.60	1.40	113.80	9.20	93.10	11.80
AD-1700780.1	3.90	2.20	6.10	1.50	35.40	11.10
AD-1720314.1	3.40	1.00	7.30	2.00	59.20	13.00
AD-1700781.1	3.00	0.70	6.90	1.90	51.30	5.40
AD-1700782.1	5.80	1.60	8.80	2.90	58.30	14.20

TABLE 8

50	CIDEB Single Dose Screens in Hep3B Cells					
	Avg % CIDEB mRNA Remaining			Avg % CIDEB mRNA Remaining		
55	Duplex ID	10 nM Mean	SD	DuplexID	10 nM Mean	SD
	AD-1686813	7.034	0.258	AD-1686057.1	5.373	0.633
	AD-1686803.1	7.037	0.549	AD-1686053.1	56.574	3.938
	AD-1686794.1	6.698	0.335	AD-1686028.1	46.840	2.177
	AD-1686783.1	9.004	0.720	AD-1686016.1	9.940	0.590
	AD-1686781.1	26.941	1.075	AD-1686007.1	18.073	1.221
	AD-1686770.1	7.394	0.857	AD-1685994.1	9.314	0.297
	AD-1686761.1	29.344	3.097	AD-1685983.1	15.368	0.551
	AD-1686751.1	75.987	4.650	AD-1685969.1	9.303	0.607
	AD-1686747.1	74.602	6.448	AD-1685956.1	24.725	7.404
	AD-1686738.1	33.637	1.984	AD-1685935.1	38.364	1.701
	AD-1686729.1	24.804	3.052	AD-1685931.1	40.894	1.620

TABLE 8-continued

CIDEB Single Dose Screens in Hep3B Cells					
Duplex ID	Avg % CIDEB mRNA Remaining		DuplexID	Avg % CIDEB mRNA Remaining	
	10 nM Mean	SD		10 nM Mean	SD
AD-1686720.1	20.335	0.768	AD-1685921.1	48.205	4.328
AD-1686709.1	10.072	0.802	AD-1685912.1	54.202	3.053
AD-1686693.1	15.320	1.441	AD-1685902.1	91.948	4.558
AD-1686680.1	8.115	0.694	AD-1685898.1	98.617	4.497
AD-1686669.1	11.933	1.964	AD-1685889.1	84.955	2.049
AD-1686658.1	9.575	0.485	AD-1685878.1	95.346	1.773
AD-1686649.1	7.356	0.570	AD-1685869.1	87.386	2.693
AD-1686639.1	11.744	0.890	AD-1685856.1	90.274	1.411
AD-1686626.1	9.617	0.753	AD-1685847.1	61.105	2.592
AD-1686616.1	10.909	0.749	AD-1685838.1	83.672	4.357
AD-1686606.1	9.636	0.540	AD-1685826.1	77.305	2.383
AD-1686597.1	7.108	0.308	AD-1685810.1	100.598	6.749
AD-1686585.1	5.715	1.579	AD-1685798.1	81.801	6.236
AD-1686576.1	8.745	0.449	AD-1685788.1	76.643	16.621
AD-1686566.1	7.504	0.442	AD-1685779.1	93.379	8.610
AD-1686565.1	51.547	2.866	AD-1685770.1	94.639	5.122
AD-1686556.1	34.239	2.082	AD-1685744.1	93.333	6.221
AD-1686543.1	50.576	2.174	AD-1685735.1	72.951	5.540
AD-1686540.1	50.093	4.978	AD-1685726.1	92.927	8.552
AD-1686531.1	56.092	3.563	AD-1685717.1	84.588	8.485
AD-1686518.1	10.338	0.356	AD-1685708.1	85.659	6.249
AD-1686508.1	56.285	2.808	AD-1685699.1	93.319	3.803
AD-1686491.1	19.182	0.216	AD-1685689.1	98.857	4.806
AD-1686475.1	52.270	2.123	AD-1685673.1	78.962	3.900
AD-1686466.1	8.819	0.655	AD-1685664.1	96.600	15.224
AD-1686451.1	12.861	0.277	AD-1685648.1	89.253	4.285
AD-1686435.1	20.744	0.806	AD-1685637.1	82.011	1.907
AD-1686426.1	15.031	0.938	AD-1685627.1	91.683	23.711
AD-1686417.1	12.149	0.855	AD-1685614.1	89.157	4.921
AD-1686403.1	15.485	1.878	AD-1685610.1	78.640	8.712
AD-1686394.1	12.687	0.710	AD-1685599.1	83.195	5.391
AD-1686377.1	15.920	0.978	AD-1685587.1	93.563	13.290
AD-1686366.1	17.513	1.255	AD-1685570.1	88.283	6.794
AD-1686354.1	18.012	0.876	AD-1685561.1	88.025	5.340
AD-1686343.1	19.232	1.476	AD-1685549.1	82.656	5.181
AD-1686334.1	24.115	1.100	AD-1685538.1	88.232	3.361
AD-1686319.1	13.232	0.799	AD-1685528.1	92.786	7.624
AD-1686286.1	22.559	0.512	AD-1685512.1	97.641	6.841
AD-1686274.1	29.551	0.741	AD-1685503.1	95.865	17.990
AD-1686265.1	70.105	5.349	AD-1685491.1	94.393	11.080
AD-1686251.1	14.695	1.084	AD-1685469.1	85.615	7.133
AD-1686242.1	4.855	0.629	AD-1685418.1	86.858	2.046
AD-1686225.1	10.479	0.390	AD-1685406.1	76.723	3.400
AD-1686216.1	15.955	1.165	AD-1685388.1	83.950	4.269
AD-1686207.1	10.015	1.533	AD-1685379.1	82.859	3.868
AD-1686197.1	4.724	1.010	AD-1685366.1	85.364	1.718
AD-1686181.1	15.356	0.366	AD-1685284.1	82.852	1.693
AD-1686171.1	19.964	0.716	AD-1685271.1	84.648	3.346
AD-1686159.1	12.973	0.587	AD-1685250.1	86.016	3.414
AD-1686147.1	6.105	1.495	AD-1685240.1	79.332	2.819
AD-1686136.1	5.802	0.945	AD-1685228.1	90.535	6.335
AD-1686126.1	11.135	1.538	AD-1685217.1	80.884	4.373
AD-1686116.1	14.156	3.058	AD-1685207.1	74.123	2.932
AD-1686098.1	22.948	0.996	AD-1685198.1	85.722	6.301
AD-1686086.1	53.797	3.239	AD-1685173.1	83.069	7.260
AD-1686075.1	34.648	2.855	AD-1685156.1	99.596	11.628
AD-1686066.1	54.370	2.364			

Example 3. In Vitro Screening Methods

A subset of the duplexes was also assessed by transfection and free uptake in primary human hepatocytes and primary cynomolgus hepatocytes

Cell Culture and Transfections:

Transfection and free uptake assays were carried out in primary human hepatocyte (PHH, BioIVT) and primary cyno hepatocyte (PCH, BioIVT). Transfection was performed by adding of Opti-MEM plus 0.1 μ L of Lipofectamine RNAiMax per well (Invitrogen, Carlsbad CA. cat #13778-150) to 5 μ L of siRNA duplexes per well into a 384-well plate and incubated at room temperature for 15 minutes. 40 μ L of in Invitrogro CP media (BioIVT, Cat #Z99029) containing $\sim 10 \times 10^3$ cells were then added to the siRNA mixture. Cells were incubated for 24 hours prior to RNA purification. Experiments were performed at 10 nM, 1 nM, 0.1 nM and 0.01 nM. Free uptake assay was performed similarly to transfection assay without using Liporectamine RNAimax and cells were incubated for 48 hours prior to the RNA purification. Experiments were performed at 250 nM, 100 nM, 10 nM, 1 nM.

Total RNA Isolation Using DYNABEADS mRNA Isolation Kit:

RNA was isolated using an Highres Biosolution integration system using DynabeadsTM mRNA DIRECTTM Purification Kit (InvitrogenTM, Catalog No. 61012). Briefly, 70 μ L of Lysis/Binding Buffer and 10 μ L of lysis buffer containing 3 μ L of magnetic beads were added to the plate with cells. Plates were incubated on an electromagnetic shaker for 10 minutes at room temperature and then magnetic beads were captured and the supernatant was removed. Bead-bound RNA was then washed 2 times with 90 μ L Wash Buffer A and once with 90 μ L Wash Buffer B. Beads were then washed with 90 μ L Elution Buffer, re-captured, and supernatant was removed. Complementary DNA (cDNA) was synthesized using High-Capacity cDNA Reverse Transcription Kit with RNase Inhibitor (Applied BiosystemsTM, Catalog No. 4374967) according to the manufacturer's recommendations. A master mix containing 1 μ L 10 $\times$ Buffer, 0.4 μ L 25 $\times$ deoxyribonucleotide triphosphate, 1 μ L 10 $\times$ Random primers, 0.5 μ L Reverse Transcriptase, 0.5 μ L RNase inhibitor, and 6.6 μ L of H₂O per reaction was added to RNA isolated above. The plates were sealed, mixed, and incubated on an electromagnetic shaker for 10 minutes at room temperature, followed by 2 hours incubation at 37 $^{\circ}$ C.

CIDEB mRNA levels were quantified by performing RT-qPCR analysis. 2 μ L of cDNA were added to a master mix containing 0.5 μ L of human or cyno GAPDH TaqMan Probe, 0.5 μ L human or cyno CIDEB probe and 5 μ L Lightcycler 480 probe master mix (Roche Cat #04887301001) per well in a 384 well plates. Real time PCR was done in a Light-Cycler480 Real Time PCR system (Roche). To calculate relative fold change, real-time data were analyzed using the Delta-Delta Threshold Cycle (Relative Quantification) ($\Delta\Delta C_q$ [RQ]) method [Schmittgen and Livak 2008] and were normalized to control assays performed using cells transfected with PBS. For all samples, CIDEB mRNA levels were first normalized to GAPDH as a reference gene. Data are expressed as percent of CIDEB mRNA remaining relative to average PBS control and error is expressed as standard deviation (SD), derived from the 4 transfection replicates.

The results of of the single dose transfection screens and free uptake screens of the dsRNA agents in PCH is shown in Table 9 and PHH cells are shown in Table 10.

TABLE 9

Transfection and Free Uptake in Primary Cyno Hepatocytes																
Duplex ID	Transfection								Free uptake							
	10 nM		1 nM		0.1 nM		0.01 nM		250 nM		100 nM		10 nM		1 nM	
	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
AD-1700826.1	9	2	16	5	17	5	25	6	36	5	47	2	73	11	60	2
AD-1699971.1	2	1	5	1	5	1	8	1	16	5	23	4	31	4	44	5
AD-1720309.1	8	1	17	2	22	3	31	3	36	6	41	5	53	12	68	7
AD-1720382.1	7	1	11	3	12	2	18	3	35	5	30	5	46	10	59	8
AD-1700355.1	6	0	14	1	15	3	22	5	25	5	35	7	42	10	55	2
AD-1720294.1	2	0	4	0	7	1	11	3	28	5	31	3	43	7	52	7
AD-1700411.1	10	1	24	2	41	6	45	7	46	7	55	4	55	2	71	7
AD-1700383.1	6	1	11	2	11	1	16	3	25	2	32	6	50	9	60	10
AD-1720304.1	11	4	19	1	29	5	36	4	40	10	48	9	49	1	58	5
AD-1700082.1	5	0	13	2	13	2	24	3	27	5	32	7	43	9	54	5
AD-1700872.1	3	1	7	1	6	1	13	2	21	4	35	1	44	4	54	9
AD-1700990.1	4	1	8	2	5	1	13	2	46	14	43	7	139	23	64	7
AD-1700558.1	7	1	17	4	23	4	25	2	35	9	42	3	53	9	64	6
AD-1700782.1	3	1	7	1	7	1	16	2	23	3	26	2	49	2	50	8
AD-1720375.1	15	3	22	4	14	2	27	5	55	5	59	4	69	7	82	5
AD-1700358.1	5	1	13	2	18	2	19	2	30	5	37	6	57	5	44	3
AD-1720314.1	16	5	10	2	9	2	14	2	23	6	37	5	40	8	51	10
AD-1700156.1	5	3	9	1	8	1	16	3	26	6	40	6	49	5	70	9
AD-1699964.1	2	0	5	1	4	1	11	3	16	2	19	2	35	6	49	9
AD-1699976.1	1	0	3	0	6	2	6	1	15	2	18	2	32	2	37	2
AD-1701007.1	7	1	9	1	10	1	16	2	28	4	41	4	56	10	53	2
AD-1700386.1	8	1	13	2	15	2	20	3	24	1	32	4	40	7	54	5
AD-1700995.1	3	1	6	1	5	0	12	1	18	2	27	3	32	3	53	8
AD-1700376.1	4	1	7	2	8	1	11	2	20	1	26	2	47	8	49	3
AD-1700915.1	7	1	14	2	16	1	25	4	37	3	32	7	43	11	54	3
AD-1700384.1	6	1	11	2	12	1	18	3	34	3	43	4	63	9	61	5
AD-1700925.1	8	2	14	3	19	3	26	6	35	8	39	4	61	11	46	0
AD-1700378.1	6	1	15	3	15	3	21	4	32	5	34	6	50	5	60	4
AD-1700870.1	4	1	10	1	14	3	24	4	29	3	36	3	54	6	57	2
AD-1700353.1	13	1	21	1	16	1	35	11	44	4	55	3	63	7	88	3
AD-1720371.1	11	2	22	5	28	8	28	7	38	4	47	3	60	10	63	2
AD-1700374.1	4	0	8	1	9	1	14	2	26	3	33	3	54	7	54	3
AD-1700825.1	12	2	22	3	18	4	27	5	40	7	46	11	58	6	71	11
AD-1720386.1	3	1	5	1	6	1	10	2	27	2	38	4	53	8	56	6
AD-1720315.1	2	0	4	1	4	1	8	1	14	3	23	2	27	6	34	6
AD-1700159.1	2	0	5	1	5	1	9	2	19	2	26	5	44	3	50	1

TABLE 9-continued

Transfection and Free Uptake in Primary Cyno Hepatocytes																
Duplex ID	Transfection								Free uptake							
	10 nM		1 nM		0.1 nM		0.01 nM		250 nM		100 nM		10 nM		1 nM	
	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
AD-1700923.1	7	1	16	2	22	4	24	4	29	1	39	7	57	4	52	7
AD-1700369.1	5	2	6	1	11	2	14	1	20	3	24	5	37	8	39	8
AD-1701016.1	5	1	14	3	12	2	20	0	29	5	33	5	43	7	60	4
AD-1700148.1	3	1	6	1	7	1	12	1	18	4	23	4	25	5	46	4
AD-1700318.1	5	1	9	1	13	4	17	1	26	4	30	5	36	8	50	9
AD-1700781.1	2	1	7	2	11	2	12	3	33	16	29	6	47	4	51	7
AD-1700905.1	10	1	15	2	16	2	22	2	40	7	49	8	54	8	57	5
AD-1720308.1	3	1	8	1	10	1	15	2	21	2	29	4	30	7	60	10
AD-1700916.1	16	3	20	2	19	3	23	3	43	3	54	9	62	11	63	9
AD-1700461.1	5	1	7	2	5	1	13	2	29	3	42	8	52	9	64	8
AD-1700871.1	2	1	6	2	12	3	11	3	16	3	19	4	41	10	46	7
AD-1720373.1	4	1	11	3	14	2	20	1	27	2	29	4	49	4	54	8
AD-1701004.1	4	1	9	0	13	2	18	1	35	5	42	4	63	12	60	8
AD-1700556.1	2	0	5	1	5	0	17	7	21	1	24	2	38	9	50	8
AD-1720283.1	14	5	20	5	16	1	30	3	41	2	49	4	54	2	77	13
AD-1700303.1	5	0	10	3	8	1	15	2	25	7	34	7	31	3	45	9
AD-1720370.1	2	0	4	1	6	1	10	1	19	6	25	5	39	5	51	11
AD-1700573.1	4	1	8	1	12	2	17	2	19	5	28	6	36	6	44	2
AD-1720372.1	3	1	9	1	12	1	16	1	26	3	36	7	48	6	59	4
AD-1700473.1	2	1	4	1	5	1	8	1	14	3	21	2	23	6	40	8
AD-1700821.1	2	0	4	1	3	0	6	1	11	1	15	4	20	1	33	1
AD-1720384.1	4	0	7	1	7	1	12	2	25	8	31	4	40	6	47	10
AD-1700314.1	5	1	10	2	9	1	16	1	20	4	25	3	41	6	48	2
AD-1700582.1	2	1	4	1	6	1	9	1	21	4	25	2	34	2	48	5
AD-1700385.1	6	1	10	1	9	1	15	3	23	5	30	6	34	7	48	8
AD-1700544.1	2	0	7	1	8	1	15	3	21	5	29	4	43	4	49	8
AD-1699970.1	1	0	2	1	2	0	4	1	11	2	13	3	29	6	31	1
AD-1700967.1	7	1	12	1	16	2	27	7	28	4	37	7	44	8	56	8
AD-1700844.1	2	0	5	0	9	1	13	2	23	4	31	4	53	7	47	5
AD-1700555.1	1	0	4	1	4	1	8	0	12	2	18	5	23	10	36	8
AD-1700880.1	5	2	6	1	10	1	12	1	24	4	41	7	43	12	53	1
AD-1700388.1	7	2	12	2	16	3	20	2	24	4	34	10	35	6	54	8
AD-1700560.1	5	1	15	2	22	3	28	4	31	6	38	3	54	4	59	4
AD-1700550.1	2	0	4	1	4	0	7	1	18	0	22	3	39	6	43	3
AD-1700180.1	1	0	3	1	5	1	10	1	22	4	30	3	43	4	46	4
AD-1700312.1	8	3	11	3	7	1	21	7	25	1	31	2	36	3	49	4

TABLE 9-continued

Transfection and Free Uptake in Primary Cyno Hepatocytes																
Duplex	Transfection								Free uptake							
	10 nM		1 nM		0.1 nM		0.01 nM		250 nM		100 nM		10 nM		1 nM	
ID	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
AD-1699985.1	2	0	7	1	10	1	13	3	29	0	33	2	48	7	49	3
AD-1701011.1	5	1	12	2	20	4	23	3	28	3	41	2	57	9	57	6
AD-1700476.1	3	1	8	1	14	2	15	1	23	3	37	9	57	11	57	9
AD-1701022.1	3	0	6	1	6	1	13	2	23	4	30	1	38	3	59	3
AD-1700554.1	3	1	5	1	5	1	14	3	18	3	21	6	41	6	63	7
AD-1699951.1	2	0	4	1	5	1	8	1	12	2	16	2	22	4	34	6
AD-1700408.1	5	1	8	2	12	2	13	4	26	6	33	2	57	9	45	5
AD-1700487.1	4	0	11	2	11	2	21	2	28	6	19	2	43	10	51	1
AD-1700943.1	5	1	9	1	9	0	18	0	33	6	43	2	50	5	68	12
AD-1700736.1	9	2	18	3	19	3	29	2	49	12	61	9	68	15	84	12
AD-1700813.1	3	1	6	1	9	2	12	2	26	6	31	4	36	8	44	3
AD-1700968.1	4	1	8	1	8	2	13	2	23	4	31	8	54	8	50	7
AD-1700780.1	4	1	3	1	5	4	8	2	9	1	14	3	19	3	31	9
AD-1700324.1	9	2	14	2	11	2	21	6	NA	NA	NA	NA	NA	NA	NA	NA
AD-1700566.1	1	0	4	0	5	1	8	1	15	4	17	1	23	5	34	6
AD-1700354.1	4	1	7	1	9	1	10	1	20	3	23	5	30	6	51	5
AD-1700365.1	8	2	14	3	22	3	29	6	33	7	45	5	69	4	61	7
AD-1700553.1	2	1	4	1	4	1	9	2	13	2	23	4	30	4	38	6

TABLE 10

Transfection and Free Uptake in Primary Human Hepatocytes																
Duplex	Transfection								Free Uptake							
	10 nM		1 nM		0.1 nM		0.01 nM		250 nM		100 nM		10 nM		1 nM	
ID	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
AD-1700826.1	13	2	24	3	28	7	32	6	33	3	34	3	77	4	108	22
AD-1699971.1	5	1	9	1	15	3	13	3	7	3	8	2	16	3	50	14
AD-1720309.1	12	0	25	4	42	7	48	6	47	10	58	11	75	9	75	22
AD-1720382.1	8	2	10	3	7	4	14	2	21	2	25	3	38	6	73	8
AD-1700355.1	12	1	17	3	25	5	29	10	28	9	32	8	54	4	98	14
AD-1720294.1	4	1	8	1	13	1	16	2	16	3	15	1	25	4	67	12
AD-1700411.1	19	2	33	4	59	10	59	14	63	4	68	14	94	5	115	15
AD-1700383.1	5	3	4	4	7	4	13	4	15	3	15	3	32	7	75	17
AD-1720304.1	17	3	30	4	41	5	41	7	54	10	60	4	73	7	102	10
AD-1700082.1	13	3	23	5	28	7	28	7	37	8	43	4	61	14	93	5
AD-1700872.1	7	1	11	3	16	2	16	2	25	6	25	3	40	4	77	15

TABLE 10-continued

Transfection and Free Uptake in Primary Human Hepatocytes																
Duplex	Transfection								Free Uptake							
	10 nM		1 nM		0.1 nM		0.01 nM		250 nM		100 nM		10 nM		1 nM	
ID	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
AD-1700990.1	4	0	7	2	8	2	8	3	5	2	8	2	18	2	33	8
AD-1700558.1	12	1	21	2	28	3	29	4	32	7	40	5	68	4	100	12
AD-1700782.1	8	1	13	1	17	2	22	3	24	2	22	4	43	3	91	19
AD-1720375.1	16	2	20	1	25	4	27	3	38	7	30	2	55	2	88	3
AD-1700358.1	11	2	22	2	28	2	31	3	37	5	36	2	61	8	82	8
AD-1720314.1	5	2	9	3	11	2	10	2	10	2	11	3	19	7	41	20
AD-1700156.1	6	1	8	1	11	2	10	2	14	5	18	3	32	3	55	8
AD-1699964.1	3	0	8	2	9	2	12	2	7	1	7	1	15	1	54	11
AD-1699976.1	6	2	11	2	12	1	13	1	8	2	10	2	24	4	60	12
AD-1701007.1	15	4	15	4	16	3	17	4	11	2	12	2	24	2	61	4
AD-1700386.1	10	0	18	3	29	4	23	2	27	2	38	8	45	9	81	6
AD-1700995.1	5	1	9	2	9	2	10	2	8	1	11	3	21	4	55	19
AD-1700376.1	6	0	13	2	16	2	15	3	9	2	14	3	32	6	70	8
AD-1700915.1	12	1	19	3	29	8	34	8	34	10	33	7	48	9	96	17
AD-1700384.1	8	2	16	2	18	2	20	2	30	4	36	8	46	8	92	2
AD-1700925.1	18	1	27	7	38	4	46	2	70	16	68	9	82	11	122	8
AD-1700378.1	11	2	19	4	28	2	27	3	34	6	42	9	62	12	97	10
AD-1700870.1	14	2	20	4	31	3	35	4	41	8	37	2	71	10	118	4
AD-1700353.1	18	1	22	5	26	4	29	4	44	5	49	4	64	13	107	1
AD-1720371.1	12	3	21	5	26	2	27	5	26	1	29	7	51	7	75	22
AD-1700374.1	8	1	15	2	18	2	18	1	15	1	18	3	37	7	77	17
AD-1700825.1	15	1	20	4	30	6	37	8	46	4	57	10	75	11	115	19
AD-1720386.1	6	1	10	2	12	2	13	2	16	2	15	2	30	5	76	6
AD-1720315.1	6	1	10	2	15	2	14	6	11	5	14	5	24	7	50	21
AD-1700159.1	5	0	12	3	13	4	16	3	15	3	20	3	32	5	91	11
AD-1700923.1	17	3	29	6	39	3	45	10	53	10	61	8	84	9	103	11
AD-1700369.1	5	0	10	1	13	1	17	1	6	1	9	1	26	2	56	16
AD-1701016.1	8	2	13	3	18	6	13	5	22	7	22	4	46	4	92	25
AD-1700148.1	7	3	13	2	18	3	17	4	18	6	17	2	38	7	69	6
AD-1700318.1	13	3	18	3	25	3	28	5	32	7	32	8	52	3	88	3
AD-1700781.1	10	1	18	3	20	2	21	3	25	4	23	3	55	9	96	24
AD-1700905.1	13	4	15	3	17	4	24	3	24	4	38	6	25	12	54	11
AD-1720308.1	10	1	17	3	20	2	23	3	19	3	24	6	42	2	78	6
AD-1700916.1	8	2	12	1	15	2	17	3	21	8	8	3	42	8	67	9
AD-1700461.1	6	2	7	3	7	3	10	4	11	3	11	5	17	5	43	18
AD-1700871.1	9	1	13	2	13	2	15	3	10	2	13	2	36	8	65	8

TABLE 10-continued

Transfection and Free Uptake in Primary Human Hepatocytes																
Duplex ID	Transfection								Free Uptake							
	10 nM		1 nM		0.1 nM		0.01 nM		250 nM		100 nM		10 nM		1 nM	
	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
AD-1720373.1	11	0	18	2	27	2	30	3	25	4	33	6	56	6	90	8
AD-1701004.1	6	1	11	1	14	2	15	2	16	3	15	4	35	6	56	12
AD-1700556.1	5	1	8	1	12	2	15	3	10	1	9	2	24	6	67	4
AD-1720283.1	12	1	14	1	21	2	16	3	33	3	31	2	38	5	78	11
AD-1700303.1	9	2	16	4	20	3	19	5	24	3	28	4	48	2	60	8
AD-1720370.1	9	1	15	2	14	3	17	2	12	2	17	4	38	4	65	10
AD-1700573.1	9	1	15	1	18	2	19	2	16	2	21	3	46	11	85	12
AD-1720372.1	10	2	14	2	20	3	22	3	15	3	21	5	48	6	92	16
AD-1700473.1	5	1	9	3	13	4	15	7	9	2	11	3	23	2	52	16
AD-1700821.1	3	0	5	1	9	2	10	1	4	1	5	1	12	3	33	17
AD-1720384.1	5	1	8	3	11	2	14	1	10	2	11	3	28	6	60	26
AD-1700314.1	6	3	7	3	15	0	21	3	11	2	19	4	39	3	83	14
AD-1700582.1	5	1	9	2	15	4	13	2	11	2	13	3	20	5	55	4
AD-1700385.1	9	1	13	2	20	3	21	6	12	5	22	4	42	7	57	9
AD-1700544.1	7	1	11	2	17	4	19	6	10	3	13	3	24	2	67	5
AD-1699970.1	3	1	6	2	8	1	8	1	5	1	5	1	10	4	24	3
AD-1700967.1	12	1	24	4	27	3	25	4	40	8	45	11	67	7	100	9
AD-1700844.1	9	1	16	2	21	1	24	1	24	6	28	7	57	8	98	10
AD-1700555.1	4	1	8	1	11	1	9	2	4	1	6	2	12	1	22	8
AD-1700880.1	5	2	10	2	12	4	10	4	9	3	19	4	35	3	42	13
AD-1700388.1	10	0	16	4	27	6	25	2	30	2	36	4	55	8	87	17
AD-1700560.1	11	2	16	2	23	2	25	2	28	5	35	8	61	11	95	9
AD-1700550.1	4	0	10	1	10	1	12	1	7	1	8	2	16	1	51	15
AD-1700180.1	7	1	11	1	16	2	16	1	18	2	14	2	33	1	64	12
AD-1700312.1	5	1	9	1	14	2	12	2	10	3	11	0	27	5	71	10
AD-1699985.1	8	1	20	4	22	4	26	5	31	9	41	9	62	5	105	9
AD-1701011.1	9	1	19	1	23	3	24	3	26	4	31	4	53	3	88	11
AD-1700476.1	12	4	18	2	27	6	31	9	41	1	45	4	59	13	109	7
AD-1701022.1	6	1	10	4	14	2	14	1	15	2	13	2	29	4	69	5
AD-1700554.1	4	1	7	1	6	3	10	2	4	1	3	1	10	3	23	11
AD-1699951.1	6	1	10	2	17	4	13	6	13	4	13	6	26	3	48	5
AD-1700408.1	9	3	15	5	22	2	22	4	29	4	31	6	53	10	95	10
AD-1700487.1	6	2	11	3	14	3	13	5	20	6	27	6	37	11	51	12
AD-1700943.1	8	1	13	2	16	3	21	8	28	4	31	4	53	3	105	17
AD-1700736.1	12	1	17	3	24	4	28	2	56	11	55	7	71	7	103	20
AD-1700813.1	9	1	14	3	14	4	16	4	23	4	25	1	45	2	67	10

TABLE 10-continued

Transfection and Free Uptake in Primary Human Hepatocytes																
Duplex	Transfection								Free Uptake							
	10 nM		1 nM		0.1 nM		0.01 nM		250 nM		100 nM		10 nM		1 nM	
ID	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
AD-1700968.1	6	2	10	1	12	3	8	3	10	3	19	3	35	11	41	29
AD-1700780.1	1	0	3	1	9	2	7	2	4	1	4	1	10	4	22	9
AD-1700324.1	8	2	9	2	10	1	12	4	18	1	20	4	33	6	44	11
AD-1700566.1	10	2	17	3	19	1	20	3	35	5	29	5	51	5	90	9
AD-1700354.1	7	1	11	2	16	2	16	1	9	2	12	3	28	4	49	8
AD-1700365.1	20	2	35	4	42	3	44	5	58	10	61	11	81	5	112	12
AD-1700553.1	2	0	5	1	6	1	6	2	3	0	5	2	8	3	30	11

Example 4. In Vivo Screening of dsRNA Duplexes

(1) Single Dose Study (3 mg/kg)

Duplexes of interest, identified from the above in vitro studies, were evaluated in vivo. In particular, at pre-dose day -14 wild-type mice (C57BL/6) were transduced with 2×10^{10} viral particles of an adeno-associated virus 8 (AAV8) vector encoding human CIDEB intravenously via retro-orbital delivery. In particular, mice were administered an AAV8 encoding a portion of human CIDEB mRNA encoding the open reading frame and 3' UTR of human CIDEB mRNA referenced as NM_001393338.1, referred to as VCAV-07736-AAV8.HsCideb-FL-trd.

At day 0, groups of three mice were subcutaneously administered a single 3 mg/kg dose of the duplexes of interest or phosphate-buffered saline (PBS). Table 11 provides the treatment design and provides the duplexes of interest. On day 7 after dosing, animals were sacrificed, and liver samples were collected and snap-frozen in liquid nitrogen. Liver mRNA was extracted and analyzed by the RT-QPCR method.

For all samples, human CIDEB Cq values were first normalized to Gapdh Cq values as a reference gene to calculate the liver CIDEB mRNA levels for each animal. For each group, the liver CIDEB mRNA levels relative to Gapdh from individual treated animals were normalized to the group mean ($\pm$ standard deviation [SD]) of relative CIDEB mRNA levels from the PBS group.

The data were expressed as percent of baseline value and presented as mean plus standard deviation. The results, listed in Table 12 and shown in FIG. 1, demonstrate that the exemplary duplex agents tested effectively reduce the level of the human CIDEB messenger RNA in vivo.

TABLE 11

Study Design for In Vivo Single Dose Study							
Test Group	Test Material	No. of Fe-males	Dose Level (mg/kg)	Dose Volume (mL/kg)	Dose Conc. (mg/mL)	Route/Regimen	
G1	PBS	3	3	10	0.3	SC/Single dose	
G2	Naïve	3	3	10	0.3	N/A	
G3	AD-1700554.3	3	3	10	0.3	SC/Single dose	
G4	AD-1700383.3	3	3	10	0.3	SC/Single dose	
G5	AD-1700374.3	3	3	10	0.3	SC/Single dose	
G6	AD-1700995.3	3	3	10	0.3	SC/Single dose	
G7	AD-1720294.3	3	3	10	0.3	SC/Single dose	
G8	AD-1720314.3	3	3	10	0.3	SC/Single dose	
G9	AD-1700556.3	3	3	10	0.3	SC/Single dose	
G10	AD-1700782.3	3	3	10	0.3	SC/Single dose	
G11	AD-1700376.3	3	3	10	0.3	SC/Single dose	
G12	AD-1700544.3	3	3	10	0.3	SC/Single dose	
G13	AD-1699964.3	3	3	10	0.3	SC/Single dose	
G14	AD-1700314.3	3	3	10	0.3	SC/Single dose	
G15	AD-1700148.3	3	3	10	0.3	SC/Single dose	
G16	AD-1700573.3	3	3	10	0.3	SC/Single dose	
G17	AD-1699971.3	3	3	10	0.3	SC/Single dose	
G18	AD-1700473.3	3	3	10	0.3	SC/Single dose	
G19	AD-1700369.3	3	3	10	0.3	SC/Single dose	
G20	AD-1699976.5	3	3	10	0.3	SC/Single dose	
G21	AD-1700555.3	3	3	10	0.3	SC/Single dose	
G22	AD-1720315.3	3	3	10	0.3	SC/Single dose	
G23	AD-1700821.3	3	3	10	0.3	SC/Single dose	
G24	AD-1699970.3	3	3	10	0.3	SC/Single dose	

TABLE 12

qPCR Results for In Vivo Single Dose Study								
Grp	Mouse #	Duplex	# message/mouse				avg/mouse	stdev/grp.
1	1	PBS	131.8	124.7	123.0	118.0	124.4	33.86
	2		123.0	122.1	125.6	123.0	123.4	
	3		65.9	67.3	63.7	64.1	65.2	

TABLE 12-continued

qPCR Results for In Vivo Single Dose Study									
Grp	Mouse #	Duplex	# message/mouse				avg/mouse	grp avg	stdev/grp.
2	4	Naive	65.9	69.2	64.1	59.8	64.7	79.3	20.60
	5								
3	6	AD-1700554.3	97.2	95.8	90.6	91.9	93.9	19.6	12.56
	7		31.2	34.1	32.7	32.7	32.7		
	8		19.2	18.2	18.8	18.2	18.6		
4	9	AD-1700383.3	8.0	7.8	7.6	7.1	7.6	66.5	1.7
	10		69.2	65.9	65.9	63.2	66.0		
	11		68.7	61.9	65.0	64.5	65.0		
5	12	AD-1700374.3	70.1	70.1	67.8	65.4	68.4	17.4	5.40
	13		12.7	12.7	13.5	12.1	12.7		
	14		16.0	15.7	15.9	17.2	16.2		
6	15	AD-1700995.3	23.8	23.0	22.8	23.6	23.3	29.0	4.39
	16		26.9	30.5	26.8	28.9	28.3		
	17		34.1	34.8	32.5	33.4	33.7		
7	18	AD-1720294.3	25.3	25.1	24.8	24.8	25.0	48.5	35.86
	19								
	20		14.4	16.1	13.3	14.5	14.6		
8	21	AD-1720314.3	37.8	40.0	36.8	39.5	38.5	35.6	12.70
	22		26.9	30.5	27.1	27.5	28.0		
	23		47.2	53.2	51.0	49.6	50.2		
9	24	AD-1700556.3	27.1	29.9	29.9	26.9	28.5	17.5	3.58
	25		14.2	16.1	16.6	16.5	15.9		
	26		14.3	14.5	16.6	14.7	15.1		
10	27	AD-1700782.3	21.9	21.7	21.3	21.6	21.6	19.7	4.09
	28		21.0	22.7	20.9	22.0	21.6		
	29		15.0	15.4	15.1	14.7	15.0		
11	30	AD-1700376.3	23.1	22.3	22.2	22.3	22.5	15.7	8.40
	31		14.5	14.1	14.4	14.5	14.4		
	32		8.2	7.8	8.1	8.1	8.1		
12	33	AD-1700544.3	23.6	26.4	26.0	22.8	24.7	22.8	11.68
	34		28.5	27.1	27.5	26.9	27.5		
	35		31.2	32.0	31.6	30.5	31.3		
13	36	AD-1699964.3	9.2	10.1	9.7	9.0	9.5	11.0	1.61
	37		12.1	12.4	12.2	12.3	12.3		
	38		8.4	9.3	9.5	9.4	9.2		
14	39	AD-1700314.3	11.6	11.6	11.3	11.1	11.4	11.3	2.31
	40		10.9	11.1	10.9	10.9	11.0		
	41		13.9	14.0	13.7	13.6	13.8		
15	42	AD-1700148.3	9.5	9.5	9.1	8.9	9.2	44.1	6.55
	43		46.0	46.3	47.6	47.9	46.9		
	44		36.6	36.6	36.3	36.8	36.6		
16	45	AD-1700573.3	48.2	48.9	47.9	49.6	48.7	25.6	4.38
	46		20.4	20.7	21.0	21.9	21.0		
	47		29.7	30.5	29.9	28.9	29.8		
17	48	AD-1699971.3	25.5	26.4	25.7	26.0	25.9	29.3	16.43
	49		43.2	44.7	40.8	41.4	42.5		
	50		11.1	11.0	11.0	10.5	10.9		
18	51	AD-1700473.3	35.3	34.8	35.6	32.0	34.4	23.5	20.90
	52		46.6	45.0	46.6	45.6	46.0		
	53		19.5	20.1	20.1	20.6	20.1		
19	54	AD-1700369.3	4.6	4.5	4.6	4.6	4.6	14.2	0.75
	55		13.9	13.2	13.3	13.2	13.4		
	56		15.1	14.8	15.0	14.7	14.9		
20	57	AD-1699976.5	14.1	13.9	14.0	15.1	14.3	24.1	14.83
	58		16.0	15.5	14.4	15.2	15.3		
	59		41.4	42.9	42.9	37.8	41.3		
21	60	AD-1700555.3	15.4	15.5	16.5	16.1	15.9	9.1	5.50
	61		3.7	3.7	3.6	3.3	3.6		
	62		16.1	15.7	13.0	13.5	14.6		
22	63	AD-1720315.3	10.2	9.1	8.8	8.7	9.2	64.9	23.44
	64		45.6	44.1	42.9	45.3	44.5		
	65		90.0	91.3	91.3	89.4	90.5		
23	66	AD-1700821.3	57.4	61.1	59.8	60.6	59.7	14.8	2.18
	67		13.3	13.8	13.9	13.6	13.6		
	68		17.1	17.3	18.2	16.9	17.4		
24	69	AD-1699970.3	13.3	13.9	13.7	13.3	13.5	19.0	7.21
	70		27.1	27.5	24.6	25.1	26.1		
	71		12.9	11.4	11.8	10.6	11.7		
	72		19.1	19.5	18.5	19.6	19.2		

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(2) Multiple Dose Study (0.75 mg/kg and 1.5 mg/kg)

Duplexes of interest, identified from the above in vitro studies, were evaluated in vivo. In particular, at pre-dose day -14 wild-type mice (C57BL/6) were transduced with 2×10^{10} viral particles of an adeno-associated virus 8 (AAV8) vector encoding human CIDEB by intravenously via retro-orbital delivery. In particular, mice were administered an AAV8 encoding a portion of human CIDEB mRNA encoding the open reading frame and 3' UTR of human CIDEB mRNA referenced as NM_001393338.1, referred to as VCAV-07736-AAV8.HsCideb-FL-trd.

At day 0, groups of three mice were subcutaneously administered a single 0.75 or 1.5 mg/kg dose of the duplexes of interest or phosphate buffered saline (PBS). Table 13 provides the study design and provides the duplexes of interest. On day 10 after dosing, animals were sacrificed, and liver samples were collected and snap-frozen in liquid nitrogen. Liver mRNA was extracted and analyzed by the RT-QPCR method.

For all samples, human CIDEB Cq values were first normalized to Gapdh Cq values as a reference gene to calculate the liver CIDEB mRNA levels for each animal. For each group, the liver CIDEB mRNA levels relative to Gapdh from individual treated animals were normalized to the group mean ($\pm$ standard deviation [SD]) of relative CIDEB mRNA levels from the PBS group.

The data were expressed as percent of baseline value and presented as mean plus standard deviation. The results, listed in Table 14 and shown in FIG. 2, demonstrate that the exemplary duplex agents tested effectively dose-dependently reduced the level of the human CIDEB messenger RNA in vivo.

TABLE 13

Study Design for In Vivo Multi-Dose Study						
Test Group	Test Material	No. of Fe-males	Dose Level (mg/kg)	Dose Volume (mL/kg)	Dose Conc. (mg/mL)	Route/Regimen
G1	PBS	3	3	10	0.3	SC/Single dose

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TABLE 13-continued

Study Design for In Vivo Multi-Dose Study						
Test Group	Test Material	No. of Fe-males	Dose Level (mg/kg)	Dose Volume (mL/kg)	Dose Conc. (mg/mL)	Route/Regimen
G2	Naïve	3	3	10	0.3	N/A
G3	AD-1700555.3	3	1.5	10	0.15	SC/Single dose
G4	AD-1700555.3	3	0.75	10	0.075	SC/Single dose
G5	AD-1699964.3	3	1.5	10	0.15	SC/Single dose
G6	AD-1699964.3	3	0.75	10	0.075	SC/Single dose
G7	AD-1700314.3	3	1.5	10	0.15	SC/Single dose
G8	AD-1700314.3	3	0.75	10	0.075	SC/Single dose
G9	AD-1700369.3	3	1.5	10	0.15	SC/Single dose
G10	AD-1700369.3	3	0.75	10	0.075	SC/Single dose
G11	AD-1700821.3	3	1.5	10	0.15	SC/Single dose
G12	AD-1700821.3	3	0.75	10	0.075	SC/Single dose
G13	AD-1700376.3	3	1.5	10	0.15	SC/Single dose
G14	AD-1700376.3	3	0.75	10	0.075	SC/Single dose
G15	AD-1700374.3	3	1.5	10	0.15	SC/Single dose
G16	AD-1700374.3	3	0.75	10	0.075	SC/Single dose
G17	AD-1700556.3	3	1.5	10	0.15	SC/Single dose
G18	AD-1700556.3	3	0.75	10	0.075	SC/Single dose
G19	AD-1699976.5	3	1.5	10	0.15	SC/Single dose
G20	AD-1699976.5	3	0.75	10	0.075	SC/Single dose

TABLE 14

qPCR Results for In Vivo Multi-Dose Study									
Grp	Mouse #	Duplex	# message/mouse				avg/mouse	grp avg	stdev/grp.
1	1	PBS	101.0	102.4	101.7	99.6	101.2	100.0	1.01
	2		103.1	101.0	97.5	96.9	99.6		
	3		100.3	101.0	97.5	98.2	99.3		
2	4	Naïve	68.5	71.9	66.2	70.9	69.4	99.7	69.67
	5		177.1	191.1	175.8	173.4	179.3		
	6		50.1	51.9	50.5	48.4	50.2		
3	7	AD-1700555.3	10.0	10.9	10.9	10.0	10.5	20.8	11.21
	8		18.9	20.4	18.5	18.9	19.1		
	9		32.2	34.0	33.1	31.5	32.7		
4	10	AD-1700555.3	43.7	46.8	45.5	42.8	44.7	42.2	2.16
	11		41.0	43.4	40.2	40.4	41.2		
	12		41.3	42.5	40.4	38.5	40.7		
5	13	AD-1699964.3	37.7	39.3	37.0	36.7	37.7	26.3	16.14
	14		15.0	15.0	14.8	14.6	14.9		
	15		67.6	67.1	64.8	67.1	66.6		
6	16	AD-1699964.3	84.9	87.3	83.8	82.6	84.6	69.6	13.85
	17		59.2	57.6	55.6	57.2	57.4		
	18								

TABLE 14-continued

qPCR Results for In Vivo Multi-Dose Study									
Grp	Mouse #	Duplex	# message/mouse				avg/mouse	grp avg	stdev/grp.
7	19	AD-	22.1	22.1	21.8	21.8	22.0	25.2	0.66
	20	1700314.3	25.1	26.9	26.1	24.6	25.7		
	21	1.5 mpk	24.2	25.4	24.6	24.7	24.7		
8	22	AD-						60.9	7.08
	23	1700314.3	57.2	56.4	54.5	55.6	55.9		
	24	0.75 mpk	67.1	67.6	63.9	65.3	66.0		
9	25	AD-	30.2	30.4	30.0	30.0	30.2	34.6	8.85
	26	1700369.3	43.7	45.2	44.9	45.5	44.8		
	27	1.5 mpk	28.2	28.2	29.4	29.6	28.9		
10	28	AD-	47.8	49.1	50.8	50.5	49.6	48.3	38.49
	29	1700369.3	9.1	9.4	9.1	9.2	9.2		
	30	0.75 mpk	86.7	89.1	87.3	81.5	86.2		
11	31	AD-	37.0	37.5	37.2	37.7	37.4	25.0	21.40
	32	1700821.3	0.3	0.2	0.2	0.3	0.2		
	33	1.5 mpk	37.5	37.2	37.7	36.7	37.3		
12	34	AD-	51.9	54.5	52.3	52.6	52.8	43.5	16.48
	35	1700821.3	24.1	24.6	24.6	24.6	24.4		
	36	0.75 mpk	53.0	54.5	54.5	50.5	53.1		
13	37	AD-	29.6	30.9	28.8	28.4	29.4	37.1	6.74
	38	1700376.3	38.5	43.4	39.3	38.3	39.9		
	39	1.5 mpk	43.1	41.9	40.7	42.5	42.0		
14	40	AD-	79.8	80.9	79.8	80.9	80.3	68.3	23.43
	41	1700376.3	42.5	42.2	41.6	38.8	41.3		
	42	0.75 mpk	82.6	84.3	83.8	82.0	83.2		
15	43	AD-	22.0	22.9	22.6	22.8	22.6	32.2	12.26
	44	1700374.3	28.2	27.1	28.8	28.2	28.1		
	45	1.5 mpk	47.4	47.8	44.0	44.9	46.0		
16	46	AD-	72.9	76.5	75.5	76.5	75.4	56.7	25.04
	47	1700374.3	29.2	28.0	27.4	28.2	28.2		
	48	0.75 mpk	68.5	66.6	64.8	65.7	66.4		
17	49	AD-	63.0	64.8	58.0	60.9	61.7	29.6	30.64
	50	1700556.3	27.6	26.5	26.9	25.2	26.6		
	51	1.5 mpk	0.7	0.6	0.6	0.6	0.6		
18	52	AD-	63.9	64.4	65.3	62.2	63.9	89.3	22.06
	53	1700556.3	98.9	102.4	102.4	96.9	100.1		
	54	0.75 mpk	105.3	101.7	103.8	104.5	103.8		
19	55	AD-	35.7	36.0	37.7	38.3	36.9	35.3	4.49
	56	1699976.5	30.0	29.8	31.3	30.0	30.3		
	57	1.5 mpk	37.5	40.7	40.4	36.7	38.8		
20	58	AD-	62.2	61.3	61.7	59.6	61.2	72.1	18.75
	59	1699976.5	97.5	95.5	93.6	88.5	93.8		
	60	0.75 mpk	62.2	61.3	61.3	60.9	61.4		

CIDEB Sequences

>NM_001393338.1 *Homo sapiens* cell death inducing DFFA like effector b (CIDEB), transcript variant 7, mRNA

SEQ ID NO: 1

CCCTTCCGGTG-
GAGCCAGCGCTGCGACCGCCTGCAGAAGGTTGACTGCGTGGTAGGGGGCCAGAGCAAGCCGAAGG

CAAGCACGATGGCGCTACCAAGCCGGCCACCCGCGCCCGTGCCGCCCCG-
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GCCAGCGTCACGCTGTAGCAGCCGAG-
CATCAGCCGAAAGGAAGCACGAAAGCGGTGAGAGTCTCCAGGCTCAGGTG

GGCGGCGGCGTGGACCGGCGACGGGTGGCACAGCTGGCATAACGCGGTCCCTC-
CACAGGTGGCGGTAGACGGCGGCCG

GGACGGCGAGCAACAGGGCGGCCAGCCA-
GACCGCCAGCAGCAGGCGGGCGGGCCAGGGCCGGGTGCGCAGCCGAGGC

GCCAGGAAGGGCGGGTGACTGCGAGGCAGCGCTGCAGGCTGAGCAGGCCGGT-
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GAGCGCGCACACGTAGTA-
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- continued

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ACAACTCCTTCTTGAACCCAGATCCCCTGCTCCCAGTCAGTTGACCTGCCC-
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CCGCCAGGAGCTGCTAGCCAAAGCATTGGAGACCCCTACTGCTGAATG-
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ACACTGCTGCAAGGCCTGGGCCATATGTTGCTGGGAATTTCTCCACCCTTCGT-
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ATCCCAGCCTGACTGCTGTACATCTAATCCCCTAC-
CAATGCCTCCTGTCCCTAAACTCCCAGCATACTGATGACA

GCCCTCTCTGACTTTACCTTGAGATCTGTCTTCATACCTTCCCCCTCAAACCTAACAAAAA-
CATTTCCAATAAAAAATA

TCAAATATTTACCACTAA

-continued

>Reverse complement of SEQ ID NO: 1

SEQ ID NO: 2

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GAAGGGTATGAAGACAGATCTCAA

GGTAAAGTCAGAGAGGGCTGTCATCAGTATGCTGGGGAGTTTAGGGACAGGAGGCAT-
TGGTAGGGGATTAGATGTAG

CAGCAGTCAGGCTGGGATCAAGATGCCTGGGGGACATCTT-
GATCTTGGCCTTTCAGGGCAAGTGGGAGGCCAGAAAG

GTGGCTAGGAAAGAACAGCATTCCTCAGGTAAGGGTATAGACTTGGGATGTGAGGCGT-
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CTGTCATAGTCTTTGCAGTGGGTCGGTTGGAATGAT-
TCTGGGGGCAGAAGCTCAGAGCCCCCTTAGTAGGAATGGAGG

CGGCCCTTCTGCTGCCACTGCTCAGCCCCCTCCACTGCATGACGAAGGGTGGAGGAAAT-
TCCCAGCAACATATGGCC

CAGGCCTTGCAGCAGTGTGGAGGTCCAACGAAGGAGCTCCCTGAGTACTTTCTTTGGGC-
CAAGTCCTTGAAAGTCAC

AACTCATAGAGTAGAGCCCGTAGAATGTGGCTTTGACATTCAGGCTGC-
CAAAGAGGTCTCGAGGGTTTTGCTTTGTAC

ACGTCAAAGGTGAATCGGGCGATGTCCTTGCTGTGCTTGGGCCTCTCCCGTCCAGGC-
CATATGACAGCACTCCACT

CCTTGTAGGGCTCCAGCTCTGACCAGACTGCAACACCATCAGGCACGTGT-
CATCCTCCAGCAGCTGGAAGAAGTCCT

CACTGTCCACTGCAGTTCATCCTCCTCTAGCACCAGGGTTAGCACTCCAT-
TCAGCAGTAGGGTCTCCAATGCTTTG

GCTAGCAGCTCCTGGCGGGTGGCAGCTGTCAGGCCTTTCGGATGGTCCGCTTGTGAT-
CACAGACACGGAAGGTCG

CTGGGGTGGTGGAGCTGAGGTCCAGACCTCCGTCCAACTCCGAGCTTATATTAGA-
TACTGACCTGAGTAAGTCAC

TGGGGTTCAGAGCTGAGAGGTACTCCATGGTGGACCGGAGAGTTCCTTCCCTG-
GAACCTCTGGGCTGGGTGGTTCTC

TCCTGTGCTGGGGCTTTAGTGGTGTCTTCTGTTACAAACCTGG-
GATCTGCTCTTCTCTCCTTGGTCGGAGGAG

GGGCTGGCTCACTGCTCTGGCTTCATTTTCCAGAGCTGCCTGCTGCAGTCACACTAGGT-
CATCTTCTCTCACTTTT

CTCCTTTTGCCGATTAGTGGACGTGACAGAGATGTGAATGGGGCAGGGATGTCCTTT-
GATGGCATCAAGACTTTAGC

TTCTGGTGCCTGTGTCCAGCTCTGATTTAGTTGCAGCCGTGATGGACAGTTGCATG-
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TGCTGCT-
CACGCCGCTCTTTGTGGCCTTCTGACCCGGCAGGCCTGGCCGCTGGCCAGGCGGGCTGCAAGGCGGTG

TACTACGTGTGCGCGCTCAGCATGTACGCCAGCGTGTGCT-
CACCGGCTGCTCAGCCTGCAGCGCTGCCTCGCAGT

CACCCGCCCTTCTGGCGCCTCGGCTGCGCAGCCCGGCCCTGGCCCGCGCTGCTGCTGGCGGTCTGGCTGGCCG

CCCTGTTGCTCGCCGTCGCCGGCGCGCTTACGCCACCTGTG-
GAGGACCGCGTATGCCAGCTGTGCCACCCGTCG

CCGGTCCACGCCGCCGCCACCTGAGCCTGGA-
GACTCTGACCGCTTTCGTGCTTCCTTCGGGCTGATGCTCGGCTG

CTA-
CAGCGTGACGTGGCACGGCTGCGGGGCGCCCGCTGGGGCTCCGGGCGGCACGGGGCGCGGTGGGCCGGCTGG

TGAGCGCCATCGTGTGCTTGCCTTCGGCTTGTCTGGGCCCCCTACCACGCAGT-
CAACCTTCTGCAGGCGGTGCGACG

CTGGCTCCACCGAAGGG

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12416006B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

We claim:

1. A double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell, or a salt thereof,

wherein said dsRNA agent, or a salt thereof, comprises a sense strand and an antisense strand forming a double stranded region,

wherein the sense strand consists of the nucleotide sequence

(SEQ ID NO: 705)

5' -AGUAUCUAAUAUAGCUCGGU-3' of

and the antisense strand consists of the nucleotide sequence

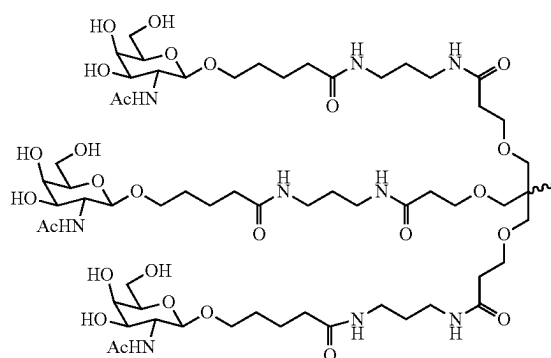
(SEQ ID NO: 1153)

5' -ACCGAGCUUAUAUAGAUACUGA-3' of

wherein all of the nucleotides of the sense strand and all of the nucleotides of the antisense strand comprise a nucleotide modification;

wherein a ligand is conjugated to the 3' end of the sense strand; and

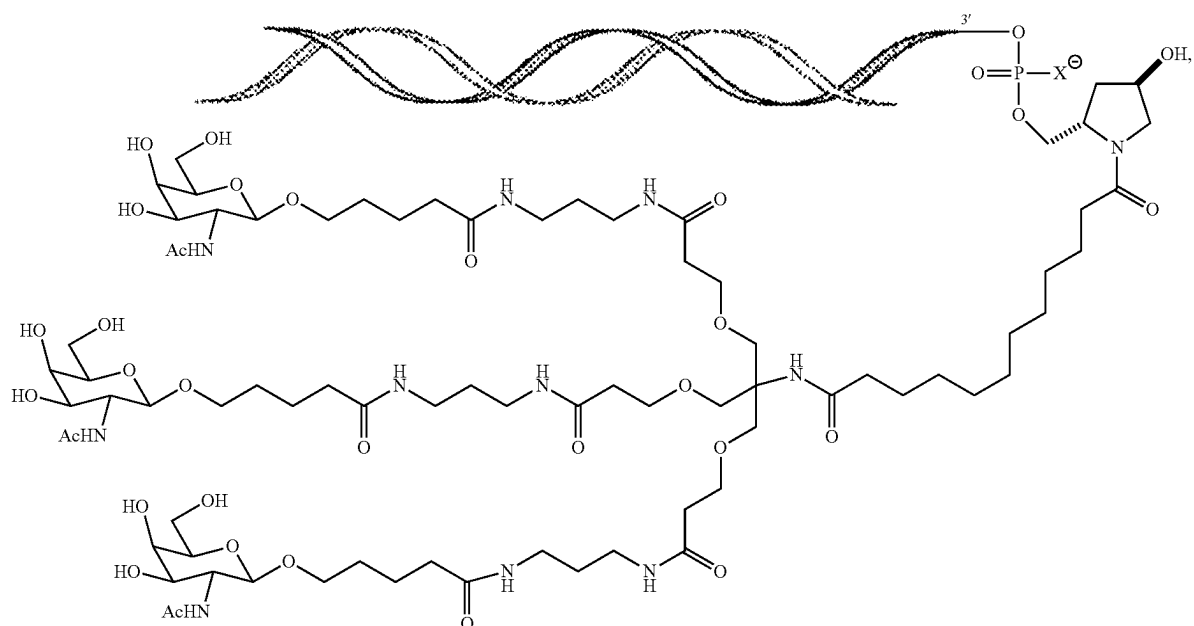
wherein the ligand is



2. The dsRNA agent, or a salt thereof, of claim 1, wherein the dsRNA agent, or a salt thereof, is conjugated to the ligand as shown in the following schematic

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wherein X is O or S.

3. The dsRNA agent, or a salt thereof, of claim 2, wherein X is O.

4. The dsRNA agent, or a salt thereof, of claim 1, wherein the dsRNA agent, or a salt thereof, further comprises at least one phosphorothioate or methylphosphonate internucleotide linkage.

5. The dsRNA agent, or a salt thereof, of claim 4, wherein the phosphorothioate or methylphosphonate internucleotide linkage is at the 3'-terminus of the sense strand or the antisense strand.

6. The dsRNA agent, or a salt thereof, of claim 4, wherein the phosphorothioate or methylphosphonate internucleotide linkage is at the 5'-terminus of the sense strand or the antisense strand.

7. The dsRNA agent, or a salt thereof, of claim 4, wherein the dsRNA agent comprises 6-8 phosphorothioate internucleotide linkages.

8. A double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell, or a salt thereof, comprising a sense strand and an antisense strand forming a double stranded region,

wherein the antisense strand comprises the nucleotide sequence 5'-asdCscgdAgdCuuaudAuUfagauacusgsa-3' of SEQ ID. NO: 2049,

wherein

s is a phosphorothioate linkage;

a, c, g, and u are 2'-O-methyl (2'-OMe) A, C, G, and U, respectively;

dA is 2'-deoxyadenosine-3'-phosphate;

dC is 2'-deoxycytidine-3'-phosphate; and

Uf is 2'-fluoro U.

9. A double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell, or a salt thereof, comprising a sense strand and an antisense strand forming a double stranded region,

wherein the sense strand consists of the nucleotide sequence, 5'-asgsuaucaAfUfAfuaagcucggu-3' of SEQ ID. NO: 1601 and the antisense strand consists of the nucleotide sequence 5'-asdCscgdAgdCuuaudAuUfagauacusgsa-3' of SEQ ID. NO: 2049,

wherein

s is a phosphorothioate linkage;

a, c, g, and u are 2'-O-methyl (2'-OMe) A, C, G, and U, respectively;

dA is 2'-deoxyadenosine-3'-phosphate;

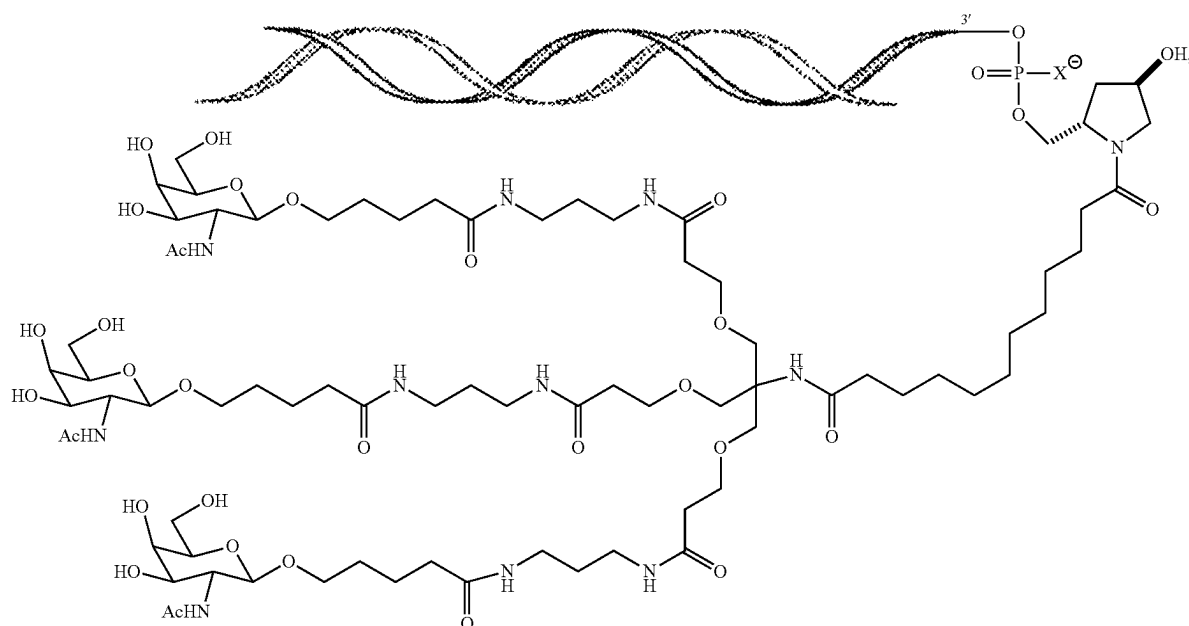
dC is 2'-deoxycytidine-3'-phosphate; and

Af and Uf are 2'-fluoro A and U, respectively, and

wherein the 3'-end of the sense strand is conjugated to a ligand as shown in the following schematic

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wherein X is O.

10. The dsRNA agent of claim 9 which is in sodium salt form.

11. A pharmaceutical composition for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEb) gene, comprising the dsRNA agent, or a salt thereof, of claim 9.

12. The pharmaceutical composition of claim 11, wherein the dsRNA agent, or a salt thereof, is present in an unbuffered solution.

13. The pharmaceutical composition of claim 11, wherein the dsRNA agent, or a salt thereof, is present in saline or water.

14. The pharmaceutical composition of claim 11, wherein the dsRNA agent, or a salt thereof, is present in a buffer solution.

15. The pharmaceutical composition of claim 14, wherein the buffer solution comprises acetate, citrate, prolamine, carbonate, or phosphate or any combination thereof.

16. The pharmaceutical composition of claim 14, wherein the buffer solution is phosphate buffered saline (PBS).

17. A double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell, or a salt thereof, comprising a sense strand and an antisense strand forming a double stranded region,

wherein the sense strand comprises the nucleotide sequence 5'-asgsuaucaAfUfAfuaagcucggu-3' of SEQ ID. NO: 1601 and the antisense strand comprises the nucleotide sequence 5'-asdCscgdAgdCuuaudAuUf-agauacugsa-3' of SEQ ID. NO: 2049,

wherein

s is a phosphorothioate linkage;

a, c, g, and u are 2'-O-methyl (2'-OMe) A, C, G, and U, respectively;

dA is 2'-deoxyadenosine-3'-phosphate;

dC is 2'-deoxycytidine-3'-phosphate; and

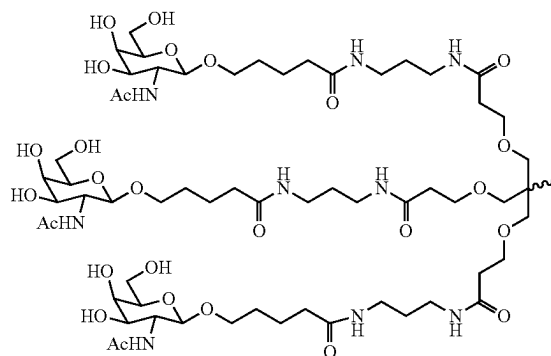
Af and Uf are 2'-fluoro A and U, respectively.

18. The dsRNA agent, or a salt thereof, of claim 17, further comprising a ligand.

19. The dsRNA agent, or a salt thereof, of claim 18, wherein the ligand is conjugated to the 3' end of the sense strand of the dsRNA agent, or a salt thereof.

20. The dsRNA agent, or a salt thereof, of claim 18, wherein the ligand comprises N-acetylgalactosamine (GalNAc).

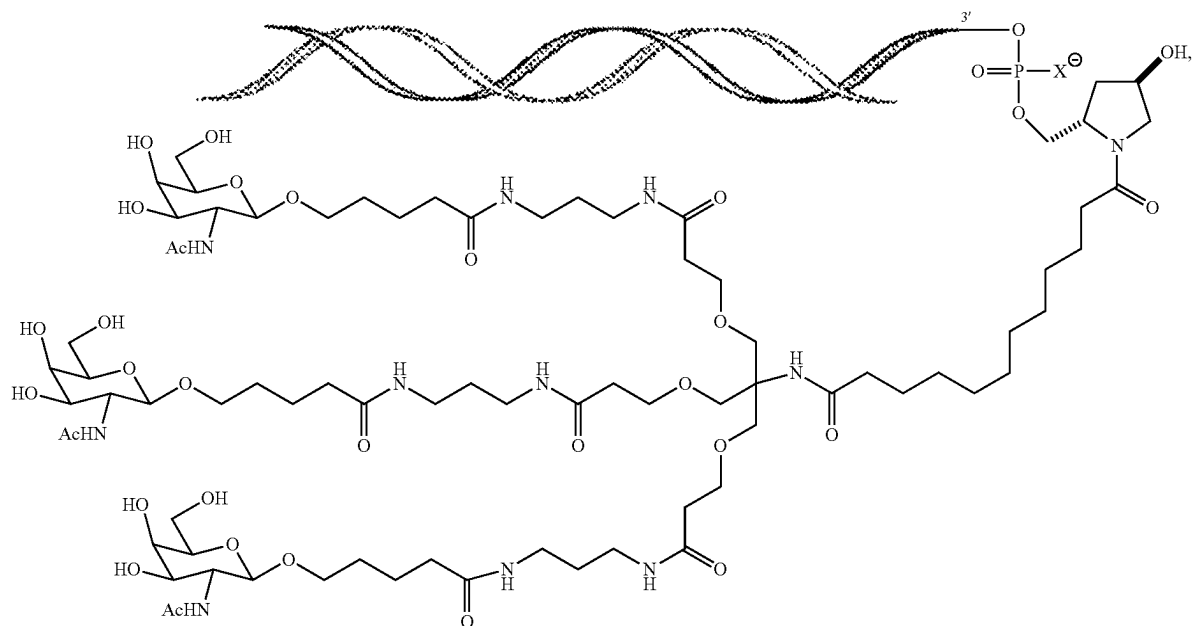
21. The dsRNA agent, or a salt thereof, of claim 20, wherein the ligand is



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22. The dsRNA agent, or a salt thereof, of claim 21, wherein the dsRNA agent, or a salt thereof, is conjugated to the ligand as shown in the following schematic



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wherein X is O or S.

23. A double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of cell death-inducing DFFA-like effector b (CIDEb) in a cell, or a salt thereof, comprising a sense strand and an antisense strand forming a double stranded region,

wherein the sense strand comprises the nucleotide sequence 5'-asgsuaucaAfUfAfuaagcucggu-3' of SEQ ID. NO: 1601 and the antisense strand comprises the nucleotide sequence 5'-asdCscgdAgdCuuaudAuUf-agauacusgsa-3' of SEQ ID. NO: 2049,

wherein

s is a phosphorothioate linkage;

a, c, g, and u are 2'-O-methyl (2'-OMe) A, C, G, and U, respectively;

dA is 2'-deoxyadenosine-3'-phosphate;

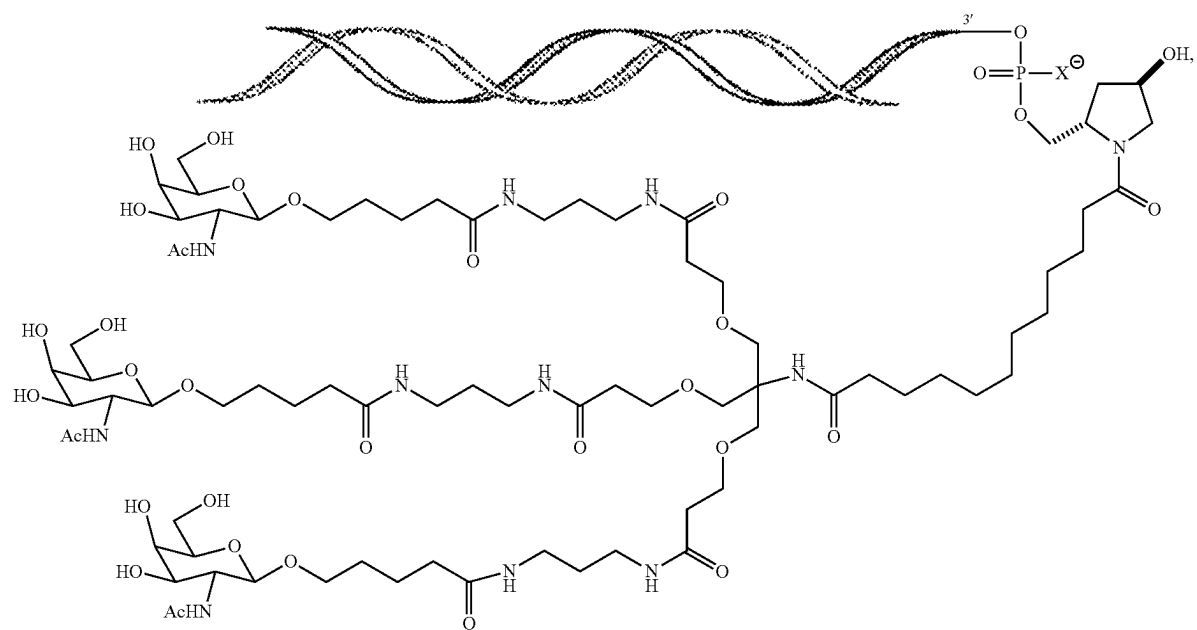
dC is 2'-deoxycytidine-3'-phosphate; and

Af and Uf are 2'-fluoro A and U, respectively, and

wherein the 3'-end of the sense strand is conjugated to a ligand as shown in the following schematic

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wherein X is O.

* * * * *